

Uniform bounds for planar polynomial limit cycles

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Abstract

For every degree, we prove that the number of isolated periodic orbits of a real planar polynomial vector field is bounded by a finite constant depending only on that degree. This establishes the uniform boundedness assertion in the second part of Hilbert’s sixteenth problem. The proof uses separation of asymptotic expansions on nested complex domains and a finite-dimensional counting argument.

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1 Introduction

Let $V = (P, Q)$ be a real polynomial vector field on $\mathbb{R}^2$. A periodic orbit is the image of a nonconstant periodic solution of

$$\dot{x} = P(x, y), \quad \dot{y} = Q(x, y).$$

It is a *limit cycle* if some open neighborhood of that image contains no other periodic orbit. We count geometric images once, without multiplicity.

Theorem 1.1 (Uniform boundedness). *For each integer $d \geq 1$, there is a finite nonnegative integer $B(d)$ such that every real planar polynomial vector field of degree at most d has at most $B(d)$ limit cycles in the whole plane.*

There is no restriction on the coefficients, the locations of the cycles, or their stability. Theorem 1.1 gives a positive resolution of the uniform boundedness assertion in the second part of Hilbert’s sixteenth problem. It does not furnish an effective formula for $B(d)$.

1.1 Historical context

Hilbert’s sixteenth problem asks, in its differential-equation part, for the maximum number and arrangement of limit cycles of planar polynomial systems [8, Problem 16, p. 465]. Two finiteness assertions must be distinguished. Individual finiteness concerns one fixed vector field; uniform boundedness asks for a bound depending only on its degree [9, pp. 301–304]. Even finiteness for every member of a family does not by itself bound the numbers occurring as its coefficients vary.

Dulac claimed a proof of individual finiteness in 1923 [4]; after a gap in that argument was identified [11], Écalle and Ilyashenko developed proofs using complex and asymptotic analysis of return maps [5, 12, 13]. Yeung identifies a coefficient-closure obstruction in a leading-term argument of Ilyashenko’s 1991 monograph [22]. This concerns that proof mechanism; it is not a counterexample to individual finiteness.

The geometric approach to uniformity studies the limiting sets of periodic orbits and bounds the number of nearby cycles in a parameter family. A foundational local result is Bautin’s sharp bound of three for cycles bifurcating from a nondegenerate weak focus or center of a quadratic vector field, under perturbations within the quadratic family [1, §3]. Roussarie develops the study of limiting periodic sets through graphics and desingularization [20]. Ilyashenko and Yakovenko prove finite cyclicity for elementary polycycles in generic finite-parameter families [10]; Kaloshin subsequently gives an explicit cyclicity bound in terms of the number of parameters [15]. The elementary hypothesis requires each singularity of the polycycle to have at least one nonzero eigenvalue. It and the genericity assumptions restrict these results. Another major partial advance concerns the infinitesimal problem: Binyamini, Novikov, and Yakovenko give explicit bounds for zeros of Abelian integrals and for cycles arising from nonsingular Hamiltonian ovals in the stated nonconservative perturbations [3]. These results illustrate both the power of local transition analysis and the extra work required at unrestricted degenerations.

There is a parallel development through definability and quasianalytic asymptotics. Kaiser, Rolin, and Speissegger construct an o-minimal expansion containing the transition maps at nonresonant hyperbolic singularities. For small analytic unfoldings that preserve the singular points and their linear parts, they obtain uniform bounds on isolated return-map fixed points near polycycles with such singularities [14]. Galal, Kaiser, and Speissegger construct Ilyashenko algebras with injective transseries asymptotics and a corresponding Hardy field [7]. They prove closure under field operations and differentiation for germs in one variable. The multivariable closure needed for a parameter argument is a separate issue. The present proof establishes separation, differentiated implicit closure, and projection bounds for its own passage class. Those properties are proved in Sections 3–5 and 11; definability of arbitrary passage maps is not an assumption.

The proof below treats coefficient variation and degeneration together. It does not use individual finiteness as a premise. This is also why cycles that move arbitrarily far from the origin and nonhyperbolic isolated cycles must be included in the last step of the argument.

1.2 Proof strategy

Choose a short transverse interval through a point of a periodic orbit, with coordinate r . Following nearby trajectories once around the orbit defines a real analytic return map $\Pi(r)$. The nearby limit cycles correspond to isolated zeros of the displacement $\Pi(r) - r$. A simple zero has $\Pi'(r) \neq 1$; such a cycle is hyperbolic. The difficulty is to bound these zeros while the coefficients vary and the passages making up the return map degenerate.

A local saddle already exhibits the scales that occur. For $\dot{x} = x$, $\dot{y} = -\lambda y$, with $\lambda > 0$, the passage from $(r, 1)$ to $(1, r^\lambda)$, $0 < r < 1$, has flight time $L = \log(1/r)$ and contraction exponent $W = \lambda L$. When $r \rightarrow 0$ and $\lambda \rightarrow 0$, the first clock can diverge while the second stays bounded or diverges at a different rate. The later passage models treat such differing scales together with their smaller correction terms.

We represent a return by equations matching the endpoints of finitely many local passages. This formulation has a predecessor in Ilyashenko and Yakovenko’s cyclic systems for elementary polycycles [10, §0.4]. Here the cuts are allowed to move along the orbit, giving continuous families of representations. Section 11 shows that, within each regular chart of a fixed matching system, representations of different hyperbolic cycles lie in different connected components of the fiber. It therefore remains to bound those components uniformly as the field coefficients and preparation parameters vary.

The counting argument reduces this uniform bound to a different assertion: each fixed finite auxiliary system has only finitely many isolated solutions when all its variables, including those later used as parameters, are allowed to vary. We call this *absolute isolated-zero finiteness*. It must hold for the differential and algebraic systems created by the counting argument, including their graph relations and multiplier equations, with the original open domains retained. The analytic and geometric parts of the proof establish these hypotheses for the matching systems.

Here is the organization of the proof.

1. Sections 2–5 prove the abstract absolute-zero criterion in Theorem 5.7. Along an alleged escaping sequence, a finite saturated flag records the ordered large scales and their exact logarithmic relations. Theorem 3.2 compares two lateral trees of asymptotic coefficients and identifies the first nonzero term. If both trees vanish, the function vanishes exactly. A differentiated implicit calculus and an induction on ordinary variables then turn this information into a local chart through each sufficiently far solution. An independent large coordinate survives in a zero chart, contradicting isolation. The differentiated implicit calculus

is stated in Theorem 4.4.

2. Sections 6–8 verify the packet and retestability hypotheses of that criterion for the analytic passage maps used later (Theorems 6.1, 7.1, and 8.1). The models include additive transfers, regular or stable transfers with one large clock, and transfers with two clocks that may be comparable or widely separated. Their arguments are treated independently before endpoint relations are imposed. In particular, exponentially small terms and the differences between lateral determinations remain part of the packet data.
3. Sections 9 and 10 give a finite geometric description for fields of bounded degree. Strong one-variable subanalytic preparation is used on the scalar fields, before passage maps are introduced. It supplies finitely many boxes and annuli (Theorem 9.7), which reduce to the preceding analytic models (Theorem 10.1). A crossing argument for periodic Jordan curves bounds the length of the resulting passage words. This finite description depends on the degree; it is fixed before any sequence is chosen for an absolute-zero test.
4. Section 11 constructs the matching equations with their exact graph relations and verifies absolute finiteness for the required closure (Theorem 11.4). The independent counting principle is Theorem 11.2. Within each regular chart, representations of different hyperbolic cycles lie in different connected components of the fiber. A suitable small signed rotation produces at least as many hyperbolic cycles as any prescribed finite collection of isolated cycles (Lemma 11.8). Spatial rescaling then gives the assertion throughout the plane.

Figure 1 records where the analytic and geometric parts meet in the counting argument.

The analytic flags and charts in an absolute-zero contradiction may depend on the chosen sequence. They are not used as a finite covering of coefficient space. The finite geometric description and the projection-counting argument supply the uniformity.

For a first reading, the theorem statements cited above and the final proof in Section 11 display the chain of implications. Definition 11.1 specifies the precise closure required at the counting interface. The preceding sections establish its analytic and geometric inputs.

1.3 Analytic ingredients

Three features of the argument deserve emphasis.

A *packet* records an actual analytic function together with two trees of successive asymptotic expansions and the domain, transition, and error estimates relating them. The two signs refer to lateral determinations on either side of the real path; higher-band raw realizations may be smooth interpolations of holomorphic determinations.

First, the separation theorem uses a *tree* of expansions on successive complex bands. It does not assume a simultaneous convergent series in all small exponentials. The hypotheses record retained arguments, permissible error costs, differentiated transition estimates, and growth on the full fringes of the domains. The proof upgrades arbitrarily high fixed exponential decay to exact vanishing by weighted Cauchy corrections and maximum principles. The domain and error conditions are indispensable: real asymptotic flatness alone gives no such conclusion.

Second, regular ordinary implicit changes must be compatible with dependent scales that can rotate rapidly under tiny complex perturbations. The construction uses high real Taylor jets and interpolation into small comparison tubes, with a fixed finite coherence order at each stage. It does not require one ordinary complex neighborhood valid for all derivative orders. The later center-lifting argument additionally keeps the power exponents controlling maps, inverses, and neighborhoods

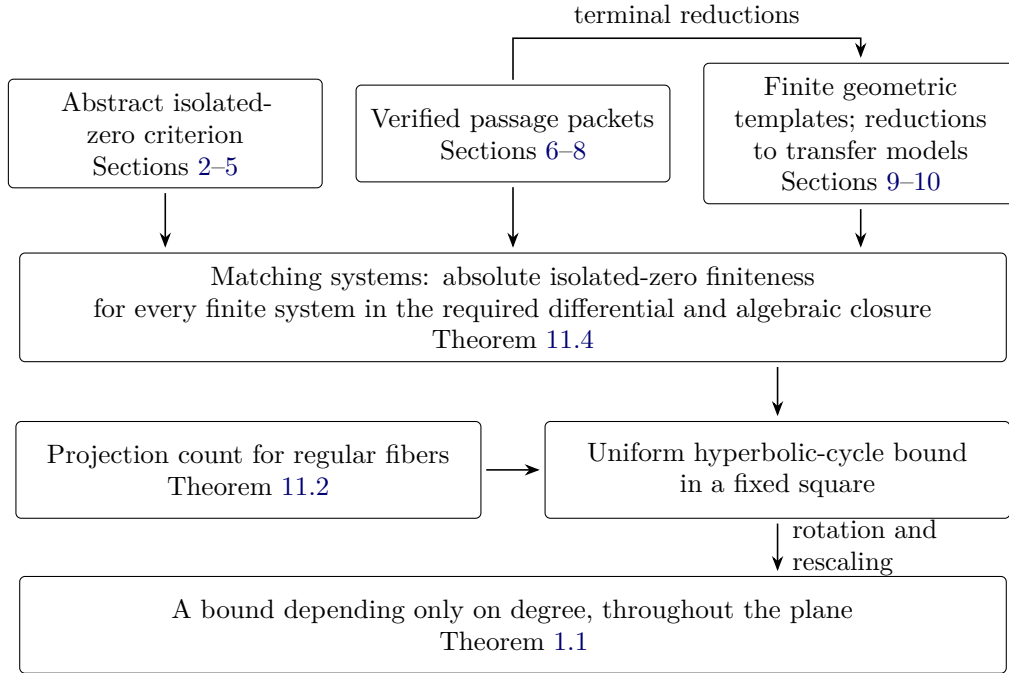


Figure 1: Main theorem interfaces. Matching combines the finite geometric description with the packet criterion, retaining the exact graph ties and open domains. The independent projection theorem converts absolute finiteness for the full required closure into uniform component bounds. The sequence-dependent flags and internal factorizations used in the absolute-zero test do not enlarge the finite geometric alphabet. The packet input to Sections 9–10 concerns the terminal reductions, not the preparation of scalar fields.

fixed while the accuracy of auxiliary records is increased. This quantitative property permits exact correction of approximate chart hits.

Third, the projection argument is formulated for an analytic class closed under the particular derivatives, graph variables, and multiplier equations used in its proof. Proper barriers and a fixed generic tilt reduce the number of parameters. The resulting absolute-to-uniform implication is independent of the particular planar passage models and can be applied to other classes for which the same hypotheses are established.

1.4 Notation and conventions

The technical constructions are local at a limiting ordinary analytic germ. An asymptotic assertion is made for a fixed finite request: a finite collection of expressions, derivative orders, Taylor accuracies, and transition estimates. Constants, sufficiently far starting points, and smaller neighborhoods may depend on that request. Whenever an exponent must be independent of a later accuracy choice, this is stated explicitly.

The notation used throughout the analytic part is summarized below; see Definitions 2.1 and 3.1. Local clock and state variables are introduced separately in each passage model.

Notation	Meaning
$Z_1 \gg \cdots \gg Z_m \gg 1$	Ordered scales of a finite flag
$f_i = \log Z_i, \quad h_i = e^{-Z_i}$	Logarithmic scales and small exponentials
S	Tuple of independent, or free, flag scales
t, t_c	Ordinary inputs and their interior limiting germ point
b_i	Bounded ordinary backgrounds in exact log relations
$\sigma(i)$	Leading later index in a dependent log relation
$s = u + iv$	Complex parameter of a selected flag path
$\Lambda_i = f'_i$	Logarithmic rate along the real path
$V_i = e^{f_i}/\Lambda_i$	Precision scale at level i
$S_\alpha^{j,\pm}$	Raw packet in band j at prefix α
$h^\nu = \exp(-\sum_i \nu_i Z_i)$	Exponential shift with lexicographic order

Lexicographic order starts at the largest scale. A positive exponent vector has positive first nonzero coordinate; later coordinates need not be nonnegative. A vanishing coefficient means a vanishing ordinary analytic germ. Complex logarithms and powers always use the specified lifted determinations. A dependent scale is evaluated at its *current* background values, including after an implicit change of ordinary variables.

All differentiations of primitive packets are in their independent arguments. Equations tying those arguments are imposed only afterward; differentiation of the resulting physical maps uses the chain rule on the tied graph. This distinction is used repeatedly in the terminal reductions and the matching equations.

2 Finite logarithmic flags and angular paths

An unbounded sequence of equation arguments can involve several widely separated scales. This section records those scales and their exact logarithmic relations in a finite flag, then constructs complex paths through the sequence anchors. Along these paths, the largest scale approaches a vertical complex direction before the smaller scales do. We construct nested contours on which their

real parts remain positive and quantitatively separated. These are the domains for the separation argument in Section 3. The final two lemmas control anchor-dependent growth and simultaneous avoidance of finitely many lattices of poles.

2.1 Sequence conventions and exact logarithmic relations

All constructions in this section concern a selected sequence of real points. Passing to a subsequence is allowed. For positive quantities, $X \gg Y$ means $X/Y \rightarrow +\infty$. A comparison involving finitely many quantities, and comparisons of a finite list at every fixed iterated exponential height, can be decided after a diagonal subsequence. None of the resulting numbers of scales or heights is asserted to be uniform over the original sequences.

The *ordinary variables* have a finite limit t_c in the interior of an analytic argument neighborhood. Bounded ratios that a formula needs are included among these arguments. Neighborhoods can be shrunk for each finite request. The endpoint inequalities defining the actual problem can nevertheless approach their boundary. An ordinary coefficient equal to zero below means the zero analytic germ at t_c , and not merely a function vanishing on the selected sequence.

Definition 2.1 (Logarithmic flag). A logarithmic flag consists of positive scales

$$Z_1 \gg Z_2 \gg \cdots \gg Z_m \gg 1, \quad f_i = \log Z_i, \quad h_i = e^{-Z_i},$$

each declared either free or dependent. The tuple of free scales is denoted by S . A dependent scale has the exact relation

$$f_i = \sum_{j>i} a_{ij} Z_j + b_i, \quad \sigma(i) = \min\{j : a_{ij} \neq 0\}, \quad a_{i,\sigma(i)} > 0. \quad (2.1)$$

The coefficients a_{ij} are fixed real numbers, and the backgrounds b_i are bounded ordinary coordinates, or their current values on an ordinary chart. The free scales are independent arguments; dependent scales are always evaluated by Equation (2.1).

The saturation condition below will ensure that bounded changes of the backgrounds preserve the ordered hierarchy. Two scales belong to the same *height block* if each is eventually bounded by a finite iterated exponential of the other. A flag is *saturated* if, in every block containing a dependent node, all dependent nodes precede all free nodes and

$$\log Z_{\text{first free}} = o(\log Z_{\text{last dependent}}). \quad (2.2)$$

An all-free block has no additional condition.

The relation defining height blocks is an equivalence relation: a composition of finitely many iterated exponentials is another finite iterate, after increasing the iterate to absorb fixed constants. Its classes are consecutive in the ordering of the scales. A dependent node and its leading child $\sigma(i)$ are in the same block, because

$$f_i = a_{i,\sigma(i)} Z_{\sigma(i)} (1 + o(1)).$$

In particular a dependency chain ends at a free node in the same block.

We use lifted logarithms and powers in complex continuations. Thus f_i continues as the expression in Equation (2.1), even if $Z_i = e^{f_i}$ winds in the plane. An exponential monomial is

$$h^\nu = \exp\left(-\sum_{i=1}^m \nu_i Z_i\right).$$

Its order is the sign of the first nonzero entry of ν , with the largest scale first. This convention does not refer to the numerical sign of the monomial. Known signs and ordinary units can be kept as separate factors. Later expansions will be trees indexed by successive prefixes of ν ; no simultaneous convergence of all these monomials is implicit in this notation.

Inserting logarithms expresses fixed real powers of large inputs as exponential monomials times bounded nonvanishing ordinary analytic factors. A later insertion of $-\log |t|$, when an ordinary coordinate t tends to zero, must also preserve the large coordinates already free at that stage. The next lemma supplies these operations.

Lemma 2.2 (Finite saturation and preservation of free coordinates). *A finite collection of large real inputs admits a flag after a fixed linear change of coordinates and the addition of bounded ordinary coordinates. One can insert the logarithm of any free scale, retaining the old scale values, by making that scale dependent and adding at most one positive free residual or one bounded ordinary residual. The free residual is $O(\log S)$ if the scale being converted is S .*

A flag admits a finite saturation by such insertions. After saturation, promoting one ordinary coordinate $t \rightarrow 0$ of known sign, by inserting $-\log |t|$, admits a finite further saturation in which every old free coordinate stays free. The number of ordinary coordinates does not increase in this promotion.

Proof. For the initial linear reduction, choose a largest absolute value among a basis of the finite span of the inputs. After a subsequence, the ratios of the other elements to it have finite limits. Subtract these limiting multiples. In the remaining span repeat the operation, changing the sign of an unbounded residual when necessary. At each step the dimension of the remaining span drops. A bounded residual is retained as an ordinary coordinate, even if its selected values vanish; every unbounded residual chosen as a new scale is strictly smaller than its predecessor. This gives a finite ordered scale basis with constant coefficients.

Apply the same elimination to $\log S$ against the existing scale basis, beginning with the largest scale. A scale much larger than $\log S$ has zero coefficient. After subtracting its limiting multiple at each comparable scale, the remainder is either bounded or, after changing sign, a new positive scale separated from every old scale. All the subtracted terms and the remainder are $O(\log S)$. Hence only one new residual is needed. Since $\log S = o(S)$, its expression uses only scales below S ; it is therefore a relation of the form Equation (2.1). Its first nonzero coefficient is positive, since $\log S \rightarrow +\infty$. The old scale values have not been changed.

Process the blocks containing dependent nodes from largest to smallest. On entering a block fix its then smallest scale T . Convert every free scale R in that block for which $\log R \neq o(\log T)$, and repeat this test only on any new residual still in the block. Every continuing chain satisfies

$$R_{q+1} \leq C_q \log R_q. \quad (2.3)$$

The initial node of a chain is bounded by $\exp^N(T)$ for a fixed N , by the definition of the block. Finitely many applications of Equation (2.3) give a residual $O(T)$ and then one $O(\log T)$. The logarithm of the latter is $o(\log T)$, so it is no longer converted. Fixed constants can be absorbed by an extra logarithmic step. A residual that has entered a lower block has already left the chain under consideration. There are finitely many initial chains and each step has at most one child; thus the procedure is finite. Residuals entering lower blocks are free until that block is processed.

Every converted node in the block has, after the comparison subsequence, $\log R \geq c \log T$. Every free node left at the end has $\log F = o(\log T)$. The original dependent nodes also have logarithms at least $\log T$. Consequently all dependent nodes precede all remaining free nodes, and Equation (2.2) holds. All-free blocks need no processing.

For a later promotion first insert $-\log |t|$ by the same linear elimination. In each old block with dependent nodes keep the old cut

$$L = \log Z_{\text{old last dependent}}.$$

Only the new residual chain is tested for conversion, with criterion $\log R \neq o(L)$. For every old free node F , $\log F = o(L)$. For each newly converted node R , the comparison subsequence gives $\log R \geq cL$; therefore

$$\frac{\log F}{\log R} \rightarrow 0, \quad \frac{R}{F} \rightarrow +\infty. \quad (2.4)$$

In particular a new residual below an old free node is never converted in this block. Thus no old free coordinate is consumed, and the new last-dependent cut still dominates the logarithms of all old free nodes. The residual-chain argument proves termination here as well. If the chain enters an old all-free block, leaving its residual free introduces no dependent node into that block. Inserting new scales cannot join two distinct old blocks: transitivity of finite-height comparability would already have put their old members in one block. The coordinate t has been consumed; at most the final step of its single chain produces a bounded residual. This proves the assertion about the number of ordinary coordinates. The inverse expression for t has the form

$$t = \text{sgn}(t) \exp\left(-\sum_i \nu_i Z_i - B\right),$$

with B bounded and with a positive first nonzero large coefficient when such a coefficient occurs. $\square$

The flag is now fixed. The next assertion concerns a different operation: we hold its free inputs fixed and vary its bounded backgrounds, always reevaluating dependent scales by their exact relations. This stability will be needed when ordinary variables are changed implicitly.

Lemma 2.3 (Stability under bounded backgrounds). *On a saturated output flag of Lemma 2.2, bounded variations of the backgrounds at fixed free coordinates preserve the ordered scale hierarchy, uniformly on each fixed bounded admissible real background set. More precisely, let*

$$H = \sum_{i=1}^m c_i Z_i + B,$$

where the c_i are fixed real numbers, their vector is nonzero, and B is uniformly bounded on that background set. If k is the first index with $c_k \neq 0$, then

$$\frac{H}{c_k Z_k} \rightarrow 1$$

uniformly under those variations; the denominator uses the current, reevaluated value of Z_k . Thus the leading index and sign are preserved. Dependent logarithms may enter such a comparison after substitution of their exact triangular formulas. No sign assertion is made when the resulting large part vanishes, or for an expression containing an uninserted free logarithm.

Proof. Process the saturated output flag's height blocks from smallest to largest. Each evaluated lower block is bounded above, uniformly on the fixed bounded real background set, by a fixed finite exponential iterate of its largest free member. Indeed it contains only finitely many triangular relations, whose coefficients are fixed and whose backgrounds are bounded. Every free member

of a higher block dominates every such fixed iterate. Thus lower-block values remain negligible compared with every free member of the block being processed.

The free members of the current block are unchanged and ordered. If the block has dependent members, let d be its last one and put $k = \sigma(d)$. The leading child lies in the same block and, by saturation, is free. Its value is therefore unchanged. All later scales have already been ordered, so the triangular relation gives

$$f_d = a_{dk} Z_k (1 + o(1))$$

uniformly on the background set. The logarithm of the first free member is $o(f_d)$ on the original sequence. It remains so uniformly, since that logarithm and Z_k are fixed. This proves the last-dependent/first-free boundary.

Now proceed upwards through the dependent members. The difference of two consecutive dependent log relations is a fixed affine combination of later scales and a bounded remainder. Its large part is nonzero, since otherwise their logarithmic gap on the original sequence would be bounded. Its first nonzero coefficient is positive on that sequence. The later scales are already ordered uniformly by the induction, so this coefficient still dominates. This proves all remaining gaps and hence the full hierarchy uniformly.

Finally divide H by $c_k Z_k$, evaluated at the current backgrounds. Every later-scale ratio tends uniformly to zero, as does the bounded remainder divided by Z_k . This proves the precise assertion about affine comparisons. A combination involving dependent logarithms first reduces to this form by exact substitution. If all large coefficients cancel, their cancellation is an identity of fixed coefficient vectors, leaving the current bounded remainder; it supplies no additional sign conclusion. $\square$

2.2 Paths through far anchors

We keep a saturated flag fixed and vary its free coordinates along a complex path, with parameter $s = u + iv$. The path must pass through the chosen anchor without depending on the ordinary backgrounds, and it must preserve scale separation on every required fixed backward interval. Its angular purpose comes from the identity

$$\Re Z_i = e^{\Re f_i} \cos(\Im f_i).$$

As $|v|$ grows, the larger scales must approach the directions $\Im f_i = \pm\pi/2$ before the smaller ones. With $\Lambda_i = f'_i$ on the real ray, the path estimates below give $\Im f_i(u + iv) \sim v\Lambda_i(u)$: the level- i direction is therefore reached at distance about $\pi/(2\Lambda_i)$ from that ray. We first construct the paths and estimate these logarithmic rates; the next subsection chooses the precise contours and their margins.

Write polylog X for a bound $C(1 + \log X)^N$, where C, N may change between occurrences and may depend on a fixed derivative order. In every use the number of derivatives and the other requests are finite. Thresholds, constants, and ordinary neighborhoods may depend on this finite request.

Fix a far anchor, write its smallest logarithm as f_m^0 , and put $u_0 = \log f_m^0$. We construct paths $s = u + iv$ through the free coordinates, defined over $u \geq u_0 - B$ for any required fixed backstep B . All tuning constants are held fixed when testing a path. The smallest node is necessarily free. Its path is

$$f_m(s) = e^s.$$

Now take $i = m - 1, \dots, 1$. Dependent nodes are evaluated from Equation (2.1). At a free node $i < m$ set $n = i + 1$ and use the following rules.

If the successor n is free, put $f_i = f_n + g_i$. A positive logarithmic gap can usually be continued by

$$g_i = x_i, \quad x_i(s) = x_i^0 e^{s-u_0}, \quad (2.5)$$

with x_i^0 equal to the anchor gap. Within a block containing dependent nodes, some free gaps must also stay small relative to the leading child of the last dependent node. If that node is d and $k = \sigma(d) > i$, cap the gap by using

$$g_i = \frac{x_i y_i}{x_i + y_i}, \quad y_i = \eta_i Z_k, \quad x_i^0 = \frac{g_i^0 y_i^0}{y_i^0 - g_i^0}. \quad (2.6)$$

Here k is free. Saturation gives $g_i^0 = o(Z_k^0)$, so one can take $\eta_i \rightarrow 0$ so slowly that $g_i^0 = o(y_i^0)$ and, for every backstep needed in the finite request,

$$\eta_i \geq Z_k(u_0 - B)^{-1/4}. \quad (2.7)$$

Then $x_i^0 \sim g_i^0 \rightarrow \infty$. The number η_i is fixed on the resulting path, though it tends to zero as anchors recede.

If the successor n is dependent, saturation puts i and n in different blocks. The new free path must dominate f_n while remaining independent of its backgrounds. For this purpose, follow leading children from n to its terminal free node k , using ℓ edges, and put

$$T = \exp^{\ell-1}(Z_k^2), \quad f_i = T^2 + x_i, \quad x_i = x_i^0 e^{s-u_0}. \quad (2.8)$$

The original distinct heights give $f_i^0 \gg (T^0)^2$, so $x_i^0 = f_i^0 - (T^0)^2 > 0$ and tends to infinity. Crucially, this rule uses only the free node k , not an ordinary input or a dependent value whose background could later change.

The real ordinary inputs can be constant, or can vary with bounded derivatives of all the fixed orders used. Independent ordinary tests are chosen real-symmetric and holomorphic, with positive-order derivatives tending to zero, and return to t_c as $u \rightarrow \infty$ on each individual path. For example, an anchor value t^0 in a smaller convex box can be tested by

$$t(s) = t_c + (t^0 - t_c)e^{-\delta(s-u_0)},$$

with $\delta > 0$ as small as needed and with slack for the fixed backstep. This also shows why smaller ordinary boxes requested later can be reached on the same ray. It is not necessary that all finite requests hold at the anchor with one common threshold.

For complex estimates, a *controlled background determination* means a local continuation with uniform Taylor control on the angular range in question. In particular it satisfies

$$b(u + iv) = b(u) + ivb'(u) + O(v^2), \quad \partial_v b(u + iv) = ib'(u) + O(|v|),$$

and the corresponding Taylor estimates through every fixed order requested at that step. The real jets are bounded. Several piecewise holomorphic determinations can have the same real jets; negligible errors arising from their smooth interpolation are retained as defects in the packet estimates below. Real boundedness alone is not used to infer these complex bounds.

Lemma 2.4 (Real and angular path estimates). *The paths just constructed pass through the given anchors and preserve the ordered scales under bounded controlled ordinary tests. Put $\Lambda_i = f'_i$ on the real ray. For each $G = f_i - f_j$, $i < j$, and for each fixed positive integer r , sufficiently far out,*

$$G \longrightarrow +\infty, \quad cG \leq G' \leq G \text{ polylog } G, \quad (2.9)$$

$$|G^{(r)}| \leq G \text{ polylog } G, \quad 0 < \frac{(f_i - f_{i+1})'}{f_i - f_{i+1}} \leq C\Lambda_{i+1}.$$

The first three bounds also hold with G replaced by f_i . In particular

$$\Lambda_1 > \dots > \Lambda_m \longrightarrow +\infty, \quad cf_i \leq \Lambda_i \leq f_i \text{polylog } f_i, \quad \frac{f_i''}{\Lambda_i^2} = o(1). \quad (2.10)$$

Fixed backsteps can reduce f_i by any prescribed fixed factor.

With holomorphic background tests, the formulas for indices $j \geq p$ have controlled holomorphic continuations on $|v| \leq C/\Lambda_p(u)$ and satisfy

$$\begin{aligned} \Re f_j(u + iv) &= f_j(u) + o(1), \\ \Im f_j(u + iv) &= (1 + o(1))v\Lambda_j(u), \\ \Im(f_j - f_l)(u + iv) &= (1 + o(1))v(\Lambda_j - \Lambda_l)(u), \quad l > j, \\ \partial_v \Im f_j(u + iv) &= (1 + o(1))\Lambda_j(u). \end{aligned} \quad (2.11)$$

These estimates also hold for controlled local determinations of the backgrounds that agree with the real path to their requisite finite Taylor orders; their antiholomorphic defects, if present, are accounted for separately. For a retained free input Z_l , every fixed path derivative is bounded by $\exp(C_\tau f_l(u))$ on its applicable angular range.

Proof. We give the induction estimates, including the cases in which a smaller scale can have a relatively large logarithmic derivative. The initial function $f_m = e^u$ has all the required estimates. If a positive function F satisfies the first three real bounds, the formula for a derivative of e^F is e^F times a polynomial in $F', \dots, F^{(r)}$. Consequently

$$|(e^F)^{(r)}| \leq e^F \text{polylog}(e^F). \quad (2.12)$$

Finite sums with a separated positive leading term obey the same upper bounds. Their first derivative has that leading positive term: the derivative ordering already known at later nodes and the exponential modulus gaps make the lower terms negligible. This handles dependent logs and differences of dependent logs. Fixed bounded derivatives of the backgrounds are lower-order terms.

An uncapped gap has $x_i^{(r)} = x_i$ and poses no difficulty. For a cap write $x = x_i$, $y = y_i$, $r_0 = x/y$. Direct differentiation gives

$$\frac{g'}{g} = \frac{1 + r_0\Lambda_k}{1 + r_0}. \quad (2.13)$$

This is positive, bounded below by a fixed positive constant, and bounded above by $1 + \Lambda_k \leq C\Lambda_n$. In addition, Equation (2.7) gives $\Lambda_k \leq \text{polylog } y$. If $x \leq y$, then $g \asymp x$, and the elementary bound

$$\frac{x}{y}(1 + \log y)^N \leq C_N(1 + \log x)^N \quad (x \leq y, \text{ } x \text{ sufficiently large})$$

absorbs each occurrence of a y derivative. To verify the bound, write $y = xe^a$ and use $e^{-a}(1 + \log x + a)^N \leq C_N(1 + \log x)^N$. If $x \geq y$, then $g \asymp y$ and the bound is immediate. Repeated differentiation of $xy/(x + y)$ gives the same alternatives: terms containing a derivative of y , after division by g , have a factor $O(x/y)$ when $x \leq y$, and otherwise have only fixed powers of logarithms of y . This proves the fixed-order bounds in g .

For the boundary rule, repeated use of the positive leading terms on the chain from n to k gives, sufficiently far out,

$$\exp^{\ell-1}(Z_k^{1/2}) \leq f_n \leq \exp^{\ell-1}(Z_k^2) = T. \quad (2.14)$$

Every fixed logarithmic derivative of T is bounded by a power of $1 + \log T$. Its first logarithmic derivative is $o(f_n)$: for $\ell = 1$ this follows from $T'/T = 2\Lambda_k = o(Z_k^{1/2})$; for larger ℓ , the numerator

is a product of lower iterates, whereas the left side of Equation (2.14) has one more exponential height. Thus $T^2 + x_i$ and its first derivative dominate f_n and f'_n . The gap $T^2 + x_i - f_n$ has the first three real estimates, and

$$\frac{(T^2 + x_i - f_n)'}{T^2 + x_i - f_n} \leq C \left(1 + \frac{T'}{T}\right) \leq C\Lambda_n.$$

The same computations give its fixed-order bounds.

There remains the dependent-to-free boundary in a saturated block. Let d be its last dependent node and $k = \sigma(d)$. The only free gaps above k are the caps in Equation (2.6), so their finite sum is $o(Z_k)$ at the anchors, with first derivative $o(Z_k\Lambda_k)$. For example, Equation (2.13) and $g \leq \eta_i Z_k$ bound the derivative by $C\eta_i Z_k(1 + \Lambda_k)$. On each individual forward ray, $x_i/Z_k \rightarrow 0$, because x_i grows as e^u and $f_k \geq e^u$ eventually. Hence the same little-oh bounds hold towards infinity on that ray, with its tuning constants fixed. The remaining free contribution is $f_k = o(Z_k)$, whose first derivative is $o(Z_k\Lambda_k)$. Therefore

$$f_{d+1} = o(Z_k), \quad f'_{d+1} = o(Z_k\Lambda_k),$$

whereas $f_d \sim a_{dk}Z_k$ and $f'_d \sim a_{dk}Z_k\Lambda_k$. This proves the real estimates across this boundary, including

$$\frac{(f_d - f_{d+1})'}{f_d - f_{d+1}} \asymp \Lambda_k \leq \Lambda_{d+1}.$$

Higher derivatives of the caps are bounded by Z_k polylog Z_k , which suffices here. Between two dependent nodes the leading scale of their difference has index at least that of the leading scale in the earlier relation. Its logarithmic derivative is at most $C\Lambda_{i+1}$: if its index is $i + 1$ this is immediate, and otherwise use the already proved ordering of later derivatives. This completes the adjacent-gap induction. A nonadjacent gap is a finite sum of adjacent positive gaps, proving its first three estimates. The lower constant can, for example, be taken to be $c = 1/2$ after going sufficiently far for any chosen fixed backstep: uncapped terms and caps have logarithmic derivative at least one, and dependent leading terms and the T terms have logarithmic derivatives tending to infinity. The discarded terms are relatively negligible. Thus a backstep $B \geq 2 \log R$ reduces every f_i by at least a prescribed factor R , once that fixed backstep is included in the construction.

For completeness, the continuation uses the explicit formulas, not an arbitrary holomorphic function having prescribed real bounds. In the downward induction, a dependent f_j is a finite affine sum of Z_k with $k > j$. The already controlled real parts of f_k make $|Z_k(u + iv)| \asymp Z_k(u)$, so differentiating this sum retains the bounds of Equation (2.12). The same argument applies directly to a dependent gap, using its first surviving coefficient. In a boundary term T , every proper exponent in its finite exponential tower and its fixed derivatives are bounded by a power of $1 + \log T$. Since $|v| \leq C/\Lambda_i \leq C'/T^2$, their variations tend to zero. Induction through that finite tower therefore preserves its moduli and the derivative estimates. An uncapped additive term has its explicit exponential continuation. In a cap, if $|x|$ and $|y|$ are separated, $x + y$ stays away from zero. If they are comparable, then $y = O(f_i)$ at the controlling node, and $|\arg y| = O(\Lambda_k/\Lambda_i) = o(1)$; indeed in this case $\Lambda_i \geq cg \asymp y$, whereas $\Lambda_k \leq \text{polylog } y$. The denominator therefore again stays away from zero. These observations prove by the same downward induction that the real derivative bounds used above remain valid, with fixed enlarged constants, along the short vertical segment.

Taylor's formula along that segment now gives

$$|\Re f_j(u + iv) - f_j(u)| \leq Cv^2 f_j(u) \text{polylog } f_j(u) = o(1),$$

since $\Lambda_p \geq cf_p \geq cf_j$. For a gap $G = f_j - f_l$ it gives an error bounded by $Cv^2 G \text{polylog } G$. Dividing by $|v|G' \geq c|v|G$ gives $O(\text{polylog } G/\Lambda_p) = o(1)$. Applying this estimate directly to the gap, rather

than subtracting two relative estimates for individual logs, proves the third line of Equation (2.11) even for nearly equal speeds. The individual imaginary and derivative assertions follow in the same way. Controlled backgrounds have the same Taylor bounds and therefore give the stated version for local determinations. Finally, the derivative polynomial for $Z_l = e^{f_l}$ is bounded by $e^{f_l} \text{polylog}(e^{f_l}) \leq e^{C_r f_l}$. $\square$

Remark 2.5. Different original height blocks need not stay different along an individual forward ray. The conclusions that persist are the ordered scale ratios, the exact logarithmic relations, and Equations (2.9)–(2.11). These are the conclusions used below. The paths of free inputs were defined without ordinary backgrounds, which permits independent ordinary tests and later regular implicit substitutions. All comparisons defining a fixed finite real regime have slack sufficiently far along the selected sequence. Continuity of the finite triangular formulas therefore permits sufficiently small absolute perturbations of the free inputs while retaining that regime and the limiting ordinary data. Their permitted sizes may depend on the anchor; no fixed neighborhood is intended.

2.3 Nested contours and background variation

At level p we use two boundaries before Z_p reaches a vertical direction. A good contour leaves $\Re Z_p$ of order $V_p = e^{f_p}/\Lambda_p$, large enough to dominate the later real parts. The outer side is closer to vertical; its defining barrier retains both that domination and an explicit polynomial angular margin. These margins also determine how accurately two background determinations must agree.

Lemma 2.6 (Good contours and buffered sides). *For a sufficiently small fixed $\varepsilon > 0$, the upper and lower p -good contours are defined by*

$$|\Im f_p| \sim \frac{\pi}{2}, \quad \cos(\Im f_p) = \frac{\varepsilon}{\Lambda_p(u)}. \quad (2.15)$$

They have collars of fixed small multiples of their cosine margin and are nested in increasing order of p . Put

$$V_p(u) = \frac{e^{f_p(u)}}{\Lambda_p(u)}.$$

On a p -good contour, for $j > p$ and $G = f_p - f_j$,

$$\Re Z_p \sim \varepsilon V_p, \quad \Re Z_j \asymp \frac{e^{f_j(u)}(\varepsilon + G'(u))}{\Lambda_p(u)} = o(V_p). \quad (2.16)$$

For large fixed A and fixed $K \geq 2$, an outer side can be placed inside the signed vertical direction by the barrier

$$L_p = Z_p - A \sum_{j>p} Z_j - \frac{Z_p}{f_p^K}, \quad \Re L_p = 0. \quad (2.17)$$

We use its inward component, where $\Re L_p > 0$. Throughout this component, with harmless fixed adjustments near its boundary,

$$\Re Z_p \geq A \sum_{j>p} \Re Z_j + c \frac{e^{f_p(u)}}{f_p(u)^{K+1}}, \quad \Re Z_p \gg f_p(u)^N \quad \text{for every fixed } N. \quad (2.18)$$

The side and good contours can be chosen with extra room before any finite subsequent narrowing. The same conclusions hold if the quantities in the barrier are replaced by holomorphic proxies with absolute $o(1)$ errors.

Uniformly between the p -good contour and its outer side, for $p < n < l$,

$$\Re Z_l = o(\Re Z_n). \quad (2.19)$$

Every retained index $j \geq p$ has a polynomial buffer of its own: for some fixed exponent C ,

$$\Re Z_j \geq \frac{e^{f_j(u)}}{(1 + f_j(u))^C}. \quad (2.20)$$

In particular $\Re Z_j / f_l(u) \rightarrow \infty$ for $l \geq j$, so fixed derivative losses $\exp(C' f_l)$ can be absorbed by any fixed positive exponential decay in $\Re Z_j$.

Proof. By Equation (2.11), $\Im f_p$, the lifted phase of Z_p , is strictly increasing with v near the positive vertical position of Z_p , which is at $v = (1 + o(1))\pi/(2\Lambda_p)$. This gives the upper contour, and real symmetry gives the lower one. If $j > p$, the relative gap formula, with $\eta_p = \cos(\Im f_p) \geq 0$, gives

$$\Re Z_j \asymp e^{f_j(u)} \left(\eta_p + \frac{\Lambda_p - \Lambda_j}{\Lambda_p} \right) \quad (2.21)$$

near the vertical direction. This also holds with the evident constant-factor interpretation when the ratio of speeds is bounded away from one. On a good contour it proves Equation (2.16), since $e^{-G}(\varepsilon + G') \rightarrow 0$ by the polylogarithmic upper bound for G' . Moreover $G' \rightarrow \infty$, so the later phase is farther from vertical than any fixed-width p collar. This proves nesting even when the two speeds have ratio tending to one. Farther inside, all relevant cosines are bounded below, and the same ordering is simpler.

To compute the buffer, write $f_p = a + i\theta$, with $a = f_p(u) + o(1)$ and $|\theta| \leq \pi/2$. Uniformly near the upper vertical direction,

$$\Re \frac{Z_p}{f_p^K} = e^a a^{-K} \left(\cos \theta + \frac{K\theta}{a} \sin \theta + O(a^{-2}) \right). \quad (2.22)$$

The second term is positive and of size $1/a$ near vertical. The lower direction gives the same sign, since $\theta \sin \theta > 0$. Away from vertical, the first term gives an even larger lower bound. Thus the real part in Equation (2.22) is at least ce^a/a^{K+1} throughout the required wedge. At the good contour this term and the later real phases are $o(\Re Z_p)$, so $\Re L_p > 0$ there. At the exact vertical direction $\Re Z_p = 0$ and both subtracted real terms are positive. The inward component therefore meets a zero side before vertical, and positivity of its barrier gives Equation (2.18). More explicitly, division of the side equation by $e^{f_p(u)}$, followed by Equation (2.21), gives, with $F = f_p(u)$,

$$\eta_{p,\text{side}} \asymp F^{-K-1} + A \sum_{j>p} e^{-(f_p-f_j)} \frac{\Lambda_p - \Lambda_j}{\Lambda_p}.$$

The terms proportional to $\eta_{p,\text{side}}$ on the right have coefficient tending to zero and were absorbed on the left. Hence $\Lambda_p \eta_{p,\text{side}} \rightarrow 0$, by the polylogarithmic gap estimates. The side therefore lies beyond every fixed good contour and still before the vertical direction. This argument uses no global injectivity of Z_p or of a proxy. An initial choice with smaller A and larger K supplies extra room; increasing A and slightly decreasing K afterwards moves the working side inward. Absolute $o(1)$ corrections are negligible compared with its diverging buffer. The identical sign argument constructs the corrected side within that extra room.

For the relative statement let $H = f_n - f_l$. From Equation (2.21),

$$\frac{\Re Z_l}{\Re Z_n} \leq C e^{-H} \left(1 + \frac{\Lambda_n - \Lambda_l}{\Lambda_p \eta_p + \Lambda_p - \Lambda_n} \right).$$

The denominator is at least $(f_p - f_n)' \rightarrow \infty$, whereas $\Lambda_n - \Lambda_l \leq H \text{ polylog } H$. The right side tends to zero uniformly, proving Equation (2.19). This is why one fixed increase of A can pay any prescribed finite collection of later phase losses at the current side.

Finally consider $j > p$. If $\Lambda_j \leq \Lambda_p/2$, Equation (2.21) gives a fixed positive cosine for Z_j . Otherwise $\Lambda_p \leq 2\Lambda_j$, and

$$f_p \leq C\Lambda_p \leq C f_j \text{ polylog } f_j.$$

The polynomial angular margin supplied by Equation (2.18) for p is therefore also an inverse power of f_j . Equation (2.21) again proves Equation (2.20). The case $j = p$ is already Equation (2.18). $\square$

We next compare the domains obtained from different determinations of the same real backgrounds. At a level- p contour, first-jet agreement preserves the separation of later phases from the vertical direction. Locating the perturbed determination's own thin side requires higher, but still finite, agreement.

Lemma 2.7 (Sensitivity to background jets). *Let $q \geq 2$ be a fixed integer. On $|v| \leq C/\Lambda_p(u)$, two controlled background determinations differing by $O(|v|^q)$, and agreeing on the real path, change any f_l , $l \geq p$, by*

$$|\Delta f_l| \leq C\Lambda_l(u)|v|^q. \quad (2.23)$$

The estimate is uniform along the short segment between the two determinations when that segment is taken with the same background slack. In particular agreement through the real first jet gives $O(\Lambda_l v^2)$. On the angular scale $|v| = O(1/\Lambda_p)$ this error is smaller than the phase gain $(\Lambda_p - \Lambda_l)/\Lambda_p$ for $l > p$. Agreement through a sufficiently high fixed jet preserves any prescribed polynomial angular buffer and good-contour collar.

Proof. A free log has zero background sensitivity. For a dependent node, differentiate Equation (2.1). By downward induction and the leading positive derivative estimate,

$$\sum_{k \geq \sigma(l)} Z_k(u)(1 + \Lambda_k(u)) \leq C\Lambda_l(u).$$

Exponential moduli remain comparable on the stated angular scale. The derivative of f_l with respect to each bounded background is therefore at most $C\Lambda_l(u)$. Integrating this derivative between the two determinations proves Equation (2.23); it also proves inductively that the intermediate determinations remain in the allowed range. For $q = 2$ the error is at most C/Λ_p , while $(\Lambda_p - \Lambda_l)/\Lambda_p \gg 1/\Lambda_p$ by Equation (2.9). For an own-contour comparison, $\Lambda_p|v|^q = O(\Lambda_p^{1-q})$. The two-sided polynomial bounds for Λ_p in f_p imply that choosing a fixed q large enough makes this smaller than each of the finitely many specified polynomial margins. No infinite-order jet matching is being asserted. $\square$

When backgrounds have slightly different band determinations, the first-jet comparison in Lemma 2.7 suffices for later phase comparisons: its $O(1/\Lambda_p)$ angular error is swallowed by the diverging gap derivative. Locating a determination's own thin side or its own good collar by a common reference uses the higher finite jet comparison instead. This distinction will be retained in ordinary implicit substitutions.

2.4 Exceptional growth and safe columns

An uncapped summand can make $\log(2 + \Lambda_p)$ too large to absorb in $\Re Z_{p+1}$. The constant D_p below records the remaining anchor dependence. The second lemma selects vertical columns that avoid finitely many real pole lattices; when D_p is large, it also preserves this avoidance throughout the collar windows needed in separation. All these assertions concern a fixed finite request.

Lemma 2.8 (The exceptional anchor constant). *For $p < m$ set $D_p = \log(2 + x_p^0)$ if f_p contains an uncapped slow additive term in Equation (2.5) or Equation (2.8), and set $D_p = 1$ otherwise. Set $D_0 = 1$. On the buffered level- n wedge, $n = p + 1$,*

$$\log(2 + \Lambda_p) \leq CD_p + o(\Re Z_n). \quad (2.24)$$

If D_p is large, then throughout the fixed-backstep range

$$\Lambda_p(u) \geq c_B e^{D_p} e^{u-u_0}. \quad (2.25)$$

Proof. If p is dependent, $f_p = O(Z_n)$ and $\Lambda_p \leq Z_n \text{polylog } Z_n$ by its triangular relation. Thus $\log(2 + \Lambda_p) = O(f_n + \log f_n)$, which is $o(\Re Z_n)$ by the polynomial buffer. A cap has size at most a later scale and its logarithmic derivative is bounded by a polylogarithm of that scale, giving the same conclusion. In the boundary construction, the height offset gives

$$\log T = o\left(\frac{Z_n}{f_n^N}\right) \quad \text{for every fixed } N;$$

this follows directly from Equation (2.14), since one additional exponential dominates all its fixed powers. It bounds the T contribution to $\log(2 + \Lambda_p)$. The only remaining contribution is

$$\log(2 + x_p(u)) \leq D_p + O(1 + |u - u_0|).$$

The second term is $o(\Re Z_n)$ on the forward ray and on each fixed backstep, because $f_n \geq f_m = e^u$ and the buffer grows exponentially in f_n . This proves Equation (2.24). An uncapped term contributes exactly $x_p^0 e^{u-u_0}$ to the positive derivative of f_p ; its other terms have positive derivatives. For large D_p , $x_p^0 \asymp e^{D_p}$, proving Equation (2.25). $\square$

The following elementary consequence is recorded for use when raw coefficients have poles at finitely many real lattices. It asserts safety for each finite request, not for an infinite set of labels.

Lemma 2.9 (Common safe columns). *Let a fixed finite list of positive real sweeps $x_a(u)$ and real lattices $c_a + h_a \mathbb{Z}$, $h_a > 0$, satisfy, uniformly far out,*

$$\begin{aligned} x_a \rightarrow \infty, \quad x'_a > 0, \quad \inf_{u \geq U} x'_a(u) \rightarrow \infty \quad (U \rightarrow \infty), \\ \left| \frac{x''_a}{(x'_a)^2} \right| \leq \frac{C_a}{x_a}. \end{aligned} \quad (2.26)$$

For a sufficiently small fixed $\delta > 0$, every fixed-length real interval far out contains a common column u_ with*

$$\text{dist}(x_a(u_*), c_a + h_a \mathbb{Z}) \geq 2\delta \quad \text{for all } a.$$

If their complex continuations obey

$$|\Im x_a(u_* + iv)| < 1 \implies |\Re x_a(u_* + iv) - x_a(u_*)| = o(1),$$

these columns stay a fixed positive distance from all the listed real poles on their whole applicable angular ranges.

Suppose additionally that on a band $j = p + 1$ the relevant sweeps satisfy $x'_a = O(Z_j \Lambda_j)$ and that at the chosen column

$$D_p > \frac{Z_j(u_*)}{f_j(u_*)^{K+3}}. \quad (2.27)$$

Then safety persists throughout every required fixed collar window

$$|f_p(u) - f_p(u_*)| \leq C_1 \log(2 + D_p)$$

in the fixed-backstep range.

Proof. For one sweep use $x = x_a$ as a coordinate and let $w(x) = 1/x'_a(u(x))$. Then

$$\frac{d}{dx} \log w(x) = -\frac{x''_a}{(x'_a)^2} = O(1/x).$$

On any fixed lattice period the ratio of the largest and smallest values of w tends to one. The time spent within distance 2δ of its lattice is consequently at most $C'_a \delta / h_a$ times the period's total time. Sum over the complete periods of a fixed u -interval. At most two incomplete end periods remain; their total time tends to zero by the lower bound on x'_a . Thus the bad proportion for this sweep is at most $C'_a \delta / h_a + o(1)$. A union bound over the finite list makes the common bad proportion less than one, by first choosing δ sufficiently small. This does not require independent or incommensurable sweeps. If the complex imaginary part has absolute value at least one, it already separates the point from the real lattice. Otherwise the assumed real shift is $o(1)$, which preserves the original detuning with slack.

Under Equation (2.27),

$$\log(Z_j \Lambda_j) = O(f_j + \log f_j) = o(D_p).$$

Equation (2.25) makes the stated f_p window have real width $O(e^{-D_p} \log(2 + D_p))$. On this window the real derivative bounds ensure the same local estimates for $Z_j \Lambda_j$; for example its logarithmic derivative has size at most $\text{polylog } Z_j = \exp(o(D_p))$ and therefore changes its logarithm by $o(1)$. Each sweep changes by at most

$$e^{-D_p + o(D_p)} \log(2 + D_p) = o(1).$$

The detuning again persists with slack. This proves the collar assertion. $\square$

3 Separation of packets

We expand an analytic function successively along a tree: each coefficient at one level is expanded again at the next. No convergence of a series in all the quantities $h_i = \exp(-Z_i)$ is asserted or used. The distinction is particularly important when a coefficient has zero formal data: decay to every fixed order at its own level must first be strengthened before that coefficient can be passed at the preceding level.

Uniqueness of an analytic function from its complete asymptotic data belongs to the quasianalytic approach developed in the limit-cycle literature [5, 13]. Transserial Ilyashenko algebras provide a related injectivity theorem for their own expansion classes [7]. Here the hypotheses are the precise band, transition, and source estimates below, and the proof establishes the needed implication directly for those data.

3.1 Data and quantifiers

Fix a saturated flag and the paths of Lemma 2.4. We use $s = u + iv$, $f_i = \log Z_i$, $\Lambda_i = f'_i$ on the real path, and

$$V_i(u) = \frac{e^{f_i(u)}}{\Lambda_i(u)}.$$

All logarithms have their lifted determinations. The symbol b_j below denotes the entire vector of bounded backgrounds in band j ; its components enter the triangular flag equations. Accordingly, $f_{k,j}$ and $Z_{k,j}$ denote the k th logarithm and scale evaluated with that vector. The subscripts on b_j do not denote the index of an individual triangular equation.

A *good contour of level i* means either of the contours of Lemma 2.6 with

$$\cos(\Im f_i) = \frac{\epsilon_i}{\Lambda_i}, \quad V_i = \frac{e^{f_i}}{\Lambda_i}, \quad (3.1)$$

where $\epsilon_i > 0$ is fixed and can be chosen arbitrarily small. Contours always have small collars. A *side of level i* is the boundary of the positive-real-part region of a holomorphic proxy for

$$L_i = Z_i - A \sum_{k>i} Z_k - \frac{Z_i}{f_i^K}. \quad (3.2)$$

Here A is sufficiently large for the finite calculation under consideration, and K lies in a fixed interval with room to decrease it finitely, for example $2 \leq K \leq 4$. Proxy construction is proved below. Before constructing the proxy, the domain has extra angular room. A later increase of A and decrease of K puts the final side strictly within that room.

The exceptional constants D_i are those of Lemma 2.8: $D_i = \log(2 + x_0)$ when f_i contains an uncapped slow summand with coefficient x_0 , and $D_i = 1$ otherwise; put $D_0 = 1$. These constants remain fixed on each chosen path. They need not remain bounded as its anchor tends to infinity.

Definition 3.1 (Packet data). Let t_c be a real ordinary germ point. A packet consists of the following data and estimates, separately for signs $+$ and $-$.

Tree and actual function. For every prefix $\alpha = (a_1, \dots, a_{j-1})$ there is a raw function $S_\alpha^{j,\pm}$ in band j , and a set $E_\alpha^{j,\pm} \subset \mathbb{R}$ of child exponents. Each such set is finite below every finite ceiling. At level $m + 1$ the functions $c_\alpha^\pm = S_\alpha^{m+1,\pm}$ are analytic ordinary germs at t_c . The two functions at level 1 are restrictions of one actual holomorphic function across the real ray. Absent exponents can be inserted with identically zero packets. Thus the supports are *iterated left-finite*, including at prefixes whose earlier coordinates need not be positive.

Independent tests. The estimates are required along the independent free paths of Section 2, with independent ordinary tests $t(s)$. These tests are holomorphic, real on the ray, bounded in their ordinary neighborhoods, and have positive-order derivatives tending to zero through every fixed requested order. They can approach any specified ordinary value on a fixed window about a varying anchor, and return to t_c , with derivatives tending to zero, at infinity on every individual path. Backgrounds have the controlled real Taylor comparisons of Lemma 2.7. Whenever a determination's own thin side must be located, the comparisons are made to sufficiently high fixed Taylor order.

Finite requests and uniformity. The following requirements apply to every finite choice of labels, normalizations, budgets, and backsteps. Ordinary neighborhoods may be shrunk and domains narrowed a finite number of times. Constants and sufficiently far starts may depend on this finite request. For a fixed such request they are uniform as anchors recede, on the full forward domains of controlled test families with common ordinary bounds and slack, common bounds for the required finite jets and Taylor errors, and common moduli in the required small-jet and flag comparisons. In

particular the finite source counts, prefactors, positive precision constants, and eventual absorption of the stated $o(U)$ losses are uniform for that family. The allowed exceptional constants D_i are retained explicitly in the growth bounds. This is uniformity for one fixed finite request; it is not uniformity over all truncation orders. Previously chosen inner functions agree on overlaps with the ones used in later requests. All further fixed-budget estimates hold eventually at infinity on the *same* path, even if their ordinary neighborhoods are smaller and their starts are farther out. The return time to a smaller ordinary neighborhood may depend on the test. There is no uniform start over all budgets or all such return times.

1. *Bands and backgrounds.* Band j extends from any sufficiently vertical preceding good contour, with a collar, to its own side (3.2). For $j = 1$ its inner boundary is the real ray; its data are genuinely holomorphic across that ray and have no antiholomorphic sources. Each band has a controlled background b_j . At a selected level- j good collar,

$$b_j - b_{j+1} = O(e^{-cV_j}). \quad (3.3)$$

For leading estimates through the entire outer fringe we additionally require, after narrowing,

$$b_j - b_{j+1} = O(e^{-a\Re Z_{j,j}}), \quad a > 0. \quad (3.4)$$

Both estimates compare determinations at the same point.

2. *Transitions.* On the outer fringe of band j , from sufficiently vertical good collars to its side, for every finite real M there is a finite cutoff $M' > M$ such that

$$S_\alpha^{j,\pm} = \sum_{\substack{a \in E_\alpha^{j,\pm} \\ a < M'}} e^{-aZ_{j,j}} S_{\alpha,a}^{j+1,\pm} + O(e^{-M\Re Z_{j,j}}). \quad (3.5)$$

The fringe satisfies $\cos(\Im f_{j,j}) \leq \epsilon/\Lambda_j$ for sufficiently small fixed ϵ , and A in (3.2) can be enlarged for this budget. New contour choices at this or later transitions do not alter the already selected inner joins.

3. *Growth and safe columns.* Throughout band j ,

$$\log^+ |S_\alpha^{j,\pm}| \leq C\Re Z_{j,j} + CD_{j-1}. \quad (3.6)$$

The D_{j-1} term is absent near the band's own side and on safe vertical columns. For every finite set of raw labels and bands, there are simultaneous safe columns in every fixed-length real interval sufficiently far along the path. If

$$D_{j-1} > \frac{Z_j(u_*)}{f_j(u_*)^{K+3}},$$

the same improvement holds on the entering level- $(j-1)$ good collars for

$$|f_{j-1}(u) - f_{j-1}(u_*)| \leq C_1 \log(2 + D_{j-1}),$$

for every fixed C_1 needed in the calculation. The constants are uniform on these safe sets. For a controlled family the columns themselves may depend on its member; simultaneity concerns the finite labels and bands within that member. The search-interval lengths and all bounds are uniform in the family.

4. *Antiholomorphic sources.* In bands $j > 1$ the data may be C^1 interpolations of local holomorphic determinations. Their $\bar{\partial}_s$ derivatives are bounded by finite sums of e^{-cU} , with sources supported where those determinations or cutoffs change. A permitted cutoff cost in band j , with r a dependent node, is

$$U = e^{\Re f_{r,j}} \quad \text{or} \quad U = e^{\Re f_{r,j}} \cos(\Im f_{k,j}), \quad r < j, \quad \sigma(r) \geq j, \quad k \geq j, \quad (3.7)$$

or any cost bounded below by a fixed positive multiple of one of these. Smoothing a level- i join has cost $U = V_i$. At the cutoff sources in (3.7), the logarithms of all loss factors in the fixed raw packets, backgrounds, interpolations, and normalizations used before the next join are $o(U)$. Cutoff derivatives obey this same rule. At a current level- i good join, a fixed normalization can instead have logarithmic loss $C\Re Z_i \leq C'\epsilon_i V_i$, which need not be $o(V_i)$. Include these finitely many losses when choosing that join: take a common sufficiently small fixed ϵ_i so that $C'\epsilon_i$ uses at most half the available positive join precision, increasing the finite transition budget when its remainder is used. The resulting join error is recorded as e^{-cV_i} with the reduced $c > 0$. Remaining later-rate, interpolation, and fixed derivative losses have logarithmic size $o(V_i)$. All these choices precede variation of the weight parameter below and are common for one controlled finite request. Already selected earlier joins remain fixed when later requests add only later-rate normalizations there. Local versions differ only by the stated errors.

The derivative strengthening used in Section 4 will impose these requirements also on each fixed finite collection of independent derivatives. No such strengthening is needed for the separation theorem itself.

Pair the two supports by inserting zero packets and order their terminal exponent vectors lexicographically, beginning with index 1. Iterated left-finiteness implies that every nonempty set of nonzero paired leaves has a first element: choose successively the least coordinate admitting a nonzero descendant. Zero formal data always means that each terminal coefficient is zero as an analytic germ, rather than just at t_c .

Theorem 3.2 (Separation). *Suppose the data of Definition 3.1 satisfy its requirements on a saturated flag. If all paired terminal coefficients vanish, the actual function is identically zero on the real flag locus sufficiently far in the regime, locally in ordinary inputs. Otherwise, let $\lambda = (\lambda_1, \dots, \lambda_m)$ be the first nonzero paired exponent. Its coefficients coincide as ordinary analytic germs, say $c_\lambda^+ = c_\lambda^- = c_\lambda$, and*

$$\exp\left(\sum_{i=1}^m \lambda_i Z_i\right) S^1 \longrightarrow c_\lambda(t) \quad (3.8)$$

locally uniformly on the real locus.

The same assertion holds for each selected leading prefix, with all earlier whole subtrees zero. In band j , use its raw packet and normalize by the remaining selected exponents, evaluating all these rates with b_j . The coupled normalized determinations have the same leading limit inside their selected contours and through the used bands. With (3.4), their leading estimates extend through the owning outer fringe for any fixed required normalization. An entire zero subtree in band $p+1$, passed at such a prefix, has size

$$O(e^{-cV_p}) \quad (3.9)$$

on more vertical level- p collars and on the subsequent portion of that fringe, where its outer determination is in use. All constants and contour choices concern fixed finite labels and requests; none is asserted uniform over the whole packet tree.

To select a leading child at level p , we multiply its transition by e^{aZ_p} , where a is the selected exponent. A child with an earlier exponent $b < a$ then carries the growing factor $e^{(a-b)Z_p}$. If that child's entire formal subtree is zero, decay to every fixed order in its own, smaller scale is not yet an estimate that absorbs this factor on the level- p collar. The essential analytic step is to prove the stronger bound $O(e^{-cV_p})$ there. Since $\Re Z_p \asymp \epsilon_p V_p$ on that collar, a sufficiently small fixed ϵ_p then absorbs all the finitely many earlier multipliers needed for this transition.

3.2 Coupling, holomorphic correction, and growth

We begin with the fixed coupled prefix used in this argument. Once the exponents $\lambda_1, \dots, \lambda_{p-1}$ have been isolated, its function in band $j \leq p$ is

$$G = S_{\lambda_{<j}}^j \exp \left(\sum_{j \leq l < p} \lambda_l Z_{l,j} \right). \quad (3.10)$$

The signs are understood on the two sides of the real ray, where the two determinations share the same actual central function. On a selected level- j join, the adjacent formulas differ by $O(e^{-cV_j})$. We interpolate across its collar. The logarithmic cost of its derivatives is polynomial-logarithmic in the controlling rates, and hence $o(V_j)$. Its $\bar{\partial}$ source therefore retains the same precision.

Each inner band $j < p$ stops at this selected good join. Only the outer band p extends to its own side. Consequently an inner source satisfies

$$\cos(\Im f_{j,j}) \geq c/\Lambda_j. \quad (3.11)$$

This lower bound will control a secondary cosine in (3.7) when its index is smaller than p . Multiplication by any further fixed later normalization preserves an old join error, with a smaller positive c , because the later real phases are $o(V_j)$ on that collar.

We establish the zero-subtree bound by induction from the terminal level toward the root. Fix the functions and joins of a coupled prefix through level n . For an entirely zero subtree we first prove, for each fixed N , the real estimate

$$|G(u)| \leq C_N e^{-NZ_n(u)} \quad (u \geq u_N).$$

The induction at deeper levels supplies this estimate by passing only finitely many children for the chosen N . Its constants and starting point may depend on N ; the already selected inner functions and joins remain fixed. The weighted maximum estimate then removes those constants and gives the suppression at the preceding transition. At level 1 there are no sources, and the same estimate gives exact vanishing. Once all zero subtrees can be passed, a finite sequence of transitions isolates the first nonzero leaf. A comparison of its two lateral limits identifies its ordinary coefficient. The full induction is assembled at the end of the proof.

Figure 2 shows this domain and its join sources. The next two lemmas correct its later rates to holomorphic functions and propagate the raw growth bound across the replaced inner bands.

Lemma 3.3 (Holomorphic proxies and Cauchy correction). *On a coupled level- p domain, the rates of indices $l \geq p$ have holomorphic proxies $\hat{Z}_l = Z_l + o(1)$, uniformly on a slightly smaller domain. Their logarithms have the prescribed lifted determinations. Where needed, f_{p-1} also has an absolute $o(1)$ holomorphic proxy. The level- p side can be defined by $\Re \hat{L}_p = 0$, with*

$$\hat{L}_p = \hat{Z}_p - A \sum_{l > p} \hat{Z}_l - \frac{\hat{Z}_p}{(\log \hat{Z}_p)^K}.$$

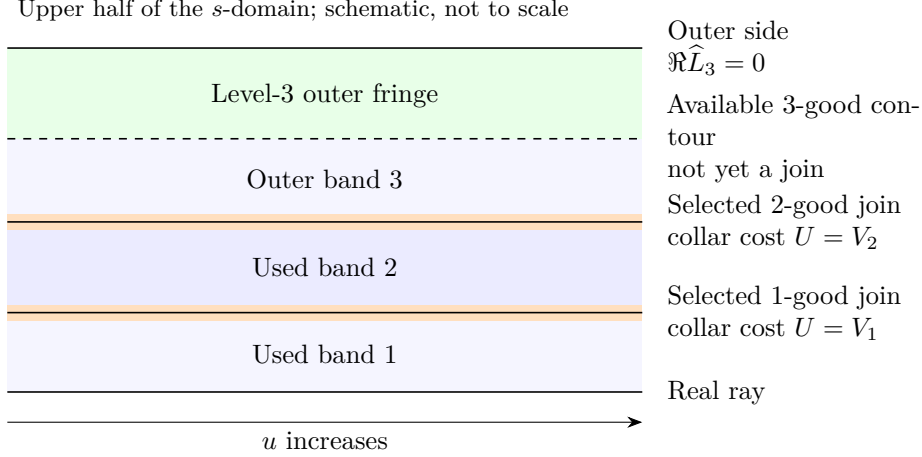


Figure 2: Coupling through level $p = 3$; the analogous lower half is omitted. Each inner band $j < 3$ ends at its selected level- j good join, whose smoothing cost is $U = V_j$. Only the outer band 3 continues through its outer fringe to the level-3 side. The dashed 3-good contour is available for the next transition and is not yet a join of this coupling. Shaded collars mark the two join sources; interior cutoff sources are not displayed.

All the angular comparisons and real-part domination estimates of Section 2 remain valid on the resulting component containing the real ray.

If $a(s)$ is a $\bar{\partial}$ source in a domain contained in a fixed-width strip and $|a(s)| \leq Me^{-u}$ for $u \geq u_b$, its Cauchy transform has norm at most CMe^{-u_b} , with C depending only on the strip width, and tends uniformly to zero as $u \rightarrow \infty$ on that fixed domain. The norm bound is also uniform over families with a common M and strip width; in particular it tends to zero when their left starts u_b tend to infinity. These statements apply as well after multiplication by a holomorphic gauge whenever its source has the displayed envelope.

Proof. Here and below sources are extended by zero when forming their area Cauchy transforms; the identity for $\bar{\partial}$ holds in the interior. For an evaluation point s , split the integral against $1/(\pi(s - \zeta))$ into $|s - \zeta| < 1$ and its complement. The first kernel has bounded integral on the unit disk and the second has modulus at most one. Thus

$$\left| \frac{1}{\pi} \iint \frac{a(\zeta)}{s - \zeta} dA(\zeta) \right| \leq C\|a\|_\infty + C\|a\|_1 \leq CMe^{-u_b}.$$

Subtracting the transform makes a C^1 function holomorphic. The small norm and decay on a fixed domain are distinct assertions. For the latter, fix $R > u_b$ and split $a = a_{\leq R} + a_{> R}$ according to the real coordinate of its argument. For $u > R + 1$, the first transform is bounded by

$$\frac{\|a_{\leq R}\|_1}{\pi(u - R)},$$

uniformly in v . The preceding near-kernel and far-kernel estimate bounds the second transform by CMe^{-R} . First let $u \rightarrow \infty$ with R fixed, then let $R \rightarrow \infty$. This proves the asserted absolute $o(1)$ on a fixed path without asserting any pointwise exponential decay rate for the correction. For varying anchors we instead use the uniform norm estimate with their far-starting domains; its constant is independent of the anchor when the source envelope and strip width are common.

We verify that the rate sources have the required small envelope. A cutoff with leading index $\sigma(r) < p$, active on an inner band j , has

$$\Re f_{r,j} \geq c \frac{Z_{\sigma(r)}(u)}{\text{polylog } Z_{\sigma(r)}(u)}. \quad (3.12)$$

Indeed, if $\Lambda_{\sigma(r)} \ll \Lambda_j$, its leading cosine is bounded below; otherwise $\Lambda_j = O(\text{polylog } Z_{\sigma(r)})$ and (3.11) gives the assertion. Later terms in its affine log relation are negligible by the real-part comparisons of Lemma 2.6. For a secondary cosine, the loss is bounded unless its rate derivative is comparable with Λ_j . In that case its negative logarithm is $O(\log(2 + \Lambda_k))$ and is absorbed by (3.12). In particular the log-cost dominates f_l for $l > \sigma(r)$; for $\sigma(r) = p - 1$ it still dominates f_{p-1} , since $f_{p-1} = \log Z_{\sigma(r)}$.

For $\sigma(r) \geq p$, positivity and the polynomial angular buffer in (3.2) give $U \gg f_p$; the more quantitative verification is included in Lemma 3.5. Join costs also satisfy $V_j \gg f_p$ for $j < p$. All these statements hold uniformly on the coupled domain; they imply in addition $U/e^u \rightarrow \infty$. Differentiating a later rate through its triangular formula introduces only a fixed finite product of later rates, of logarithmic size $O(f_p)$, as well as the allowed background losses. These factors can therefore be absorbed into e^{-cU} . The $\bar{\partial}$ derivatives of all rates $l \geq p$ are bounded by arbitrarily small multiples of e^{-u} far enough out. The preceding integral constructs their absolute $o(1)$ proxies.

The continuous lifted logarithm of Z_p and the estimate $\hat{Z}_p/Z_p = 1 + o(1)/Z_p$ give a continuous logarithm of $\hat{Z}_p$; since $\hat{Z}_p$ is holomorphic and nonzero, this logarithm is holomorphic. Its difference from f_p is exponentially small in $f_p(u)$. If f_{p-1} is free it is already holomorphic. Otherwise its exact affine expression in later rates and a bounded background admits the same correction.

Construct these corrections first on the piecewise domain with extra room. Near vertical, the inward rotation in Z_p/f_p^K has positive real part of size at least $c|Z_p|/f_p(u)^{K+1}$. This tends to infinity faster than every fixed power of $f_p(u)$ and overwhelms the absolute corrections. The plane comparisons of Lemma 2.6 show that $\Re \hat{L}_p$ reaches zero within the available room after increasing A and decreasing K . Its positive component has the asserted sides and comparisons. Neither this argument nor the maximum principles below requires a globally one-to-one map $s \mapsto \log \hat{Z}_p$. $\square$

Lemma 3.4 (Propagation of coupled growth). *Suppose the first $p - 1$ transitions have been isolated and coupled. Their coupled function has the level- p bound*

$$\log^+ |G| \leq C \Re Z_p + C D_{p-1}, \quad (3.13)$$

with the exceptional term absent on the outer side and on simultaneous safe columns. If a level- p term is isolated on a good collar with exponentially small error and coupled to its child, the corresponding level- $(p + 1)$ function has the analogous bound, after a fixed backstep. At a terminal transition it is bounded. All old joins and source types are preserved.

Proof. The assertion starts with the raw estimates. Write $n = p + 1$. On the chosen p collar the new coupled function equals its raw outer child up to $O(e^{-cV_p})$. Divide it holomorphically by $\exp(C' \hat{Z}_n)$, choosing C' to absorb the child's $\Re Z_n$ growth. Old sources, including derivatives of fixed normalizations, remain exponentially small. Subtract their Cauchy transform. The lateral boundary logarithm is then at most $C D_p + O(1)$, while the old interior bound gives logarithmic growth $O(\Re Z_p)$ plus a path constant.

Choose a common safe left column u_b in a fixed backstep. There the old bound costs $O(e^{f_p(u_b)})$. Put $X + iY = \log \hat{Z}_p$. On the good-contour domain the positive harmonic functions

$$\delta \Re(\hat{Z}_p e^{a_0 s}), \quad C_0 e^{f_p(u_b)} e^{-\kappa(X - f_p(u_b))} \cos(\kappa Y) \quad (3.14)$$

are respectively an infinity barrier and a left boundary barrier. Take $0 < \kappa < 1$. The small $a_0 > 0$ is chosen relative to the good-contour constant: $a_0|v|$ is smaller than half the angular margin ϵ_p/Λ_p , so the first real part is positive. It dominates the growth at infinity for every fixed $\delta > 0$. Apply the logarithmic maximum principle on bounded truncated components, then let the right boundary tend to infinity and δ decrease to zero. The remaining left contribution tends to zero if a fixed backstep was chosen with $f_p(u) > Bf_p(u_b)$ for a sufficiently large fixed B ; such backsteps exist by Lemma 2.4. Restoring the divisor and the small Cauchy correction proves (3.13) with p replaced by n .

The raw outer part already has the safe-column improvement. To carry it through the replaced inner wedge, if D_p is not absorbed by the new rate bound, use the safe collar on the box $|X - f_p(u_*)| \leq R$, where $R = C_1 \log(2 + D_p)$. The end boundary loss CD_p is majorized by a constant times

$$D_p \frac{\cosh(\kappa(X - f_p(u_*))) \cos(\kappa Y)}{\cosh(\kappa R)}.$$

Choose the fixed C_1 so that this is bounded at $X = f_p(u_*)$. The lateral collar has the improved bound by hypothesis; sources remain small. The box lies in the available range because $R = o(f_p)$. The maximum principle therefore removes D_p on the column. The outer side improvement is raw. At $p = m$ omit the divisor and use bounded ordinary terminal coefficients; the same proof gives boundedness. A finite number of steps uses only finitely many safe columns and labels, as required. $\square$

3.3 The source comparisons

The growth bound is now available for a fixed coupled domain. To improve a zero subtree, we will multiply its function by $\exp(H\hat{L}_n - C\hat{Z}_n)$, with $H > 0$ and fixed C . Write $x = \Re\hat{L}_n$. A source of precision e^{-cU} then has size at most $\exp(Hx - cU)$ after the fixed losses are absorbed. Thus the ratio U/x controls how large a weight H the sources permit.

For a target with value $x_a > 0$, separate sources with $x \geq \rho x_a$ from those with $x < \rho x_a$, where $\rho > 0$ is fixed and small. A lower bound for U/x on the first set pays the weight there; the second set costs at most $\rho H x_a$. The following lemma gives the required lower bounds in the two possible slope regimes. It also proves a common integrable envelope for a spare factor $e^{-cU/2}$. This second estimate makes the constant in the Cauchy correction bound independent of H .

Lemma 3.5 (Source ratios). *In a coupled level- n domain put $x = \Re\hat{L}_n$. For $n > 1$ write*

$$i = n - 1, \quad g = f_i - f_n, \quad r_0 = \frac{g'(u_a)}{\Lambda_n(u_a)}, \quad g_0 = g(u_a).$$

At a target on a sufficiently vertical level- i collar, and through its further fringe,

$$x_a \asymp \Re Z_n(s_a) \asymp e^{f_n(u_a)} \frac{r_0}{1 + r_0}. \quad (3.15)$$

For every fixed $\rho > 0$ and every fixed upper bound on r_0 , every coupled source satisfying $x \geq \rho x_a$ obeys

$$\frac{U}{x} \geq c \frac{e^{g_0}}{g'(u_a)}. \quad (3.16)$$

When r_0 is bounded below by a positive constant this also holds for central targets with $x_a \asymp e^{f_n(u_a)}$.

For large r_0 and central targets with $x_a \asymp e^{f_n(u_a)}$, one has instead

$$\frac{U}{x} \geq e^{cg_0} \quad \text{on sources with } x \geq \rho x_a. \quad (3.17)$$

For a fixed coupled calculation, $U/e^u \rightarrow \infty$ and $U/x \rightarrow \infty$ uniformly on its sources at infinity. These comparisons are unchanged by the absolute proxy errors.

Proof. All estimates can first be made with the piecewise rates; their buffer tends to infinity, so the $o(1)$ corrections have no effect. Write $t = f_n(u)$ and $\lambda = \Lambda_n(u)$, with a subscript 0 for the target. From Lemma 2.4,

$$ct \leq \lambda \leq t \text{polylog } t, \quad \left| \frac{d \log \lambda}{dt} \right| \leq \frac{\text{polylog } t}{t}. \quad (3.18)$$

At the target the angle of f_n differs from that of f_i by the gain g'/Λ_i . Since $g' \rightarrow \infty$, the fixed good-contour margin is negligible in comparison. Later real phases are negligible against $\Re Z_n$. This proves (3.15); the same is true if the level- i angle is moved farther toward vertical. Background jet errors are swallowed by that gain, by Lemma 2.7. In particular this gain is larger than the polynomial buffer needed for the level- n side.

Bounded target slopes and join sources. Suppose first that $r_0 \leq R$ with R fixed. The source constraint and $x \leq Ce^t$ give

$$t \geq t_0 - O(1) - \log(1 + 1/r_0). \quad (3.19)$$

Since $g'_0 \geq cg_0 \rightarrow \infty$, $r_0 \geq c/\lambda_0$, so the possible backward interval has length $O(\log \lambda_0) = O(\log t_0)$. On that interval $\lambda \asymp \lambda_0$ by (3.18). The gap estimates also give

$$\left| \frac{d \log g'}{dt} \right| = \left| \frac{g''}{g' \lambda} \right| \leq \frac{\text{polylog } t_0}{t_0},$$

because $g \leq g_0 \leq C_R \lambda_0$ in a backward interval. Consequently $g' \asymp g'_0$, and its integrated loss satisfies

$$g(u) \geq g_0 - O_R(1), \quad (3.20)$$

since $r_0 \log(1 + 1/r_0)$ is bounded on $(0, R]$. At forward points,

$$\frac{d}{du} \log \frac{e^g}{g'} = g' - \frac{g''}{g'} > 0$$

eventually, using $g' \geq cg$ and the polynomial-logarithmic bound on g''/g' . Thus in both directions $e^g/g' \geq c_R e^{g_0}/g'_0$ whenever the source constraint can matter.

A join at $j < n$ has $U = V_j$. With $J = f_j - f_n$, the collar comparisons give

$$x \leq C e^{f_n} \frac{1 + J'}{\Lambda_j}, \quad \frac{U}{x} \geq c \frac{e^J}{J'}.$$

Write $J = g + \Delta$, $\Delta = f_j - f_i \geq 0$. If $j = i$, $\Delta = 0$; otherwise $\Delta' \leq \Delta \text{polylog } \Delta$. Since $g' \rightarrow \infty$, exponential growth in Δ absorbs the possible Δ' loss, giving

$$\frac{e^J}{J'} \geq c \frac{e^g}{g'}.$$

This proves (3.16) for joins.

Cutoff sources at bounded target slopes. Consider a cutoff source, with earlier label $r < j \leq n$, $F = f_r(u)$ and $\sigma = \sigma(r)$. If $\sigma < n$, estimate (3.12) applies on its own inner band. Because $\sigma \leq i$

and $f_i \leq \log Z_\sigma$, its log-cost exceeds f_i by an unbounded factor. The secondary cosine loss described there does not change this conclusion. It is therefore stronger than both required ratios.

Suppose $\sigma \geq n$. There are four estimates to check. If $F \leq t^{3/2}$, the angular scale is at most C/λ , and Taylor's formula together with the fixed derivative bounds gives

$$\Re f_{r,j} = F + O\left(\frac{F \text{polylog } F}{\lambda^2}\right) = F + o(1).$$

For a simple cost, or a mixed cost with $k \geq n$, division by x therefore gives

$$\frac{U}{x} \geq ce^{F-f_n}. \quad (3.21)$$

For $k < n$, the source lies on an inner band $j \leq k$ ending at its good join. Its secondary cosine is at least c/Λ_k , and the angular comparison yields

$$\frac{\cos(\Im f_{k,j})}{\cos(\Im f_{n,j})} \geq \frac{c}{1 + \Lambda_k - \Lambda_n}. \quad (3.22)$$

Since $r < j \leq k \leq i$, we have $F \geq f_k$. Set $\Delta = f_k - f_i \geq 0$. The resulting ratio is at least

$$c \frac{e^{g+\Delta}}{1 + g' + \Delta'} \geq c \frac{e^g}{g'}.$$

Together with the backward and forward comparisons already proved, this settles the small- F case.

If $t^{3/2} < F \leq t^3$, the leading affine relation gives $F \asymp Z_\sigma$ and $f_\sigma = \log F + O(1) = O(\log t)$. Hence $\Lambda_\sigma/\lambda \rightarrow 0$, the angle of its leading rate is small, and

$$\Re f_{r,j} \geq cF. \quad (3.23)$$

If $F > t^3$ and $\sigma > n$, the leading cosine is at least c/λ ; later affine terms are negligible. Thus

$$\Re f_{r,j} \geq cF/\lambda. \quad (3.24)$$

Finally, if $\sigma = n$, the large A in the side equation absorbs the later affine terms and leaves the buffer:

$$\Re f_{r,j} \geq cx + c \frac{e^t}{t^{K+1}}. \quad (3.25)$$

This also holds on the relevant inner contours.

For bounded target slopes, $g_0 \leq C_R \lambda_0$. At backward sources the possible drop in t is only $O(\log t_0)$; at forward sources $g_0 \leq C_R \lambda_0 \leq t \text{polylog } t$. Each of (3.23)–(3.25) therefore dominates $t + g_0 + \log(2 + F)$. In (3.24) use $F/\lambda \geq t^2/\text{polylog } t$ and $\log F \leq t + O(1)$ when $\sigma > n$. The negative logarithm of a secondary cosine is either bounded or $O(\log(2 + F))$: if $k < n$, use its inner-band lower bound; if $k \geq n$, use the side's polynomial buffer and ordered angles. All those losses and division by $x \leq Ce^t$ are absorbed. This proves (3.16) for all cutoff costs, uniformly even when $r_0 \rightarrow 0$.

Large target slopes. Now take a large target slope and a central target. The source constraint implies $t \geq t_0 - O(1)$. The adjacent-gap bound $g'/g \leq C\lambda$ implies

$$g(u) \geq cg_0 \quad \text{whenever } t \geq t_0 - O(1). \quad (3.26)$$

The join calculation and the small- F calculation above give $U/x \geq e^{cg_0}$: their derivative denominators are only polynomial-logarithmic in the corresponding diverging gaps. For $\sigma = n$, the target relation

gives $g_0 \leq f_i(u_a) \leq f_r(u_a) \leq Ce^{t_0}$. Thus (3.25) and the source constraint $x \geq \rho x_a \asymp e^{t_0}$ leave a fixed positive multiple of g_0 in the log-cost. The buffer term absorbs t and the secondary cosine losses, giving the same ratio.

It remains to justify the extra angular loss when $F > t^{3/2}$ and $\sigma > n$. Put

$$J = f_n - f_\sigma, \quad d = \frac{J'}{\lambda} = 1 - \frac{\Lambda_\sigma}{\lambda}, \quad \eta = \cos(\Im f_{n,j}), \quad q = \min(1, \eta + d).$$

The leading real contribution satisfies

$$\Re f_{r,j} \geq cF(\eta + d). \quad (3.27)$$

Moreover $d \geq c/\lambda$, and so $L = \log(1/q) = O(\log t)$. The gap derivative bounds give

$$\left| \frac{d \log d}{dt} \right| = \left| \frac{J''}{J'\lambda} - \frac{f_n''}{\lambda^2} \right| \leq \frac{\text{polylog } t}{t}. \quad (3.28)$$

Decrease t to $t - L$ and denote the corresponding real position by u' . The quantity d changes by a bounded factor (in fact by $1 + o(1)$). Since $q \geq d$, $dL \leq d \log(1/d)$ is bounded. Integrating $df_\sigma/dt = 1 - d$ now gives

$$Z_\sigma(u') = Z_\sigma(u) \exp\left(-L + \int_{t-L}^t d(\tau) d\tau\right) \asymp q Z_\sigma(u). \quad (3.29)$$

Because $x \leq Ce^t \eta \leq Ce^t q$ and $x \geq \rho x_a$, one has $t - L \geq t_0 - O(1)$. The indices $r < j \leq n = i + 1$ imply $r \leq i$; hence

$$g(u') \leq f_i(u') \leq f_r(u') \asymp Z_\sigma(u').$$

Combining this with (3.26) and (3.29) proves $F(\eta + d) \geq cg_0$. Independently, (3.23) or (3.24) gives $\Re f_{r,j} \gg t + \log(2 + F)$. These two lower bounds, each used with a sufficiently small fixed fraction of the log-cost, absorb division by x and all secondary cosine losses, leaving $\log(U/x) \geq cg_0$. This proves (3.17).

The common integrable envelope. The same estimates without a target show the asserted uniform growth at infinity. For joins, $U/x \geq ce^{f_j - f_n} / (f_j - f_n)' \rightarrow \infty$. For cutoffs with small F , the bound is of this form with a larger gap; the other cases have a log-cost dominating f_n or its relevant angularly diminished value. All costs dominate e^u uniformly over the controlled finite-request families in Definition 3.1. More explicitly, on a join

$$\log U = f_j - \log \Lambda_j \geq \frac{1}{2} f_j \geq \frac{1}{2} e^u$$

eventually. For a cutoff, the small- F estimate leaves $F - O(\log(2 + F)) \geq cf_n$ after the secondary cosine loss; the intermediate and large- F estimates and the buffer in (3.25) leave at least cf_n as well. The early-index estimate is stronger. Thus, after decreasing a fixed $c > 0$ for the finite source list,

$$U \geq \exp(ce^u). \quad (3.30)$$

Every threshold here depends only on the finite quantitative controls specified in the definition. In particular, for any fixed source precision constant $c_* > 0$, the residual factor $e^{-c_* U}$ has a common envelope Me^{-u} on the full forward domains, with M independent of their anchors. The stipulated uniform $o(U)$ absorption permits the same conclusion after all fixed losses. These constants are also independent of any subsequently chosen weight parameter. $\square$

3.4 The zero-subtree estimate

Lemma 3.6 (Weighted maximum estimate). *Let G be coupled through level n with the growth bound of Lemma 3.4. Suppose that on the real ray*

$$|G(u)| \leq C_N e^{-N Z_n(u)} \quad (u \geq u_N), \quad \text{for every fixed } N > 0. \quad (3.31)$$

The constants C_N, u_N can depend arbitrarily on N . Fix the coupled functions, its joins, a safe left column, and the full domain $x = \Re \hat{L}_n > 0$. There are fixed $C, C_0, c > 0$ such that for every $H > 0$ and every point of that domain,

$$\log |G| + Hx - C \Re \hat{Z}_n \leq C_0 + \max \left\{ H \sup_{\text{left}} x, \sup_{\text{sources}} (Hx - cU/2), 0 \right\}. \quad (3.32)$$

In particular C_0 is independent of H, C_N , and u_N . When $n = 1$ there are no sources, and (3.31) implies exact vanishing.

Proof. Correcting the weighted sources. Choose C greater than the fixed coefficient in the coupled growth estimate on safe columns and the side. Multiply G by the holomorphic gauge

$$\exp(H \hat{L}_n - C \hat{Z}_n).$$

Its side modulus is bounded independently of H because $x = 0$; its left boundary logarithm is at most $H \sup_{\text{left}} x + O(1)$. With constants decreased to absorb the fixed losses, its sources have modulus at most a finite sum of $\exp(Hx - cU)$. Set

$$K_H = \max \left\{ 0, \sup_{\text{sources}} (Hx - cU/2) \right\}.$$

The source-ratio lemma makes K_H finite for every fixed H . Crucially, keep half the negative cost:

$$e^{Hx - cU} \leq e^{K_H} e^{-cU/2}. \quad (3.33)$$

The last factor is bounded by a fixed multiple of e^{-u} and is integrable on the bounded-width domain, because $U/e^u \rightarrow \infty$. Lemma 3.3 thus gives a Cauchy correction of norm at most $C_1 e^{K_H}$, where C_1 is independent of H . Let F_H be the resulting holomorphic function. By (3.30), C_1 is uniform for the controlled finite-request families as well; the bound uses their full forward domains, not just a window about the target. Their possibly different return times to smaller ordinary neighborhoods do not enter this constant.

Boundedness for each fixed weight. For each fixed H , preliminary real decay makes the weighted function bounded on the real ray: choose N larger than the fixed multiple of $H + C$ needed there, and include the finite interval before u_N in its bound. This bound can depend on H in an uncontrolled way; we do not use it in the final estimate. To see that F_H is bounded on each half-domain, write $X + iY = \log \hat{Z}_n$. Its logarithmic growth is $O_H(e^X)$ plus a path constant. On the upper half use the positive harmonic barrier

$$\delta e^{aX} \cos(a(Y - \pi/4)), \quad 1 < a < 2,$$

and on the lower half use $\delta e^{aX} \cos(a(Y + \pi/4))$. The side angles tend to $\pm\pi/2$, so a sufficiently far start makes both barriers positive on their respective half-domains. They dominate the infinity growth. The maximum principle on finite truncations and then $\delta \downarrow 0$ proves boundedness on each half, hence on the full domain.

Removing the preliminary constants. Now apply the bounded maximum principle to that *full* domain. The real ray is no longer a boundary. One can justify the infinite-domain principle by $\delta e^{a'X} \cos(a'Y)$, $0 < a' < 1$, before letting $\delta \downarrow 0$. Only the side and the fixed safe left column remain. After restoring the Cauchy correction their bounds give (3.32), with an additive constant depending on C_1 and the fixed data alone. This explains why the arbitrary constants in (3.31) disappear.

For $n = 1$, choose a fixed backstep so that the target's positive x_a is larger than $\sup_{\text{left}} x$. There is no source term, and (3.32) gives

$$\log |G(s_a)| \leq C_0 + C \Re \widehat{Z}_1(s_a) - H(x_a - \sup_{\text{left}} x).$$

Letting $H \rightarrow \infty$ proves $G(s_a) = 0$. Every sufficiently far real target permits this construction, giving the asserted identity on the real locus. $\square$

Lemma 3.7 (Upgrade at the preceding transition). *For $n > 1$, under the hypotheses of Lemma 3.6, the outer packet of G satisfies $|G| \leq C e^{-cV_{n-1}}$ on sufficiently vertical level- $(n-1)$ collars and their further fringe.*

Proof. Put $i = n - 1$ and use the notation of the source-ratio lemma. A fixed backstep can make

$$\sup_{\text{left}} x \leq \rho x_a \tag{3.34}$$

for any chosen small fixed $\rho > 0$. Indeed, f_n on that column can be made a fixed factor smaller than $f_n(u_a)$, whereas $x_a \geq c e^{f_n(u_a)} / \Lambda_n(u_a)$ at the target. Safe columns exist in the needed fixed intervals.

Bounded slopes. If r_0 is bounded, choose

$$H = c_1 \frac{e^{g_0}}{g'_0}$$

with $c_1 > 0$ sufficiently small. On sources with $x \geq \rho x_a$, (3.16) gives $Hx - cU/2 \leq 0$; on the remaining sources it is at most ρHx_a . Thus the maximum on the right of (3.32) is at most ρHx_a . At the target $\Re \widehat{Z}_n \asymp x_a$ and $H \rightarrow \infty$, so the subtracted fixed growth term is negligible. Finally,

$$Hx_a \asymp \frac{e^{g_0}}{g'_0} e^{f_n(u_a)} \frac{g'_0 / \Lambda_n(u_a)}{1 + g'_0 / \Lambda_n(u_a)} = \frac{e^{f_i(u_a)}}{\Lambda_i(u_a)} = V_i(u_a). \tag{3.35}$$

This proves the result for bounded slopes, uniformly down to $r_0 = 0$ as a limiting regime. When r_0 stays in a fixed compact subinterval of $(0, \infty)$ the same proof applies to central level- n targets.

Large slopes: decay in the central region. For large slopes, first use central level- n targets. Take $H = \exp(c_2 g_0)$ with c_2 smaller than the constant in (3.17). Exactly the preceding argument gives

$$|G(s_a)| \leq \exp\{-\exp(c_3 f_i(u_a))\} \tag{3.36}$$

for some fixed $c_3 > 0$, since $Hx_a \geq c \exp(f_n(u_a) + c_2 g_0)$ and $f_i = f_n + g_0$. This estimate is weaker than the desired one at the preceding transition, so a second maximum principle is needed.

Large slopes: transfer to the preceding fringe. Use the holomorphic coordinate function $X + iY = \widehat{f}_i$. Work first in the local tube $|v| < 2W / \Lambda_i(u)$, with fixed slack at its real end columns, and take the component meeting the real ray of the inverse image, within this tube, of the rectangle

$$X_0 - 1 < X < B_1 X_0, \quad |Y| < W, \quad X_0 = f_i(u_a),$$

where W is a large fixed number. Stop at a right real column if Λ_i/Λ_n first falls to a fixed $M_0 \gg W$. For a sufficiently large starting ratio, its short left extension retains a large ratio by the derivative bounds. On the tube's two angular walls the path estimates and the absolute proxy correction give $Y = \pm 2W + o(1)$ and $X = f_i(u) + o(1)$. The selected component, where $|Y| < W$, cannot meet those walls. There are therefore no additional angular boundary pieces to estimate. All its points remain in this local tube and are central for level n : their f_n angle is at most $O(W/M_0)$. The level- i target, with $|Y| \sim \pi/2$, is inside it.

Every source encountered in this rectangle has

$$U \geq c \frac{e^X}{\Lambda_i}. \quad (3.37)$$

For joins this is the gap calculation above. A cutoff with $\sigma < n$ has the stronger bound (3.12). If $\sigma \geq n$, its leading angle is small. With $F = f_r$ and $T = f_i$, the case $F \leq T^{3/2}$ gives $\Re f_{r,j} = F + o(1)$ by the Taylor calculation on the scale W/Λ_i . Since $r \leq i$, this pays e^X ; a secondary cosine with $k \leq i$ costs at worst Λ_k , and Λ_k/Λ_i is absorbed by e^{F-f_i} using the same gap calculation as (3.22). If $F > T^{3/2}$, use that every phase of index at least n has size $O(W/M_0)$ throughout this rectangle. The leading cosine is bounded below, later affine terms remain negligible, and $\Re f_{r,j} \geq cF$. This exceeds $T + \log(2 + F)$ by an unbounded factor, paying every such secondary loss. This proves (3.37).

Take $\kappa = \pi/(2W) < c_3$, and let w be the positive harmonic function

$$w(X, Y) = c_4 \frac{e^{X_0}}{\Lambda_i(u_a)} \sinh(\kappa(X - X_0 + 1)) \cos(\kappa Y). \quad (3.38)$$

It is the real part of a holomorphic function of $\hat{f}_i$ and vanishes on the left and horizontal sides. The rate derivative bound implies

$$\left| \frac{d \log \Lambda_i}{dX} \right| \leq \frac{\text{polylog } X}{X} = o(1). \quad (3.39)$$

Consequently, throughout the rectangle,

$$\frac{w(X, Y)}{e^X/\Lambda_i(X)} \leq C c_4 \exp(-(1 - \kappa - o(1))(X - X_0))$$

apart from the harmless one-unit left extension. Choosing c_4 small makes the weight at most a fixed small fraction of every source cost in (3.37).

On a full right edge $X = B_1 X_0$, the logarithm of the size of w is at most $[1 + \kappa(B_1 - 1)]X_0 + O(\log \Lambda_i(u_a))$. Choose

$$B_1 > \frac{1 - \kappa}{c_3 - \kappa}. \quad (3.40)$$

Then (3.36) pays the whole weight there. If the domain was stopped at the bounded-ratio column, the bounded-slope result (3.35) pays it instead, after reducing c_4 using (3.39). The other edges have zero weight, and (3.36) bounds G there.

Multiply G by the holomorphic gauge with real logarithm w . Its sources retain an integrable negative half-cost, so their Cauchy correction is bounded independently of the target. The maximum principle on the bounded component therefore bounds the weighted function by a fixed constant. At the target, the cosine in (3.38) is bounded below and $\sinh \kappa > 0$, so $w(s_a) \geq c V_i(u_a)$. This proves the desired $e^{-c V_i}$ bound. The argument uses only positive harmonic functions pulled back by $\hat{f}_i$ and works without global coordinate injectivity. It applies unchanged to the farther level- i fringe. Nominal pointwise cutoff variants differ there by errors with the same lower cost, so satisfy the same bound. $\square$

3.5 Finite isolation and terminal comparison

Lemma 3.8 (One finite transition). *Suppose a prefix is coupled through level p . Fix a child exponent a , and assume that every earlier child has an entire zero subtree and satisfies the upgrade of Lemma 3.7 at level p . Then a sufficiently vertical fixed good collar can be chosen so that*

$$e^{aZ_{p,p}} S_{\lambda_{<p}}^p = S_{\lambda_{<p},a}^{p+1} + O(e^{-cV_p}) \quad (3.41)$$

on both sides, with zero insertion if necessary. It can be coupled to that child while preserving all preceding joins.

Proof. Only finitely many children precede a by left-finiteness. For an earlier exponent $b < a$, its outer packet is $O(e^{-c_b V_p})$ beyond its previously selected join. On a more vertical good collar, $\Re Z_p \asymp \epsilon_p V_p$. Choose $\epsilon_p > 0$ small enough that every finite multiplier $\exp((a-b)\Re Z_p)$ is absorbed by half of the corresponding $c_b V_p$. This is a finite restriction. The collar may also be chosen beyond every earlier child's selected join and within the allowed inward extension of the chosen child.

Take a transition budget larger than a and use (3.5). For every included later exponent $b > a$, its positive gap $b - a$ suppresses its raw packet: on the p collar

$$\Re Z_{p+1} = o(V_p), \quad D_p = o(V_p),$$

and there are only finitely many such terms. The first estimate is the contour hierarchy. For the second, any uncapped coefficient contributes at most $O(\log \Lambda_p)$ on a fixed backstep, and $\log \Lambda_p = o(V_p)$; the same conclusion holds along the ray. The tail error is also $O(e^{-cV_p})$ after increasing the budget if necessary. These observations give (3.41).

In the inner bands multiply the existing coupled function by $\exp(aZ_{p,j})$, then switch on this collar to the selected outer child. On an old j join, $j < p$, the new rate has real part $o(V_j)$, so old seam errors persist. Background changes alter this fixed multiplier by a relatively exponentially small amount there. The source hypotheses handle all derivatives of the interpolation and normalization. The growth bound for the new coupling follows from Lemma 3.4. No new transition at an earlier inner band has been requested. $\square$

Lemma 3.9 (Comparison of terminal coefficients). *Suppose a complete exponent vector has been isolated on the two sides and coupled, with all earlier subtrees passed as in Lemma 3.8. Its two terminal ordinary coefficients are the same analytic germ. The fully normalized coupled function tends to this germ on the real ray and through the corresponding inner wedge and used bands.*

Proof. By terminal growth propagation the normalized function is bounded on the whole inner domain after a fixed backstep. For a target family with real anchors $u_a \rightarrow \infty$, choose safe left columns u_b in fixed backstep intervals. The finite raw, background, and source data used by this terminal coupling have common bounds on their full forward domains, by the finite-request hypothesis. In particular (3.30) supplies a common Me^{-u} envelope for their corrected sources. Their Cauchy corrections therefore have norm at most $CMe^{-u_b} \rightarrow 0$, uniformly over the target family. On a single fixed path the compact-left and tail argument in Lemma 3.3 gives the corresponding vanishing at infinity. Denote the corrected functions by $\tilde{G}$; they are holomorphic and uniformly bounded. On the two outer collars their values approach, respectively, $c_+(t)$ and $c_-(t)$, by the terminal transition.

We give the argument that these two boundary limits cannot differ. Fix an ordinary value t_* in a sufficiently small real neighborhood and take a family of tests approaching it on every fixed

u window about the varying anchor. Such tests can return to t_c at infinity: for example, on the needed right half-strip use

$$t(s) = t_c + (t_* - t_c)e^{-\delta_a(s-u_a)}, \quad \delta_a \downarrow 0,$$

with the fixed backward range and ordinary neighborhood chosen with slack. Its fixed positive derivatives tend to zero and it returns to t_c on every chosen path. Taking $0 < \delta_a \leq \delta_0$ gives common finite-jet bounds on the entire forward domains, and $\delta_a \rightarrow 0$ gives a common small-jet modulus and convergence to t_* on each fixed target window. The time of return to a smaller box may diverge; that time is used only in the preliminary real estimates, whose constants have already been removed by Lemma 3.6.

Write $X + iY = \log \hat{Z}_m$. For a fixed small $\eta > 0$, the boundary values on a window about the target are within η of the corresponding constants, once the anchor is sufficiently far. The X distances from the target to both ends of a fixed u window tend to infinity by Lemma 2.4. On that window apply the logarithmic maximum principle to the difference from $c_+(t_*)$. Its upper boundary is bounded by $\log \eta$, its lower boundary by a fixed constant $\log M$, and its two ends by the bounded growth estimate. An affine function of Y interpolates the lateral logarithmic bounds. The actual angles tend to $\pm\pi/2$; adding arbitrarily small relative angular slack and decaying cosine barriers at the two ends yields, at the real target,

$$\limsup \log |\tilde{G} - c_+(t_*)| \leq \frac{1}{2} \log \eta + \frac{1}{2} \log M.$$

For completeness, the end barriers may be taken as fixed multiples of $e^{-\kappa(X-X_-)} \cos(\kappa Y)$ and $e^{-\kappa(X_+-X)} \cos(\kappa Y)$ with $0 < \kappa < 1$. Their values at the target tend to zero. First let the target go to infinity, then let the angular slack and η go to zero. The real-ray limit is $c_+(t_*)$. Applying the same argument from the lower boundary gives the limit $c_-(t_*)$. Thus these two values coincide for every such real t_* . The identity theorem for analytic germs makes the coefficients equal.

With their common value on both lateral boundaries, apply the same maximum principle to their difference from $\tilde{G}$. The lateral bound is now η on both sides, so the estimate holds uniformly across the entire inner wedge, including targets approaching either boundary. Restoring the vanishing Cauchy correction changes nothing. Uniformity on smaller ordinary neighborhoods follows either from these estimates with common finite slack or by applying the construction to a putative sequence of violating ordinary inputs. The same argument uses the corresponding coupled determinations in the already used bands, and proves their asserted convergence. $\square$

Proof of Theorem 3.2. We describe the induction explicitly to keep its finite and infinite requests separate. A coupled prefix includes fixed functions on the earlier inner bands, fixed selected joins, the growth bounds, and the finite list of their source types. An assertion about that prefix may request more accuracy *at infinity on its existing path*; it may not change those already fixed functions or joins.

Induct downward on the level n . The assertion to prove is: for every coupled prefix whose entire level- n subtree has zero terminal data, the preliminary estimates (3.31) hold for all fixed N ; they imply the preceding-transition upgrade when $n > 1$ and exact vanishing when $n = 1$.

At $n = m$, all ordinary child coefficients are zero. For each fixed budget the transition formula therefore gives arbitrary exponential decay on a sufficiently vertical m collar. Equivalently insert a zero terminal child at any exponent $a > N$, choosing the remainder budget greater than a . The proof of growth propagation, applied after multiplication by e^{aZ_m} , gives a bounded function in the inner wedge after a fixed backstep. On the real ray this is $O(e^{-aZ_m})$, and hence implies (3.31) for the requested N . No uniformity over N has been used. The weighted lemma and the upgrade lemma now establish the induction assertion at level m .

Suppose it is known strictly below level n and consider an entirely zero subtree coupled through level n . Fix one finite requested real order N . Choose an exponent $a > N$ at this coordinate and insert a zero child there if necessary, enlarging the transition budget beyond it. There are finitely many children earlier than a . Pass them in increasing order. To pass one child, first isolate it on its own sufficiently vertical good contour using Lemma 3.8 and the upgrades already obtained for its earlier siblings. Its coupled subtree at level $n + 1$ is zero, so the induction hypothesis below n gives its preliminary real estimates and its e^{-cV_n} upgrade. This permits the next sibling to be isolated. Only finitely many siblings are needed before reaching the inserted zero child a .

After isolating that child, the growth propagation lemma gives on the real ray

$$|G(u)| \leq C \exp(-aZ_n(u) + CZ_{n+1}(u) + CD_n)$$

when $n < m$, and the corresponding bound without the last rate at $n = m$. Since $Z_{n+1} = o(Z_n)$ and D_n is a constant on this path, this is the requested $O(e^{-NZ_n})$ estimate sufficiently far out. Alternatively one may continue the zero child down to a terminal zero; both constructions use only finitely many transitions for this request. We have proved all fixed preliminary orders at level n , and can now apply Lemma 3.6 and Lemma 3.7.

There is no circular use of a stronger estimate at the current level. Preliminary orders at level n invoke upgrades only at level $n + 1$; the height decreases each time. Finite breadth can grow with N , but the height is at most m . Moreover all joins strictly preceding the child under consideration remain fixed. Higher requests need new labels only at that child or deeper, and any new fixed later multiplier costs $o(V_j)$ on an old j join. Thus the full domain in the weighted lemma has fixed source estimates independent of its multiplier H . The preliminary real estimates may start arbitrarily farther out as H varies; their constants disappear in (3.32). This completes the downward induction and, at $n = 1$, proves the exact-zero assertion.

If the paired array is nonzero, choose its first nonzero exponent λ . At the first coordinate pass only the finitely many earlier subtrees, all of which are zero by definition of λ . The established upgrades and Lemma 3.8 isolate λ_1 . Repeat at its selected child, then at the next one. This finite process reaches the paired ordinary coefficients at λ . Lemma 3.9 proves their equality and the real leading limit (3.8). It also proves the corresponding limits for the coupled normalized determinations inside their selected contours and along the bands already used.

It remains to justify the owning-fringe extension in the statement. Beyond a selected leading join the outer packet is the one whose convergence was just proved. In the finite transition (3.5), every earlier term still has its e^{-cV_j} suppression on that more vertical portion, and every later term has a strictly positive current exponent. Choose the fringe constant sufficiently small for the finitely many earlier multipliers and choose A large enough that $\Re Z_j$ dominates the finite later real-phase losses. The transition remainder can be requested to any fixed needed depth. Thus the same normalized leading estimate holds throughout that fringe.

In this last comparison rates in the two backgrounds must also be compared. Estimate (3.4) is sufficient: triangular differentiation of a fixed later rate Z_k , $k > j$, has logarithmic sensitivity at most Cf_{j+1} in these variables, whereas $\Re Z_j$ dominates every fixed power of f_j on the buffered side. Hence the absolute change in each such later rate is $o(1)$ after any fixed needed normalization. The extracted current factor is always kept at its own inner background; it is not reevaluated at the next background. On selected good joins the stronger e^{-cV_j} comparison already gave the same conclusion. This proves the stated fringe extension and finishes the theorem. $\square$

3.6 Uniformity for the differentiated calculus

The preceding proof also supplies the uniform estimates needed when ordinary tests vary with their anchors. We collect their precise scope here for the differentiated calculus in Section 4.

Lemma 3.10 (Uniformity for a controlled finite request). *Fix finitely many packet labels, transition budgets, coupled joins, and normalizations, and a family of tests with the common quantitative controls in Definition 3.1. The fixed labels used on the coupled domain must stay in their common ordinary ranges throughout that domain. Deeper labels used only for preliminary real estimates may have smaller ranges entered at a time depending on the member of the family.*

The proxy, growth, source-ratio, and preceding-transition estimates above have constants uniform in this family. If each member has the preliminary estimates (3.31), no uniformity of their C_N, u_N or of their return times is needed for the weighted estimate or its upgrades. Terminal comparison is uniform at receding anchors when the tests have a common slow-variation modulus on each fixed window about those anchors. The same statements apply to any fixed finite collection of derivative packet trees satisfying the differentiated hypotheses.

Proof. First choose common sufficiently vertical collars and residual positive join precisions after the finite current-level losses have been absorbed as specified in Definition 3.1. Take the largest of the finitely many raw-growth constants, the smallest positive source-precision constant c_* , and a common bound on source counts and prefactors. The uniform $o(U)$ condition permits a common start after which every fixed source loss is at most $c_*U/4$. Equation (3.30) then supplies a common integrable envelope Me^{-u} for the remaining negative cost. Hence the unweighted Cauchy corrections have norm at most CMe^{-u_b} on the full forward domains. For the weighted correction the same calculation, with the spare half-cost retained, gives $C_1e^{K_H}$ with one C_1 independent both of H and of the test. This argument uses the full-domain quantitative bounds for the fixed labels; pointwise eventual bounds on unrelated paths would not suffice.

The proxy side buffers diverge uniformly and absorb these absolute corrections. The source-ratio estimates use only the common finite real and angular flag constants, so their constants and finite thresholds are common as well. In the growth argument choose one sufficiently large fixed backstep using the common lower bound $f'_p \geq cf_p$. Its left boundary contribution is bounded by a common multiple of

$$\exp((1 + \kappa)f_p(u_b) - \kappa f_p(u_a)),$$

which tends to zero uniformly with this choice. The possible D_p loss stays explicit; choosing one $R = C_1 \log(2 + D_p)$ with $\kappa C_1 > 1$ makes $D_p / \cosh(\kappa R)$ bounded uniformly. The uniformly available safe columns and entering collars supply the required boundary estimates. Their numerical locations may vary with the test within the fixed backstep intervals; one column for a continuum of tests is not required. No bound on the time of return to a smaller ordinary box has entered these calculations.

For each fixed test and H , its own preliminary real estimate establishes half-domain boundedness. The subsequent full-domain maximum principle uses only the common side and left estimates and the common $C_1e^{K_H}$ bound. It therefore gives one C_0 in (3.32) for the whole family, regardless of the preliminary constants. The source-ratio lemmas then choose the same small multipliers in $H = c_1e^{g_0}/g'_0$ or $H = e^{c_2g_0}$, and the same wide-strip constants W, M_0, B_1, c_4 , throughout each of the stated slope regimes. This proves uniformity of the upgrades. When $n = 1$, the limit $H \rightarrow \infty$ is taken separately at each target; a common preliminary start for all H is unnecessary.

For terminal comparison, the common finite-window modulus makes each lateral error smaller than any fixed $\eta > 0$ after a common far anchor. The Cauchy errors are uniformly $O(e^{-u_b})$, and the fixed-window X distances to both ends tend uniformly to infinity. The end cosine barriers therefore disappear uniformly. At the real targets the affine interpolation has weights tending uniformly to

$1/2$, and bounds the difference from either terminal coefficient by a fixed bounded constant times a positive power of η . Letting $\eta \downarrow 0$ identifies the two coefficients uniformly there. Once this equality is known, the errors on both lateral boundaries are at most η relative to their common coefficient. The constant lateral bound, together with the vanishing end barriers, gives convergence throughout the inner wedge, even for targets approaching a side where one of the two interpolation weights tends to zero. This proof needs slow variation near the target and common fixed-label estimates on the full domain, but no common time at which every deeper ordinary box is entered. $\square$

4 Differentiated packet calculus and regular ordinary charts

The separation theorem is an assertion about a full tree of successive expansions. Its application to equations requires a calculus of those trees, including a calculus after solving regular equations in the bounded variables. We establish that calculus here. In particular, no simultaneous convergence of the full exponential series will be used.

The need to verify closure separately from asymptotic injectivity is substantive: Yeung's analysis of a coefficient class in the classical finiteness argument exhibits a failure at a differential closure step [22]. We therefore specify the independent arguments and all differentiated estimates before making an ordinary implicit substitution. The ensuing closure theorem applies to the packet hypotheses stated here.

4.1 The differentiated primitive contract

Throughout this section a flag, its lifted logarithms, and its positive and negative angular determinations are fixed. Write S for the tuple of free nodes and t for the independent ordinary variables. A derivative in an ordinary variable always holds S fixed. A derivative in a retained free variable holds the other independent variables fixed; dependent nodes are then differentiated by their triangular equations. For a complex input z write

$$D_z = \partial_{\Re z}, \quad \mathcal{A}_z = \partial_{\Im z} - i\partial_{\Re z}.$$

Thus $\mathcal{A}_z$ is zero on a holomorphic function. The two lateral determinations are differentiated separately.

A *finite request* specifies finitely many packet labels, a finite derivative order, finitely many transition budgets, and a finite Taylor accuracy. Its constants, ordinary neighborhoods, angular rooms, and starting points may depend on the request. On a selected path every further fixed request must be available sufficiently far towards infinity on that same path. This does not require a common starting point for all requests. Ordinary tests are real-symmetric holomorphic paths, return to the fixed ordinary germ at infinity, and have the required positive-order real jets tending to zero. Fixed backsteps and slightly larger argument ranges are available. The following definition specifies the uniformity needed when the test itself changes with the target.

Definition 4.1 (Controlled families for one finite request). Fix a finite request, a compact ordinary box contained with positive distance in the argument domains of its finitely many labels, and fixed angular rooms and backsteps. A controlled family has common bounds for the finitely many real jets and Taylor-error constants required by this request, throughout each member's full forward domain beginning at the prescribed backstep. The finitely many flag inequalities used there have common constants and common moduli for their vanishing relative errors. For convergence near moving anchors, the positive-order ordinary jets also have a common modulus tending to zero on every fixed window about those anchors.

By *uniform packet estimates on this controlled family* we mean that raw growth, side and safe improvements, transitions, and differentiated source estimates have common constants. This includes the number of source terms, their positive precision constants, and the thresholds at which their specified losses are absorbed. Uniform safe-column estimates mean that each member has a suitable column in the prescribed fixed-length interval; the column itself may depend on the member. A common far threshold means thresholds in the finitely many numerical rate and gap inequalities used in the request; it need not be a threshold in the smallest rate alone. The estimates hold on each full forward domain once those inequalities and its stipulated ordinary range hold there. The exceptional constants D_i remain explicit allowed losses.

Every member still returns individually to the fixed germ. A further request may require a smaller ordinary box and a different finite set of controls, reached at a member-dependent later time. Neither a common return time to all smaller boxes nor uniformity over infinitely many requests is part of this definition.

The common constants in Definition 4.1 are a quantitative requirement on primitive estimates. They do not follow from eventual validity on each separate test. Below we prove that separation and the calculus preserve this requirement, and verify that the interpolation tests belong to such a family. The primitive estimates in Sections 6–8 supply this requirement from norms on common argument boxes. After those constructions, Proposition 8.12 collects their quantitative bounds.

Assumption 4.2 (Differentiated primitive data). The primitive packet trees satisfy Definition 3.1 and the following additional requirements.

1. The initial backgrounds are ordinary coordinates or regular analytic functions of them. A raw label in band j is a local smooth function of ordinary inputs and free inputs whose indices are at least j . Its local formula, including lifted branches and any choice of cutoff, is the same on overlapping tests with the same retained determinations. It has no local dependence on a discarded free input. Band 1 functions are holomorphic.
2. Raw growth, its improvement on the band's own side and on safe columns and their stipulated entering collars, and transitions with arbitrarily deep owning-fringe errors hold after any fixed number of independent derivatives. Transitions are differentiated on both sides, including their extracted exponential factors. Every fixed derivative of $\mathcal{A}_z$ applied to a label or a background has the source bounds in Definition 3.1, with a possibly smaller positive constant in the negative exponent. These bounds hold locally up to the edges where both formulas are used. Pole losses satisfy the same growth and source bounds.
3. The preceding assertions also hold on controlled bounded ordinary input paths: their real jets are bounded, their required complex values agree with the corresponding real Taylor continuations to sufficiently high fixed order, and all argument ranges have fixed spare room. Slow real inputs may have their positive-order jets tending to zero. Safe columns and the required entering collars can be chosen simultaneously for every finite collection of labels and derivatives on these paths. Smooth changes of cutoff are made locally in the retained variables, consistently on overlaps; their derivative losses have logarithm $o(U)$ when their precision is e^{-cU} . For a fixed finite request these estimates hold with the controlled-family uniformity of Definition 4.1, on the full forward domains, not only at individually selected target points. These are bounds for the primitive and its independent derivatives evaluated at the controlled inputs. A composed path is used in a source calculation only when its inputs are holomorphic or already have the differentiated source defects of the previous chart depth. Finite Taylor matching by itself gives no exponential bound on a path's anti-CR defect.

4. Supports are iterated left-finite. More explicitly, the possible first coordinates form a set finite below each finite ceiling; at each fixed prefix the possible next coordinates have the same property. Each fixed label and its required derivatives extend to every sufficiently vertical preceding good contour. There is room for finitely many nested polynomial angular buffers and for an increase of the side parameter.

These are hypotheses on actual primitive evaluations, not on a formal series alone. In particular they include safe columns for controlled real detunings after an ordinary chart substitution.

On a current chart let b_j denote the entire background vector in band j , and put

$$Z_{k,j} = Z_k(S, b_j), \quad W_j = e^{-Z_{j,j}}. \quad (4.1)$$

The background invariant is a differentiated transition

$$b_j = b_{j+1} + \sum_{a>0} W_j^a b_{j,a}, \quad (4.2)$$

where coefficients belong to the next band. An equality of this kind means finite truncations with arbitrarily deep errors on the owning fringe. The background has controlled real jets and complex Taylor realizations; at a selected join its difference from the next background is $O(e^{-cV_j})$. On the whole owning fringe the difference is $O(e^{-a_0 \Re Z_{j,j}})$ for some $a_0 > 0$, after narrowing. Terminal backgrounds are analytic ordinary germs.

In Equation (4.2) the extracted factor W_j is evaluated at b_j . It is never silently replaced by its value at b_{j+1} . For example, if

$$Q(S, x) = \exp(\exp(S + x)), \quad x = e^{-S-B},$$

then

$$\log \frac{Q(S, x)}{Q(S, 0)} = e^S (e^x - 1) \longrightarrow e^{-B}.$$

Smallness of x alone therefore does not permit freezing the background of the extracted Q -shift. Only the rates of indices strictly later than the current transition are translated below.

Definition 4.3. A packet expression is *positive in series order* if, in each lateral formal array, every coefficient with lexicographically negative exponent is the zero analytic germ. A *strict positive increment* has a lexicographically positive exponent. These terms refer to exponent order, not to the numerical sign of a real function. The exponent-zero ordinary germ of a positive expression is denoted by K_0 .

Theorem 4.4 (Packet calculus). *Assume the flag estimates of Section 2, the packet separation theorem, and Assumption 4.2. Start with the primitive data and the initial ordinary backgrounds. The following operations can be iterated a finite number of times.*

1. *Raw finite sums, products, and any fixed independent derivatives have differentiated packet trees with iterated left-finite supports. So does multiplication by any fixed real exponential monomial*

$$h^\gamma = \exp \left(- \sum_i \gamma_i Z_i \right).$$

2. *Positive expressions admit cleaned zeroth-prefix realizations K_j . Every fixed ordinary derivative is bounded and converges to the corresponding derivative of K_0 throughout the usable bands. If a mixed independent derivative has earliest differentiated free index l , then for $j \leq l$ its value*

on K_j , multiplied by $e^{Cf(u)}$, tends to zero for every fixed C . Differentiated owning transitions satisfy $K_j - K_{j+1} = O(e^{-c\Re Z_{j,j}})$. Real local uniform boundedness of a raw expression implies positivity; its real limit, when specified, identifies K_0 .

3. A regular analytic operation on finitely many positive expressions, whose limiting tuple belongs to its analytic domain, is positive and has the corresponding differentiated tree. This includes inversion of a unit whose limiting value is nonzero.
4. Let $K(S, t, x)$ be a real square vector of positive expressions on a previous chart. If

$$K_0(t_c, x_c) = 0, \quad \det D_x K_0(t_c, x_c) \neq 0,$$

then the actual regular branch $x = \chi(S, t)$ near (t_c, x_c) and every old raw expression pulled back to that branch have differentiated packet trees. The new backgrounds obey Equation (4.2). The ordinary limiting root is analytic. Band realizations are smooth regular sheets, holomorphic in band 1, with controlled jets, differentiated source bounds, safe-column estimates, and differentiated seam bounds. They retain local independence from discarded free inputs.

For each fixed finite stack of ordinary implicit charts there is a finite coherence jet order, depending on the stack and fixed angular buffers but independent of subsequent derivative orders and transition budgets. Tests on the same free path with the same complex ordinary target and enough common real jets give the same local sheet. Every finite request can then use a larger working Taylor order and farther or narrower working domains; on overlaps these constructions agree. All assertions for a fixed finite request are uniform on controlled families in the sense of Definition 4.1.

All formal operations are performed separately in the two signs, by finite coefficient rules at successive prefixes. They commute with differentiation, regular analytic substitution, and the stated regular implicit solution. Identically zero terminal coefficient germs remain zero under these rules.

The proof occupies the rest of this section. Notice that the theorem makes a local assertion on lifted determinations. It asserts no global single-valuedness over different free-path histories.

4.2 Support algebra and the current shifts

Lemma 4.5 (Finite coefficient rules). *Finite unions and Minkowski sums of iterated left-finite subsets of $\mathbb{R}^m$ are iterated left-finite. Each coefficient in their convolution is a finite sum. If $A \subset \mathbb{R}^m$ is iterated left-finite and consists of strict positives, the semigroup of finite sums of elements of A is iterated left-finite. A fixed exponent has only finitely many ordered representations using nonzero factors from A .*

Proof. For two supports A, B , their first-coordinate sets have lower bounds a_*, b_* . In a sum with first coordinate at most M , the first coordinate from A is at most $M - b_*$ and the one from B is at most $M - a_*$. There are finitely many such choices. After a first-coordinate sum is fixed, apply the same argument to each of these finitely many pairs of fibers in dimension $m - 1$. Induction proves both assertions about convolution.

For the semigroup, induction on m is again appropriate. A positive first-coordinate factor has first coordinate at least some $\delta > 0$, since the set of such coordinates is left-finite. With first-coordinate budget M there are at most $\lfloor M/\delta \rfloor$ such factors, and their first coordinates have finitely many possibilities. Fix one of these finite patterns. By the preceding convolution argument their tails form an iterated left-finite set, bounded below at the next coordinate. All remaining factors have first coordinate zero and strictly positive tails. By the induction hypothesis their tail semigroup is iterated left-finite and has finite ordered multiplicities at any fixed tail. Convolve it

with the bounded-length tail support already obtained. There are finitely many interleavings of two fixed finite lists. This gives both left-finiteness and finite ordered multiplicities. The empty sum represents zero, and zero factors have been excluded, so it does not acquire infinitely many repetitions. $\square$

Consequently an analytic Taylor expansion in strict positive increments is defined coefficient by coefficient. Its constant term is evaluated as an ordinary analytic operation; it is not repeated as a zero factor in the semigroup argument. At one transition a positive local seed has a smallest positive exponent. A fixed owning-coordinate budget therefore requires only finitely many seed factors and finitely many Taylor derivatives. Iterated left-finiteness does not assert finiteness of lexicographic initial segments: for instance $\{(0, n) : n \geq 1\} \cup \{(1, 0)\}$ has an infinite initial segment below $(1, 0)$. Every finite rule above is taken at successive fixed prefixes, as the packet construction requires.

Here is the numerical estimate behind these formal operations. Let a transition for F and one for G be given on a common j -fringe, with remainders R_F, R_G . A fixed raw label on a sufficiently narrowed fringe obeys $|F| + |G| \leq e^{C\Re Z_{j,j}}$. Thus

$$|R_F G| \leq e^{-(M_F - C)\Re Z_{j,j}}, \quad |R_G R_F| \leq e^{-(M_G + M_F)\Re Z_{j,j}}.$$

Choose the finite truncations deep enough for the requested budget and the lower bounds of the supports. Only after the finite list of resulting outer labels is known, narrow the fringe so that each of their logarithmic losses $C'\Re Z_{j+1,j+1} + C'D_j$ is at most $\eta\Re Z_{j,j}$, with any prescribed fixed $\eta > 0$. This is allowed by the owning-fringe geometry and the path separation. There is thus no iteration in which a newly generated outer-label constant forces an endlessly deeper first-coordinate truncation. The product rule gives the same estimate for a fixed number of derivatives, after taking extra depth. Sums are simpler. Their sources are still of the permitted kind because multiplying an e^{-cU} error by factors of logarithmic size $o(U)$ only decreases c . The distinction between cutoff errors and current-join errors in these products is recorded in Lemma 4.7 below.

We next translate later rates. At a free node $k > j$ the value does not change with the background. At a dependent node its exact logarithmic relation gives

$$Z_{k,j} - Z_{k,j+1} = Z_{k,j+1} \left[\exp \left(\sum_{l>k} a_{kl}(Z_{l,j} - Z_{l,j+1}) + b_{k,j} - b_{k,j+1} \right) - 1 \right]. \quad (4.3)$$

Work downwards in k . The bracket has strictly positive local j -order; its value and any fixed derivatives are exponentially small compared with each fixed product of retained later rates. Taylor expansion of the exponential therefore gives its differentiated packet transition. The only numerical scale powers generated by this formula are powers of dependent nodes. They reduce exactly to later shifts, since

$$Z_k^p = e^{pb_k} \prod_{l>k} h_l^{-pa_{kl}}. \quad (4.4)$$

Thus the coefficients in this downward induction are already legitimate outer packet expressions. Applying Taylor's formula once more yields

$$e^{-\gamma Z_{k,j}} = e^{-\gamma Z_{k,j+1}} (1 + \text{strict positive local } j\text{-increments}), \quad k > j. \quad (4.5)$$

Taylor remainders satisfy the preceding product estimates and their differentiated versions. At the current index the factor W_j^γ is simply extracted. Induction from the last transition now constructs the tree of every h^γ and every analytic background factor. The same calculation shows that changes in a retained raw logarithm appearing in an earlier cutoff are absolutely tiny at a seam; its numerical retirement has not been postponed.

Lemma 4.6 (Differentiation of the symbols). *Every fixed derivative of a raw packet expression has the tree obtained by formal differentiation. The series for $D \log W_i = -DZ_{i,i}$ has zeros before coordinate i and has nonnegative local i -exponents. Its later exponents may be negative. In addition, $D_{S_l} W_i = 0$ if $l < i$. Formal differentiation preserves positivity and cannot move a strict positive exponent to a negative or zero exponent.*

Proof. For a free Z_i , its independent derivative is a constant or zero. For a dependent Z_i , triangular differentiation expresses its derivatives as finite sums of products of dependent powers, background derivatives, and lower derivatives. Equation (4.4) handles the powers. There is a possible apparent recursion: differentiating a background transition differentiates the extracted W_j as well. Resolve it first by increasing total derivative order d , and at a fixed d by descending from the later bands. The terms $D^d Z_{j,j}$ are linear in the highest derivatives $D^d b_j$, with coefficients made out of lower derivatives and dependent powers. In the d -fold differentiated form of Equation (4.2), these unknown highest derivatives occur on the right only after differentiation of a factor W_j^a with $a > 0$. The resulting finite system has the form

$$(I - E_j)D^d b_j = H_j,$$

where E_j has strictly positive local order and H_j is already known at this stage. On a sufficiently far narrowed fringe $\|E_j\| < 1/2$. The inverse $(I - E_j)^{-1} = \sum_{n \geq 0} E_j^n$ is a numerical bounded unit and a formal series with finite coefficients by Lemma 4.5. Truncating it gives the required differentiated remainder, since the original background transition was differentiated before being truncated. This proves the derivative-symbol rule without assuming it in the background recursion.

For a full monomial the formal rule is

$$D(c_\lambda h^\lambda) = (Dc_\lambda)h^\lambda + c_\lambda h^\lambda \sum_i \lambda_i D \log W_i, \quad (4.6)$$

read successively at the owning transitions. If p is the first nonzero coordinate of $\lambda > 0$, terms with $i < p$ have multiplier $\lambda_i = 0$; terms with $i > p$ do not alter coordinate p ; and terms with $i = p$ add a nonnegative coordinate there. Hence positivity is preserved. A zero coefficient is an analytic germ identity, so its ordinary derivatives and all its contributions to Equation (4.6) vanish. Differentiating each finite actual transition gives exactly these rules; products and the extra transition depth already established control the remainders. The claim follows for higher derivatives by induction. $\square$

4.3 Negligible defects and cleanup of positive expressions

Lemma 4.7 (Propagation of differentiated source errors). *Fix a finite derivative and composition request. Suppose that each primitive defect and each input-sheet defect in its chain rules has one of the permitted differentiated source bounds, and that every remaining factor has logarithmic size $o(U)$ at the corresponding cutoff source of cost U . Then the compositions have the same source bounds, with smaller positive precision constants. Holomorphic inputs contribute no input-sheet defect.*

At a level- j good join, include all fixed current-level losses in $C\mathbb{R}Z_j$, where $\mathbb{R}Z_j \asymp \varepsilon_j V_j$. For a transition remainder $e^{-M\mathbb{R}Z_j}$, choose the finite budget $M > C$ with the required output margin before fixing the contour. The product then has precision $e^{-(M-C)\mathbb{R}Z_j}$, hence e^{-cV_j} for a positive fixed c on that contour. For an independently established error $e^{-c_0 V_j}$ with $c_0 > 0$ fixed independently of the new contour choice, choose ε_j sufficiently small that $C\mathbb{R}Z_j \leq c_0 V_j/2$. In either case the remaining later-rate and fixed derivative losses of logarithmic size $o(V_j)$ can be absorbed afterwards. For a controlled family, assume that the defect bounds and the $o(U)$ and $o(V_j)$ controls are common for this finite request. Then all these choices are common to the family.

Proof. For an input map $\Psi = (\Psi_a)$, Wirtinger differentiation in any one source coordinate gives

$$\bar{\partial}(F \circ \Psi) = \sum_a (\partial_a F \circ \Psi) \bar{\partial} \Psi_a + \sum_a (\partial_{\bar{a}} F \circ \Psi) \bar{\partial} \bar{\Psi}_a.$$

Here the derivatives of F are in its independent input coordinates. Thus every term contains an input-sheet defect or an ambient primitive defect. After any fixed further derivatives there are finitely many such terms, and each still contains a differentiated defect. Its other factors multiply e^{-cU} by $\exp(o(U))$, which decreases the positive constant c but preserves the allowed precision. The two stated join estimates are the corresponding product estimates with the current loss kept explicit. A fixed $\varepsilon_j V_j$ need not be $o(V_j)$, so reducing the collar alone cannot compensate for a transition budget $M \leq C$. Finite Taylor agreement of an input supplies geometric control only; its differentiated source bounds must already be available for this chain-rule argument. $\square$

We record how the differentiated source errors are used at a transition. For every finite M , after the permitted narrowing and increase of the side parameter,

$$e^{-cU} \exp(o(U)) = O(e^{-M\Re Z_{j,j}}) \quad (4.7)$$

for each source active on either side of the j -transition, including any fixed path-derivative losses. Here is the comparison of scales. For a source from the next band with $f_r \leq f_j^{3/2}$, the angular estimate of Lemma 2.4 gives an absolute $o(1)$ loss in the logarithm of the cost. If a secondary index $k > j$ occurs, its cosine is bounded below by a constant times $(\Lambda_j - \Lambda_k)/\Lambda_j$; the gap estimates pay its logarithm. More explicitly, on a fringe with $\cos(\Im f_j) \leq \varepsilon/\Lambda_j$, the simple and mixed costs respectively satisfy

$$\frac{U}{\Re Z_j} \geq ce^{f_r - f_j} \frac{\Lambda_j}{\varepsilon}, \quad \frac{U}{\Re Z_j} \geq ce^{f_r - f_j} \frac{\Lambda_j - \Lambda_k}{\varepsilon}.$$

The gap derivative tends to infinity; these quotients therefore tend to infinity even when $r = j$. If $f_r > f_j^{3/2}$, the logarithm of the cost dominates f_j even with this loss. On the inner side, a cost with $\sigma(r) = j$ is made sufficiently strong by increasing the side parameter. When its secondary index is j , the quotient of its scale by $\Re Z_{j,j}$ contains the diverging factor $e^{f_r - f_j}$ with the same cosine on numerator and denominator. Earlier leading indices give still greater suppression. These are precisely the allowed source cases. Finally, a fixed derivative of a retained free path of index $l \geq j$ is bounded by $e^{Cf_l(u)}$ by Lemma 2.4; these losses have logarithm $o(U)$. This proves Equation (4.7). In particular a finite smooth Taylor computation may discard its anti-CR terms at any prescribed owning-coordinate precision, but must first bound those terms as above.

Lemma 4.8 (Differentiated cleanup). *Let K be positive. Keep its raw zero-prefix packets $K_j = S_{0,\dots,0}^j$, inserting a zero packet if necessary, and in each of their transitions delete all negative local exponents. The resulting transitions still hold with arbitrarily deep errors and all requested independent derivatives. Their zero-prefix derivative packets are exactly the derivatives of K_j .*

Proof. Fix a finite transition and derivative request. A negative prefix through level j has an entirely zero formal subtree, and every earlier subtree is also zero. The zero-subtree conclusion of Theorem 3.2 therefore bounds its raw outer coefficient by $O(e^{-cV_j})$ on a sufficiently vertical j -collar and the corresponding fringe. Lemma 4.6 says the same about the formal derivative tree of K .

To obtain a bound for actual derivatives of an individual deleted coefficient, retain the raw labels before canceling zero terminal leaves and use induction on derivative order. In the formal expansion of a derivative of K , the contribution at its negative prefix contains that derivative of the coefficient itself. Every other contribution has fewer derivatives on a coefficient at a negative

prefix; its remaining factors arise from differentiated shifts. Positive prefixes cannot contribute, by Lemma 4.6, and the all-zero prefix has no shift derivative. There are finitely many relevant terms at the fixed budget. The induction hypothesis gives e^{-cV_j} for their coefficient derivatives, whereas the remaining factors have logarithmic size $o(V_j)$ on this fringe. Subtracting these terms from the derivative packet proves the assertion for the next derivative order. The anti-CR parts are covered by Equation (4.7).

For the finitely many deleted local exponents $a < 0$, choose the good contour sufficiently vertical that $|a|\Re Z_{j,j}$, the requested $M\Re Z_{j,j}$, and all fixed losses are smaller than, say, $cV_j/2$. Multiplying the deleted coefficients by W_j^a then leaves an $O(e^{-M\Re Z_{j,j}})$ error with the requested derivatives. The coupled leading estimates and the original transition extend this comparison through the usable owning fringe. This proves the cleaned transition. A strict positive earlier exponent cannot feed into a zero prefix on differentiation, so the zero-prefix derivative packet is the actual derivative of K_j , as claimed. $\square$

Lemma 4.9 (Uniform separation for a controlled family). *For one controlled family and one fixed finite coupled calculation, the coupled growth and zero-subtree upgrade constants of Section 3 are common to the family. Leading limits are uniform at its moving targets. Deeper preliminary estimates used to prove a zero-subtree upgrade may have member-dependent constants and starting points, including dependence on the return time to the germ.*

Proof. Definition 4.1 supplies precisely the common finite-label, source, and flag controls in Lemma 3.10. Apply that lemma to the fixed coupled calculation and its finitely many derivative packet trees. Its weighted estimate removes the individual preliminary constants and return times; its terminal comparison uses the common slow-variation modulus near the moving targets. This gives the stated common constants and uniform convergence at moving targets. $\square$

Lemma 4.10 (Bounds, limits, and free derivatives). *The cleaned realizations of a positive expression satisfy all the bounds in part (2) of Theorem 4.4. Conversely, real locally uniform boundedness of a raw expression on an anchored ordinary neighborhood implies positivity.*

Proof. Apply the coupled leading conclusion of Theorem 3.2 with zero prefix to the cleaned expression and its ordinary derivatives. If its exponent-zero coefficient is absent, insert it as zero. This yields boundedness and convergence to the terminal exponent-zero germ K_0 inside the chosen contours; the cleaned transition gives the same result to the polynomial side after increasing its side parameter. There is a strict gap to the next positive local exponent, so that transition and all its fixed derivatives give

$$K_j - K_{j+1} = O(e^{-c\Re Z_{j,j}}).$$

Equality of the two exponent-zero germs follows from separation. All these statements are uniform on the controlled families of Definition 4.1, by Lemma 4.9. In particular this controls families of different tests at different anchors; pathwise eventuality alone is not being used for this step.

Now differentiate in a free input and let l be the earliest index of any free input differentiated. A term whose first active index is after l has no dependence on S_l , because its coefficients use only retained inputs. A terminal coefficient also has no free dependence. Thus every nonzero formal derivative term has a strict positive first index at or before l . In the cleaned zero-prefix packet at level $j \leq l$, the first such index lies between j and l . Normalize by its leading positive shift and use separation. On the relevant domain $\Re Z_k/f_l \rightarrow \infty$ for $j \leq k \leq l$, by Equation (2.20) in Lemma 2.6. Hence that shift absorbs e^{Cf_l} for every fixed C . If there is no nonzero term in this interval, request an arbitrary positive dummy exponent at level l in the zero-subtree estimate. Positive exponents at earlier levels are later in lexicographic order than this dummy request and cause no obstruction.

This proves the asserted decay, also after further ordinary and free derivatives. A product of finitely many free-path derivative factors is at most $e^{C'f_l}$, so the same estimate pays all chain-rule losses.

For the converse, if a negative coefficient is nonzero, iterated left-finiteness supplies a first nonzero negative exponent λ . Choose an ordinary point at which its nonzero analytic coefficient is nonzero. Separation gives $K = h^\lambda(c_\lambda + o(1))$ there. The first nonzero coordinate of λ is negative, and the ordered real scales imply $|h^\lambda| \rightarrow \infty$. This contradicts local uniform boundedness. Once positivity is known, its real limiting germ is its zero coefficient by the first part of the proof. $\square$

This converse is useful after a normalization: multiplication by a monomial is first a raw operation, and its positivity can then be checked by actual real bounds. No positivity assumption on the expression before division is required.

Lemma 4.11 (Regular analytic operations). *Let H be analytic near the limiting tuple of finitely many positive expressions. Applying H to their actual values produces a positive expression with the formal analytic substitution and the differentiated packet estimates. This includes a matrix inverse at an invertible limiting matrix.*

Proof. In a band the cleaned tuple stays in a fixed smaller neighborhood of the limiting tuple by Lemma 4.10. At a transition write it as the outer tuple plus a strict positive increment δ . Taylor's formula with integral remainder gives

$$H(y + \delta) = \sum_{|\alpha| < N} \frac{\partial^\alpha H(y)}{\alpha!} \delta^\alpha + O(\|\delta\|^N).$$

All derivatives of H required by a finite request are bounded on the smaller neighborhood. Since δ and its fixed derivatives are exponentially small on the owning fringe, take N large enough and apply the product estimates to control the differentiated remainder. The coefficients are outer packet operations, to which the same procedure applies recursively. Lemma 4.5 gives the formal finite coefficient rule. Source errors retain their precision under bounded derivatives of H , and finite intersections of the primitive safe sets suffice. The actual result is bounded on the real neighborhood and therefore positive by Lemma 4.10. $\square$

4.4 The real implicit branch and its trial inputs

It remains to prove regular implicit closure. A dependent exponential may rotate many times on a fixed complex disk in an ordinary variable, so we cannot assume that such a disk is an admissible implicit domain. We will first solve the real equation, then use its Taylor jets to construct complex roots on smaller domains.

Proceed by induction on the number d of regular ordinary charts already constructed. At depth d assume there is an integer r_d , fixed by the stack and its angular rooms, independently of later derivative orders and transition budgets, such that tests on the same free path, with a common complex ordinary target and common real input jets through r_d , give the same local stack of solved sheets. On each test the solved variables agree with their real Taylor jets to every prescribed finite order on sufficiently far working subdomains. They have all the required fixed path and independent derivative bounds, differentiated sources, and exponentially small seams. These assertions are uniform for every fixed finite request on controlled families. At depth zero they are the primitive local-determination and controlled-path hypotheses.

Consider $K(S, t, x) = 0$ as in Theorem 4.4, and put $J_0 = D_x K_0(t_c, x_c)$. Shrink the ordinary neighborhood so that $D_x K_0(t, x)$ stays uniformly close to the invertible matrix J_0 . By Lemma 4.10,

the real equation for sufficiently far real inputs has a unique nearby branch $x = \chi(S, t)$. For example, the map $x \mapsto x - J_0^{-1}K(S, t, x)$ contracts a fixed sufficiently small ball and maps it into itself, because K and $D_x K$ converge to K_0 and $D_x K_0$. The real analytic implicit-function theorem then gives the actual branch. Differentiating its exact equation yields bounded fixed ordinary derivatives. Along a slow real test its positive-order path jets tend to zero: each chain-rule term either contains a small ordinary path derivative or a free derivative paid by Lemma 4.10.

To continue this branch in band j , we will solve $K_j = 0$ near a high Taylor value of χ . Before estimating that equation, we need the old chart at every trial input in a small tube about the Taylor value. The following interpolation keeps the lower real jets which identify the old sheet, while attaining each complex trial input exactly.

Lemma 4.12 (Interpolation with fixed lower jets). *Fix an integer r and a finite higher jet order. Let $v \neq 0$ be small. Suppose a bounded ordinary input has prescribed real jets at u through that higher order, with its positive-order jets small, and the proposed complex target differs from the associated high Taylor value at $u + iv$ by $O(|v|^R)$, where $R > r + 2$. After initially matching sufficiently many jets, the target can be attained exactly by a bounded real-symmetric holomorphic test that keeps all jets through order r . The test returns to the prescribed ordinary germ on the real ray. Its additional coefficients tend to zero as $v \rightarrow 0$. For a family with common finite jet bounds, target-error constants, and ordinary slack, the interpolants have common bounds for every fixed requested number of derivatives on a full forward strip. If the prescribed positive real jets and $|v|$ tend uniformly to zero, these derivative bounds have a common vanishing modulus. Their times of return to a smaller neighborhood of the germ need not be uniform.*

Proof. It suffices to treat one scalar component. On a fixed narrow strip put

$$B(s) = 1 - \tanh^2(c_0(s - u)), \quad \phi_n(s) = B(s) \tanh^n(c(s - u)),$$

where $c > 0$ is fixed and small, and $c_0 > 0$ can be made still smaller. Both functions are holomorphic on a strip with fixed extra room, real-symmetric, bounded there, and $\phi_n(s)$ tends to zero as $s \rightarrow +\infty$ on the real ray. Its Taylor coefficient of degree n at u is $c^n \neq 0$, and lower coefficients vanish. A finite triangular linear system therefore matches any prescribed finite real jet. A real value offset from the germ is supplied by B ; its positive derivatives are made small by choosing c_0 small. The coefficients matching the remaining positive-order jets are small. Taking the ordinary neighborhood smaller preserves the argument range, with slack.

Let n_e, n_o be the first even and odd integers greater than r ; then $\max(n_e, n_o) \leq r + 2$. At $s = u + iv$, B is real and $\tanh(civ) = i \tan(cv)$. Thus $\phi_{n_e}(u + iv)$ is real nonzero and $\phi_{n_o}(u + iv)$ is purely imaginary nonzero. Two real coefficients therefore correct independently the real and imaginary target errors, without changing the prescribed lower jets. Their sizes are at most

$$C|v|^{R-n_e} + C|v|^{R-n_o} \leq C'|v|^{R-r-2} = o(1).$$

If required, start with a still higher matched jet so its own Taylor remainder is covered by this target error. All corrections decay along the ray, so the return to the germ is unchanged. The construction applies componentwise to vector inputs.

We check the full-domain quantitative assertion. Fix a strip $|\Im(s - u)| < \sigma$ strictly inside the poles for the chosen fixed c , and keep $0 < c_0 \leq c$. For every fixed pair of integers L, k , the functions $B \tanh^n(c(s - u))$, $n \leq L$, and their first k derivatives are bounded there by constants depending only on L, k, c, σ . These constants are independent of u and c_0 . Derivatives of positive order of B alone are bounded by $C_k c_0$. The diagonal entries of the finite jet matrix are $n!c^n$, independent of c_0 ; its other entries are bounded. Its inverse therefore has a common bound for $0 < c_0 \leq c$.

Write a_0 for the value offset from the germ and let $a_1, \dots, a_L$ be the prescribed positive-order real jets. The initial jet-matching coefficients satisfy

$$\max_{1 \leq n \leq L} |b_n| \leq C_L \left(\max_{1 \leq n \leq L} |a_n| + c_0 |a_0| \right).$$

The two final correcting coefficients have the uniform bound already proved, since $|\phi_n(u+iv)| \geq c_n |v|^n$ for sufficiently small $|v|$, uniformly for $0 < c_0 \leq c$. Consequently the final interpolant T obeys

$$\sup_{\substack{\Re s \geq u - B_* \\ |\Im s| < \sigma}} |\partial_s^k T(s)| \leq C_{L,k} \left(c_0 |a_0| + \max_{1 \leq n \leq L} |a_n| + |v|^{R-r-2} \right), \quad k \geq 1. \quad (4.8)$$

Here B_* is any fixed backstep, and L is first chosen large enough for the fixed jet and remainder request. The value bound is common as well. On the real line the value-offset term lies on the segment from the germ to the prescribed value; on the narrow strip its excess and the remaining small terms are absorbed by the fixed ordinary slack. Taylor's integral remainder and the next derivative bound in Equation (4.8) give common complex Taylor-error constants. Choosing c_0 with a common modulus tending to zero proves the claimed vanishing derivative modulus on the entire forward domain, while every individual bump still decays to zero at infinity. This proves membership in Definition 4.1 whenever the free paths satisfy its common flag controls.

No derivative of the interpolation coefficients with respect to the complex target is used. Those derivatives can contain powers of $|v|^{-1}$. The estimates concern derivatives of the test as a function of s ; intrinsic derivatives on the old graph are instead obtained from its already established packet or implicit equations. $\square$

Band domains can be prescribed before the new complex root is known. Use the actual free paths and sufficiently high Taylor jets of the real backgrounds as reference backgrounds. By Lemma 2.7, on $|v| \leq C/\Lambda_i(u)$ a perturbation $\Delta b = O(|v|^q)$ changes a relevant lifted logarithm by at most

$$|\Delta f_i| \leq C \Lambda_i (1 + \log(2 + \Lambda_i))^C |v|^q. \quad (4.9)$$

The estimate holds throughout the short perturbation segment. Its triangular proof uses that the derivative contributions from the children of a dependent node are dominated by its first child $Z_{\sigma(i)} \Lambda_{\sigma(i)}$; a free node has no background variation. Choosing q large makes the right side smaller than the polynomial angular buffer and the fixed good-collar width. We may use several strictly nested rooms. This makes the reference domains independent of an unknown implicit continuation.

The interpolation lemma supplies a controlled family of trial inputs, rather than separate tests with unrelated eventual starting points. At depth d , its fixed lower-jet agreement identifies one old graph throughout the trial tube. The construction below preserves this controlled-family assertion using the finite coefficient rules, uniform separation, and common regular inverse bounds.

The coherence order identifies the local sheet. A larger working Taylor order will control the accuracy of a particular finite calculation. Increasing that order must leave the identified sheet unchanged.

4.5 Complex roots and coherence

First choose a broad comparison-tube order R using only r_d , Equation (4.9), and first-derivative control of the finitely many old equations. Choose it with a finite margin, in particular larger than the jet orders needed for interpolation at the old depth. This choice will determine the next

coherence threshold. Only now, for a given finite request, choose an arbitrarily larger working Taylor order N and put

$$p_N(v) = \sum_{a=0}^{N-1} \frac{\chi^{(a)}(u)}{a!} (iv)^a, \quad (4.10)$$

where these are real path derivatives along the actual composite real branch. The old chart can be evaluated at every point of $\|x - p_N(v)\| \leq C_0|v|^N$ and on the broader comparison tube of order R . Indeed interpolation gives the required input tests, and old coherence puts them on the same old graph. The older solved variables have common lower Taylor centers since their real equations have the prescribed common real jets. The old regular Jacobians give one graph throughout these nested tubes, including the segments between trial points. This statement uses no new root. For a controlled family the prescribed real solution jets have the common bounds and vanishing moduli just obtained by real implicit differentiation. Equation (4.8) therefore puts all these trial inputs, for all its anchors and targets, in one controlled family at the old depth. The old derivative estimates and limiting Jacobian comparison apply uniformly to the entire trial tube. Only the finite request fixes N and the numerical far thresholds; the individual interpolating test does not change those choices.

Consider the residual on that old graph,

$$R_j(v) = K_j(S(u + iv), t(u + iv), p_N(v)).$$

Its derivatives in v through order N are bounded on each used band, with one bound for the controlled family at this fixed request. The ordinary chain-rule factors are bounded, and every free-path factor is absorbed by the independent free-derivative estimate of Lemma 4.10. In the first band the residual is holomorphic near zero, and its first N Taylor coefficients vanish because the real equation and the jets in Equation (4.10) agree. At a selected join v_i , the residual and all its requested derivatives have exponentially small jumps, even after the finite path-factor losses.

For clarity, repeated integration on the successive intervals gives the following elementary piecewise Taylor estimate. If $R^{(a)}(0) = 0$ for $0 \leq a < N$, $\|R^{(N)}\| \leq C$, and its derivative jumps at v_i are $\Delta_{i,a}$, then

$$\|R(v)\| \leq \frac{C|v|^N}{N!} + \sum_{i: |v_i| \leq |v|} \sum_{a=0}^{N-1} \frac{\|\Delta_{i,a}\| |v - v_i|^a}{a!}. \quad (4.11)$$

Each seam bound is $o(|v_i|^{N-a})$, because exponential seam precision beats every fixed power of its angular location. Since $|v - v_i| \leq |v|$ on a fixed lateral vertical segment, each summand is $O(|v|^N)$. There are finitely many joins. Consequently

$$R_j(v) = O(|v|^N) \quad (4.12)$$

throughout the band up to its used side. This argument uses high real jets and bounded derivatives, not a Cauchy estimate on a fixed ordinary disk.

Identify $\mathbb{C}^{\dim x}$ with a real vector space, and write $J_0^{\mathbb{R}}$ for the real representation of J_0 . On the entire broad comparison tube the ordinary derivative converges uniformly to the corresponding derivative of K_0 , and the anti-CR part is uniformly negligible. The difference from J_0 is the sum of this convergence error and $D_x K_0(t, x) - J_0$; the second is uniformly small on a sufficiently small fixed ordinary neighborhood. Consequently a common far threshold gives

$$\|I - (J_0^{\mathbb{R}})^{-1} D_x^{\mathbb{R}} K_j\| \leq \frac{1}{2}. \quad (4.13)$$

The map $T_j(x) = x - (J_0^{\mathbb{R}})^{-1}K_j(x)$ contracts there. By Equation (4.12), for a sufficiently large fixed C_0 it maps the closed ball of radius $C_0|v|^N$ about $p_N(v)$ into itself. It has exactly one root x_j in that ball. Furthermore any two roots in the broad tube coincide, by integrating Equation (4.13) on their connecting segment. The first-band root is holomorphic by the holomorphic implicit-function theorem.

At any nonzero band point the ordinary smooth implicit-function theorem gives a local smooth continuation of this root. Such a neighborhood may be arbitrarily small. The real jet centers vary continuously with the base point; the available lower-order tube has slack. The local continuation therefore remains in the same broad tube and agrees with the tested root there by uniqueness. This proves that the constructed objects form smooth local sheets, rather than unrelated pointwise roots. Near $v = 0$ the actual real and first-band implicit branches give the same conclusion. At the terminal edge the limiting analytic equation can be evaluated first at x_m , and its ordinary regular inverse yields x_{m+1} ; no extension of an old band sheet past that edge is required.

To check coherence, take two tests on the same free path, with the same complex ordinary target and enough common real input jets. Their real implicit solution jets agree to the corresponding order, since successive differentiation of the regular equation determines those jets uniquely. Choose this fixed order high enough that both roots lie in the same broad tube of order R . Old coherence and interpolation identify the previous sheets there, including on the segment between the roots. Equation (4.13) then identifies the new roots. This defines a finite r_{d+1} using only r_d , the angular margin, and first-derivative regularity. A later construction with larger N lies in the same broad tube sufficiently far out, so it gives the same root. Thus branch identification does not consume the arbitrary derivative accuracy requested afterwards.

4.6 Differentiated sheets and their transition series

We give the remaining estimates for the new graph explicitly. For an independent real coordinate y , its exact first derivative satisfies

$$D_y x_j = -(D_x^{\mathbb{R}} K_j)^{-1} D_y K_j. \quad (4.14)$$

For a higher multi-index α , differentiating the equation gives

$$D_x^{\mathbb{R}} K_j D^\alpha x_j = -D_y^\alpha K_j - \mathcal{P}_\alpha,$$

where $\mathcal{P}_\alpha$ is a finite sum of products of old derivatives and lower derivatives of x_j . The inverse is uniformly bounded by Equation (4.13). Induction proves bounded ordinary derivatives. If a free derivative with earliest index l occurs, each term contains either an old derivative with such a free index or a lower graph derivative with that index; induction and Lemma 4.10 give decay sufficient to pay e^{Cf_l} for any fixed C . Applying $\mathcal{A}_y$ to the exact equation gives the same invertible real linear system with right side consisting of old anti-CR defects. Higher differentiated defects follow from the same finite induction: each term contains an old differentiated defect, and its other factors have the established absorbable losses. Lemma 4.7 therefore preserves the old source precision after decreasing its exponent constant. Replacing the real Jacobian by its complex-linear part changes these formulas only by errors of that precision.

At a j -join compare the two equations on their common old graph. Evaluate the outer one at the inner root. The cleaned transition gives

$$K_{j+1}(x_j) = K_{j+1}(x_j) - K_j(x_j) = O(e^{-c\Re Z_{j,j}(x_j)}).$$

The segment from x_j to x_{j+1} lies in the comparison tube. Its Jacobian is uniformly regular, so

$$d_j := x_j - x_{j+1} = O(e^{-c\Re Z_{j,j}(x_j)}). \quad (4.15)$$

For derivatives, subtract Equation (4.14) and its higher versions on the two sides. Differences of old derivatives at the same input are bounded by their differentiated transition; their differences at x_j and x_{j+1} are bounded by the next old derivative integrated on the segment. The inverse difference is $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. Induction therefore proves Equation (4.15) with every requested independent derivative, using extra transition depth to pay fixed triangular losses. This is a differentiated equation argument; it does not differentiate a pointwise smallness assertion. The resulting path-derivative seam bounds and Equation (4.11) also give arbitrary fixed Taylor accuracy for the derivatives of the new sheets.

At band j , all the equations on the old graph use only retained free inputs. Differentiating in a discarded free direction has zero right side in Equation (4.14); uniqueness therefore gives zero derivative of the new graph in that direction. This proves local retained-input independence. A discrete continuation choice may have used the free path, but it introduces no local derivative and is fixed on the lifted germ.

We now construct the transition series, keeping

$$W'_j = W_j(S, t, x_j)$$

as the current extracted symbol. All new outer coefficients are evaluated at x_{j+1} . Taylor-expand the old outer coefficients along the segment from x_{j+1} to $x_j = x_{j+1} + d_j$. Smooth complex Taylor expansion uses D_x ; the terms containing anti-CR derivatives have arbitrarily deep fringe precision by Equation (4.7). The segment is exponentially short. Its rate and retained cutoff logarithms are stable by the triangular estimates, so fixed losses are absorbable relative to $\Re Z_{j,j}(x_j)$. Let $J = D_x K_{j+1}(x_{j+1})$. The finite Taylor equation, to any specified budget, has the form

$$0 = Jd_j + \sum_{h \geq 2} B_{0,h}[d_j^h] + \sum_{a > 0, h \geq 0} (W'_j)^a B_{a,h}[d_j^h]. \quad (4.16)$$

Brackets denote the corresponding symmetric multilinear evaluations in the vector case; both sums are truncated as required, with a remainder whose independent derivatives have the desired precision. This follows by applying the old differentiated transition at x_j and Taylor's integral remainder on the common segment, using the already proved differentiated smallness of d_j .

Solve Equation (4.16) formally for

$$d_j = \sum_{a > 0} (W'_j)^a d_{j,a}. \quad (4.17)$$

Multiply by J^{-1} . Every term other than the linear term either contains a positive seed exponent or contains at least two factors of d_j . The positive seed support has a smallest positive local exponent and is left-finite. At a given owning budget, its number of factors and the depth of the recursion are bounded. Thus Lemma 4.5 gives a unique finite coefficient calculation for each $d_{j,a}$. Every such coefficient uses finitely many old outer coefficients and derivatives, and the bounded unit J^{-1} . The inverse itself has an outer transition with the same limiting inverse and strict positive increments by Lemma 4.11.

To verify that the formal answer describes the actual graph, truncate Equation (4.17) at a sufficiently deep finite order and call its value d_* . Crucially, evaluate d_* at the same current numerical W'_j as the exact equation. Formal cancellation and the finite remainder estimates show that d_*

satisfies the finite polynomial equation with an error of the requested differentiated precision. The exact d_j satisfies it with the same precision. Subtracting yields

$$(J + E)(d_j - d_*) = R, \quad \|E\| \rightarrow 0,$$

where R has that precision with all requested independent derivatives. The bounded inverse and repeated differentiation of this identity give the same precision for $d_j - d_*$. Fixed derivatives of W'_j incur only the previously established triangular losses; taking additional initial depth pays them. This proves the actual differentiated transition series without assuming that pointwise asymptotics can be differentiated.

For an arbitrary old raw packet F , apply its old transition at x_j and Taylor-expand its outer coefficients at x_{j+1} using Equation (4.17). Its negative exponents only require finitely many extra orders at any fixed prefix. The resulting outer coefficients again use the same finite grammar. In particular $b'_j = b_j(S, t, x_j)$ has the strict positive background transition. Terminal roots solve the common regular analytic equation and therefore give analytic coefficient germs.

Finally, the new raw growth, sources, and safe estimates follow from these finite formulas and the strengthened primitive condition. At the bottom of a formula its non-unit raw factors are primitive evaluations on the bounded controlled coordinates of the stack. The real coordinates are slow, the complex coordinates have the just-proved high real-jet control, and finite simultaneous safe columns are available by Assumption 4.2. Positive factors and inverses are uniformly bounded on the comparison tubes by interpolation and the regular Jacobians. Fixed derivative losses have already been paid in the source comparison. This completes the induction on chart depth and the proof of Theorem 4.4.

4.7 Formal compatibility and refinement

For precision we collect the formal conclusion used in the zero argument. At every finite prefix, addition and multiplication are finite convolution rules, differentiation is Equation (4.6), analytic substitution is Taylor expansion in strict increments, and a regular ordinary implicit substitution is the uniquely solved recursion in Equation (4.16). These descriptions hold separately in the two signs and coincide with the actual differentiated transitions. The ordinary exponent-zero coefficients are evaluated on the unique regular analytic limiting branch. Hence they respect analytic germ identities, including identities obtained by differentiation. Applying the same well-defined operations to both sides preserves a formal identity. In particular differentiation, multiplication, and pullback preserve a formally zero expression, and Theorem 3.2 identifies the resulting actual zero. This statement concerns the full packet tree; it does not identify unrelated numerical realizations by discarding a flat error.

Lemma 4.13 (Retesting a finite chart recipe). *Enlarge a saturated flag by finitely many permitted insertions without converting its old active free nodes. Suppose the primitive data can be retested on this flag with Assumption 4.2, the same terminal analytic germ recipes and lateral determinations, and identity transitions at unused new nodes. Then any previously constructed finite ordinary chart recipe can be retested on the refinement. Its real expressions on the old anchored inputs are unchanged, and its positivity, normalizing divisions, regular implicit steps, and formal compatibility remain valid.*

Proof. Retest the old primitives before making a new ordinary substitution. Initially the bounded parameters introduced by the refinement are passive ordinary inputs. Old free values at the real anchor are unchanged, and the actual real formulas and their ordinary derivatives are the same.

Proceed through the finite old recipe. A raw operation is available by Theorem 4.4. Every previously positive normalization is still bounded on its real ordinary neighborhood, so Lemma 4.10 proves its positivity on the refined flag. Its old ordinary limit identifies its exponent-zero germ. Thus a previously regular analytic or implicit step retains its limiting Jacobian and can be performed by the same theorem. The additional passive inputs are held fixed in these comparisons. Induction proves the assertion for the entire recipe. Unused inserted levels contribute only identity transitions, and later pullback introduces their actual dependence in the prescribed order. Preservation of the old free nodes is supplied by Lemma 2.2 in the applications. $\square$

The hypothesis about primitive terminal germs in Lemma 4.13 is substantive. It is verified below by fixed formal recursions, compatible bounded-charge coefficients, or canonically normalized infinite-anchor germs. The chart calculus preserves that property; it cannot manufacture it from an unknown exponentially small ambiguity.

5 Ordinary elimination and absolute isolated zeros

We now turn separation into a finiteness statement for isolated zeros. Given an escaping sequence of zeros, we construct charts through its anchors on which the equations vanish and every original free flag input survives. A sufficiently small change of a recoverable free input then gives a different solution near the original one, within the same retained open domain. The final step retests the zero formal data on the perturbed sequence; Theorem 5.7 states the resulting criterion, including the ordinary analytic alternative and the required closure operations.

The induction decreases the number of ordinary variables or, at fixed dimension, the order of a limiting zero. Its main difficulty is that restricting to a derivative center need not preserve a zero of the original field. Already for $G(y, z) = z^2 + c(y)$, the derivative center $\partial_z G = 0$ is $z = 0$, but a zero with $z \neq 0$ has nonzero center value $G(y, 0) = c(y)$. We therefore prove the exact-zero assertion together with two quantitative alternatives for a center field that need not vanish: one normalizes a value that is at least a fixed power of a chosen small scale, and the other produces a chart on which the field is bounded by a prescribed power of that scale. These alternatives allow the induction to return from the center to the original anchors.

The return to an anchor must be exact. The first two subsections give the estimates for lifting a center chart and correcting its approximate hit. Their key requirement is that the powers controlling derivatives, inverses, and neighborhoods be fixed before the requested approximation accuracy is increased. Proposition 5.5 then carries these estimates through the ordinary-variable induction.

Throughout this section the large variables are independent flag inputs. In particular, a parameter of an original equation is allowed to vary when testing whether a zero of the full system is isolated. Every monomial is evaluated at the *current* physical backgrounds of the chart. The regular implicit calculus of Theorem 4.4 and the separation theorem are used after any necessary retest of the primitive packets on an enlarged saturated flag.

5.1 Quantitative chart conventions

Fix a sequence of real anchors, pass to subsequences when making finite comparisons, and write (S, t) for free large inputs and ordinary inputs with an interior limiting value. A subscript A labels the selected sequence of anchors. The separately requested record accuracy is $\mathcal{A}$; constants and ordinary germ neighborhoods may depend on $\mathcal{A}$, whereas power exponents declared uniform do not. A chart recipe is a finite sequence of regular ordinary implicit changes, monomial scalings, and the promotions defined below. Widths used in these substitutions are current monomials of

strict positive order times bounded regular units with positive limiting value. Equations defining its regular implicit changes have positive series and an invertible exponent-zero Jacobian. All analytic evaluations are on bounded arguments with regular limiting values. The starting chart and its positive expressions, with their derivatives, are regarded as given. Estimates for a recipe concern only the changes it adds to that starting chart.

An ordinary function will be called *positive* when its two formal series have no nonzero coefficient at a negative lexicographic exponent; this terminology does not prescribe its numerical sign. Its fixed ordinary derivatives are bounded and converge to their ordinary exponent-zero germs by Theorem 4.4. Conversely, boundedness on an anchored real regime and an ordinary neighborhood proves positivity. We use this converse for normalized quotients; numerical boundedness is not, by itself, an assertion of analyticity at a zero rate.

For a positive numerical scale $r \rightarrow 0$, a finite recipe is *power moderate in r* if every inverted monomial d satisfies

$$|d| \geq r^C \quad (5.1)$$

for some fixed C , and its finite required forward derivatives, inverse Jacobian, and inverse neighborhoods at its hits have power bounds in r . Different finite requests may have different C . Thus a source ball of radius cr^N , an image ball of radius $c'r^{N'}$, and an inverse Jacobian norm at most $C'r^{-C}$ are permitted. Constants and starting points along the sequence may depend on the request. Comparisons of monomials are evaluated at the augmented reference anchor or at points proved sufficiently close to it. A monomial need not have bounded ratios at different ordinary inputs in a fixed ordinary box.

There are two kinds of additional coordinates. Added free scales and bounded residuals which enter active changes are recorded as outputs R of the chart. Temporarily independent bounded residuals will be called passive records. An old coordinate is consumed when its replacement is made; it and its replacement are not both counted as active recursion variables. Exogenous inputs e describe a later flag extension or a pending thickness. They are held fixed in the recursion, and their own coordinates are not included among the records R whose dependence on e will be made small. This designation is relative to the pending center recipe; it does not remove an ordinary argument of the field at the start of a recursive call from the active count. After a completed lift, every replacement ordinary input on which the field depends is counted in the next call. When taking local inverse Jacobians at fixed original free inputs, new free coordinates are counted among the coordinates of that inverse, using absolute increments.

Lemma 5.1 (Promotion and logarithmic sensitivities). *A nonzero ordinary coordinate $x_A \rightarrow 0$ can be replaced, after a saturated flag extension, by*

$$x = \varepsilon \rho, \quad \rho = h^\nu e^{-B} > 0, \quad \varepsilon \in \{1, -1\}, \quad (5.2)$$

where h^ν has strictly positive order. A bounded log remainder B is retained whenever it is the sole replacement degree of freedom, even if its values at the anchors are zero. It may be omitted only when a free residual already supplies the replacement coordinate. The replacement has a single new degree of freedom ξ , either a free large variable or a bounded ordinary residual. It preserves the old free inputs and does not increase the number of active ordinary inputs. With physical ordinary variables and all other records held fixed, and $M = |\log \rho|$,

$$1 \leq |\partial_\xi \log \rho| \leq C(1 + M)^C. \quad (5.3)$$

Let m be a fixed exponential monomial satisfying $\rho^C \leq |m| \leq \rho^{-C}$ for some fixed C at the reference anchors. Every fixed derivative of $\log |m|$ in physical ordinary backgrounds, bounded records, or

absolute free increments has a power bound in $1+|\log \rho|$ on sufficiently small neighborhoods preserving the saturated comparisons, using the bounded background jets supplied by the packet calculus. These are the independent physical or prefix coordinates in which those jets are bounded; pullback through a power-conditioned prefix may introduce its stated power loss in ρ . In particular,

$$\rho \partial_x \log \rho = o(1). \quad (5.4)$$

Equation (5.2), with current backgrounds depending on x , is therefore a regular implicit equation in x .

Proof. Expand $-\log |x_A|$ against the ordered old large scales, subtracting successive limiting multiples. The residual is either bounded or one new independent positive large scale, with coefficient 1 or -1 . Convert only the new residual's chain of logarithms until it meets the old cuts. Lemma 2.2 proves finite termination, preservation of old free inputs, and the asserted ordinary count. In the bounded case take the residual as B . If a new free residual survives, take its final free member as ξ .

At a direct bounded or unconverted residual the derivative in (5.3) has absolute value 1. Each logarithm conversion replaces that derivative by its product with the positive residual scale and a sign. The other summands are old scales held fixed for this explicit derivative. There is only one new chain, so its derivative has no competing summands. Every scale encountered is $O(M)$ at the anchor. The finite product is therefore at least 1 and at most a fixed power of $1 + M$.

For the sensitivity assertion, a fixed two-sided moderate monomial satisfies $\log |h^\alpha| = -\sum \alpha_i Z_i$. Its largest nonzero scale controls this sum on a saturated ordinary neighborhood. If it and its inverse are moderate in ρ , that largest scale is $O(1 + |\log \rho|)$. Triangular differentiation of $\log Z_i = \sum_{j>i} a_{ij} Z_j + b_i$ introduces only finite products of scales at or below the largest scale involved, and bounded derivatives of the current backgrounds. The same assertion follows inductively for each fixed higher derivative. A nonnegative-order monomial with a power-moderate inverse satisfies the required two-sided bounds; any bounded nonvanishing ordinary factor is handled by its bounded jets. This proves the stated bound for the small moderate factors and their reciprocals. Absolute increments of a free scale have derivative 1 at that scale, and cause no additional loss in the triangular induction. Shrinking to an absolute power-sized box preserves all comparisons. Equation (5.4) follows because ρ beats every fixed power of $|\log \rho|$. The derivative of $x - \varepsilon \rho(S, t, x)$ in x tends to 1; its right-hand side is positive in the series sense. Theorem 4.4 applies. $\square$

The logarithmic sensitivity estimate is used only for monomials satisfying two-sided power bounds at the scale under consideration. A positive function or monomial much smaller than every power of that scale is handled as one positive expression with bounded fixed derivatives, without separating the large factors in a formal product rule.

Lemma 5.2 (A quantitative local inverse estimate). *Suppose a map Φ has, on a ball about q_0 , bounds*

$$\|D\Phi(q_0)^{-1}\| \leq Cr^{-c}, \quad \|D^2\Phi\| \leq Cr^{-b},$$

and the available source ball has radius at least $c_0 r^a$. There are source and image balls of fixed power radii on which Φ is invertible. Their exponents depend only on a, b, c , and a discrepancy $O(r^T)$ within the image ball is corrected by a source displacement $O(r^{T-c})$. For inverse derivatives through an order $k \geq 1$, assume in addition power bounds for the forward derivatives $D^j \Phi$ through order $\max(k, 2)$ on that source ball. Then those inverse derivatives have fixed power bounds. A perturbation has the same conclusions if it is sufficiently power close in $C^{\max(k, 2)}$ on the same available source ball.

Proof. Choose $N > a$ and $N > b + c$. On a ball of radius ηr^N , for a sufficiently small fixed η , the norm of $D\Phi(q_0)^{-1}(D\Phi(q) - D\Phi(q_0))$ is at most $1/2$. The map

$$q \mapsto q + D\Phi(q_0)^{-1}(p - \Phi(q))$$

is a contraction preserving that ball whenever $|p - \Phi(q_0)| \leq c_1 r^{N+c}$. Its fixed point gives the inverse and the displacement estimate. Differentiating the inverse equation expresses each fixed derivative in finite products of the inverse Jacobian and forward derivatives through the requested order. The additional hypotheses give fixed power bounds for these products. The inverse perturbation identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ transfers the estimates to a sufficiently close map. This argument also supplies explicit slack for smaller source and image balls. $\square$

We call a local map *power conditioned in a width r* if its requested finite forward derivatives and inverse Jacobian have power bounds in r , it is defined on an available source neighborhood of power radius, and its restriction there has an inverse on an image neighborhood of power radius, as in Lemma 5.2. The coordinates and absolute free increments are those fixed above.

5.2 Lifting a finite center recipe

Lemma 5.3 (Taylor replay at current backgrounds). *Let an ambient chart have ordinary variables (y, w) near $w = 0$. Suppose a full-dimensional regular recipe on $w = 0$ has been constructed and is power moderate in a width ρ , with its records included among its outputs. All necessary flag extensions and passive records may be included before this recipe is used. Then the recipe can be lifted to either $w = pu$, for bounded ordinary u , or a specified promotion $w = \varepsilon\rho$ with replacement coordinate ξ . At common independent inputs the lifted lower map and record outputs approximate the center recipe to $O(\rho^T)$ in any specified finite C^k norm, for arbitrary T .*

For a thickness substitution, the resulting full map is power conditioned. For a promotion the same holds if the inverse center records at fixed physical y have arbitrarily power-small derivatives in its exogenous promotion coordinates. Calibrated approximate hits whose errors are sufficiently power small can be corrected to exact hits. Only finitely many monomial denominator powers are used for each choice of T, k .

Proof. Include every flag node and passive residual needed for the finite recipe from the start. An old recipe not using these additions is revalidated in its original order by the real-bound criterion in Theorem 4.4; old free inputs remain independent. First solve the equation for w with its current backgrounds, using Lemma 5.1. Replay the old coordinate replacements subsequently in their chronological order.

Here is the finite algebra used in this replay. Expand every needed active-coordinate derivative by the chain rule and every derivative of a solved coordinate by its implicit Jacobian formula. The result uses a finite jet of base positive functions and monomials, previous solved coordinates, regular Jacobian inverses, and the finite list of moderate monomial denominators. For a base positive function A replace its center value by the reverse Taylor polynomial

$$\mathcal{T}_J A(S, y, w) = \sum_{j=0}^{J-1} \frac{(-w)^j}{j!} \partial_w^j A(S, y, w). \quad (5.5)$$

The derivatives here are in the ambient independent coordinate before substitution. On $|w| = O(\rho)$, Taylor's formula with integral remainder and the bounded derivative calculus gives $\mathcal{T}_J A - A(S, y, 0) = O(\rho^J)$; after a prescribed finite number of independent derivatives and substitutions the loss is only a fixed power, paid by increasing J .

Throughout the replay and the common-domain argument below, first absorb the reciprocal of every negative-order inverted monomial, together with all its required jets, as a whole base positive expression. This is an exact rearrangement of the expression and leaves the recipe and its actual map unchanged. Every divisor requiring a relative Taylor correction, and every generator retained for such relative comparisons, may therefore be taken to have nonnegative order and a power-moderate inverse. Each satisfies the two-sided bound in Lemma 5.1. A zero-order monomial is constant in the exponential scales; any accompanying bounded nonvanishing regular factor is handled by its bounded jets.

For such a two-sided moderate divisor d , factor its Taylor correction as the current d times a unit. To justify the factorization relatively, put $M = 1 + |\log \rho|$. At common other inputs, the logarithmic sensitivities of d and ρ are $O(M^C)$, and their logarithms therefore change by $O(\rho M^C) = o(1)$ on the segment from w to 0. A continuity argument, begun at either endpoint, keeps ρ comparable throughout that segment. Thus $d(S, y, 0)/d(S, y, w) = 1 + o(1)$. Taylor remainder estimates divided by the moderate lower bound give arbitrary power accuracy for this unit and its required derivatives. Its inverse is a regular positive operation. Bounded monomials without a moderate inverse are treated as whole base positive functions together with their finite jets. Passive records occurring in such monomials are independent arguments until the stage of the replay which consumes them.

This flattening is finite even for a recipe that itself resulted from earlier lifts: a previous Taylor formula, derivative formula, or unit inverse is simply another finite operation. Fix the finite formulas first, list all denominator losses, and then take the base Taylor depth larger than their sum and the requested T, k . A corrected expression approximating a derivative need not be the derivative of another corrected expression; it has its own formula (5.5) and differentiated error bound.

Suppose the replay is complete through one stage. Compare the next defining field on the two preceding charts at the same independent inputs, including its unknown. The flattened formulas have the same regular ordinary limit and Jacobian, and differ by arbitrarily high power error. Their analytic arguments remain in a smaller common neighborhood of their regular limits. Real boundedness proves their positivity before the implicit operation is taken. At the center root the new field has an arbitrarily small residual. Its regular inverse gives the root with that accuracy, and implicit differentiation gives the specified differentiated comparison. Evaluations at these nearby roots preserve moderate monomial comparisons. Induction proves the asserted approximation of the lower map and all its record outputs. There is no evaluation of an extracted symbol at a frozen background in this argument.

For clarity, use the *completed* replay outputs $y = Y(q, \xi)$, $R = R(q, \xi)$, and $w = W(q, \xi)$, after all chronological substitutions including the first w equation. Thus their derivatives already include every dependence through w . At fixed physical y , eliminating just the lower block gives the inverse-record derivative

$$K = R_\xi - R_q Y_q^{-1} Y_\xi. \quad (5.6)$$

The inverse perturbation identity and the finite derivative bounds show that the replay changes this expression by $O(\rho^{T-C})$ for a fixed loss C once the center recipe is chosen. In a thickness lift the center map does not use u , so its inverse records have zero u -derivative. Hence the replayed inverse records have arbitrarily power-small u -derivatives. At fixed physical y , the equation $w = \rho u$ consequently has effective derivative comparable to ρ in u : the record contribution is small by logarithmic sensitivity, and direct current- w feedback is a unit by (5.4).

For a promotion, regard $L = \log \rho$ as a function of physical (y, w) , records R , and the explicit co-ordinate ξ . The exact identity $W = \varepsilon \exp L(Y, W, R, \xi)$ holds on the completed map. Differentiating it after eliminating Y_q gives

$$W_\xi - W_q Y_q^{-1} Y_\xi = \frac{w}{1 - w L_w} \sigma, \quad (5.7)$$

where the logarithmic Schur factor is exactly

$$\sigma = L_\xi + L_R(R_\xi - R_q Y_q^{-1} Y_\xi) = L_\xi + L_R K. \quad (5.8)$$

The derivatives L_ξ, L_w, L_R in these formulas hold the other physical variables and records fixed; K , by contrast, uses the total derivatives of the completed outputs. Thus no term from record feedback through w has been omitted. By hypothesis the center counterpart of K is arbitrarily power small, and the replay error is $O(\rho^{T-C})$. The entries of L_R cost only powers of $1 + |\log \rho|$, while $|L_\xi| \geq 1$. Thus $|\sigma| \geq 1/2$ eventually, and $1 - wL_w = 1 + o(1)$ by (5.4). The physical Schur complement in (5.7) has absolute value at least $c\rho$. All other blocks have power bounds. Block inversion therefore bounds the *entire* inverse matrix by a power of $1/\rho$, using only the inverse of Y_q before this step. In particular, the smallness of K was proved without assuming the inverse of the full (y, w) map.

The bounds also hold for any specified finite derivatives on smaller absolute power-sized boxes. Such boxes preserve the scale regime; the finite-order estimates can equivalently be retested on every sequence of points in them. Lemma 5.2 therefore supplies local image neighborhoods and exact hit correction.

The calibration preceding this correction matters. For a desired physical point (y_A, w_A) and reference records R_A^0 , set $u = w_A/\rho(S, y_A, w_A, R_A^0)$, or choose the promotion residual by expanding $-\log |w_A|$ at that same point and those records. Do not form this ratio at an imprecisely displaced center. The lower map and its records then differ from the reference hit by arbitrary power errors. Logarithmic sensitivity makes the resulting error in w arbitrarily power small as well. The quantitative inverse corrects both y and w exactly, with unchanged limiting ordinary parameters. The correction changes records by another fixed power loss, which is absorbed in the previously chosen accuracy. Finitely many nested lifts multiply power losses and remain power moderate. $\square$

Lemma 5.4 (Common domains for accuracy refinements). *Fix a width r equal to a strict positive-order current monomial times a bounded regular unit with positive limiting value. All power bounds and moderation in this lemma refer to r . Consider a family of center recipes with a fixed chronological list of coordinate replacements, a fixed active flag, and fixed scaling exponents. Refining its accuracy changes finite Taylor depths and their finite formulas, but not that list. Suppose the regular equations and analytic arguments at each stage have the same ordinary limiting germs throughout the family. Suppose also that, for every chronological prefix, its actual map and record outputs, the finite forward derivatives required below, its inverse where used, and its available source neighborhood have power bounds with exponents fixed before the accuracy request. Constants and ordinary germ neighborhoods may depend on accuracy. The inverted monomials are, after the preceding whole-reciprocal preprocessing, products of powers of a fixed finite list of two-sided moderate generators; the powers in a flattened formula may depend on accuracy. Here a fixed monomial generator means a fixed exponent vector evaluated at the current backgrounds, not a frozen numerical function.*

Fix $J_0 > 0$ and a finite derivative request, and use the new transverse width $\rho = r^{J_0}$. The Taylor replay in Lemma 5.3 can then be made, through every chronological stage, on source balls $B(q_A, c_A r^N)$ with N independent of the accuracy request A . The same holds for graph tubes used in its regular implicit stages, with fixed power exponents. On these domains the comparison can be made arbitrarily power accurate in the requested differentiated norm. The resulting family has the same prefix properties, including power-sized image neighborhoods whenever the Schur hypotheses of Lemma 5.3 hold.

Proof. We first justify replay on actual ordinary neighborhoods; estimates on a power tube alone would not justify a positivity test. A two-sided moderate generator d satisfies $d \geq r^C$ at the reference anchors. Enlarging C to $C' > C$ makes $r^{C'}/d$ a strict positive-order monomial times a bounded

regular unit. The flag hierarchy is preserved under independent bounded ordinary variations, so $d \geq r^{C'}$ eventually on a smaller ordinary neighborhood. This argument concerns orders at each input. It does not compare the numerical values of d or r at different inputs. The same assertion for d^p has exponent pC' ; p may depend on the chosen finite recipe.

Fix one member of the center family and flatten its old finite formulas before selecting the new replay Taylor depth. All their denominator powers and chain-rule losses are now a finite fixed list. The new Taylor depth introduces higher base jets and more summands, but no new monomial divisor: a base jet is kept as a whole positive function, and differentiating the new finite sum to the fixed requested order loses only that order and the previously listed prefix powers. Factoring a divisor correction uses the same old divisor and a regular unit. Thus the loss to be beaten does not grow with the new Taylor depth. The finitely many base jets finally needed at that request are defined and bounded on some ordinary box $U_{\mathcal{A}}$ with radius independent of r , by Theorem 4.4. Its radius and bounds may depend on $\mathcal{A}$. Use a smaller box with spare room. At every input in this box, the segment of length $O(r^{J_0})$ in the new transverse coordinate stays in the ambient base box eventually. The current and center widths are comparable along that segment by their logarithmic sensitivities. The reverse Taylor remainder and the moderate generator bounds therefore hold on the entire smaller ordinary box, pointwise in its current width.

Proceed through the chronological stages. By the preceding stages, all their solved ordinary outputs are defined on ordinary boxes. Flatten the next defining field and all its analytic arguments on that previous chart. A finite denominator power loses a finite power of r ; increase the new Taylor depth after that loss is known. Thus each corrected field or argument differs from its center counterpart by an arbitrarily small power on a genuine ordinary box. Its real limit is the center's limiting ordinary germ. Analytic argument tuples stay in the domain of evaluation with spare ordinary room. The real-bound criterion now proves positivity on this box, before the next analytic evaluation or implicit inversion is performed. The defining equation has the same regular limiting Jacobian as its center equation. Theorem 4.4 supplies the regular branch on a smaller ordinary box and bounded finite jets of its ordinary outputs. This proves the stage induction on ordinary boxes. Their radii can shrink with the finite request, but no radius in this argument is a power r^{-A} depending on the accuracy. In particular, an analytic argument that is bounded only on such a shrinking tube would not pass this construction.

We next choose common power boxes. There are finitely many structural stages. Record the preassigned source exponents and forward derivative exponents of their center prefixes, taking all independent coordinates and the unknowns of later regular equations as parameters when needed. If an intermediate target source ball has exponent a_j and the map into it has derivative loss b_j , require $N > a_j + b_j + 1$; include $a_j = 0$ for an ordinary box. There are only finitely many such requirements. On a source displacement of size $c_{\mathcal{A}}r^N$, each center ordinary argument then moves by $O_{\mathcal{A}}(r^{N-b})$ for one of these fixed losses b , and so remains inside any of its accuracy-dependent ordinary boxes for sufficiently far anchors.

The same choice, enlarged by a fixed amount if necessary, keeps every two-sided moderate generator comparable to its reference value. Indeed in the independent coordinates of a prefix its logarithmic derivative is bounded by a fixed power of $1 + |\log r|$. Pullback by a prefix costs at most r^{-b} with b already recorded. Integration on the source ball gives

$$|\Delta \log d| + |\Delta \log r| \leq C_{\mathcal{A}} r^{N-b} (1 + |\log r|)^C = o(1).$$

This estimate is continued from the center as long as the ratios stay, say, between $1/2$ and 2 ; it prevents the first exit, and hence proves comparability on the whole ball. Replacing d by a power d^p multiplies this logarithmic estimate by the fixed number p for that request. It changes $C_{\mathcal{A}}$ and the

starting point, not the chosen exponent N . Nonmoderate positive expressions are never inverted and retain their bounded-jet estimates as whole functions. Saturation preserves the remaining scale comparisons under these small changes.

For completeness, carry out the regular implicit stages on these preassigned boxes, rather than extracting a new unspecified box after each solve. For the next stage let $v_0(q)$ be the center root. Choose a trial tube

$$|v - v_0(q)| \leq c_{\mathcal{A}} r^E$$

whose exponent E exceeds, with spare room, the sum of each preceding target-domain exponent and the derivative loss of the map into that domain. Include earlier trial-tube exponents among these target-domain exponents. For the contraction also require $E > b + c + 1$ if the normalized equation has second-derivative loss b and inverse-Jacobian loss c . Choose these trial exponents in chronological order; every requirement at a new stage uses exponents already fixed at earlier stages. These losses are supplied by the prefix bounds; in the equation's own ordinary coordinates its limiting Jacobian is a unit and its fixed jets are bounded. Ordinary argument slack can again be absorbed into $c_{\mathcal{A}}$ and a later start. On this tube, use the ordinary-box comparison just established, after pullback by the preceding prefixes. If the finite formula at this request loses a power $C'_{\mathcal{A}}$, choose its Taylor depth so that the residual and its required derivatives are $O(r^{T-C'_{\mathcal{A}}})$. For a desired output comparison $O(r^P)$, choose explicitly

$$T - C'_{\mathcal{A}} > \max\{E + c, P + H\} + 1,$$

where H bounds the sum of the fixed inverse, derivative, and composition losses for the requested differentiated output. Thus even after multiplication by the inverse norm r^{-c} the correction is smaller than the trial radius r^E . At each q the fixed-Jacobian contraction used in Lemma 5.2 then gives a root strictly inside the trial tube. The field's Jacobian and its first variation stay within the chosen contraction bounds there. Uniqueness makes the roots at neighboring q agree, so they define a graph on the whole preassigned source ball. Implicit differentiation gives the requested comparison on that same ball.

This argument is sequential: before the next stage, the previous ones are already defined with spare room on its trial inputs. Increase N and the finitely many trial exponents at the outset to meet all prefix requirements. Each requirement involves only the fixed structural list, width exponents, center derivative bounds, and finite derivative order. Higher Taylor depths affect only constants, the error order to be requested next, and the far-enough start. They add no coordinate replacement or new domain condition. Thus these power exponents are independent of record accuracy.

Finally, the differentiated comparison transfers the center prefix forward bounds to the actual replayed prefixes on these common boxes. The regular implicit stages have unit inverses in their solved directions; width stages use the Schur estimates of Lemma 5.3. All other inverse blocks have the preassigned prefix bounds. The local inverse lemma, now with its available source radius explicitly supplied and its required higher forward bounds, produces image radii with fixed exponents. This proves the family and prefix conclusions. $\square$

5.3 The simultaneous elimination invariant

A numerical floor is a sequence $\delta_A > 0$ tending to zero; it is not initially a flag variable or a function on the chart. After a subsequence, a numerical value is *visible* if its absolute value is at least δ_A^K for some fixed $K > 0$, and *negligible* if it is smaller than every fixed positive power of δ_A . A floor realization is a saturated extension, preserving active free inputs, and a positive width r equal to a

strict positive monomial times an optional bounded positive log-remainder unit, such that

$$\log(1/r_A) \asymp \log(1/\delta_A) \quad (5.9)$$

at the augmented reference input. Its additional bounded coordinates are exogenous, not active recursion variables.

The exact-zero assertion is proved together with two quantitative alternatives because a restriction to a derivative center need not remain an exact-zero problem. The center quantity H introduced in Equation (5.14) may be visible or negligible even when the original field vanishes. A visible center supplies a normalization; a negligible center supplies a chart accurate enough to lift and correct to the required exact hit. The conditioning powers below are fixed before record accuracy, so increasing that accuracy can make the correction smaller than the already available inverse neighborhood.

Proposition 5.5 (Elimination with a deferred floor). *Let F be a real positive scalar field on an active chart with n ordinary inputs. The following statements hold simultaneously.*

1. *In exact mode, along anchors where $F = 0$, finitely many regular changes, promotions, and exact ordinary restrictions produce a chart through those anchors on which F has identically zero formal data and hence vanishes on its real regime. All old free inputs survive.*
2. *If F_A is visible relative to a numerical floor, there is a full-dimensional regular chart with an exact hit and*

$$F = \tau U, \quad (5.10)$$

where τ is a nonnegative-order monomial, possibly 1, and U is positive with nonzero exponent-zero value at the limiting hit. All added divisors, widths, and local conditioning costs are power moderate in $|F_A|$. The active ordinary count does not increase.

3. *If F_A is negligible, active flag additions and their reference records can be chosen without realizing the floor. For every eventual realization r , field tolerance M , finite derivative request k , and record accuracy A , there is a full-dimensional regular chart hitting the input, of active ordinary count at most n , on which*

$$|F| \leq r^M. \quad (5.11)$$

Its added divisions and widths are power moderate in r . Its records at the hit differ from the references by $O(r^A)$, and their inverse first derivatives in the exogenous inputs, at fixed physical inputs, are $O(r^A)$ on a power-sized neighborhood.

The third assertion has the following stronger order of quantifiers. For fixed M, r, k , the thickness exponents, forward bounds for the actual map and its record outputs through order at least $\max(k, 2)$, inverse-Jacobian powers, and source and physical-image neighborhood exponents can all be fixed before A is requested. Constants and sufficiently far starting points may depend on A . The denominator powers in an individual finite flattened formula may depend on A . Completed active charts remain fixed as A increases; only pending thickness recipes and their finite Taylor replays are refined.

The construction also has the following prefix invariant. For the fixed request before choosing A , there is a fixed finite chronological list of coordinate replacements and scaling exponents. Every prefix has the same ordinary limiting regular equations and analytic argument germs throughout the accuracy family. Its maps and records, with the unknown of a subsequent equation still an independent ordinary input, have preassigned finite derivative and source-domain powers; its inverse and image powers are preassigned where an inverse is used. Its monomial divisors belong to powers of a fixed finite list of moderate current monomials and their triangular derivative factors. Only

their multiplicities inside the finite formulas can grow with accuracy. Thus the family meets the hypotheses and conclusions of Lemma 5.4 at every pending lift.

Proof. We use induction on n , and, at fixed n , on the nonzero directional order of the ordinary leading germ at the limiting hit. Every step is performed at the exact hit of completed preceding steps. We verify the quantitative third assertion, including its prefix invariant, in the same induction, not after constructing an uncontrolled family of charts. For an identity exit the structural list is empty. For the single thickness exit its one exponent is fixed by the field tolerance; all ordinary limiting germs are independent of accuracy and its direct inverse costs that fixed power.

Normalization and immediate exits. If both formal data arrays are zero, separation gives $F = 0$ on the real regime. This proves the exact assertion and the coarse assertion with the identity chart and no new records. Otherwise separation and leading normalization give

$$F = aG, \quad G \longrightarrow f, \quad (5.12)$$

where a is a nonnegative-order monomial and f is a nonzero real analytic germ. The normalized expression G has raw packet trees, and its locally uniform real convergence makes it positive by the real-bound criterion in Theorem 4.4. That theorem then gives the differentiated convergence in (5.12). If a is negligible in floor mode, the identity chart already gives (5.11) for every fixed M after any realization: the valuation of a exceeds every floor power and G is bounded. No division by a is used in this output. If a is visible, it is a moderate divisor and visibility or negligibility is unchanged on passing from F to G . If f is nonzero at the limiting hit, G is a regular unit, proving the visible assertion; exact zeros are impossible. These observations also prove the induction when $n = 0$.

A regular derivative coordinate. At a zero of f , choose an ordinary direction z , complementary coordinates y , and $m \geq 1$ such that

$$\partial_z^j f = 0 \quad (0 \leq j < m), \quad \partial_z^m f \neq 0$$

at the limiting point. This is possible by taking a direction where the first nonzero homogeneous Taylor term does not vanish. Define

$$w = \partial_z^{m-1} G. \quad (5.13)$$

Its z derivative has nonzero limit, so it is a regular ordinary coordinate with $w_A \rightarrow 0$. In exact mode, if $w_A = 0$ along a subsequence, restrict exactly to $w = 0$ and apply the smaller- n assertion. If $m = 1$, then $G = w$: a visible nonzero w is promoted, whereas negligible w is replaced by $r^J u$ with J selected from the field tolerance. The latter hits with $u \rightarrow 0$, has no new active records, and has inverse cost r^{-J} independent of record accuracy. Together with the exact restriction this proves the $m = 1$ step.

Assume $m > 1$. On the center $w = 0$ let

$$c_\ell = (\partial_z^\ell G)|_{w=0} \quad (0 \leq \ell \leq m-2), \quad H = \sum_{\ell=0}^{m-2} c_\ell^{2L/(m-\ell)}, \quad (5.14)$$

where L is a positive common multiple of $1, \dots, m$. The exponents in this sum are even integers; hence $H \geq 0$ and its smallness controls each c_ℓ . At the center projection with the same (S, y_A) these coefficients tend to zero. Apply the smaller- n assertion to H . Its numerical floor is

$$\theta_A = |w_A| \quad \text{if } w_A \neq 0 \text{ is visible, or in exact mode;} \quad \theta_A = \delta_A \quad \text{otherwise.} \quad (5.15)$$

In the latter case w_A is negligible, with $w_A = 0$ allowed.

A visible center and bounded w/s . Suppose H is visible in its center call. Write its visible output as $H = s^{2L}U_H$, with U_H a positive numerical unit and s a strict positive monomial. Taking a fixed real power of the output monomial is allowed. Then $s_A \geq \theta_A^K$ for some K , all center costs are moderate in s , and

$$|c_\ell| \leq Cs^{m-\ell}. \quad (5.16)$$

The quotients in this inequality are positive by real boundedness.

If w_A/s_A stays bounded, use Lemma 5.3 with $w = su$. Choose replay accuracy larger than every power needed below and correct its hit exactly. At common lower parameters, the lifted y and records differ from their center values by arbitrary power errors, and the current and center s have ratio tending to 1. Taylor expansion in the original independent z at fixed current (S, y) gives

$$z - z_0 = bsu + O(s^2), \quad b = (\partial_z^m G(S, y, z_0))^{-1}, \quad w(S, y, z_0) = 0.$$

The coefficient of $(z - z_0)^{m-1}$ in G is exactly zero. Dividing the Taylor expansion by s^m and using (5.16) shows uniform boundedness, and gives the ordinary limit

$$\frac{G}{s^m} \longrightarrow P(u) = \sum_{\ell=0}^{m-2} \frac{\gamma_\ell b_0^\ell}{\ell!} u^\ell + \frac{g_m b_0^m}{m!} u^m, \quad \gamma_\ell = \lim \frac{c_\ell}{s^{m-\ell}}. \quad (5.17)$$

Here $g_m = \lim \partial_z^m G \neq 0$ and $b_0 = g_m^{-1}$. The coefficients are ordinary germs in the remaining limiting parameters and are independent of u . Errors caused by replacing $z - z_0$ with bsu are $O(s)$ after normalization, by (5.16). At least one γ_ℓ is nonzero at the limiting remaining parameter, since H/s^{2L} has nonzero limit. Real boundedness proves that G/s^m is positive with exponent-zero germ P .

Every zero of P has multiplicity less than m . Indeed an m -fold root u_0 would give $P(u) = c(u - u_0)^m$. Its absent degree- $m-1$ coefficient forces $u_0 = 0$, which would make all lower coefficients zero, a contradiction. At a limiting nonzero value take the unit exit. Otherwise apply the same- n induction at the smaller directional order in u . In floor mode the factor as^m is visible, so the inherited numerical floor suffices. If the final output is visible, all preliminary costs are moderate in $|F_A|$, because $|F_A| \leq Ca_A s_A^m$ and both a, s are bounded above. If the final output is negligible, these completed active changes have floor-moderate costs and are fixed before the later accuracy request. All their chronological prefixes are fixed recipes. Their ordinary limiting germs and analytic argument slack are consequently fixed, and they have a finite list of monomial generators and finite derivative and domain powers in any later floor realization. Compose the smaller-order family's prefix bounds with these fixed prefixes. A source displacement r^N stays in each required intermediate source box by taking N larger than the sum of its fixed radius exponent and the preceding derivative losses. This choice is independent of the subsequent accuracy. The inverse derivative formulas and the local inverse lemma give the corresponding fixed inverse and image powers. This proves preservation of the prefix invariant through the smaller-order branch; it does not require rebuilding an active change when the pending accuracy is increased.

A visible center and unbounded $|w|/s$. If $|w_A|/s_A \rightarrow \infty$, the center floor in (5.15) must be $|w_A|$: a negligible w_A could not dominate visible s_A . Promote w against the center's already extended flag, as an exogenous addition to that recipe, calibrated at the desired physical point and reference records. Put $\rho = |w|$. The center recipe is independent of the new coordinate, $s \geq \rho^K$, and $s/\rho \rightarrow 0$. Lift it by Lemma 5.3. The same Taylor formula now gives

$$\frac{G}{\rho^m} \longrightarrow \frac{g_m b_0^m \varepsilon^m}{m!} \neq 0. \quad (5.18)$$

Indeed each lower term is bounded by $C(s/\rho)^{m-\ell}$ after normalization. This is the visible output with moderate conditioning. It excludes exact zeros in this case.

A negligible center relative to nonzero w . Suppose H is negligible and its floor is $\theta_A = |w_A|$. First finish the center's active flag additions and select their reference records. Then realize this one pending floor by promoting w , expanding $-\log |w_A|$ at the desired physical input and the reference records, and converting only its new residual chain. The result is a permitted realization $\rho = |w|$: changing a bounded background from that physical point to its center projection costs $O(|w_A|)$, so the logarithmic comparisons of all moderate scales are unchanged, by Lemma 5.1.

Request a center field tolerance sufficiently large that every c_ℓ will be $o(\rho^{m-\ell})$, including the losses of the lift. Fix its map, record-output, inverse, and neighborhood powers using the strengthened smaller- n assertion at that tolerance and realization. Only now request reference-record and inverse-record accuracy beyond those powers. Choose the Taylor replay even more accurately. After eliminating the lower physical y block, the logarithmic promotion Schur factor is exactly (5.8). The inverse center records have arbitrarily small derivatives in ξ by that assertion; the replay preserves them to any further fixed precision. Hence $|\sigma| \geq 1/2$, all inverse powers are controlled, and the initial calibration's record discrepancy yields a physical hit error inside the fixed image neighborhood. Correct the hit by Lemma 5.2.

The center estimates, arbitrary replay accuracy, and bounded fixed derivatives give $c_\ell = o(\rho^{m-\ell})$ at the actual Taylor centers. Therefore (5.18) again holds. This case immediately returns a visible output; it cannot occur for exact zeros and does not pass an older numerical floor further down the recursion.

The coarse case and its uniform conditioning powers. The only remaining case has center floor δ_A , negligible H , and negligible w_A . Retain the center's active flag extension and reference records. Moving from (y_A, w_A) to $(y_A, 0)$ is smaller than every floor power in bounded backgrounds, so it preserves the floor comparisons. Fix an eventual realization r , a field tolerance M , and a derivative request k .

Choose a thickness exponent J_0 from M and a center field tolerance M_H large enough for Taylor's formula to imply (5.11) when

$$w = r^{J_0} u, \tag{5.19}$$

with bounded u . These choices do not involve record accuracy. Explicitly, $H \leq r^{M_H}$ implies $|c_\ell| \leq r^{M_H(m-\ell)/(2L)}$; choose each resulting exponent larger than M with a fixed slack, and choose $J_0 m > M$ with slack. Bounded Taylor coefficients and a then give the desired bound after a sufficiently accurate lift.

Apply the smaller- n assertion, including its prefix invariant, to preassign bounds for the center maps $Y^{[A]}$, records $R^{[A]}$, and their derivatives through a sufficiently high fixed order, inverse Jacobians, and source and image neighborhoods. Their exponents do not depend on the requested record accuracy $\mathcal{A}$. The floor realization r is a strict positive monomial times a bounded regular unit with positive limiting value. Lemma 5.4 therefore applies to this center family with the already fixed new thickness exponent J_0 . It gives a source exponent and regular-stage trial-tube exponents before $\mathcal{A}$ is chosen. On those same domains its replay can be made arbitrarily accurate in the fixed differentiated norm, increasing the finite Taylor depth only after the losses in the selected flattened formula are known. All new defining equations and analytic arguments have been checked on genuine ordinary boxes in that lemma, before positivity or a regular inverse is invoked. Consequently the *actual* lifted maps and records inherit the preassigned power bounds on a preassigned power-sized

domain. Larger powers inside a finer formula neither reduce its available source-radius exponent nor worsen these actual-map bounds.

Before lifting, the lower map is independent of u . At fixed physical y the lifted inverse records thus have u -derivatives $O(r^{T-C})$, by (5.6); their exogenous derivatives differ from the arbitrarily small center derivatives by an error of the same kind. Write $L = \log r$ and use the completed maps $Y(q, u), R(q, u), W(q, u)$, as in the lift lemma. With $K_u = R_u - R_q Y_q^{-1} Y_u$, their exact physical Schur complement is

$$W_u - W_q Y_q^{-1} Y_u = r^{J_0} \frac{1 + u J_0 L_R K_u}{1 - w J_0 L_w} = r^{J_0} (1 + o(1)). \quad (5.20)$$

Here L has no explicit dependence on the newly added u , and its displayed derivatives hold physical variables and records fixed. All implicit record feedback is already in K_u . Its smallness and the logarithmic sensitivity bounds give the last equality. The full inverse therefore loses only a fixed power determined by J_0 and the center bounds.

The same fixed-power conclusion holds for the forward derivatives: they are those of the replayed center maps and (5.19), with the current- w equation initially solved by its regular unit Jacobian. Record outputs are either active free coordinate projections or positive ordinary expressions governed by Theorem 4.4, so the required bounds include their derivatives. They are not arbitrary outputs with uncontrolled growth. Lemma 5.2 now supplies source and image radii with exponents fixed from these bounds. Thus all conditioning and neighborhood powers have been chosen before the final record accuracy.

Given any requested $\mathcal{A}$, take the lower record accuracy and the replay depth large enough to absorb these fixed inverse and output-derivative losses. At the desired physical point choose $u = w_A / r^{J_0}$ using the reference records, as in the lift lemma. This tends to zero because w_A is negligible. Calibration, inverse correction, and the record derivative bounds give an exact hit with record error $O(r^A)$. Solving at fixed (y, w) costs only the same fixed powers. More explicitly, for an exogenous coordinate e put $K_e = R_e - R_q Y_q^{-1} Y_e$, using completed derivatives at fixed u . Differentiating the thickness identity at fixed physical (y, w) gives

$$\partial_e u|_{y,w} = -\frac{u J_0 (L_e + L_R K_e)}{1 + u J_0 L_R K_u}, \quad \partial_e R|_{y,w} = K_e + K_u \partial_e u|_{y,w}.$$

The first expression has a fixed power bound, while K_e, K_u can both be made smaller than any prescribed power on the common domain. Thus the second is $O(r^A)$ after requesting the lower accuracies beyond that fixed loss. Earlier completed charts have no dependence on the pending exogenous inputs: at fixed original physical input their intermediate physical inputs are fixed as well. Composing their power-conditioned inverses preserves the required smallness. This proves the entire strengthened coarse assertion.

To make the prefix part explicit, the new structural list consists of the fixed thickness equation and the smaller-dimensional list replayed in its former order. The limiting thickness equation is $w = 0$ with unit w -Jacobian, and each replayed regular equation or analytic argument has its center's limiting germ. Lemma 5.4 supplies its prefix domains and finite derivative bounds. The only new monomial generators are the monomial factor of the fixed width r^{J_0} and the finitely many triangular factors needed when differentiating it; repeated differentiation changes their powers. Its inverse block is controlled by (5.20). Thus the new list and all prefix exponents are fixed before the record accuracy, as required.

Deferred floors and denominator accounting. There is only one outstanding numerical floor in a negligible output. A negligible center relative to visible $|w_A|$ realizes that floor immediately

and exits visibly. A visible center chart is lifted and its exact hit completed before further recursion. Consequently neither kind of active decision depends on a future realization of an inherited floor. Along a chain which still passes the inherited floor into centers, every removed w_A is negligible relative to it. Old free scales remain fixed and bounded backgrounds move by negligible powers. Put $M = |\log \delta_A|$. At a fixed threshold $f_i = \log Z_i = O(M^q)$, the triangular relation for f_i has largest participating child scale $O(M^q)$. The finite triangular differentiation used in the proof of Lemma 5.1 therefore bounds every fixed background derivative by $O(M^D)$ for some fixed D . This argument uses the polynomial scale-log threshold directly; it does not require the corresponding exponential monomial to be power moderate in δ_A . Integration shows that a background displacement negligible relative to every power of δ_A cannot carry a scale log across a fixed comparison range about that threshold. Thus scale logs of at most a fixed power of M retain their comparisons, and a much larger log cannot enter their comparison range. After realization these are equivalently lexicographic monomial comparisons in the extended saturated flag.

Reference records of completed charts are taken at exact hits. Those of a pending center output are taken at its center projection; negligible center displacements and the eventual arbitrary power-accurate hit preserve their comparisons. Pending floor recipes add only thickness parameters replacing consumed ordinary coordinates, not active promotions tied to an extra recursion variable. Thus the induction really decreases n or, at fixed n , decreases m .

Finally keep a list of inverted monomials at each step. Outside an immediate negligible exit the leading a is moderate. The multiplicity step adds only s and its fixed powers, moderate by visibility and the inequalities already proved. Coarse outputs use only floor-moderate scales. The flattening and all fixed derivative formulas introduce finitely many further powers of these same denominators. Each recipe is therefore power moderate in its required scale. The family assertion is stronger only for actual maps and their neighborhoods, whose uniform exponents were proved by finite differentiability comparison above. This completes both inductions. $\square$

5.4 The absolute zero theorem and its closure

The final retest requires a property of the primitive library, in addition to the estimates used in Theorem 4.4. We state it explicitly to separate a local formal identity from a claim about arbitrary changes of asymptotic regimes.

Definition 5.6. A packet library is *retestable for elimination* if it satisfies Theorem 4.4 on each saturated extension used above and has the following germ invariance. Arbitrarily small absolute perturbations of retained old free inputs which preserve the affine flag relations, regime gaps, limiting ordinary data, and fixed sector determinations can be tested with the same terminal ordinary formal coefficient germs. Refining by unused nodes inserts identity transitions before pullback. Numerical anchor and cutoff choices must disappear by the specified formal coefficient recursions or normalized limiting jets before terminal coefficients are reached. The property concerns the selected local lifted germs and does not assert single-valuedness over all free-input histories.

Theorem 5.7 (Absolute isolated-zero finiteness). *Let a finite real equation system be specified on an open domain with its original strict inequalities and graph ties. Regard all coordinates, including coefficient parameters and auxiliary coordinates, as variables. Suppose every sequence of distinct solutions has a subsequence with one of the following descriptions:*

1. *the equations extend as ordinary real analytic functions to a neighborhood of its finite limit; or*
2. *after finitely many regular graph constructions and coordinate charts, it is an anchored saturated flag regime for a library retestable in the sense of Definition 5.6, with at least one old free*

input recoverable from the original solution tuple.

In the second description every equation admits the finite normalizations and positive operations of Theorem 4.4. Then the system has finitely many isolated real solutions in its full domain.

The assertion applies separately to every finite system in the following closure whenever these same sequential hypotheses hold: independent derivatives of equations and domain margins; finite sums, products and regular analytic evaluations; inverses on their specified nonzero domains; denominator clearing with those domains retained; fixed real specializations; addition of real parameters, multiplier variables, polynomial equations, and regular graph ties, including finite derivative graphs. No uniform bound over infinitely many generated systems, and no assertion after forgetting their original domains, is part of this theorem.

Proof. Suppose there is an infinite sequence of distinct isolated solutions. In the first alternative a real analytic zero set has locally finitely many connected components [2, Corollary 2.7], so its isolated points cannot accumulate at an interior point of an analytic extension neighborhood. Intersecting with an open original domain cannot turn a positive-dimensional local analytic branch at one of its interior points into an isolated point.

In the second alternative, multiply each equation by an appropriate nonvanishing monomial so that it is positive, and replace the finite system by the sum of the squares of these real normalized equations. The zero set on the retained domain is unchanged. The sum is positive by the calculus. Apply exact mode of Proposition 5.5. At every sufficiently far anchor the resulting chart satisfies identically zero formal data. Ordinary restrictions may have reduced its ordinary dimension, but they have never removed an old independent large input.

Take an isolation neighborhood about each alleged isolated solution. In its final chart choose a nonzero absolute perturbation of a recoverable old free input small enough that the mapped point remains inside that neighborhood, inside all strict domains and chart neighborhoods, and in the same saturated regime. The perturbations can be successively smaller than every finite list of required bounds; they preserve the limiting ordinary data and exact affine flag ties. Recoverability makes their solution tuples different from the original ones.

Retest on this perturbed sequence. The primitive terminal germs are the same by Definition 5.6. Every later formal operation also gives the same germ: algebra and derivatives act on the same ordinary identities, finite Taylor replays use the same formulas, and regular implicit equations select the same ordinary root germs. In particular, both all-zero formal arrays remain zero. Separation therefore says that the actual normalized equations vanish at all sufficiently far perturbed points. These are distinct solutions in the proposed isolation neighborhoods, a contradiction.

The closure statement means applying this argument to each particular finite generated system. Every allowed operation is a finite calculus operation or a regular graph construction; polynomial relations in newly added variables are ordinary analytic relations. A derivative or multiplier system must still meet the sequential hypotheses, with every original strict domain and graph tie retained. Clearing a denominator on its nonzero domain preserves its zero set. Thus the argument applies to that system with all its coordinates variable. It does not replace this hypothesis by finiteness of the original unaugmented equation set. $\square$

Remark 5.8. The concrete primitive constructions below verify the retest property by coefficient recursions, compatible finite-charge quotients, or canonically normalized infinite-anchor jets. Merely discarding an unknown flat difference would not verify the equality of terminal germs needed in the last proof. The sequence-dependent flags and finite internal recipes used here likewise do not assert a uniform number of those recipes before a sequence is selected.

6 The two additive passage models

The two scalar passages below have an action given by a linear function of the passage variable plus a logarithmic correction. One describes a damped transverse state; the other describes a residual state driven by layers near the two endpoints. The common small quantity is the exponential of the negative action difference. We construct packet expansions for both passages, including the independent derivatives and retestability required by Sections 4 and 5.

The coefficients and ordinary parameters are analytic on neighborhoods slightly larger than the boxes used below. Amplitudes are chosen smaller than a fixed constant determined by these boxes, independently of any asymptotic order. Constants for a fixed coefficient, charge budget, or derivative order may depend on that request.

We first construct the actual analytic passage. Its finite expansion in the transfer charge will have coefficients built from slow sectorial solutions and their primitives. We retain their exact inner endpoint values, then expand each charge coefficient successively at the flag indices where its small arguments first occur. These are three distinct choices: a finite charge cutoff, an optimal slow Gevrey cutoff, and finite Taylor degrees for the later flag transitions. No convergence of the entire packet tree is required.

6.1 Models, independent inputs, and actual continuation

The real inputs satisfy

$$q \geq q_* > 1, \quad Q/q \longrightarrow \infty. \quad (6.1)$$

Here q_* is chosen once from the field and its boxes. Either q tends to infinity or it belongs to an ordinary neighborhood of a fixed positive value. Put

$$\gamma = \delta q, \quad A(w) = w + \gamma \log(w/q), \quad A(p) = Q, \quad P = e^{-(Q-q)}, \quad Z = \gamma/w, \quad (6.2)$$

where δ is small and all logarithms have the stated lifted determination. Thus p is the outer endpoint in the w coordinate, whereas Q is its action value. Since $A(q) = q$, the action difference is $Q - q$ and its decaying exponential is P . The two endpoints are related by

$$p = QU, \quad U + \delta(q/Q)\{\log(Q/q) + \log U\} = 1. \quad (6.3)$$

The solution U near 1 is analytic in the small ratio and ratio-logarithm arguments appearing here. We use a finite vector X of profiles

$$X_j = \chi_j(w/q)^{r_j} \quad (r_j < 0), \quad X_j = \chi_j(w/Q)^{r_j} \quad (r_j > 0), \quad (6.4)$$

with fixed rational exponents and small amplitudes. The profile Z may be included among the negative profiles. Redundant amplitudes and all ordinary field parameters may first be taken as independent arguments.

The two models are

$$(A) \quad v' = \sigma(1 + Z)v + w^{-1}g(X, v), \quad \sigma \in \{-1, 1\}, \quad g(0, v) = 0, \quad (6.5)$$

$$(B) \quad z' = G(X, E, H, z), \quad G(X, 0, 0, z) = 0, \quad (6.6)$$

where prime denotes d/dw . Model (A) is traversed from q to p when $\sigma = -1$ and from p to q when $\sigma = 1$; its input lies in a small disk about 0. Since $A'(w) = 1 + Z$, the homogeneous multiplier in

either prescribed direction is exactly P . Model (B) is traversed from q to p and has its input in a smaller disk about any chosen ordinary reference state. Its fast profiles are

$$E_s = e_s(w/q)^{\beta_s} e^{-a_s(A(w)-q)}, \quad H_s = d_s(w/Q)^{\nu_s} e^{-b_s(Q-A(w))}. \quad (6.7)$$

The exponents β_s, ν_s are rational, the amplitudes are small, and the positive weights a_s, b_s belong to one fixed discrete charge lattice. We measure charge in its positive unit and write the weights as positive integers; with another unit every occurrence of P is replaced by the corresponding fixed power of P .

Write η for the ordinary field parameters suppressed in the equations. The output maps have the independent arguments

$$\begin{aligned} \text{(A)} \quad & (q, Q, \delta, \chi, v_{\text{in}}, \eta) \longmapsto v_{\text{out}}, \\ \text{(B)} \quad & (q, Q, \delta, \chi, e, d, z_{\text{in}}, \eta) \longmapsto z_{\text{out}}. \end{aligned}$$

The endpoint p is determined by $A(p) = Q$. Relations between amplitudes, clocks, and physical endpoints are imposed only in the applications in Section 10; they are not imposed when differentiating these primitive families. A flag test expresses the independent passage arguments in free flag coordinates and ordinary coordinates.

For flag tests, Q and any large q are affine combinations of flag nodes and bounded coordinates, with fixed real coefficients on the large nodes and positive leading coefficients. Convert every participating primary basis node to a dependent log node. Also convert any free node in the resulting log formulas that will be used as a polynomial factor. Saturate as in Section 2. Consequently every required power of a primary node, and every finite polynomial in its log formula, is a finite sum of shifts times bounded analytic factors. Subsequent finite flag refinements are permitted.

Theorem 6.1 (Additive primitive packets). *Under these assumptions, the output maps of (6.5) and (6.6) satisfy the packet conditions of Definition 3.1 and the differentiated primitive conditions of Assumption 4.2. They have no exceptional column loss: every working column is safe. Their only nonholomorphic defects above the first band have simple costs $e^{\Re f_i}$ and, when q is large, $e^{\Re f_l}$, where i, l are the leading indices of Q, q . Each cost is used only while its log formula is retained. The complete terminal coefficient germs, on each lateral determination, are unchanged by inserting unused nodes and by retesting a fixed finite construction on a permitted refinement preserving the old active free nodes, with the same lifted relations and limiting ordinary data. In particular this primitive library is retestable for elimination in the sense of Definition 5.6, with the fixed local lateral determinations.*

We prove the theorem in several steps. Up to and including band i the packet is the actual passage, with zero prefixes at earlier transitions. The flag estimates give $\Re Q \gg \log |Q|$ and $\Re(Q - q) \gg \log |Q|$; on the i fringe, after increasing the side dominance constant,

$$\Re(Q - q) \geq c\Re Z_i. \quad (6.8)$$

On the upper determination the endpoints lie in slightly enlarged first quadrants and $|q/Q|$ is small; the lower determination is reflected. For bounded q use its ordinary neighborhood near the positive ray.

In the A plane join q horizontally to $q + S$, then straight to $Q - S$, then horizontally to Q , where $S = C \log |Q|$ and C is a sufficiently large fixed number. The real part increases. On this polygon $|A| \gtrsim |q|$, and on its middle segment $|A|$ bounds below a fixed multiple of the convex combination of the endpoint moduli. The implicit formula for $w(A)$ gives

$$w(A) = A(1 + O(|\delta|)), \quad |dw/dA| \leq C,$$

because $(q/A) \log(A/q)$ is bounded there. In particular

$$\int |X| |dw/w| \leq C \max_j |\chi_j|, \quad (6.9)$$

$$\sup(|E| + |H|) + \int (|E| + |H|) |dw| \leq C \max_s (|e_s|, |d_s|). \quad (6.10)$$

Include $|\delta|$ in the first maximum. For example, a positive slow power integrates as $\int_0^1 t^{r-1} dt = 1/r$; a negative power has the corresponding inner-endpoint estimate. On an endpoint horizontal segment each active fast profile is an exponential in horizontal distance times a fixed polynomial. On the middle segment the margins S pay every fixed power and the length. These observations prove the two inequalities on all three segments, with constants depending only on the fixed exponents and boxes.

The exact homogeneous propagator in (A) is $e^{\sigma(A(w)-A(w'))}$; its modulus is at most one in the prescribed direction. The integral equations, (6.9), (6.10), and smallness keep solutions strictly inside their state boxes. The same estimates hold with parameter room. Locally freeze the horizontal lengths and vary the endpoints. Analytic ODE dependence gives holomorphy; changing lengths by fixed factors gives the same solution by a homotopy through the bounded paths. This supplies one continuation from the real analytic family, including across the real ray. No sector normalization has yet been introduced.

The sectorial constructions below normalize auxiliary solutions, while the actual passage just constructed is fixed. A simple instance of (A) shows why their two endpoint contributions must be kept together. Take $\delta = 0$, $g(X, v) = X = \chi q/w$, and the damped direction. A particular solution V of $V' + V = \chi q/w^2$ has the formal expansion

$$\widehat{V}(w) = \chi q \sum_{j \geq 0} (j+1)! w^{-j-2}.$$

Two lateral particular solutions can differ by Ke^{-w} , whereas the actual transfer is

$$V(p) + P\{v_{\text{in}} - V(q)\}. \quad (6.11)$$

Here $p = Q$, so changing V by Ke^{-w} changes the displayed value by $Ke^{-p} - PKe^{-q} = 0$. The exact inner value $V(q)$ must therefore remain in the coefficient of P . Replacing it independently by its formal series would discard the term that cancels the outer lateral change. The general construction retains inner endpoint values for this reason.

6.2 Lateral domains and the slow equations

After the i transition, Q can acquire a large lifted log phase. The slow equations will use its lifted powers while q remains in its lateral quadrant, or near its bounded positive value. Freeze a representative of $|Q|$, take B a sufficiently large fixed multiple of that representative, and use the upper polygonal domain

$$\Omega_B^+ = \{x + iy : -B < x < B, |y| < M(B - |x|), y > c_0|q| - \varepsilon_0|x|\}. \quad (6.12)$$

Here $0 < c_0 \ll \varepsilon_0$, M is fixed and large, and admissible limiting anchor vertices are $-B$ and B . Choose the constants so that a slightly enlarged half-annulus

$$|q|/2 \leq |w| \leq 2|Q|, \quad -\eta_0 \leq \arg w \leq \pi + \eta_0$$

has distance at least $\varepsilon|w|$ from the edges. Increasing $B/|Q|$ provides any fixed additional outer room. The lower domain is its reflection. Each has a fixed logarithm, $|w| \gtrsim |q|$, and all the profiles stay in a small polydisk if their amplitudes were chosen small enough, also on the larger domain.

Every point of (6.12) can be reached from either vertex by a graph monotone in x with $|dw| \leq C|dx|$. Indeed the upper and lower boundaries are Lipschitz graphs with bounded slopes; their vertical intervals collapse at the vertices, and interpolation between boundary graphs gives the claimed paths. Nearby graphs are homotopic through such graphs. Volterra solutions with the appropriate damping direction therefore continue to one analytic function on the domain: locally vary the final path segment and use uniqueness to compare overlapping continuations. Starting at a vertex can equivalently be defined by a limit from a full interior graph; the field is nonsingular at that nonzero-radius point and analytic with room beyond it. The finite predecessors in a coefficient recursion extend to the same vertices in this fashion. Parameter changes with anchors frozen give local holomorphic choices.

For (A) let v_0 solve

$$v'_0 - \sigma(1 + Z)v_0 = w^{-1}g(X, v_0) \quad (6.13)$$

with zero data at the decaying vertex. Define $b = g_v(X, v_0)$ and $\mathfrak{a} = \sigma(1 + Z) + b/w$. The fiber change and its equations are

$$v = T(w, u) = v_0(w) + u + \sum_{m \geq 2} t_m(w)u^m, \quad u' = \mathfrak{a}u, \quad (6.14)$$

$$t'_m + (m - 1)\mathfrak{a}t_m = w^{-1}[g(X, T) - g(X, v_0) - b(T - v_0)]_{u^m}. \quad (6.15)$$

The t_m have zero data at the opposite vertex. This opposite choice is forced by the sign of their diagonal.

For (B), let m, n be the multi-indices of E, H , respectively, and put

$$a = \sum_s a_s m_s, \quad b_f = \sum_s b_s n_s, \quad \Delta = b_f - a, \quad \kappa = \beta \cdot m + \nu \cdot n. \quad (6.16)$$

Thus a, b_f are scalar total charges. Seek $z = u + t(w, E, H, u)$ with t supported on $\Delta \neq 0$ and $u' = \mathcal{C}(w, E, H, u)$ with $\mathcal{C}$ supported on $\Delta = 0$, without a constant term. Coefficient comparison gives

$$\mathcal{C} = [G(X, E, H, u + t)]_{\Delta=0}, \quad (6.17)$$

$$t'_{mn} + [\Delta(1 + Z) + \kappa/w]t_{mn} = [G(X, E, H, u + t) - \partial_u t \mathcal{C}]_{mn}, \quad \Delta \neq 0. \quad (6.18)$$

There is no balanced term in $\partial_u t \mathcal{C}$: adding a balanced charge to a nonzero imbalance preserves that imbalance. These equations are triangular in total fast degree. Each imbalance has zero data at its decaying vertex, the left vertex when $\Delta > 0$. For a balanced monomial, the w -dependent fast exponentials cancel: the corresponding term of the balanced equation is

$$P^j e^m d^m q^{-\beta \cdot m} Q^{-\nu \cdot n} w^\kappa \mathcal{C}_{mn}(w, u), \quad j = a = b_f \geq 1. \quad (6.19)$$

This is the drift that survives the removal of the imbalanced terms.

A fixed charge budget K needs only coefficients modulo $\min(a, b_f) > K$. The retained region in Figure 3 is a finite union of fixed-side tails, not a finite total-degree truncation. The next lemma proves convergence of each such tail on a common fast radius. The actual remainder after omitting the other charges is estimated later in (6.52).

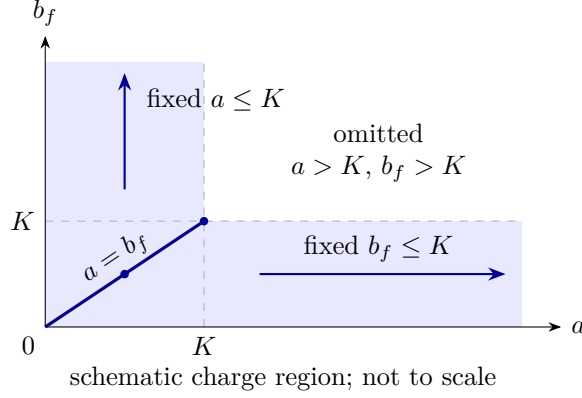


Figure 3: A finite request retains $\min(a, b_f) \leq K$, a finite union of convergent fixed-side tails; its balanced part is finite. The excluded region has both charges larger than K , so its monomials carry the corresponding power of the transfer charge on the actual passage. Not every schematic lattice position need occur. No coefficient bound uniform in the fixed side, or convergence of the unrestricted mixed series, is asserted.

Lemma 6.2 (Slow coefficients on a common fast radius). *The graph, fiber transformation in its convergent state-disk norm, and b in (A) are bounded on the lateral domains. The transformation is close to the identity and has a regular inverse on a smaller disk covering the inputs. In (B), for each fixed m the full H tail converges on one fixed fast radius; the corresponding assertion holds with the sides interchanged. These radii cover the actual fast amplitudes with fixed slack and do not decrease with m, n , or K . State, profile, and spatial neighborhoods have common positive remaining margins. The finite balanced lists have the same estimates.*

Each such fixed request has a formal series in $h = 1/w$, whose coefficients are analytic in independent profile and ordinary arguments on fixed smaller polydisks and satisfy

$$\|c_j\| \leq C^{j+1} j!. \quad (6.20)$$

On padded interiors, for $1 \leq J \leq c|w|$,

$$\left\| c(w) - \sum_{j < J} c_j(X) w^{-j} \right\| \leq C^{J+1} J! |w|^{-J}; \quad (6.21)$$

a sufficiently small proportional optimal cutoff gives $O(e^{-c|w|})$. For unscaled positive-profile coefficients $\tau_j = \chi_j Q^{-r_j}$, so that $X_j = \tau_j w^{r_j}$, the same estimates for ∂_τ^α acquire the factor $|w|^{r \cdot \alpha}$. Fixed ordinary derivatives are allowed. Neighboring anchor choices of size comparable to $|Q|$ differ by $O(e^{-c|Q|})$ on the common working interior, including at its inner radius, with fixed polynomial derivative losses allowed. At $\tau = 0$ the finite positive-profile jets have canonical infinite-anchor determinations with these estimates.

Proof. We first construct the actual coefficients on the lateral domains. For (B), this is an induction in the left degree: at each step a finite list of opposite degrees precedes one convergent high opposite tail. A fixed request involves only finitely many such blocks. We then obtain formal Gevrey bounds and compare their optimal truncations with those actual coefficients. The common domain choices are made before this second induction, so its slow order can grow without exhausting the margins. Weighted derivatives and anchor comparisons complete the proof. Throughout, a common fast

radius does not assert coefficient bounds uniform in the left degree or convergence when both fast degrees grow.

1. *Convergent lateral coefficients.*

For $\mathcal{L} = \partial_w + \Delta(1 + Z) + \kappa/w$, its zero-data inverse has kernel

$$\exp[-\Delta(w - w') - \Delta\gamma \log(w/w')](w'/w)^\kappa. \quad (6.22)$$

On an admissible monotone graph, $|\gamma/w| \leq C|\delta|$. If $|\kappa/\Delta|$ is universally bounded, small δ and large fixed q_* give kernel decay $Ce^{-c|\Delta||x-x'|}$ and inverse norm $C/|\Delta|$. For a fixed exceptional pair the last factor in (6.22) is bounded by $C(1 + |x - x'|)^{C\kappa}$: the log angle is bounded and both radii are bounded below. Its exponential moment is finite. Thus exceptional pairs only change their constants; they do not require increasing q_* as the requested charge order increases. Fixed profile weights commute through the kernel with the same polynomial ratio estimate.

The graph equation is a contraction on a small sup-norm ball, because $g = O(X)$ and its state derivative is $O(X)$. For the fiber coefficients use the norm $\sum_{m \geq 2} R^m \|t_m\|$ on a small state disk. The inverse gains $C/(m - 1)$, compensating the degree factor in analytic composition. More explicitly, expand g in a disk strictly inside its analytic state box, majorize its coefficients by the corresponding absolutely convergent positive Taylor series, and solve successive coefficient truncations in a small ball. The $1/w$ multiplier is at most C/q_* ; terms containing mb/w have uniformly small norm after division by $m - 1$. The majorant map has small Lipschitz constant. Its bound is independent of the truncation, so the increasing sums converge, and Cauchy on a smaller state disk makes $T_u - 1$ small. The inverse follows by contraction.

The pure E and pure H transformations in (B) have no drift interaction. Their right sides have a fast factor. A Taylor majorant on small fixed fast disks and a larger state disk is therefore a contraction with small state derivative. Now fix $m \neq 0$ and write

$$T_m(H) = \sum_n t_{mn} H^n, \quad A_0(H) = G_z(X, 0, H, u + T_0(H)).$$

Here $A_0(0) = 0$. At the current left order, the unknown T_m enters G only as the linear convolution $A_0 T_m$; every nonlinear remainder uses smaller componentwise left orders. Also $\mathcal{C}_{0,*} = 0$, so $\partial_u t \mathcal{C}$ never differentiates the current unknown T_m . Current balanced coefficients multiply only $\partial_u T_0$; they form a finite list with $b \cdot n = a \cdot m$ and are determined in increasing opposite degree before the high tail.

Choose a fixed fast radius ρ strictly inside the pure-side radius and use the weighted absolute coefficient norm on a chosen lateral domain Ω and state disk $D_R = \{|u - u_c| < R\}$,

$$\|F\|_{\rho,R} = \sum_n \rho^{|n|} \sup_{w \in \Omega, |u - u_c| < R} |F_n(w, u)|. \quad (6.23)$$

It is a Banach algebra without any loss of ρ . To make the uniformity explicit, put $a_{\min} = \min_s a_s$, $b_{\min} = \min_s b_s$, $\beta_{\max} = \max_s |\beta_s|$, and $\nu_{\max} = \max_s |\nu_s|$. For $N = N(m) \geq \max(1, 2a/b_{\min})$ and $|n| > N$,

$$\Delta_{mn} \geq \frac{1}{2} b_{\min} |n|, \quad |\kappa_{mn}/\Delta_{mn}| \leq \beta_{\max}/a_{\min} + 2\nu_{\max}/b_{\min}.$$

In particular the high-tail kernel constants and the initial choice of q_* are independent of m . After solving the finite initial list, its high tail U satisfies

$$U = \mathcal{L}_m^{-1} \Pi_{>N} (A_0 U + F_m), \quad \|\mathcal{L}_m^{-1} \Pi_{>N} F\|_{\rho,R} \leq \frac{C_0}{N} \|F\|_{\rho,R}. \quad (6.24)$$

The forcing F_m consists entirely of known lower-left tails and the finite initial list. Taking $N > 2C_0\|A_0\|_{\rho,R}$ yields

$$\|U\|_{\rho,R} \leq (2C_0/N)\|F_m\|_{\rho,R}.$$

This proves convergence on the same ρ at every left order. Large mixed coefficients do not violate the field box: their Taylor construction is coefficientwise in the left variables about the small pure-side base, rather than an evaluation of the whole mixed series.

2. *Domains shared by all slow orders.* The preceding induction gives the actual coefficients on a common fast radius. To compare them with formal slow expansions, we must also reserve state, profile, and spatial room at every finite dependency step. We make those choices now, independently of the slow truncation order. Fix a finite coefficient and derivative request and close its dependencies under smaller left orders, opposite initial indices, and lower jets. Choose positive limiting profile and state radii x_∞, R_∞ strictly inside the pure-side construction, and a spatial padding ζ_∞ strictly inside the available geometric room. Choose $\eta_k = c_1/(k+1)^2$, $k \geq 1$, with total sum smaller than each of the three available margins, and define

$$x_k = x_\infty + \sum_{h>k} \eta_h, \quad R_k = R_\infty + \sum_{h>k} \eta_h, \quad \zeta_k = \zeta_\infty - \sum_{h>k} \eta_h. \quad (6.25)$$

The fixed amplitudes can be chosen so that actual profiles lie in $|X| < x_\infty/2$ on the whole larger lateral domain. State coefficient functions at degree k are constructed on $|u - u_c| < R_k + \eta_k/4$ and are estimated, with arbitrary fixed state jets, on D_{R_k} . Lower-left coefficients are available at R_{k-1} ; in particular, for a fixed derivative order d ,

$$\|\partial_u^d F_{m'}\|_{\rho, R_k + \eta_k/4} \leq d! \left(\frac{4}{3\eta_k} \right)^d \|F_{m'}\|_{\rho, R_{k-1}}, \quad |m'| < k. \quad (6.26)$$

The same estimate applies to predecessor differences in a comparison. The current unknown is never differentiated in u by its defining equation. The reserved $\eta_k/4$ gives its requested output state derivatives after it has been constructed.

There are finitely many left indices of degree k . Take a common threshold N_k for their high opposite tails, first larger than all $2a/b_{\min}$ and then large enough to absorb the pure multiplier in both the actual and formal norms below. List the finite opposite degrees in increasing order, followed by the high tail. Include the finitely many lower-jet comparison steps required for the present request. Call the number of resulting steps s_k . Each infinite same-sign tail counts as one step, not as infinitely many indices. For $0 \leq v \leq s_k$, use the spatial domains

$$\Omega_{k,v} = \left\{ w \in \Omega_B^\pm : \text{dist}(w, \partial\Omega_B^\pm) > \left(\zeta_{k-1} + \frac{v\eta_k}{s_k} \right) |w| \right\}. \quad (6.27)$$

These are estimates on restrictions of the same actual coefficient functions, not newly prescribed boundary data. A target in $\Omega_{k,v}$ has, in either horizontal direction, a segment of length $b_k|w|$ contained in $\Omega_{k,v-1}$, with all radii comparable to $|w|$, where one may take $b_k = \eta_k/[8s_k(1 + \zeta_\infty)]$. This follows from the 1-Lipschitz property of distance and of modulus. All orders end on the same positive remaining profile/state margins and on a spatial padding less than ζ_∞ . Increasing a finite jet request changes s_k, b_k and the constants, without changing those limiting margins or the fast radius ρ .

3. *Formal Gevrey bounds.* On the domains just chosen, write

$$\partial_w = h(D_X - h\partial_h), \quad D_X = \sum_j r_j X_j \partial_{X_j}.$$

Divide the recursions by their nonzero diagonal at $h = 0$, a nonzero integer charge times $1 + Z$. Work modulo h^{J+1} , $J \geq 1$, put $\theta = \varepsilon_k/(J+1)$, and, for $0 \leq j \leq J$, set

$$x_{k,j}^{(J)} = x_k + \frac{\eta_k}{4} + \frac{\eta_k}{4} \left(1 - \frac{j}{J+1}\right), \quad (6.28)$$

$$\|F\|_{k,J} = \sum_{j=0}^J \theta^j \sum_n \rho^{|n|} \sup_{\substack{|X| < x_{k,j}^{(J)} \\ |u-u_c| < R_k + \eta_k/4}} |F_{jn}(X, u)|. \quad (6.29)$$

For a finite scalar coefficient omit the sum over n . Every current profile disk lies at least $\eta_k/2$ inside the predecessor target disk $|X| < x_{k-1}$, and at least $\eta_k/4$ outside the current target disk $|X| < x_k$. Thus there is both inherited room and fixed residual room. In a product the target disk at degree $a + b$ lies in both input disks, so this norm is an algebra. The disk gap for a single slow degree is $\eta_k/[4(J+1)]$. Cauchy's inequality and the factor θ give

$$\|hD_X\| \leq C\varepsilon_k/\eta_k, \quad \|h^2\partial_h\| \leq C\varepsilon_k, \quad \|\Delta^{-1}h\kappa\| \leq C\varepsilon_k \quad (6.30)$$

on the high tail; the finite exceptional list has its own constants.

To spell out the induction inequality, let $M_{k,v}^{(J)}$ denote the norm of the current formal coefficient block, and let $M_{\prec}^{(J)}$ be the maximum of the norms already constructed that enter its forcing. Fixed left-degree extraction of G is a finite Taylor calculation about the pure-side base. Its terms have bounded analytic multilinear coefficients and finitely many predecessor factors. The drift contributes products with a lower-left state derivative, bounded by (6.26). Hence there are fixed D_k, d_k and constants $C_{k,v}$, independent of J , for which that forcing is bounded by

$$\Phi_{k,v}(M_{\prec}^{(J)}) := C_{k,v}(1 + \eta_k^{-d_k})(1 + M_{\prec}^{(J)})^{D_k}.$$

The exponents are finite because the left Taylor degree and the jet request are finite; no bound as k increases is required. Dividing by the diagonal, and moving only its slow differential correction to the right, gives for an exceptional coefficient

$$M_{k,v}^{(J)} \leq \Phi_{k,v}(M_{\prec}^{(J)}) + C_{k,v}\varepsilon_k(1 + \eta_k^{-1})M_{k,v}^{(J)}. \quad (6.31)$$

For the high tail it gives the same inequality with the last coefficient replaced by

$$\frac{CM_A}{N_k} + C_k\varepsilon_k(1 + \eta_k^{-1}), \quad (6.32)$$

where M_A bounds the pure-side multiplier in its larger-disk formal norm, uniformly in J . Balanced blocks satisfy the forcing inequality without a current unknown on the right. The graph, fiber disk norm, and pure-side blocks obey the corresponding analytic majorant inequality with their already chosen small Lipschitz constant; their slow differential correction has the same bound.

First choose N_k to make $CM_A/N_k \leq 1/4$. Induct through the finite list, taking ε_k smaller than every preceding admissible choice and so small that all the differential coefficients in (6.31)–(6.32) are at most $1/4$. Absorption yields $M_{k,v}^{(J)} \leq 2\Phi_{k,v}(M_{\prec}^{(J)})$. Its finite recursion supplies a bound independent of J . Shrinking ε_k only decreases predecessor norms, so it does not invalidate a prior bound. This proves the simultaneous formal induction on the specified disks. At $J = j$ it implies

$$\|c_j\| \leq C_0((j+1)/\varepsilon_k)^j \leq C^{j+1}j!,$$

using $j^j \leq e^j j!$. Fixed formal profile or state derivatives are taken on the reserved target disks and change only their fixed Cauchy constants.

4. *Comparison with optimal truncations.* The formal norm now bounds the substitution residual. To turn this into an estimate for the actual coefficient, we use a last horizontal comparison segment of length $\delta_m |w|$ in the appropriate damping direction, lying in a padded interior, with radii comparable to $|w|$. Substitute the formal polynomial with J a small multiple of $|w|$. Test that polynomial and all analytic operations at a dummy h radius larger than the actual $|1/w|$ by a fixed factor. Formal cancellation below degree J and the just proved sum bound give an exponentially small residual in the norm appropriate to its diagonal. A diagonal or κ factor can contribute $1 + |n|$ to this residual. More precisely introduce the forcing norm

$$\|F\|_{\rho, R, -1} = \sum_n \frac{\rho^{|n|}}{1 + |n|} \sup_{\Omega \times D_R} |F_n|.$$

The residual is $Ce^{-c|w|}$ in this norm, whereas the inverse maps this forcing space boundedly into (6.23). Indeed apply the inverse coefficientwise before summing: on the high tail

$$(1 + |n|)/|\Delta_{mn}| \leq C. \quad (6.33)$$

Thus there is no hidden fast-radius loss in this step.

Here is the actual comparison inequality, including its starting term. For a target of modulus R , take the segment from (6.27), oriented in the current damping direction and parametrized by real distance $t \in [0, L]$, $L = b_k R$. Let E be actual coefficient minus its common formal polynomial. The actual coefficients are uniformly bounded by the preceding Volterra construction, and the formal sum norm bounds that polynomial on the segment. Thus $\sum_n \rho^{|n|} \sup_{D_{R_k}} |E_n(0)| \leq B_{k,v}$, with no dependence on R or J . Normalize by $R^{r-\alpha}$ for a fixed positive-profile jet; moduli on this segment are comparable to R , so the same assertion holds with a different fixed constant.

Use the coefficient-sum weighted norms

$$\|E\|_\omega = \sum_n \rho^{|n|} \sup_{\substack{0 \leq t \leq L \\ u \in D_{R_k}}} e^{\omega t} |E_n(t, u)|, \quad \|F\|_{\omega, -1} = \sum_n \frac{\rho^{|n|}}{1 + |n|} \sup_{\substack{0 \leq t \leq L \\ u \in D_{R_k}}} e^{\omega t} |F_n(t, u)|. \quad (6.34)$$

The sum is outside each coefficient supremum. This convention is essential for the inverse from the weak forcing norm to the strong norm; no interchange with a single supremum of the weak sum is used. The high-tail kernels are bounded by $Ce^{-\lambda(1+|n|)(t-s)}$. For $0 < \omega < \lambda/2$, coefficientwise integration therefore gives a bounded map from the second norm to the first, and a bound C/N_k from the strong forcing norm to the first. Finite exceptional kernels have bounds $C_{k,v} e^{-\lambda_{k,v}(t-s)}$ and the analogous scalar assertion for $\omega < \lambda_{k,v}/2$.

At a current high tail the difference equation has the form $\mathcal{L}_m E = A_0 E + F$, where F is the formal substitution residual plus predecessor differences. Precisely, if P_J is the current polynomial, A_J, F_J are the polynomial predecessor evaluations, and $\mathcal{R}_J = \mathcal{L}_m P_J - A_J P_J - F_J$, then

$$F = (A_0 - A_J)P_J + (F_{\text{actual}} - F_J) - \mathcal{R}_J.$$

The finite Taylor operations defining the forcing are Lipschitz on their bounded coefficient balls; their Lipschitz constants are finite derivatives of $\Phi_{k,v}$. Inherited state derivatives use (6.26). Consequently, by the previous comparison stages and the residual estimate, on this whole segment

$$\|F\|_{0, -1} \leq C_{k,v} e^{-a_{k,v} R}$$

for some $a_{k,v} > 0$. Predecessors may all use the current smaller proportional cutoff: subtracting the finite Gevrey tails preserves an exponential error. In the exceptional list there is no current multiplier; for a graph or pure side its place is taken by the small linearized coefficient. Variation of constants now gives

$$\|E\|_\omega \leq C_{k,v} B_{k,v} + \frac{1}{2} \|E\|_\omega + C_{k,v} e^{\omega L - a_{k,v} R}. \quad (6.35)$$

The factor $1/2$ is ensured by the tail threshold, or the original pure-side smallness, after the harmless weighted-kernel change. Choose additionally $\omega b_k \leq a_{k,v}/2$. At the endpoint this proves the explicit estimate

$$\|E(L)\|_{\rho, R_k} \leq C_{k,v} (B_{k,v} e^{-\omega b_k R} + e^{-a_{k,v} R}). \quad (6.36)$$

It has a positive exponential rate at every finite step. Alternating signs use the successive domains in (6.27); one infinite high tail uses one segment, with its whole coefficient sum. Thus the finite induction closes without any fast-radius loss or exhaustion of the prescribed state, profile, or spatial margins. This proves the optimal error; subtracting the omitted Gevrey terms proves (6.21) at smaller orders. Remaining bounded radii for a fixed J are absorbed in its constant.

5. Weighted profile derivatives. Differentiating at fixed anchors in τ gives polynomial weights $w^{r \cdot \alpha}$. The highest derivative of a graph or pure side has the same small linearization, and the highest derivative of a high tail has the same absorbable convolution. The other terms use lower derivatives. Apply the unweighted resolvent first and then commute a fixed polynomial weight through its decaying kernel; the resulting constant is an exponential moment $\int_0^\infty e^{-ct} (1+t)^{C_\alpha} dt < \infty$. One need not demand that this weighted constant remain small. More explicitly, $A_0(0) = 0$ means that each convolution step increases the opposite degree. In the iterated Volterra resolvent, a column starting at degree n_0 is therefore bounded in weighted absolute coefficient sum by

$$C e^{-\lambda(1+|n_0|)t} \sum_{v \geq 0} \frac{(C M_A t)^v}{v!} \leq C e^{-\lambda'(1+|n_0|)t},$$

after the chosen high-tail threshold is increased once, if necessary. For a pure side the same inequality follows from its original smallness. Along a monotone path, the ratio of either endpoint modulus to the other is at most $1 + Ct/q_*$. Multiplying this resolved kernel by a fixed power of that ratio gives a finite exponential moment and still an inverse bound $C_\alpha/(1 + |n_0|)$. Thus the global weighted jet bounds, and in particular the normalized starting bounds in (6.35), follow before any jet comparison. Adding the finitely many lower-jet comparisons to the spatial schedule then proves the differentiated errors with the same remaining domains.

6. Anchor comparison and limiting jets. Compare neighboring anchors using a common inner polygon with horizontal extremes fixed large multiples of $|Q|$, leaving the working target region $|\Re w| \leq C_1 |Q|$ strictly inside. At each of the finitely many comparison steps divide its allotted horizontal margin by s_k and choose a starting line on the appropriate side. The monotone graphs used in constructing the lateral domain connect that line to every target, with bounded slope and real distance at least $b'_{k,v} |Q|$, where $b'_{k,v} > 0$. These paths can pass above the inner obstacle; they need not be horizontal across it. Graph and pure-side differences have homogeneous absorbable equations. At a later step the difference forcing is bounded by the derivative of the same finite polynomial majorant times predecessor differences. The weighted argument (6.35), now with $R = |Q|$, therefore gives

$$E_{k,v} \leq C_{k,v} |Q|^{p_{k,v}} \left(e^{-\omega b'_{k,v} |Q|} + \max_{\prec} E_{\prec} \right).$$

Here $p_{k,v}$ accounts for the fixed jet weights on the whole path; without such jets it can be zero. Finite induction proves $E_{k,v} \leq C'_{k,v} e^{-c_{k,v} |Q|}$ with $c_{k,v} > 0$. The starting distance is of outer-scale

size also when the target is near q , so the estimate holds uniformly down to the inner radius. At $\tau = 0$ positive-growth profiles disappear; their finite jets solve equations with fixed polynomial weights. Compare consecutive dyadic outer anchors and sum their exponentially small differences on each fixed interior. The locally uniform limit gives the asserted analytic, canonically normalized infinite-anchor jets and all their estimates. $\square$

6.3 Slow primitives and their normalization

The linear coordinate in (A) needs a primitive I satisfying

$$DI = b(w), \quad D = w\partial_w. \quad (6.37)$$

In (B), every balanced drift term in (6.19) has positive degree in P . A fixed charge coefficient of its flow therefore uses only finitely many iterations of the integral equation. These give nested integrals of entries $w^{d_e}c_e(w)$ in logarithmic time, with fixed rational d_e . The factor changing dw to dw/w is included in d_e . The functions c_e are balanced slow coefficients or fixed derivatives in an ordinary state argument, which is held independent at this step. For each required word use a strictly upper triangular matrix having its successive entries on the first upper diagonal. A unit upper triangular fundamental matrix Y has entries satisfying

$$DY_e = w^{d_e}c_e(w)Y_{\text{tail}}, \quad Y_\emptyset = 1. \quad (6.38)$$

Include all consecutive subwords. Its transfer is $Y(p)Y(q)^{-1}$, a polynomial in these finitely many entries. This representation uses no exponential bound on an unrestricted matrix flow over a long logarithmic interval.

Lemma 6.3 (Normalized slow primitives). *There are particular formal primitives*

$$\hat{Y}_e = w^{d_e} \sum_{j \geq 0} w^{-j} B_{e,j}(X, L), \quad d_e = \sum_{e \in e} d_e, \quad L = \log(w/q), \quad (6.39)$$

where $B_{e,j}$ is polynomial in L of degree at most the word length and its analytic coefficient norm is at most $C^{j+1}j!$. The analogous assertion holds for $\hat{I}$ with $d = 0$ and length one.

Normalize the actual primitives by a sufficiently small proportional optimal truncation at a positive w_* comparable to $|Q|$, in the common padded interior. Then I is bounded on the needed paths, and on a matching outer ray the normalized outer values are approximated by their formal optimal truncations with error $O(e^{-c|Q|})$. The inner kernels

$$I(q), \quad q^{-d_e}Y_e(q) \quad (6.40)$$

are bounded, have Gevrey expansions in $1/q$ obtained by setting $L = 0$ in (6.39), and obey the differentiated estimates of Lemma 6.2. In particular a positive-profile jet of order α is bounded after division by q^{r^α} . Neighboring finite normalizations change these kernels by $O(e^{-c|Q|})$, allowing fixed polynomial losses.

At zero positive profiles there are canonical infinite-anchor primitive jets. Their normalized inner values have the same Gevrey estimates. For each fixed jet and each fixed N , finite and infinite normalized jets differ by $O_N(|Q|^{-N})$. This last assertion is power accuracy, and does not claim exponential accuracy in $|Q|$.

Proof. 1. Formal primitives and outer normalization. Multiply the already constructed formal series and argue by word length. The factorial inequality $\sum_{k=0}^j k!(j-k)! \leq Cj!$ preserves the asserted Gevrey order. On a profile monomial X^μ the primitive operation is inversion of

$$r \cdot \mu + d_e - j + \partial_L. \quad (6.41)$$

All its nonzero constant parts have absolute value at least $1/D_0$ for one fixed common rational denominator D_0 . On polynomials of degree at most s , for $c \neq 0$ its inverse is the finite expression

$$(c + \partial_L)^{-1} F = \sum_{k=0}^s (-1)^k c^{-k-1} \partial_L^k F.$$

At $c = 0$ integrate in L , choosing zero integration constant. These operators have bounds depending only on s, D_0 . Use absolutely convergent profile coefficient norms on smaller disks to sum them. Induction gives (6.39) with degree at most the word length. In particular the formal choices have no undetermined constants.

Freeze w_* and the truncation order on each modulus choice. Define the actual triangular solution by those truncated values at w_* ; define I in the same way. For its leading coefficient, $\hat{b}_0 = g_v(X, 0)$ has no constant profile monomial. A resonant monomial at slow order zero must therefore involve both positive and negative profile weights. If the total positive weight is $\rho > 0$, its evaluation contains $(q/Q)^\rho$, and these positive weights have a fixed positive rational minimum. Thus the coefficient of L at order zero pays $\log(w_*/q)$. Higher slow orders carry at least w_*^{-1} , which pays this logarithm as well. The normalized value of I at w_* is bounded. Since $b = O(X)$ on the actual graph, integrating along radial and bounded-angle arc paths gives a bounded variation by the nonzero profile powers, as in (6.9). This proves the boundedness needed for its later exponential.

If p lies on the matching Q ray, connect it to w_* using paths with all radii comparable to $|Q|$. A common smaller optimal truncation has exponentially small substitution residual, by Lemma 6.2. Triangular integration loses only fixed powers and fixed-degree logarithms; these are absorbed by decreasing the exponential constant. This proves the outer endpoint assertion. No actual evaluation of $I(p)$ at a multiply wound Q is required.

2. Uniform propagation to the inner endpoint. The outer normalization alone does not bound the inner kernels when $|Q|/|q|$ diverges. We compare the actual and formal primitives along the full inward path. The triangular equations preserve their explicit power weights, and a sufficiently high slow order makes those weights integrable with constants independent of the endpoint ratio.

Fix the finite word and jet list, closed under consecutive subwords and lower jets. Put $q_0 = |q|$, $R = w_*$, $p_\alpha = r \cdot \alpha$, $\beta_{v,\alpha} = d_v + p_\alpha$, and $\ell(t) = 1 + \log(t/q_0)$. Fixed ordinary derivatives may be included in the jet list with weight zero. First compare on the radial part $q_0 \leq t \leq R$, and then on a bounded-angle arc to q . For a word $v = e v'$ define

$$P_v^J(w) = w^{d_v} \sum_{j < J} B_{v,j}(X, L) w^{-j}, \quad c_e^{<L_0}(w) = \sum_{a < L_0} c_{e,a}(X) w^{-a}.$$

Formal coefficient comparison gives the exact finite identity

$$DP_v^J = w^{d_v} \sum_{a+b < J} c_{e,a} B_{v',b} w^{-a-b}.$$

Consequently the local substitution residual is exactly

$$\mathcal{R}_v^J := w^{d_e} c_e P_{v'}^J - DP_v^J = w^{d_v} \sum_{b < J} w^{-b} B_{v',b} (c_e - c_e^{<J-b}). \quad (6.42)$$

Thus its estimate does not presuppose an estimate for an actual primitive remainder. On the fixed smaller profile disks the formal coefficient bounds and Cauchy's inequality give

$$|\partial_\tau^\eta B_{v',b}| \leq A_\eta C_0^{b+1} b! |w|^{p_\eta} \ell(|w|)^S,$$

with one fixed S covering all log degrees in the request. Lemma 6.2 gives, for $1 \leq L_0 \leq c|w|$,

$$|\partial_\tau^\eta (c_e - c_e^{<L_0})| \leq A_\eta C_0^{L_0+1} L_0! |w|^{p_\eta - L_0}.$$

The derivatives are at fixed q, w_* and other independent arguments, so they commute with D . After Leibniz differentiation of (6.42), each term has power $d_v - b + p_{\alpha-\eta} + p_\eta - (J - b) = \beta_{v,\alpha} - J$. Using $\sum_{b < J} b!(J - b)! \leq 3J!$ and absorbing the finitely many jet binomial factors proves

$$|\partial_\tau^\alpha \mathcal{R}_v^J(w)| \leq C_\alpha^{J+1} J! |w|^{\beta_{v,\alpha} - J} \ell(|w|)^S. \quad (6.43)$$

The initial normalization tail has the very same weight. Indeed choose $K_R = \lfloor \varepsilon R \rfloor$ on larger parameter disks such that $C_0 K_R / R \leq \theta < 1$. Fixed derivatives cost Cauchy prefactors, while the coefficient-growth constant C_0 can be chosen on those larger disks. For $J \leq c q_0 \leq K_R$, the geometric tail estimate and $Y_v(R) = P_v^{K_R}(R)$ give

$$\left| \partial_\tau^\alpha (Y_v - P_v^J)(R) \right| \leq \frac{A_\alpha}{1 - \theta} C_0^{J+1} J! R^{\beta_{v,\alpha} - J} \ell(R)^S. \quad (6.44)$$

The anchors and integer orders are frozen in these derivatives.

Write $E_{v,\alpha}^J = \partial_\tau^\alpha (Y_v - P_v^J)$. Its differentiated equation is

$$D E_{v,\alpha}^J = w^{d_e} \sum_{\eta \leq \alpha} \binom{\alpha}{\eta} (\partial_\tau^\eta c_e) E_{v',\alpha-\eta}^J + \partial_\tau^\alpha \mathcal{R}_v^J. \quad (6.45)$$

Since $|\partial_\tau^\eta c_e| \leq C_\eta |w|^{p_\eta}$, the power of each predecessor error on the right is exactly $d_e + p_\eta + \beta_{v',\alpha-\eta} - J = \beta_{v,\alpha} - J$. Choose $J > 1 + \max_{v,\alpha} \beta_{v,\alpha}$, the maximum being over the entire finite system. For any such β and $q_0 \leq t \leq R$,

$$R^{\beta-J} \ell(R)^S + \int_t^R s^{\beta-J} \ell(s)^S \frac{ds}{s} \leq C_S t^{\beta-J} \ell(t)^S. \quad (6.46)$$

For the integral put $s = te^x$, use $\ell(te^x) \leq \ell(t)(1+x)$, and bound $\int_0^\infty e^{-x}(1+x)^S dx$. The boundary term follows from the bounded supremum of the same function. Both bounds are uniform in t, R, q_0, J ; in particular the initial tail cannot accumulate a factor depending on R/q_0 .

For explicit triangular bookkeeping, let

$$X_{v,\alpha}^J = \sup_{q_0 \leq t \leq R} \frac{|E_{v,\alpha}^J(t)|}{C_0^{J+1} J! t^{\beta_{v,\alpha} - J} \ell(t)^S}.$$

Equations (6.43)–(6.46) give

$$X_{v,\alpha}^J \leq B_{v,\alpha} + C_S \sum_{\eta \leq \alpha} B_{e,\eta} X_{v',\alpha-\eta}^J, \quad (6.47)$$

after choosing the common C_0 to dominate the residual constants. All B 's depend only on the fixed finite request and domain margins. The right side contains strictly shorter words. Finite triangular induction bounds every $X_{v,\alpha}^J$ independently of J, R, q_0 . The bounded-angle arc has radius

comparable to q_0 and bounded lifted logarithm; integrating the same finite triangular system there changes only these constants. At $w = q$, divide by $q^{d_v+p_\alpha}$ to obtain $C^{J+1}J!q_0^{-J}$.

To cover all requested orders $J \geq 1$, choose a fixed integer $B > 1 + \max_{v,\alpha} \beta_{v,\alpha}$ and first make the comparison at $J + B$. Decrease the target proportional constant so that $J + B \leq c_0 q_0$ on the unbounded range. The apparent extra factorial is canceled by the extra radius power:

$$\frac{(J+B)!}{J!} q_0^{-B} = \prod_{s=1}^B \frac{J+s}{q_0} \leq c_0^B. \quad (6.48)$$

Add back the fixed B formal terms of degrees J through $J + B - 1$. Their Gevrey bounds give the same $C^{J+1}J!q_0^{-J}$ estimate, with an enlarged constant. The remaining bounded range of q_0 is handled by first integrating down to a fixed radius above this threshold and then over the remaining compact radial interval and arc. There are only finitely many allowed J on that range, and $q_0 \geq q_* > 1$; all constants are finite and uniform there. This changes no original smallness condition or initial q_* . The case I is the length-one version of the same calculation.

3. Finite and infinite normalizations. For two neighboring finite normalizations compare first near their outer normalization points. Their formal optimal values and the nonprimitive coefficients differ by $O(e^{-c|Q|})$. Propagate inward using the triangular difference equations. Every word and fixed jet loses at most a fixed power of $|Q|$, since q is bounded below, so the inner difference remains exponentially small.

At $\tau = 0$ use the canonical nonprimitive infinite-anchor jets from Lemma 6.2. Subtract a differentiated formal truncation of sufficiently high fixed order J and solve the residual equations by integration from positive-real infinity, then by an interior arc when required. Convergence follows from (6.46). Indeed a fixed derivative ∂_τ^α at zero selects exactly the positive-profile multi-degree α ; all remaining profile powers are negative. Its j th formal term is therefore bounded by $C_{\alpha,j}|w|^{d_e+r\cdot\alpha-j}(1+\log(|w|/|q|))^s$. Taking J beyond all these fixed power costs makes the subsequently omitted terms tend to zero at infinity. Increasing J gives the same solution: the difference solves the homogeneous differentiated triangular recursion, and its first nonzero entry is constant with limit zero at infinity; inductively every entry is zero. The same construction at $J \leq c|q|$ gives its Gevrey estimate. Locally uniform convergence gives analytic parameter germs. At w_* compare this infinite solution and the finite normalization at any sufficiently large fixed J . Their difference is smaller than any prescribed power of $|Q|^{-1}$, after allowing the fixed polynomial/logarithmic losses. Propagate to q by the triangular equations, using the exponentially small nonprimitive differences on the common path. Choosing J still larger pays the finite propagation losses and proves the claimed $O_N(|Q|^{-N})$ estimate. $\square$

In particular every bounded inner kernel K just constructed admits a finite Taylor expansion in the positive profiles at their current values $\tau_j = \chi_j Q^{-r_j}$. For a prescribed finite total Taylor degree N , the integral Taylor remainder and the weighted jet bounds give

$$K(\tau) = \sum_{|\alpha| < N} \frac{\chi^\alpha}{\alpha!} (q/Q)^{r \cdot \alpha} [q^{-r \cdot \alpha} \partial_\tau^\alpha K(0)] + O((|q|/|Q|)^{r_{\min} N}), \quad (6.49)$$

where $r_{\min}$ is the smallest positive profile weight. For normalized word kernels K already includes q^{-d_e} . The Taylor segment remains in the original independent profile boxes. Replacing each finite zero-profile jet in this finite sum by its infinite-anchor jet has arbitrary additional $|Q|^{-N'}$ accuracy. If there are no positive profiles, the Taylor step is vacuous: retain the single zeroth jet, with the same finite-to-infinite replacement.

Lemma 6.4 (Bounded analytic Gevrey operations). *Finite sums and products, analytic operations at bounded arguments with fixed room in their convergence boxes, and regular inverses preserve the normalized Gevrey and jet conclusions above. For two independent slow variables the resulting formal coefficients have bounds*

$$\|c_{ab}\| \leq C^{a+b+1} a!b!. \quad (6.50)$$

Rectangular optimal evaluation has separate errors $O(e^{-c|Q_0|}) + O(e^{-c|q_0|})$; changing only one optimal cutoff has the accuracy of its own variable. These conclusions remain uniform when other bounded analytic arguments vary independently on smaller fixed disks, and under fixed derivatives on those disks.

Proof. For finite budgets J, N , test the truncated inputs at dummy radii $\varepsilon/(1+J)$ and $\varepsilon/(1+N)$. Their nonconstant increments are small in absolute coefficient norm and their constant values stay strictly inside the analytic boxes. Absolute Taylor majorants therefore bound the output independently of J, N . Extracting its (a, b) coefficient with $J = a, N = b$ gives (6.50), using $a^a \leq e^a a!$ and the analogous inequality for b . For a regular inverse the analytic implicit theorem applies on the dummy disks, uniformly on the smaller argument boxes; the resulting majorant is bounded there. Actual inverses and their approximants differ by the residual times a bounded inverse Jacobian, with the same finite differentiated bounds.

At sufficiently small proportional optimal orders the dummy radii exceed the actual values by a fixed factor. Terms omitted in one variable have geometrically small absolute sum in that variable, uniformly in the other dummy disk. This gives the two separate exponential errors and proves the assertion when one cutoff alone is changed. At a replacement of a bounded kernel, ordinary Cauchy bounds first pay its actual error; the formal calculation is then performed on the replacement. Fixed derivatives use smaller disks with room. For infinite-anchor jets the ordinary finite chain rule commutes with the locally uniform limiting construction, so these are the same canonical jets as those computed after the operation. $\square$

6.4 Finite expansions in the transfer charge

For (A), use T and I from one fixed lateral construction at both endpoints. Exact conjugacy gives the terminal value

$$T\left(w_{\text{out}}, Pe^{I(w_{\text{out}})-I(w_{\text{in}})}T^{-1}(w_{\text{in}}, v_{\text{in}})\right). \quad (6.51)$$

The exact linear coordinate stays in its disk along the passage because of homogeneous damping and the bounded small variation of $\int b dw/w$. Expanding the final evaluation of T at $u = 0$ to any fixed degree in P has a remainder $C_K|P|^{K+1}$. Keep the actual bounded inner kernels at q ; replace outer kernels at p by their optimal formal values, with exponential error in $|Q|$.

Inside $e^{\pm I(p)}$ the formal logarithm deserves care. Write its coefficient at slow order j as $R_j(X)L(p)$. The order-zero resonant sum is $(q/Q)^\rho$ times an analytic function of small rational-power ratios, for some fixed $\rho > 0$, by the opposing-profile argument in Lemma 6.3. Treat $(q/Q)^\rho L(p)$ as a small analytic argument. For $j \geq 1$ factor one $Q^{-1}L(p)$ and shift the remaining slow series by one fixed index, retaining the bounded factors of p/Q . This avoids an unbounded logarithm as an analytic input or a shift of Gevrey index depending on the requested order. Lemma 6.4 then applies to the exponential.

For (B), retain the coefficients of (6.17)–(6.18) with $\min(a, b_f) \leq K$ and denote the result by $t^{[K]}, \mathcal{C}^{[K]}$. The two fixed-side parts of $t^{[K]}$ are genuine analytic sums on their opposite fast disks, as established for the region in Figure 3. Their pure parts t_L, t_R are small, including their state derivatives. Although the other coefficient norms may depend on K , every mixed retained part

evaluated on the actual integration polygon is $O(C_K|Q|^{C_K}|P|)$, with every requested fixed state derivative.

Here is the estimate used repeatedly. For an analytic fast sum all of whose monomials have $\min(a, b_f) \geq k$, select from each side a sub-multi-index of charge in $[k, k + c_*]$, where c_* is the largest generator weight. Since both $\Re(A - q)$ and $\Re(Q - A)$ are nonnegative, the exponential product of those selected factors is at most $|P|^k$. Their fixed rational powers cost at most $C_k|Q|^{C_k}$. The remaining fast factors sum on the common disk, using the fixed amplitude slack. Thus

$$|F(E(w), H(w))| \leq C_k|Q|^{C_k}|P|^k\|F\|_\rho \quad (6.52)$$

for that restricted support, and likewise on fixed smaller disks for the needed derivatives. No unrestricted double mixed series is used.

Invert $u \mapsto u + t^{[K]}(q, E, H, u)$ to impose the initial condition; the pure-side derivative is small and the mixed part is exponentially small by (6.52). Solve $u' = \mathcal{C}^{[K]}$. Every term has at least one charge P , so this is a small perturbation over the polygon despite fixed polynomial length losses. Then $z = u + t^{[K]}(w, E, H, u)$ has residual

$$O(C_K|Q|^{C_K}|P|^{K+1}). \quad (6.53)$$

To verify it with possibly large mixed coefficients, Taylor-expand G about $u + t_L + t_R$ in the mixed part through degree K . The actual Taylor remainder already has (6.53). The finitely many products are analytic sums on smaller common radii. Their coefficients of charge grade at most K cancel by (6.17)–(6.18). Transport differentiation preserves the charges already present; unused terms therefore come only from the analytic composition and $\partial_u t \mathcal{C}$ products. Bound them by (6.52). Termwise differentiation is valid on the smaller disks. Finally, $G_z = O(E, H)$ gives a bounded integral of the state Lipschitz constant by (6.10). The integral comparison with the exact passage multiplies (6.53) by at most another fixed polynomial factor in $|Q|$.

The balanced terms have the exact form (6.19). The endpoint fast arguments are exactly

$$E(q) = e, \quad H_s(q) = d_s(q/Q)^{\nu_s} P^{b_s}, \quad (6.54)$$

$$E_s(p) = e_s(p/q)^{\beta_s} P^{a_s}, \quad H_s(p) = d_s(p/Q)^{\nu_s}. \quad (6.55)$$

Replace P here and in (6.19) by a dummy variable. For $|P_{\text{dummy}}| \leq |Q|^{-C_K}$, with C_K sufficiently large, the initial inverse, balanced evolution, and final transformation are analytic and bounded: all mixed terms and integrated balanced terms are small, and the pure parts remain regular. Cauchy's formula therefore gives a Taylor expansion through degree K with error $C_K|Q|^{C_K}|P|^{K+1}$ at the actual exponentially small P .

Each coefficient in this expansion has a finite slow construction. At degree zero the initial inverse is the pure-left inverse, a bounded kernel $H_0(q, z_{\text{in}})$ with regular endpoint estimates. Higher inverse jets use fixed derivatives and the convergent fixed-side tails. Taylor–Picard coefficients of the balanced equation are finite nested integrals of balanced coefficients and their state derivatives at the constant $u = H_0$. Splitting products of integrals by the finitely many orderings of their integration variables writes them as finite sums of the triangular transfer entries $Y(p)Y(q)^{-1}$. First evaluate these entries with u independent; then substitute H_0 . The final endpoint transformation uses the same fixed-side coefficients and finite derivatives. All power prefactors in q, Q, p are explicit and finite at each charge order.

Proposition 6.5 (Finite charge expansion). *On any sufficiently tight i fringe, through its buffered side, both actual passages have an expansion to an arbitrary fixed charge cutoff, with remainder*

$$C_K|Q|^{C_K}|P|^{K+1} + C_K e^{-c_K|Q|}. \quad (6.56)$$

Each coefficient is a finite expression in explicit power and fixed-degree logarithmic prefactors, bounded normalized inner kernels at q , and outer formal Gevrey series in a variable of size Q^{-1} . Their analytic operations have bounded independent arguments with fixed margins. Outer coefficients may be evaluated at bounded inner kernels such as H_0 . Every fixed derivative has the same description with its own constants and polynomial losses.

Proof. The constructions above prove the expansion. Formula (6.3) writes $X(p)$, p/Q , and $1/p$ in analytic units of the ratio and $(q/Q) \log(Q/q)$. The formal word coefficients have only fixed-degree log polynomials, which remain outside analytic operations when unbounded. The only exponential of a primitive is controlled by the separate small arguments described after (6.51). Thus the factorial estimates apply to actual slow series indices; no estimate uniform in the charge budget or in the finite explicit prefactors is being asserted. For a fixed M , choose K using (6.8). Its polynomial loss is negligible relative to $\Re Z_i$. Then tighten the fixed good contour until $c_K |Q|$ dominates $M \Re Z_i$ throughout that fringe; proceeding to the buffered side only improves this ratio. The error is consequently $O(e^{-M \Re Z_i})$. The differentiated assertion follows from the proved kernel derivatives, analytic operations on smaller disks, and larger initial charge budgets. $\square$

6.5 From charge coefficients to packet trees

Write the leading affine decompositions exactly as

$$Q = Q_0(1 + \Theta_Q), \quad Q_0 = A_0 Z_i, \quad q = q_0(1 + \Theta_q), \quad q_0 = A_1 Z_l, \quad (6.57)$$

where $A_0, A_1 > 0$ are fixed. For bounded q put $q_0 = q$ and $\Theta_q = 0$, and regard l as later than every transition. Each correction is a finite sum of strict positive shifts times bounded analytic coefficients, with first index greater than i , or greater than l , respectively. This uses the stipulated conversion of polynomial scale factors to log nodes. Set

$$h_Q = Q_0^{-1}, \quad h_q = q_0^{-1} \quad (q \text{ large}), \quad s_0 = q_0/Q_0, \quad M_0 = \log(Q_0/q_0). \quad (6.58)$$

The first index of h_Q is $s = \sigma(i)$. Let r be the first index of M_0 , which has a positive leading coefficient. Comparing the exact log formulas gives

$$r \geq s, \quad \sigma(l) \geq s \quad (q \text{ large}); \quad s \leq l \implies r = s. \quad (6.59)$$

For bounded q , $r = s$. Indeed no term of either log can occur before s : a first term of f_l before s would contradict $f_i > f_l$. If $s \leq l$, the first index of f_l is strictly later than l , so it cannot cancel the leading term of f_i at s . Consequently

$$p_0 = \min(l, r) \leq s. \quad (6.60)$$

All bounded formal outer arguments are analytic in finitely many strict positive shifts with bounded coefficients. Besides the Θ arguments, these include positive rational powers of s_0 , the components of $s_0^\rho M_0$ for fixed $\rho > 0$, and those of $h_Q M_0$. To see this last assertion, every unbounded term of M_0 is a node Z_k with $k \geq r \geq s$. Replacing this polynomial factor by its log formula introduces only shifts with first index strictly later than k . Thus multiplication preserves the positive first index r of s_0^ρ , or s of h_Q . Logarithms of $1 + \Theta$ and of p/Q are analytic near their regular values. Explicit unbounded power and log prefactors outside analytic operations are finite sums of shifts times analytic units in the same small arguments.

In contrast to those analytic substitutions, the charge factor is always extracted exactly:

$$P^n = \exp[-n(Q - q)]. \quad (6.61)$$

Its exponent is the original affine combination of nodes and bounded coordinates, with leading term nA_0Z_i . It is never evaluated by Taylor-altering the exponent through Θ_Q or Θ_q .

Retiring an anchor or cutoff means removing its numerical value from subsequent coefficient formulas. It does not identify its remainder with zero. Each retirement is performed only at a transition where the stated remainder pays the requested exponential budget. A finite-anchor kernel, its optimal slow polynomial, and a fixed formal coefficient are distinct objects; the exact lateral information has already been retained in the charge construction, as illustrated by (6.11).

Proposition 6.6 (Dispersal and retirement). *Every coefficient furnished by Proposition 6.5 admits, on each lateral determination, a packet tree using only retained free inputs and ordinary inputs. The numerical outer anchor for an inner kernel is removed at p_0 in (6.60); the numerical optimal outer variable is removed at s . When q is large, its optimal variable is removed at $\sigma(l)$. All transitions have arbitrary fixed exponential accuracy on their full owning fringes. The support is iterated left-finite and all terminal leaves are analytic ordinary germs.*

The indices l, r, s and $p_0 = \min(l, r) \leq s$ were defined in (6.57)–(6.60). The retirement order is as follows.

Case	Finite outer anchor in the inner kernel	Outer optimal h_Q	Inner large- q variable
$l \leq r$	Removed at l by optimal inner q replacement	Removed at s	Optimal h_q removed at $\sigma(l)$
$r < l, q$ large	Removed at $r = s$ by positive-profile Taylor extraction and infinite-anchor jets	Removed at s	Kernel replaced at l ; optimal h_q removed at $\sigma(l)$
q bounded	Removed at $r = s$ by the same Taylor and jet construction	Removed at s	None; ordinary infinite-anchor germs remain

At coincident indices the formal replacement precedes extraction of the current-index arguments; in particular, when $l = s$ the inner replacement precedes extraction of h_Q . In the last two rows the finite-to-infinite primitive-jet comparison has arbitrary fixed power accuracy in Q , as in Lemma 6.3. It is used at $r = s$, where those powers pay the transition budget.

Proof. First set aside the finite explicit shift prefactors; a fixed shift translates a packet tree. We describe the remaining bounded coefficient expression. Keep its outer Gevrey series as an optimal polynomial in h_Q until index s is extracted. Rewrite analytic unit factors into its coefficients, preserving $C^{a+1}a!$ bounds. Keep the actual finite-anchor inner kernels through the band p_0 , using the respective upper or lower construction.

This prescription includes an independence requirement. Before p_0 , put

$$Q_{\text{eval}} = Q_0(1 + \Theta_Q), \quad \tau_j = \chi_j Q_{\text{eval}}^{-r_j} \quad (r_j > 0),$$

using the ratio arguments currently retained or independently varied in a Taylor calculation. Freeze the anchor modulus and order using Q_0 , independently of these varied ratios. Those ratios can vary on fixed small complex disks: endpoint scales and profile sizes change only by fixed factors, and the constructions have analytic room for these changes. Other artificial small arguments of the outer formal coefficients need not vary the inner kernel. Although the lifted phase of Q_0 may wind, its needed logarithm is retained, and $|Q_0| \gg |q|$. All relevant powers have their fixed lifted branches. At the first, i -matching evaluation these independent arguments take their actual values.

At each transition, Taylor-expand every analytic small argument whose first index is current to an arbitrary fixed budget. Leave the other small arguments independent; set an extracted one to

zero in subsequent coefficient evaluations. This is permitted already before p_0 . Cauchy estimates on the original argument disks ensure that derivatives created by such an earlier Taylor step retain the uniform kernel, positive-profile, and Gevrey estimates needed at p_0 . In particular an earlier Taylor expansion of a Q -ratio differentiates the bounded holomorphic coefficient formula while Q_0, q and the anchor convention are held fixed. It does not differentiate a varying anchor or reimpose an identity between independent arguments.

There are two possible first retirements.

Case $l \leq r$. At the l fringe replace every bounded inner kernel by its optimal h_q expansion. Its error $O(e^{-c|q|})$ pays any fixed l -transition budget after the contour is tightened; split past prefactors into their remaining shift tails before making this comparison. The formal coefficients use $X(q)$ and regular endpoint arguments. Positive profiles at q carry powers of s_0 and hence are among the allowed strict positive shifts. Numerical finite-anchor dependence has now disappeared. If an outer coefficient was evaluated at an inner kernel, use the rectangular series of Lemma 6.4, with coefficient bounds $C^{a+b+1}a!b!$ in $h_Q^a h_q^b$. Its separate optimal precisions permit the two variables to be extracted at their different first indices. At index l first perform this formal replacement and then expand arguments whose first index is l ; this may include h_Q , whereas h_q has first index $\sigma(l) > l$.

Case $r < l$, including bounded q . By (6.59), here $r = s$. At that fringe use (6.49) to Taylor-remove the positive profiles and replace each retained zero-profile jet by its normalized infinite-anchor jet. Arbitrarily high fixed powers of q/Q and Q^{-1} pay arbitrary depths at this transition: their logarithms M_0 and $\log Q_0$ have positive first index r . Keep the explicit $s_0^{r-\alpha}$ factors and analytic units outside the normalized jets. At this same index extract the required finite powers of h_Q . One may do that finite h_Q truncation first, uniformly in the bounded inner arguments, then perform the inner Taylor and jet replacements. Every resulting coefficient contains only bounded infinite-anchor q -sector kernels and finite analytic operations. It no longer uses Q_0 numerically. If q is large, retain these kernels to the l fringe and then replace them by their optimal h_q series. If q is bounded, they are already ordinary analytic germs on the chosen lateral determination.

For either case, a current-first optimal variable needs only a finite degree at a fixed transition budget. From its Gevrey bound and a sufficiently small proportional cutoff,

$$\left| \sum_{j=J}^{J_{\text{opt}}} c_j h^j \right| \leq C_J |h|^J$$

for fixed J . Analytic Taylor tails satisfy the corresponding power bound. The flag side estimate makes every strict positive monomial small throughout its available band, and at its owning fringe gives a gain $e^{-c\Re Z_j}$. Hence larger fixed J gives every prescribed transition precision. Variables with later first index retain their optimal evaluation uniformly on the independent parameter disks. Split the finitely retained monomials at the current index, extract their current powers, and leave later components as numerical prefactors for the next band. This recursively defines the packets. The preceding argument applies separately to both angular signs.

This order of operations removes each numerical anchor or cutoff before its controlling log formula becomes unavailable. In the first case the Q anchor disappears at $l \leq s$, and its remaining formal optimal variable at s . In the second case both disappear at $r = s$. An inner optimal q variable survives only until $\sigma(l)$. The common-disk estimates in Lemma 6.4 prove that each cutoff change retains its own exponential precision, even in a rectangular composition. No error of merely power accuracy in Q was used before the transition $r = s$ where such powers pay the budget.

Finally the charge set at index i is a nonnegative discrete lattice. At a fixed charge there are finitely many prefactor translations, followed by Taylor powers of a finite set of lexicographically strict positive shifts. At a fixed current-coordinate budget only finitely many positive current-first

factors can occur; zero current-first factors are treated at the next coordinate. After a prefix is fixed, the remaining negative translations form a finite list. Induction on coordinates proves left-finiteness and finite collision multiplicity, as in the support argument of Theorem 4.4. The finite Taylor and jet operations at each later extraction preserve this property. At the terminal edge all numerical large arguments have retired. Formal coefficients and bounded- q infinite-anchor kernels are analytic ordinary germs, so every leaf is such a germ. There is no assertion that the unrestricted two-sided fast series or the full packet tree is simultaneously convergent. $\square$

6.6 Source precision and independent derivatives

We finish by verifying the quantitative parts of the packet interface. Choose anchors and proportional optimal orders deterministically on fixed-factor bins of $A_0 e^{\Re f_i}$ and, where needed, $A_1 e^{\Re f_l}$. Locally freeze the representative and integer order. Choose normalization systems for a fixed finite formula simultaneously on larger parameter disks. If an earlier coefficient has a different small proportional convention, the overlapping comparisons permit its fixed convention with only the already proved exponential error. Smooth the local evaluations by partitions on overlapping retained log-modulus bins. All branches use the same lifted log relations.

Neighboring finite-anchor kernel evaluations differ by $e^{-c|Q_0|}$ before their retirement. Neighboring optimal polynomials differ by $e^{-c|Q_0|}$ or $e^{-c|q_0|}$ while the corresponding variable is active. Since $|Z_k| = e^{\Re f_k}$, these are the simple source costs

$$U_i = e^{\Re f_i}, \quad U_l = e^{\Re f_l}. \quad (6.62)$$

They are used in a band j only if the originating node $k < j$ still has $\sigma(k) \geq j$. Infinity jets and their bounded analytic evaluations require no numerical anchor or cutoff and create no such source. The first band is the actual holomorphic family and is not smoothed.

For completeness, the simple costs dominate all the losses even at the buffered side. Let $k < j$, $\sigma(k) \geq j$, write $F = f_k(u)$, $L = f_j(u)$, and put $U = e^{\Re f_{k,j}}$ on band j . We claim

$$U/|Z_j| \longrightarrow \infty \quad (6.63)$$

uniformly to that side after increasing its finite dominance constant. The following four cases use exactly the real, angular, and side estimates of Section 2.

If $F \leq L^{3/2}$, the angular Taylor loss in the retained affine log formula is $O(F \text{ polylog } F / \Lambda_j^2) = o(1)$, since $\Lambda_j \geq cL$. Thus $U/|Z_j| = e^{F-L+o(1)} \rightarrow \infty$. If $L^{3/2} < F \leq L^3$, then $\sigma(k) > j$, for a leading term at j would make F comparable to e^L . The leading node in that log has logarithm $\log F + O(1) = O(\log L)$, and angular speed bounded by a fixed power of $\log L$. Its phase on the j band tends to zero. The leading positive term consequently retains a fixed fraction of its real size, and $\Re f_{k,j} \geq cF \gg L$. If $F > L^3$ and $\sigma(k) > j$, the later-angle gain and side comparison give

$$\Re f_{k,j} \geq cF / \Lambda_j \geq cF / (L \text{ polylog } L) \gg L.$$

Indeed the phase deficit of the leading later node is at least a constant times $1/\Lambda_j$, by the diverging gap derivative; its later terms are negligible relative to that real leading term. Finally if $\sigma(k) = j$, the exact log begins with a fixed positive multiple of Z_j . Increasing the side dominance constant absorbs its finitely many later terms and gives

$$\Re f_{k,j} \geq c \Re Z_j \geq ce^L / L^{K+1} \gg L.$$

These cases prove (6.63).

After explicit prefactors have been set aside, the raw bounded pieces have bounded size on every working band. For example a resonant outer logarithm is multiplied by its specified small positive shift; an unbounded explicit log polynomial was kept outside analytic operations and converted to finite shift prefactors. A fixed remaining shift uses indices at least j , so its log modulus is at most $C\Re Z_j$ by side dominance. Thus raw growth is

$$\log^+ |S_\alpha^j| \leq C_\alpha \Re Z_j \quad (6.64)$$

throughout the band. There are no sine poles or other exceptional column factors. Every column, and every needed entering collar, has the safe improvement. By (6.63), all these losses have log-size $o(U)$ at a source; products, interpolation, and any fixed later normalization preserve $e^{-c'U}$ precision.

We verify derivatives in the independent retained free and ordinary variables, rather than only along a path. On band j , all numerical node values of index at least j and their finite triangular sensitivities are bounded by $e^{Cf_j(u)}$. Any earlier retained log is used through its exact affine/background expression in these nodes. With anchors and orders fixed, local Cauchy neighborhoods of radius

$$e^{-Cf_j(u)} \quad (6.65)$$

therefore preserve the small-argument disks, relative-modulus slack, and angular room; enlarge C for a fixed requested derivative order. One first uses a slightly wider polynomial buffer and smaller dominance constant, and then restricts to the working band. Cauchy differentiation loses at most $e^{C_D f_j(u)}$. Since the side buffer gives $\Re Z_j \gg f_j(u)^n$ for every fixed n , this loss is paid by taking larger transition budgets. At a source it has log-size $o(U)$ by (6.63). Derivatives of smooth partitions and of their costs have the same property: fixed powers of U and finite triangular sensitivity factors have logarithms $o(U)$. Thus fixed derivatives of the antiholomorphic defects retain their allowed source precision. The difference $\partial_{\Im z} - i\partial_{\Re z}$ vanishes for frozen holomorphic choices and equals just these smoothing defects for the chosen packets.

The statements are stable under the controlled bounded-input tests in Assumption 4.2. To be explicit, use the actual real controlled backgrounds for the real comparisons and their common Taylor jets at the complex test point. The triangular sensitivities of a retained log have only a fixed power loss relative to its leading size on the angular scale. Taking a sufficiently high fixed shared jet order makes the induced error smaller than the available angular and polynomial side buffer. In the small- F case of (6.63) take enough jets to leave an absolute $o(1)$ log error; in the other three cases a sufficiently small relative error preserves the displayed lower bound. These pointwise tests retain the local Cauchy neighborhoods with slightly more slack. The derivative, source, and safe-column assertions hence hold simultaneously for any fixed finite collection of labels and derivatives. Starts, boxes, and constants may depend on that finite request; every fixed further request is eventually available on the same path returning to the ordinary germ. No uniformity over all budgets at one starting point is used.

Finite Taylor agreement locates the arguments in the common domains and preserves the rate comparisons; it gives no source estimate for an arbitrary smooth test. For holomorphic inputs, or for a sheet already having differentiated source bounds, apply Lemma 4.7. The primitive's intrinsic defects have the simple costs in (6.62); Equation (6.63) pays their fixed losses. Inherited sheet defects retain their own permitted costs, which may include a cosine factor. The general source comparisons accompanying (4.7) pay the same Cauchy and triangular losses of logarithmic size $O(f_j)$, as well as the fixed raw and normalization losses of size $O(\Re Z_j)$. The chain-rule factors therefore have logarithmic size $o(U)$ at the corresponding cutoff sources in both cases. The composed primitive retains the required differentiated source bounds.

At a newly chosen level- p good collar, a fixed current-scale normalization has logarithmic loss at most $C\Re Z_p \leq C'\varepsilon V_p$; fixed later-scale and derivative losses are $o(V_p)$. The join part of Lemma 4.7

therefore applies. Its finite transition budget is chosen before the contour, and errors e^{-cV_p} with contour-independent $c > 0$ allow the stated small- ε choice. This verifies the join precision separately from the $o(U)$ estimates for active cutoff sources.

6.7 Uniformity and canonical terminal germs

We have obtained growth, transition, and differentiated source bounds. We now track their dependence on the controlled input family, then show that the terminal coefficients do not depend on the numerical anchors and cutoffs used along the way. Together with extension to preceding contours, these properties permit the retests used in ordinary elimination.

Lemma 6.7 (Uniformity over a fixed controlled family). *The additive constructions satisfy the common-constant requirement of Definition 4.1. In particular, fix one finite set of labels, derivatives, Taylor orders, and transition budgets, a compact ordinary argument box with common positive analytic slack, and common bounds for the finite real jets, Taylor errors, angular rooms, and flag inequalities used by that request. Then the additive growth, transition, source, derivative, and safe-column estimates have common constants and common numerical rate and gap thresholds throughout every member's full forward domain where those inequalities and argument ranges hold. These constants and thresholds are independent of the member's return time to the terminal germ.*

Proof. First fix the field sup norms on the common larger complex boxes, their distances to the working boxes, the amplitude bound, q_* , the profile exponents and charge lattice, the polygon slope bounds, and the finite requested coefficient and jet lists. Include the positive lower bounds for $|1 + Z|$ and the regular inverse Jacobians used in that finite construction. These are finite quantitative data on the stipulated compact boxes. Every constant in the w -integral equations is a function of these data: the parameters of the ordinary test are frozen at its current evaluation while the w -equation is solved. Its earlier or later values do not enter the Volterra equation.

In particular the reservoirs (6.25), opposite thresholds N_k , finite counts s_k , and polynomial majorants $\Phi_{k,v}$ can all be chosen from these fixed data, before choosing an ordinary test or its anchor. Equations (6.31)–(6.32) then give common ε_k and formal bounds; Equations (6.35)–(6.36) give common positive comparison exponents. The weighted resolvent uses only the same coefficient norms and fixed polynomial jet powers. For primitives, $\beta_{v,\alpha}$, S , the constants in (6.47), and the fixed shift B in (6.48) are likewise fixed by that finite list. Thus their remainder bounds and all analytic operations have constants uniform over the independent argument boxes. The large- Q and large-ratio qualifications used in this asymptotic model are common numerical thresholds. The bounded- q alternative uses its stipulated ordinary neighborhood; the compact radial intervals used for small orders in the fixed- J argument have the uniform continuation bounds already proved.

Only passage from those independent arguments to a complex flag band uses the ordinary test's real jets. We give the quantitative check. Suppose its background Taylor discrepancy satisfies $|\Delta b| \leq H_N |v|^{N+1}$ with the same H_N for the family, and $|v| \leq C/\Lambda_j$ in band j . The triangular sensitivity bounds of Section 2, on the common enlarged angular domain, imply for retained $l \geq j$

$$|\Delta f_l| \leq CH_N \Lambda_j^{-N}. \quad (6.66)$$

For an earlier retained log f_k with $\sigma(k) \geq j$, use its exact affine formula in Z_l , $l \geq j$. The sum of its term moduli is bounded by a fixed multiple of $F = f_k(u)$, using its positive leading term and the common gap thresholds. Therefore

$$|\Delta f_k| \leq CH_N F \Lambda_j^{-N}. \quad (6.67)$$

Both estimates persist on the segment between the reference and test backgrounds after the common angular room is narrowed, so their use does not assume the desired test-domain conclusion.

Write $L = f_j(u)$. In the small-log case $F \leq L^{3/2}$, (6.67) is an absolute $o(1)$ bound uniformly in the family when $N > 2$, since $\Lambda_j \geq cL$. When $\sigma(k) > j$, compare it with the available lower bound cF/Λ_j : its relative size is at most $CH_N\Lambda_j^{1-N}$. When $\sigma(k) = j$, compare instead with the side lower bound cF/L^{K+1} : its relative size is at most $CH_NL^{K+1}\Lambda_j^{-N}$. Choosing the one fixed jet order $N > K + 2$ makes both ratios tend to zero with a common modulus. Equation (6.66) similarly preserves the later-node angle slack and the polynomial side buffer. This proves the controlled-test versions of the domain conditions and all four cases of (6.63) using only common jet and flag constants.

The Cauchy neighborhoods (6.65) can now use one constant C for the finite request. The local bin partitions can be taken from one fixed smooth partition in the retained log moduli; their finite derivative bounds are numerical constants. Multiplying by the common triangular sensitivity bounds shows that all partition and Cauchy losses have uniformly negligible logarithm relative to the active simple costs. There are only finitely many labels and costs in this request, so their common minimum positive precision constant is still positive. Every column is safe, with the already uniform bound (6.64).

All arguments are pointwise quantitative implications of the finite box, jet, rate, gap, and slack inequalities. They apply at every point of every full forward domain on which those common inequalities hold, and not merely on a window selected separately for each test. No step uses how soon a test returns to the germ, or makes a compactness argument over such return times. A later finite request can require a smaller ordinary box and different controls; its constants are then chosen from those new data, as permitted by Definition 4.1. $\square$

Each fixed label, with its fixed derivative request, extends inward to every sufficiently vertical preceding good contour. Indeed its local restrictions are small-argument inequalities and the bounded-angle condition on its next still numerical endpoint. The flag angular comparison puts that endpoint in the slightly enlarged lateral quadrant throughout this inward range; the remaining optimal expressions use lifted logs with no further angle restriction. All argument inequalities have the fixed spare room used above. The threshold for the preceding contour may depend on the label and derivative order, as allowed by the primitive contract.

It remains to record why the terminal germs are invariant in the retesting used by ordinary elimination. At a fixed charge, all formal slow coefficients are fixed by the algebraic recursions with nonzero diagonals; the formal primitive constants were fixed by (6.41). Charge coefficients of two different finite budgets agree up to their common degree, because the same triangular coefficient equations, regular initial inverse, and finite Taylor–Picard recursion determine them. Where a bounded inner kernel survives, its zero-positive-profile jets are the canonical infinite-anchor solutions fixed by decay of the subtracted remainder at infinity. Ordinary analytic compositions and regular inverses commute with these finite coefficient and jet recursions.

Inserting an unused flag node therefore inserts an identity transition with exponent zero. A permitted refinement preserving old active free nodes pulls back the monomial description, these same fixed recursions, and the same ordinary germ functions; only the grouping of intermediate coefficients changes. Tiny free perturbation sequences that preserve the affine log ties, regime gaps, limiting ordinary data, and chosen lateral determinations consequently give the same complete terminal coefficient arrays after this identification. Apply Theorem 4.4 successively for any fixed finite ordinary-chart recipe, always keeping its extracted current symbol at the current inner background. The terminal operations are the same analytic operations and regular limiting root equations, so the terminal germs remain the same. This is a local statement for the fixed lifted determinations; it

claims no global monodromy independence. Nor does it identify a flat error with zero: the charge construction, as (6.11) illustrates, retains that information before retirement.

The full-fringe transition estimates of Propositions 6.5 and 6.6, the source and derivative estimates just proved, and actual first-band holomorphy establish every assertion of Theorem 6.1.

7 One-rate regular and stable passages

We use the scalar-state convention of Section 6. All holomorphic functions below are bounded on fixed complex neighborhoods of the closed slow interval $[0, 1]$, the state disk, and the ordinary parameter boxes. All evaluations use smaller neighborhoods with room. An amplitude bound, once chosen, is fixed independently of subsequent charge and asymptotic requests.

The proof follows the three stages of Section 6: actual continuation, a finite charge expansion, and dispersal of its coefficients into packets. Here the slow coefficient construction uses one inverse large variable and has no surviving inner endpoint kernel.

Let q be a positive large affine combination of flag nodes and bounded ordinary coordinates, with fixed coefficients on the large nodes. Write i for its leading index. As in the additive construction, convert every participating primary basis node to a dependent node, and saturate the flag. Thus powers of these nodes are exact shift monomials times bounded analytic factors. The statement also permits the flag extensions of Section 4.

Theorem 7.1 (One-rate passage packets). *The following two passage outputs satisfy the packet conditions of Definition 3.1 and the differentiated primitive conditions of Assumption 4.2.*

1. A regular base flow with endpoint layers,

$$\begin{aligned} z_s &= V(s, z) + qW(s, E, H, z), & W(s, 0, 0, z) &= 0, \\ E_\nu &= e_\nu e^{-a_\nu q s}, & H_\nu &= d_\nu e^{-b_\nu q(1-s)}. \end{aligned} \tag{7.1}$$

The weights are positive multiples of a fixed charge unit, and the amplitudes are sufficiently small. The base flow from the chosen input disk is analytic and regular throughout $[0, 1]$, with room for a small perturbation.

2. A stable linear generator with bounded slow remainder,

$$v_s = -qv + g(s, v), \tag{7.2}$$

with input in a sufficiently small disk about zero.

Ordinary parameters may occur analytically in all the displayed data. The estimates hold for every fixed number of independent retained-free and ordinary derivatives, and for the controlled input tests of Assumption 4.2. No exceptional unsafe-column loss is required. For each fixed finite request the estimates have uniform constants and far-regime thresholds on the controlled families of Definition 4.1, independently of the individual tests' return times to the ordinary germ. This primitive library is retestable for elimination in the sense of Definition 5.6, with its fixed local sector determinations.

A passage may first be reversed to put it in the stable direction in Equation (7.2). The model in Equation (7.1) may also be applied on finitely many smaller slow intervals. Rescaling an interval multiplies q by its fixed length and changes the endpoint amplitudes accordingly. A sufficiently short fixed interval has the required regular base flow on any bounded analytic state box.

7.1 Actual passage maps

Write $z = \Phi_s(y)$, where Φ_s is the base flow in Equation (7.1) and $\Phi_0 = \text{id}$. Regularity and the available room give an analytic inverse near the compact family under consideration. Consequently

$$y_s = q\widetilde{W}(s, E, H, y), \quad \widetilde{W} = (\partial_y \Phi_s)^{-1} W(s, E, H, \Phi_s(y)),$$

and $\widetilde{W}(s, 0, 0, y) = 0$. The endpoint operation Φ_1 is analytic on a larger disk than will be used. It therefore suffices to treat $V = 0$; we again call the transformed field W .

On the flag sectors through band i , the side estimates give

$$\Re q \geq c\Re Z_i, \quad \Re q \gg \log |q|, \quad (7.3)$$

with the first inequality on the owning fringe and the corresponding right-half-plane dominance on earlier bands. Put $D = qs$ and use the polygon

$$0, \quad S, \quad q - S, \quad q, \quad S = C \log |q|. \quad (7.4)$$

Here C is a sufficiently large fixed constant. Real parts increase, and D/q stays within $O(S/|q|)$ of $[0, 1]$. On each horizontal end the nearby layer is integrable with bound proportional to its amplitude. On the middle both endpoint layers have an exponential margin, giving, for sufficiently large C ,

$$\sup_{\gamma} (|E| + |H|) + \int_{\gamma} (|E| + |H|) |dD| \leq C_0 \max_{\nu} (|e_{\nu}|, |d_{\nu}|). \quad (7.5)$$

For example, its length is $O(|q|)$ and its left-layer contribution is at most $O(|q|e^{-a_{\min} S})$; the right layer is identical. Opposite layers on the end segments are controlled by $\Re q - S \gg S$. The vanishing of W at $(E, H) = 0$ gives the same bound for the integral of its state Lipschitz constant. The integral equation for $y_D = W$ thus keeps the solution in the prescribed smaller state disk, with a fixed margin, by choosing the amplitudes small.

Locally freeze S , vary the endpoints analytically, and solve the same integral equation. This proves holomorphic dependence. Two permissible lengths give the same solution: deform their polygons into one another through polygons obeying the same bounds, and use analytic continuation and uniqueness. This continuation agrees with the real passage and is holomorphic across the real ray. No cutoff is involved in this actual map.

For Equation (7.2), on real $0 \leq s \leq 1$ the integral equation is

$$v(s) = e^{-qs} v_{\text{in}} + \int_0^s e^{-q(s-r)} g(r, v(r)) dr.$$

The forcing contribution is bounded by $\sup |g|/\Re q$ while the solution remains in its box. A continuation argument gives existence and a uniform smaller disk for large q and small fixed input, and ordinary analytic ODE dependence gives holomorphy on these sectors. These arguments apply uniformly on smaller ordinary parameter disks.

7.2 The finite charge expansion

We first use unit charge, so a_{ν}, b_{ν} are positive integers and $P = e^{-q}$. For a general charge unit, replace P by the corresponding power and change the constants below. The next lemma gives the expansion that will be dispersed into a packet tree.

Lemma 7.2 (Finite charge description). *For every integer $K \geq 0$, the layer output has, on the owning fringe, a description*

$$z_{\text{out}} = \sum_{k=0}^K P^k \sum_{\ell=0}^k q^\ell F_{k\ell}^{[J]}(1/q) + O((1 + |q|)^{C_K} |P|^{K+1} + e^{-c_K|q|}). \quad (7.6)$$

Each $F_{k\ell}$ is a formal Gevrey-1 series, analytic in the ordinary inputs. Here $[J]$ denotes truncation at a locally frozen integer $J \asymp \delta|q|$, with $\delta > 0$ sufficiently small for the fixed request. The stable output has the same conclusion with only $\ell = 0$. Estimates persist under fixed ordinary derivatives on smaller disks. Individual constants and optimal proportions may depend on K and the fixed coefficient request.

To obtain this expansion, separate the layer monomials that remain constant along the passage from those that decay towards one endpoint. For layer multiindices put

$$A = a \cdot m, \quad B = b \cdot n, \quad \Delta_{mn} = B - A.$$

Along the actual profiles,

$$E^m H^n = e^m d^n P^B e^{(B-A)qs}.$$

The terms with $A = B$ are therefore constant in s apart from their slow coefficients. Their accumulated drift supplies the powers of q in Equation (7.6). The other terms can be removed successively using the nonzero charge gap. We implement this separation by the layer derivation

$$\mathcal{D}_{\text{lay}} = - \sum a_\nu E_\nu \partial_{E_\nu} + \sum b_\nu H_\nu \partial_{H_\nu},$$

which has eigenvalue Δ_{mn} on $E^m H^n$. Let Π denote its zero-eigenvalue projection. For the stable model, a formal change about its slow graph leaves a linear fiber equation; its endpoint damping supplies the factor P without an accumulated balanced drift. This will give the same charge expansion with $\ell = 0$.

7.2.1 Formal normalization

Lemma 7.3 (Slow-strip formal normalization). *Put $h = 1/q$. For the layer model there are unique formal series in h and the fast variables of the form*

$$y = u + t(s, E, H, u; h), \quad \Pi t = 0, \quad u_D = \mathcal{C}(s, E, H, u; h), \quad \Pi \mathcal{C} = \mathcal{C},$$

determined by

$$\mathcal{C} = \Pi W(s, E, H, u + t), \quad (h\partial_s + \mathcal{D}_{\text{lay}})t = (1 - \Pi)(W(s, E, H, u + t) - t_u \mathcal{C}). \quad (7.7)$$

There is no constant balanced coefficient. For each fixed left multiindex the full right tail converges in a common fixed fast radius, and conversely with left and right interchanged. On any fixed finite collection of such tails and balanced coefficients, the coefficient of h^j has norm at most $C^{j+1}j!$. The pure transformations and their state derivatives are small on smaller fixed fast disks.

For the stable model there are formal series

$$T(s, u; h) = v_0(s; h) + u + \sum_{m \geq 2} t_m(s; h) u^m, \quad b(s; h) = g_v(s, v_0(s; h)),$$

with a convergent common u -disk norm and the same Gevrey bound, satisfying

$$h\partial_s v_0 = -v_0 + hg(s, v_0), \quad (7.8)$$

$$[h\partial_s + (m-1)(-1 + hb)]t_m = h[g(s, T) - g(s, v_0) - b(T - v_0)]_{u^m}, \quad m \geq 2. \quad (7.9)$$

Here v_0 and all t_m start at positive h -order. All these statements allow any fixed ordinary and state derivatives on smaller disks. Constants may depend on the finite request; the fast amplitude and input bounds need not shrink with that request.

Proof. Differentiating $y = u + t$ gives

$$y_D = \mathcal{C} + ht_s + \mathcal{D}_{\text{lay}}t + t_u\mathcal{C}.$$

Multiplication by a balanced monomial preserves imbalance, so $\Pi(t_u\mathcal{C}) = 0$. This proves Equation (7.7), including its signs. Every monomial of W contains a layer factor. The equations are therefore triangular in total fast degree, even at h -order zero; nonzero imbalances have the fixed gap $|\Delta_{mn}| \geq 1$. The coefficient with $m = n = 0$ vanishes and stays zero under the recursion.

Here are the norm details needed in the presence of infinite tails. Use the sum of coefficient sup norms times fixed fast radii, with sup norms on slightly larger slow and state domains. For a pure side the drift is absent. Its composition operator has small Lipschitz norm because W and W_y have a fast factor. Charge division has norm at most one, so a small fixed fast radius gives its analytic majorant and small state derivative on a smaller state disk.

Fix a nonzero left multiindex m and induct on its total degree. The only dependence of W on the current-left tail is linear convolution with

$$W_y(s, 0, H, u + t_{0,*}),$$

which has no constant H coefficient. At current left order, a differentiated factor in $t_u\mathcal{C}$ has lower left order, except that the known pure derivative $t_{0,*,u}$ may multiply a current balanced coefficient. That balanced coefficient is determined on the finite list $B = A$, in increasing right degree, before it can occur in the higher right tail. Thus no derivative of the unknown current tail occurs. First solve the finite list $B \leq N$. For N large relative to A , the remaining diagonals satisfy $|\Delta_{mn}| \geq c(1 + |n|)$. The inverted convolution on this tail has norm at most C/N , hence is absorbable. Its forcing consists of convergent lower-left tails and the finite initial list. Fix one fast radius strictly inside the pure radius throughout; increasing the threshold, rather than decreasing that radius, absorbs every such convolution. The argument with the sides interchanged is identical. State derivative margins can be allotted summably by total one-side degree, so the common remaining state disk is nonempty.

To control the formal slow derivatives, work modulo h^{J+1} , with weighted norm

$$\sum_{j=0}^J (\varepsilon/J)^j \|F_j\|_j \quad (J \geq 1),$$

where the slow domains decrease linearly with j across one fixed allotted margin. Product norms are submultiplicative. Cauchy's estimate across one step of the slow margin bounds ∂_s by CJ , so $h\partial_s$ has norm $O(\varepsilon)$. Choose ε small after the finite coefficient request has been fixed. It is then absorbed after each nonzero charge division. The pure small equation and the current-left high-tail convolution have the same bounds uniformly in J . Lower-order state derivatives use the already allotted margins, and the balanced coefficients are obtained algebraically. This gives a bound independent of J for each requested weighted norm. Taking $J = j$ gives $C(j/\varepsilon)^j \leq C_1^{j+1} j!$. Margins

for the slow strip, like the state margins, may be allotted summably over successive fixed-side degrees. This proves all the asserted layer bounds without asserting convergence when both fast indices grow without restriction.

For the stable model, the desired conjugacy equation is

$$hT_s + (-1 + hb)uT_u = -T + hg(s, T). \quad (7.10)$$

Its constant coefficient is Equation (7.8), its linear coefficient is the definition of b , and its higher coefficients give Equation (7.9). The graph equation is a small formal fixed-point equation in the same weighted norm: $h\partial_s$ is small and hg has small analytic Lipschitz norm. After the graph is known, divide the fiber equations by $m - 1$. This diagonal inverse is bounded on the absolute u -coefficient norm for $m \geq 2$; it cancels the factor $m - 1$ in the hb term. Taylor composition about v_0 is bounded and Lipschitz on a fixed sufficiently small u disk, with range strictly inside the field disk. The nonlinear remainder has no constant or linear fiber term. Thus the small h multiplier and the small weighted slow derivative give a convergent majorant for all fiber degrees at once. At $h = 0$ the graph and nonlinear fiber terms vanish. The weighted estimate again gives the factorial bound. Cauchy estimates on remaining parameter and state margins give the stated fixed derivatives. $\square$

7.2.2 Optimal truncation and comparison with the actual flow

For a fixed finite request choose a sufficiently small constant $\delta > 0$ and truncate every series needed for it at a common order $J \asymp \delta|q|$. Locally this integer is frozen. The weighted proof above applies at a dummy h radius c/J larger than the used $|h|$ by a fixed factor. All operations in the proof are bounded on that dummy disk. Formal cancellation through degree J and Cauchy's estimate therefore bound the substitution residual by $O(e^{-c|q|})$ on smaller domains. A layer diagonal factor can be absorbed by one final fixed shrink of the fast radius, since $|\Delta_{mn}|$ grows at most linearly in the fast degree. This observation also controls fixed derivatives of the residual. A smaller proportional cutoff, or two overlapping fixed-factor choices, changes each retained expression by $O(e^{-c'|q|})$.

Proof of Lemma 7.2. For the layer model, retain the normalization modulo the monomial ideal generated by $E^m H^n$ with $A, B \geq K + 1$. This consists of finitely many fixed-one-side tails, so Lemma 7.3 applies. Write $t = t_L + t_R + t_{\text{mix}}$ in this quotient. Along the polygon, a mixed monomial obeys

$$|e^{-AD-B(q-D)}| \leq |P|^{\min(A,B)}, \quad 0 \leq \Re D \leq \Re q. \quad (7.11)$$

Hence $t_{\text{mix}} = O_K(P)$ in modulus, whereas $t_L + t_R$ is uniformly small by the fixed amplitude choice. Taylor-expand $W(s, E, H, u + t_L + t_R + t_{\text{mix}})$ in its last mixed increment through degree K . Its actual remainder is $O_K(|P|^{K+1})$. This finite Taylor calculation is legitimate on the larger slow and state domains; it does not require t_{mix} to be small on an unrestricted fast bidisk.

All its low-charge formal coefficients cancel in the normalization equation, giving residual $O(e^{-c_K|q|})$ after optimal truncation. Any remaining analytic fast sum whose monomials have $A, B \geq K + 1$ has actual value $O_K(|P|^{K+1})$. To see this directly, choose from each such monomial componentwise subindices with left and right charge at least $K + 1$ and minimal for that property. Their charges are bounded by $K + 1$ plus the largest weight, so only finitely many subindices occur. Extracting them gives $|P|^{K+1}$ by Equation (7.11); the remaining absolute series is bounded by the fixed fast norm. This also applies to the finite mixed Taylor operations just used.

The retained balanced drift is exactly

$$\mathcal{C}(s, E, H, u; h) = \sum_{k=1}^K P^k C_k^{[J]}(s, u; h)$$

on the actual layer trajectory, since every nonconstant balanced monomial has $A = B = k \geq 1$. Invert the input transformation and solve this approximate drift. The input inverse is regular: its pure part is close to identity and its mixed part tends to zero. The drift moves u by $O_K(|q||P|)$, keeping it in its disk. Along the actual polygon the Lipschitz integral for W is bounded by Equation (7.5). Comparing the transformed approximate solution with the true solution by the integral inequality therefore costs at most a constant times the polygon length $O(|q|)$. Polynomial losses are absorbed by reducing the exponential constant or by the displayed power in Equation (7.6).

At the two endpoints the fast arguments are respectively

$$(E, H) = (e, (d_\nu P^{b_\nu})_\nu), \quad (E, H) = ((e_\nu P^{a_\nu})_\nu, d).$$

Treat P temporarily as independent. On $|P| \leq c_K/(1 + |q|)$ the endpoint inverses and the slow drift $u_s = q \sum_{k=1}^K P^k C_k^{[J]}(s, u; h)$ remain in fixed bounded boxes. Their output is bounded and analytic there. Cauchy's estimate gives a Taylor remainder bounded by $C_K(1 + |q|)^{K+1}|P|^{K+1}$ on the actual exponentially smaller P . Each coefficient of degree k uses finitely many endpoint inversion jets and at most k nested drift integrals, because every drift insertion has positive P degree. Changing dD to $q ds$ places its q factor outside the coefficient integral. Induction on k thus gives a polynomial in q of degree at most k , with coefficients formed by bounded analytic operations and integrals over fixed subintervals of $[0, 1]$. The final Φ_1 evaluation has the same property. The Gevrey composition and regular-inverse estimates of Lemma 6.4, or its one-variable dummy-disk proof, apply to these finite operations. Integration on a fixed interval preserves the same coefficient bound. This proves Equation (7.6). In particular no large q occurs inside an order-zero inverse: the entire drift has positive charge.

For the stable model, take optimal polynomials $T^{[J]}, b^{[J]}$ from Lemma 7.3, using a common sufficiently small proportion. The transformation is $\text{id} + O(h)$ and has a regular inverse on the fixed smaller input disk. Set

$$u(s) = \exp\left(-qs + \int_0^s b^{[J]}(r; h) dr\right) (T^{[J]}(0, \cdot; h))^{-1}(v_{\text{in}}).$$

Since $b^{[J]}$ is bounded, damping and a fixed input margin keep $u(s)$ in the fiber disk. The residual of $v_{\text{app}}(s) = T^{[J]}(s, u(s); h)$ in the D equation is $R = O(e^{-c|q|})$. If L bounds $|g_v|$ on the working disk, variation of constants and the scalar integral inequality give

$$|v(1) - v_{\text{app}}(1)| \leq \frac{|q|}{\Re q - L} \sup_{0 \leq s \leq 1} |R(s)|.$$

The logarithm of this loss is $O(\log |q|)$ in Equation (7.3), hence it is absorbed in the exponential error. The approximate endpoint is

$$T^{[J]} \left(1, P \exp \left(\int_0^1 b^{[J]}(s; h) ds \right) (T^{[J]}(0, \cdot; h))^{-1}(v_{\text{in}}; h) \right). \quad (7.12)$$

Taylor expansion in P takes place on a fixed disk and uses only bounded operations with regular inverses. The same Gevrey rules give the desired coefficients, with no outside power of q . Ordinary derivative estimates in both constructions follow on smaller fixed parameter disks from these uniform analytic estimates. $\square$

7.3 Dispersal and independent derivatives

We complete the proof of Theorem 7.1, making the retirement of numerical truncations explicit. Write

$$q = q_0(1 + \Theta), \quad q_0 = A_0 Z_i, \quad A_0 > 0, \quad h_0 = q_0^{-1}. \quad (7.13)$$

By the dependent-node assumptions, Θ is a finite sum of strict-small shift monomials with first index greater than i , times bounded analytic factors. The exact formula for f_i uses only nodes of index at least $\sigma(i) > i$ and the bounded backgrounds. Thus h_0 is itself a strict-small shift monomial with first index $\sigma(i)$.

Up to band i use the actual holomorphic output and zero prefixes on preceding transitions. At the i fringe take the charge description to any sufficiently large fixed K . Since $\Re q \geq c\Re Z_i$ and $\log |q| = o(\Re Z_i)$, its charge remainder pays every prescribed fixed depth by increasing K . Its optimal error pays every such depth too: on that fringe $\Re Z_i/|Z_i| \rightarrow 0$, while $|q| \asymp |Z_i|$. Keep the affine exponential $P^k = e^{-kq}$ exactly as its flag shift, including its bounded factor. In particular changing a ratio argument in a later Taylor operation does not change this already extracted shift.

For a coefficient in Equation (7.6), factor $q^\ell = q_0^\ell(1 + \Theta)^\ell$. The first factor is an explicit shift translation. Absorb the unit into the bounded coefficient and rewrite

$$\sum_{a \geq 0} F_a q^{-a} = \sum_{a \geq 0} F_a (1 + \Theta)^{-a} h_0^a.$$

On one fixed small complex disk of the ratio arguments this changes the Gevrey bound by at most a factor C^a . Freeze the optimal order using a fixed-factor representative of $|q_0|$, independently of those ratio arguments. It follows that the coefficient and all its fixed ratio derivatives have a uniformly bounded optimal description on smaller ratio and ordinary disks.

Proceed through the later flag indices in order. At index j , Taylor-expand precisely those analytic small arguments whose first index is j , to a sufficiently large fixed degree for the requested budget. Set these arguments to zero in subsequent coefficient evaluations, retaining every other argument. Their remainders pay arbitrary j -depth because each current-first monomial costs at least $e^{-c\Re Z_j}$ on its owning fringe. Until $j = \sigma(i)$, keep the h_0 series optimally evaluated. Its numerical log formula uses only retained inputs throughout those bands. At $j = \sigma(i)$ keep only a fixed sufficiently large h_0 degree: the Gevrey estimate and the small optimal proportion give

$$\left| \sum_{a=J}^{N_{\text{opt}}} F_a(\Theta) h_0^a \right| \leq C_J |h_0|^J$$

for each fixed J . This pays arbitrary fixed current depth. Thereafter the coefficient uses finite formal degrees only; its numerical optimal order has disappeared. Every shift is split at the current index and its remaining tail is passed to the next band. This is the single-variable instance of Proposition 6.6, with no inner kernels.

These recursive operations also verify the support condition. Charges at i are nonnegative integers; at each fixed charge there are finitely many explicit translations. The further supports are Taylor powers of a finite collection of lexicographically strict positive shifts and the powers of h_0 , each dispersed at its first index. Only finitely many current-first degrees contribute below a given ceiling. Repeating this observation at a fixed prefix proves iterated left-finiteness; coincident exponent tuples are added with finite multiplicity. At the last level the coefficients are analytic ordinary germs. No simultaneous convergence of the full tree has been used.

For completeness consider cutoffs, raw growth, and derivatives. Choose the representative moduli for $|q_0| = A_0 e^{\Re f_i}$ in overlapping fixed-factor bins. On each bin use a frozen integer cutoff and hence a holomorphic expression in the retained inputs. Neighboring expressions differ, with each fixed derivative on smaller domains, by

$$O(e^{-ce^{\Re f_i}}). \tag{7.14}$$

Smoothly interpolate on overlaps as in the packet definition. These sources occur only in bands $i < j \leq \sigma(i)$, where f_i is retained in its exact triangular formula; their cost $U = e^{\Re f_i}$ is an allowed

simple source cost. The flag cost estimates give $\Re Z_j + f_j = o(U)$ there. Thus all fixed shift normalizations, raw factors, and interpolation derivative losses have logarithmic size $o(U)$. The actual expression on band 1 has no such source.

With explicit shift factors removed, a fixed coefficient is bounded on smaller analytic argument disks. Remaining shifts have modulus at most $\exp(C\Re Z_j)$ on band j by side dominance. This proves raw growth on the whole band with no D_{j-1} term. In particular every admissible column is safe, including all the entering collars required by Definition 3.1.

To differentiate in independent coordinates, freeze local cutoff choices and take complex neighborhoods of radius $\exp(-Cf_j(u))$ in the retained free and ordinary inputs. The finite triangular formulas bound the nodes of index at least j and their fixed sensitivities by $\exp(C'f_j(u))$. An earlier active logarithm such as f_i is evaluated by its exact linear and bounded-background formula and has the same sensitivity bound. Its exponential q_0 need not have that absolute bound: keeping the change in f_i bounded instead keeps q_0 within a fixed factor and preserves the frozen optimal-cutoff margin. By increasing C , and beginning on a slightly wider polynomial buffer, these neighborhoods preserve the slow, state, small-ratio, and angular margins used above. Cauchy's estimate costs only $\exp(C_r f_j(u))$ for each fixed derivative order r . Since $\Re Z_j \gg f_j(u)^N$ for every fixed N up to the side, deeper initial transition budgets absorb these losses. At a cutoff source they are absorbed by U in Equation (7.14); derivatives of the interpolating partitions have the same permitted logarithmic loss. Applying the same argument on both sides of each transition proves the differentiated transitions and all fixed differentiated anti-CR defect bounds.

7.4 Controlled inputs and uniformity

Lemma 7.4. *For each fixed finite request, the preceding primitive estimates are uniform on a controlled family of Definition 4.1. Finite real-Taylor agreement supplies the geometric comparisons for evaluation of the independent-variable estimates. Under pullback, differentiated source bounds hold for holomorphic tests and for the smooth chart sheets with the differentiated source bounds supplied by Theorem 4.4.*

Proof. Fix the request, the flag and rate recipe, and the common controlled family data. In particular fix a compact ordinary box inside the larger analytic domains, with a common positive margin, on the full forward domain of the tests. The field sup norms, base-flow inverse margin, state and fast disks, and all ordinary Cauchy constants can be fixed on these domains. The polygon integral and the stable variation-of-constants bound use only these norms, the fixed weights, and the displayed sector inequalities. Hence their constants and required lower bounds on the large rates are common to the family.

For the formal estimates, the pure-side contraction constants and the high-tail convolution bound use those same common analytic norms. For each of the finitely many one-side indices required, choose its finite initial list and high-tail threshold once. Choose the weighted slow-derivative constant ε and the reserved slow and state margins for this finite collection once as well. The stable divisors $m - 1$ and the nonzero layer charge gaps are fixed. Thus the Gevrey constants, optimal proportions, residual bounds, endpoint inverse margins, and finite charge remainder constants are uniform over the ordinary box. These are estimates of analytic functions at the current ordinary arguments; none waits for a test to return closer to its limiting ordinary point.

The common finite real-jet bounds and real-Taylor error constants, with the common small-jet modulus where a slow estimate requires it, give the same flag comparisons and angular room used in the dispersal proof. Its ratio disks, side constants, and Cauchy-loss exponents can consequently be chosen for the whole family. After the finite request has fixed all powers to absorb, the inequalities

between the flag scales give common far-regime thresholds for those absorptions, expressed in the finitely many numerical rate and gap inequalities in the request. This uses the controlled family's comparisons on the full forward domain, rather than eventual entry of each test into a smaller ordinary neighborhood. The same argument applies to cutoff overlaps and their fixed derivatives. The fixed-factor bins have bounded overlap, so the number of local source terms is uniform as well. For a coefficient already constructed, retain its chosen convention on a larger argument disk; any smaller comparison cutoff used in a new finite request is reconciled with it by Equation (7.14). No uniform choice over all requests is asserted, nor is a uniform rate of return of the ordinary values to their germ point needed.

For antiholomorphic derivatives, use the source estimates of the frozen holomorphic formulas and their explicit cutoff overlaps. Finite real-Taylor agreement preserves their domains; it does not supply a source estimate for an arbitrary smooth input. A holomorphic ordinary test contributes no input defect. A smooth chart sheet supplied by Theorem 4.4 already has the required differentiated source bounds.

The remaining factors in the pullback chain rule are the primitive raw and Cauchy factors bounded above and the controlled sheet derivatives. At an intrinsic cutoff source their logarithmic losses are $o(U)$ by Equation (7.14) and the comparisons above. For an inherited sheet source, use its permitted source cost and the comparisons of Definition 3.1 and Equation (4.7); no restriction to a simple cost is imposed on that source. At a level- p join the later-rate and Cauchy losses are $o(V_p)$; a fixed current normalization is retained as $C\mathfrak{R}Z_p$. Thus Lemma 4.7 applies, with a transition budget larger than the fixed current losses when the defect comes from a transition, or a sufficiently vertical collar for an independently established e^{-cV_p} defect. Band 1 uses holomorphic primitives and holomorphic chart sheets and has no source. This proves the source statement with the same controlled-family uniformity. $\square$

After any first-index extraction all formulas use only the retained free inputs and ordinary variables, and the fixed local determinations agree whenever those inputs agree.

7.5 Retesting the coefficient germs

We verify the additional assertion of Definition 5.6. The formal normalizations in Lemma 7.3 are uniquely determined by the fixed analytic fields and weights: imbalance division and balanced projection determine the layer coefficients, while the graph and fiber recursions determine the stable coefficients. No numerical large input or integration anchor is an initial datum in these formal recursions. At fixed charge and formal degree, Lemma 7.2 then uses only their fixed coefficients, analytic endpoint inversion at the same regular root, and integration over the fixed slow interval. The resulting ordinary analytic germs are consequently independent of the numerical anchor along a flag test.

More explicitly, perturb the retained old free inputs by arbitrarily small absolute amounts while preserving the affine flag relations, regime gaps, limiting ordinary data, and chosen sector determinations. The exact affine shifts and the ratio formulas in Equation (7.13) retain the same coefficient expressions. Every finite Taylor extraction therefore uses the same formal coefficient germ. The numerical optimal order may change, but on overlaps its effect has the precision in Equation (7.14); at $\sigma(i)$ it is removed altogether, leaving only fixed formal degrees. It thus cannot appear in a terminal coefficient. There are no additional anchor-dependent kernels in this construction. Refining the flag by unused nodes inserts zero-prefix identity transitions; necessary exact log substitutions merely rewrite the same monomials before pullback. On any saturated extension the same finite-request proofs above apply to that rewritten affine rate and its retained

log formulas. Theorem 4.4 supplies the stated calculus on those extensions, and the coefficient argument just given supplies its terminal-germ invariance. This concerns only the selected local lifted determinations. It proves retestability and completes Theorem 7.1.

8 Passages with two logarithmic scales

Let L, W be positive inputs tending to infinity, and write $c = W/L$. An occurrence of c in a layer or in the homogeneous part of the equation always denotes this exact ratio. Other bounded parameters of the field are independent ordinary parameters, even if the original matching problem subsequently ties them to a function of this ratio. We consider the following scalar equations on $0 \leq \theta \leq L$:

$$v' = -cv + g(X, Y, v), \quad g(0, 0, v) = 0, \quad (8.1)$$

$$z' = G(X, Y, z). \quad (8.2)$$

In Equation (8.1), the left and right layers are

$$X_\ell = e_\ell e^{-a_\ell \theta}, \quad Y_\ell = d_\ell e^{-b_\ell (L-\theta)}, \quad a_\ell, b_\ell \in \mathbb{Q}_{>0}.$$

In Equation (8.2), their rates have the form $\alpha_\ell + \beta_\ell c$, with rational α_ℓ, β_ℓ and $c \rightarrow c_0 < \infty$. If $c_0 > 0$, all limiting rates are positive and $G(0, 0, z) = 0$. If $c_0 = 0$, a rate either has $\alpha_\ell > 0$, or has $\alpha_\ell = 0, \beta_\ell > 0$; in this case G vanishes when all layers of the first kind are set to zero. We call those layers fast. Thus every field monomial contains a fast layer. For positive c_0 every layer can be designated fast.

For prescribed input $v(0) = v_{\text{in}}$ or $z(0) = z_{\text{in}}$, the output is the state at $\theta = L$. The clocks, layer amplitudes, input state, and ordinary field parameters are independent arguments of this output; the occurrences of c in the layers and homogeneous generator are evaluated by the exact formula W/L . Endpoint and coefficient ties are imposed later on these arguments.

Fields are analytic and bounded on larger complex boxes than those used for evaluation. The amplitudes have sufficiently small fixed modulus; in Equation (8.1) the input state is also small, whereas in Equation (8.2) it may range over a compact subbox of the state domain. All parameter statements below hold on smaller ordinary complex neighborhoods, uniformly on their closures.

Theorem 8.1 (Mixed passage packets). *Suppose that the primary inputs are affine combinations of flag scales with constant coefficients and bounded analytic parts, and that all large basis nodes in those combinations have dependent logarithms. For an unbalanced limit, put $P = \max(L, W)$ and $R = \min(L, W)$ on the reference sequence, and include a primary input $C_{\text{aux}} \rightarrow \infty$ tied there to P/R . Assume the saturated flag and controlled tests of Sections 2 and 4. Then the outputs of Equations (8.1) and (8.2) satisfy the primitive hypotheses of Assumption 4.2, including fixed independent derivatives, simultaneous safe columns, and refinement invariance of terminal analytic germs. They are retestable for elimination in the sense of Definition 5.6.*

The amplitude and input smallness is independent of expansion depth. For a family of positive limiting slopes it can also be chosen independently of the slope, provided the ordinary boxes and field bounds are common and every field monomial contains a designated layer whose limiting rate has a common positive lower bound. The other rates need only be positive at each particular limiting slope. Constants and starting points for a fixed packet label may depend on that slope. The two unbalanced cases have common smallness for each fixed rate list.

The three ratio regimes use different coefficient constructions. For comparable scales write $H = W - c_0 L$; for an unbalanced passage write $C = P/R$, with P, R as in the theorem. Here P is the larger primary action, whereas in the preceding two sections it denoted a small transfer charge.

Ratio regime	Construction	Intermediate coefficient data
$c \rightarrow c_0 \in (0, \infty)$	Integration trees and residues	Analytic cluster functions when H is bounded; separated exact shifts with explicit power prefactors and analytic coefficient jets when H is unbounded
$c \rightarrow 0$	Integration trees, then cutoff retirement	Meromorphic ratio coefficients, then Gevrey series in $1/C$
$c \rightarrow \infty$ (model (8.1))	Compatible input and output charts	Meromorphic chart coefficients, then Gevrey series in $1/C$

Each route is local to its chosen limiting regime. The ratio in the layers and homogeneous generator remains the exact W/L ; other ordinary field parameters remain independent until their argument ties are imposed.

We first prove the angle estimates used by all the constructions. The pole estimates enter with the unbalanced coefficient formulas. In this section a constant attached to a fixed charge, coefficient, or derivative order need not be uniform as that finite request increases.

8.1 Primary combinations and ratio geometry

Write the flag-path parameter as $s = u + iv_{\text{ang}}$, distinguishing its imaginary coordinate from the state variable v .

Lemma 8.2 (Ratio geometry). *If a positive primary combination J has leading index p , then*

$$J = A_J Z_p (1 + \Theta_J), \quad A_J > 0, \quad (8.3)$$

where Θ_J is a finite sum of strict-small monomials, of first index greater than p , times bounded analytic factors. On every usable band $j \leq p$, including its required side,

$$|J| \asymp |Z_p|, \quad \Re J \asymp |Z_p| \cos(\Im f_p) \gg (\log |J|)^q \quad (q \text{ fixed}). \quad (8.4)$$

For an unbalanced passage, let i, n, k be the leading indices of P, R, C_{aux} , respectively. Then $n, k > i$, possibly $n = k$, and there is an exact identity of flag expressions

$$f_i = f_n + f_k + b_*, \quad C := \frac{P}{R} = \frac{A_P}{A_R} e^{b_*} Z_k \frac{1 + \Theta_P}{1 + \Theta_R}. \quad (8.5)$$

Here b_* is bounded background data, $\Lambda_i = \Lambda_n + \Lambda_k + O(1)$, and $\sigma(i) > \min(n, k)$.

Let $C_\bullet$ denote the same ratio formula with any already dispersed small arguments set to zero in its units. While it is used meromorphically, on bands of index at most k ,

$$|C_\bullet| \asymp |Z_k|, \quad \arg C_\bullet = (1 + o(1)) v_{\text{ang}} \Lambda_k, \quad (8.6)$$

$$|\Im C_\bullet| \gtrsim |Z_k| \min(1, |v_{\text{ang}}| \Lambda_k). \quad (8.7)$$

The lifted argument is in $[-\pi/2 - o(1), \pi/2 + o(1)]$, and is transverse to the real axis at the k fringe. If $|v_{\text{ang}}| \Lambda_k \ll 1$, then

$$\Re C_\bullet = C_\bullet(u) (1 + O((v_{\text{ang}} \Lambda_k)^2)). \quad (8.8)$$

Through the i band the actual positive actions P, R have increasing arguments, $0 < \arg R < \arg P < \pi/2$ on the positive side. At its fringe,

$$\frac{\cos \arg R}{\cos \arg P} \rightarrow \infty, \quad |\sin(\arg P - \arg R)| \gtrsim \frac{\Lambda_k}{\Lambda_i}. \quad (8.9)$$

The negative side has the reflected assertions.

Proof. Divide the affine formula for J by its leading term. Ratios of primary nodes, and the reciprocal of the leading node multiplying a bounded term, have exact logarithms in later nodes. Their first nonzero exponents are positive, which proves Equation (8.3). On the real ray their moduli tend to zero. On a j band the flag angle estimates control their real parts as well: if $l > p$ and the p cosine is small, the possible relative cosine gain is at most a constant times

$$1 + \frac{\Lambda_p - \Lambda_l}{\Lambda_j \cos(\Im f_j) + \Lambda_j - \Lambda_p}.$$

The logarithmic-gap derivative estimate absorbs this factor in $|Z_l/Z_p|$. If the p cosine is small, $\Lambda_j = O(\Lambda_p)$; the polynomial side buffer therefore remains a polynomial buffer in $f_p(u)$. This proves Equation (8.4), including the bounded affine part. The same estimates after differentiation show that the argument of the unit in Equation (8.3) is $o(|v_{\text{ang}}|)$.

On the reference sequence the assumed tie makes $f_i - f_n - f_k$ bounded. Each of these logarithms has an exact triangular affine expression. A nonzero coefficient of its largest remaining large scale would make their difference unbounded, so every large coefficient cancels. Their difference is exactly a bounded background combination b_* . Neither f_n nor f_k uses a node at or before $\min(n, k)$, proving the assertion about $\sigma(i)$. Differentiation gives the assertion about Λ_i . This argument uses the auxiliary tie on the reference sequence; it does not impose that tie at independent ambient test points.

The triangular angular formulas applied to the surviving expressions in Equation (8.5) prove Equations (8.6) and (8.7). Deleting an already dispersed argument preserves the proof: every remaining strict-small term of first index $l \geq j$ is $O(e^{-a\Re Z_l})$, and its fixed angular derivatives lose only $\exp(O(f_l(u)))$. For controlled ordinary backgrounds use their real Taylor jets before this comparison. Downward differentiation bounds the sensitivity of f_l by $f_l(u)$ polylog $f_l(u)$ on the relevant angular scale; a sufficiently high fixed common jet thus preserves any specified polynomial angular margin. Real symmetry removes the linear term from the real part, giving Equation (8.8). In particular $|\Im C_\bullet| < 1$ implies $\Re C_\bullet - C_\bullet(u) = O(|C_\bullet(u)|^{-1}) = o(1)$.

Finally $\Lambda_i - \Lambda_n = \Lambda_k + O(1)$, with a diverging positive gap. The good-to-side angular range gives Equation (8.9). The flag side dominance also makes $\Re R/\Re P$ smaller than any prescribed fixed positive constant after its dominance constant is chosen. All comparisons have slack under the smaller neighborhoods subsequently used. $\square$

8.2 The actual analytic passage and integration trees

Lemma 8.3 (Continuation in the leading band). *Let i be the leading index of the larger primary action, or the common leading index of L, W when they are comparable. The actual passage is bounded and holomorphic throughout band i , with the parameter and state room needed for fixed independent derivatives.*

Proof. For comparable scales put $H = W - c_0 L$ and $\delta = H/L$; H is bounded ordinary data or a signed primary combination with later leading index. On the i band, $\delta \rightarrow 0$. At its fringe put $\rho = \Re L/|L|$ and denote the side dominance constant by $\mathcal{A}$. The exact identity for $\Im(H/L)$ gives

$$\frac{|\Im \delta|}{\rho} \leq \frac{|H|}{|L|} + \frac{|\Re H|}{\Re L} \leq \frac{C_H}{\mathcal{A}} + o(1). \quad (8.10)$$

Indeed H has only later affine nodes and bounded terms, whereas the side inequality makes $\Re Z_i$ dominate $\mathcal{A}$ times the sum of the later real parts. Increasing $\mathcal{A}$ once also makes $\Re L \geq (A_L/2)\Re Z_i$. The first term tends to zero by the modulus hierarchy, and the bounded terms are negligible by the

polynomial buffer. These estimates are uniform for the controlled families of Definition 4.1; C_H depends on the fixed affine formulas and common ordinary bounds, not on the moving anchor. The last quantity need not tend to zero for fixed $\mathcal{A}$.

Fix $0 < \kappa \leq 1/4$ before choosing any expansion budget. For the finite rate list let K_0 bound each generator's absolute detuning coefficient divided by its positive limiting rate, including the homogeneous generator when present. Choose $\mathcal{A}$ so that $2K_0C_H/\mathcal{A} < \kappa/4$, and then impose the numerical far thresholds $K_0|\delta| < 1/4$ and the corresponding small bound on the $o(1)$ in Equation (8.10). Every fixed positive limiting action then retains a fixed positive fraction of its real action: for a generator with limiting rate $q_h > 0$ and detuning coefficient β_h ,

$$\Re[(q_h + \beta_h\delta)L] \geq q_h\Re L - |\beta_h| |\Re H| \geq \frac{1}{2}q_h\Re L.$$

For any sum of generator costs $\nu + B\delta$ one also has $|\arg(\nu + B\delta)| \leq \kappa\rho/2$. The cone parameter is fixed independently of the later budget; the side constant and thresholds may depend on the limiting slope.

For comparable scales and for $c \rightarrow 0$, integrate on the polygon

$$0, S, L - S, L, \quad S = K \log |L| > 0.$$

Fast real rates are bounded below on the horizontal ends. Along the middle every left and right fast profile has an exponential margin from the corresponding end; choosing K large pays its length. The integral of the sum of the absolute fast profiles is therefore at most a constant times the amplitude size, uniformly in L . Pure slow profiles stay bounded. Homogeneous c actions, when present, have increasing real part: $\Re c \geq 0$ and $|L| \cos \arg W \gg S$. Signed corrections to fast rates are absorbed by $\Re W/\Re L \ll 1$ in the small-slope case, and by the preceding δ estimate in the comparable case.

For $c \rightarrow \infty$, use the straight segment when $\cos \arg L$ has a positive fixed lower bound, and the same polygon otherwise. In the latter case, including a switching overlap, $\Lambda_n \gtrsim \Lambda_i$, so the real path estimates give $f_i \leq \text{poly}(f_n)$. The buffer for W then gives $|L| \cos \arg W \gg S$, as required for monotone damping. The integral of $|g_v|$ is small in all cases, because each field monomial has a fast factor; the stable propagator has modulus at most one. Picard iteration on these paths is a contraction in a fixed state box, after the amplitudes are reduced once. The same estimate applied to the forcing keeps the state inside a smaller box. Ordinary parameter Cauchy margins give the claimed room.

Local deformations of the polygons preserve their estimates and give holomorphic dependence. On overlaps uniqueness identifies the solutions, including an interpolation to straight segments in the switching region. Their union is the continuation of the real passage. Before index i use this actual function with zero prefixes. $\square$

For comparable or small slopes, expand the integral equation by analytic substitution. A planar rooted integration tree has a field monomial at each vertex, and therefore at least one fast factor there. In model (8.1), homogeneous input leaves and the linear edge propagators are included. The free term is respectively $e^{-W}v_{\text{in}}$ or z_{in} . Let r be the number of vertices and M_0 the total layer degree plus homogeneous-input leaf count. Then $M_0 \geq r$, and the sum of vertex arities is $O(M_0)$. Cauchy bounds for the field, the finite number of layer generators, and the exponential count of planar trees give total weights bounded by η^{M_0} after a fixed exponential factor is absorbed into a sufficiently small η . Integrating one fast factor per vertex on the polygons of Lemma 8.3 proves absolute convergence and the equation by substitution. In model (8.2) expand about the bounded input, using its fixed state margin, so it does not have to be small.

Each tree integration domain is the union of at most $r!$ ordered simplexes. On one such simplex the $r + 1$ successive gaps have costs $C_j = A_j + cB_j$. Its coefficient, after removing the monomial weight, is the convolution of the $r + 1$ exponentials, and hence

$$\sum_{s=-C_j} \operatorname{Res} e^{sL} \prod_{j=0}^r (s + C_j)^{-1}. \quad (8.11)$$

For distinct costs this follows by partial fractions; the integral and the total residue are entire in all costs, which proves the identity also at collisions. Straight paths can be used for this identity because the simplex integrands are entire.

Example 8.4 (Collision of two costs). For one event the convolution is

$$\int_0^L e^{-C_0 t} e^{-C_1(L-t)} dt = \begin{cases} \frac{e^{-C_0 L} - e^{-C_1 L}}{C_1 - C_0}, & C_0 \neq C_1, \\ L e^{-C_0 L}, & C_0 = C_1. \end{cases}$$

The separate residue terms have a denominator at a cost collision, whereas their sum extends analytically there. The cluster integral below keeps this cancellation before the corresponding charge is dispersed.

Lemma 8.5 (Ordered cost and contour bounds). *For comparable slopes put $\nu_j = A_j + c_0 B_j$. These values belong to a finitely generated positive-weight semigroup and satisfy*

$$|B_j| \leq K_0 \nu_j, \quad \nu_j + \nu_{j'} \geq a|j - j'|. \quad (8.12)$$

For small slopes, A_j is in a nonnegative rational lattice, B_j in a signed rational lattice, and

$$A_j + A_{j'} \geq a|j - j'|, \quad B_j \geq -K_0 A_j, \quad |B_j| \leq K_0 M_0. \quad (8.13)$$

The l th main cost in increasing order is at least $a(l - 1)/2$. For every large fixed N , the aggregate high-residue sum at the leading fringe has modulus at most

$$\frac{K_N K_*^{r+1}}{r!} e^{-bN\Re L}, \quad (8.14)$$

where $b > 0$ and the exponential base K_ are independent of N . In the comparable case “high” means main costs above a threshold in $[2N, 3N]$. In the small-slope case it means projected costs $p_j = A_j + \mu B_j$ above such a threshold, where $\mu = \Re W / \Re L$.*

Proof. Every event between gaps j and j' supplies a fast rate to one of their costs: a left layer contributes before its event and a right layer after its event. Nonnegative homogeneous damping contributes additional cost. This proves the pairwise lower bounds. For comparable slopes each generator has a positive limiting weight, so its B contribution is bounded by a fixed multiple of that weight. For small slopes the negative B contributions occur only in fast rates and are bounded by their A contribution; pure slow rates and homogeneous damping have nonnegative B . The total degree bounds the absolute sum of B contributions. If l costs are at most t , their first and last original indices differ by at least $l - 1$, whence $2t \geq a(l - 1)$. This proves the sorted bound.

We give the contour estimate since it keeps the smallness of the analytic inputs independent of N . Reflecting if necessary, take the positive fringe and write $\rho = \Re L / |L| \rightarrow 0$; $\Re L \gg \log(1/\rho)$. Fix the same $0 < \kappa \leq 1/4$ as in Lemma 8.3. In the comparable case Equation (8.10), with its

specified choice of side constant, puts every pole within angle $\kappa\rho/2$ of the negative real ray. Choose $N' \in [2N, 3N]$ separated from the finitely many base semigroup values there. Enclose the high poles by a cap $\Re s = -N'$ and rays of angles $\pi \pm \kappa\rho$. Since $\kappa < 1$, both rays have a fixed positive exponential-decay margin. Distances on the cap have a positive fixed lower bound for the finitely many low costs. On the rays at modulus x , at most $K(x+1)$ sorted indices require the distance bound $b\rho x$; all others have distances bounded below by a fixed multiple of their sorted index. The factorial of those initial indices is bounded by $K^m x^m$ if their number is m . Consequently

$$r! \prod_j |s + C_j|^{-1} \leq K_*^{r+1} (K_*/\rho)^{K_*(x+1)}.$$

On the cap the last factor can be replaced by $K_N \rho^{-K_N}$. Along the rays $\Re(sL) \leq -bx\Re L$, so the latter factors are absorbed by half the decay sufficiently far out. Integration over the rays and cap proves Equation (8.14). The same estimate makes the closing arcs tend to zero; all poles of each finite product are included before the limit is taken.

For small slopes, side dominance makes μ small, whereas Equation (8.9) gives $\mu/|c| \rightarrow \infty$. Thus $p_j \geq A_j/2$. The number of projected costs below $3N+1$ is bounded in terms of N , so for each simplex there is a threshold $N' \in [2N, 3N]$ a fixed distance from all these projected costs. The poles lie in the cone between angles $\pi/2$ and $\pi + \kappa\rho/2$: when $B_j < 0$ use $|c| \sin(\arg L - \arg W) \leq \mu\rho$ and $|B_j| \leq K_0 A_j$. Use the cap $\Re(sL) = -N'\Re L$ and rays of angles $\pi + \kappa\rho$ and $\pi/3$. The cap has length and modulus $O(N')$, distances at least $b_N \rho$ for its finitely many small costs, and factorial distances for the large A_j 's. On the first ray the preceding estimate applies. On the second ray, starting at modulus comparable to $N'\rho$, the exponential decays as $e^{-bx|L|}$ and angular separation gives distances at least bx . For $x < 1$ only a fixed number of sorted indices require that bound, so a fixed inverse power of ρ suffices; for $x \geq 1$ use the same factorial argument. This proves the stated estimate also in this case. All low projected costs have $A_j \leq 2N'$ and, since $\mu/|c| \rightarrow \infty$, $B_j < \varepsilon|Z_k|$ for every fixed positive ε sufficiently far out. $\square$

8.3 Comparable slopes and low residues

Fix a main comparable charge ν and put $J = \{j : \nu_j = \nu\}$. Lemma 8.5 bounds $|J|$ in terms of ν . The factor

$$K_\nu(t, \delta) = \prod_{j \notin J} (t + \nu_j - \nu + B_j \delta)^{-1} \quad (8.15)$$

is analytic on a fixed small (t, δ) bidisk, with bound $K^{(\nu)} K_*^{r+1}/r!$. Indeed finitely many low distinct values have a positive separation, and every sufficiently high value pays a fixed multiple of its sorted index. This also bounds any fixed jet after a disk shrink.

If $H = W - c_0 L$ is bounded, the whole cluster contributes $e^{-\nu L} L^{|J|-1}$ times

$$\frac{1}{2\pi i} \int_\Gamma \frac{e^z K_\nu(z/L, \delta)}{\prod_{j \in J} (z + B_j H)} dz, \quad (8.16)$$

where a fixed circle encloses these finitely many bounded poles uniformly on a smaller ordinary neighborhood. The integral is an analytic function of its small and ordinary arguments, with the same factorial bound. If H is unbounded, split the cluster into its distinct B values, a finite set for fixed ν . Direct residue differentiation gives exact shifts $e^{-\nu L - BH}$, boundedly many powers of L and δ^{-1} , and analytic values or jets of Equation (8.15).

The $r!$ possible linear extensions are canceled by the factorial gain, and the remaining K_*^{r+1} is paid by the original geometric weights. Thus summing these coefficients over trees is normally

convergent on fixed smaller argument boxes. The $K^{(\nu)}$ and disk sizes may depend on the charge but occur only once per fixed low-charge expression, rather than once per vertex. The error in Lemma 8.5, summed in the same norm, pays arbitrary depth at the i fringe by increasing N .

Every remaining power or unit, including H/L , is an exact monomial times an analytic unit in strict-small arguments. A bounded H has only ordinary terms. Split the affine exponential shifts exactly and Taylor-disperse each small argument at its first index, using the calculus of Section 4. Taylor remainders pay arbitrary finite depth after fixed prefactor losses; the support is iterated left-finite. This constructs the coefficient functions and transitions in the comparable case. Their differentiated bounds and terminal-germ invariance are collected with the unbalanced cases below.

Under the common fast-rate hypothesis in Theorem 8.1, use that fast layer at each event in the path integral and sorted-index estimate. The constant a and the exponential base K_* can then be fixed uniformly. Shrink the detuning neighborhood, separately for each slope, to keep the large-cost distances above a common multiple of their base values. Only the low-charge constants and radii change. The same original amplitude choice therefore works throughout the asserted family. More explicitly, fix the cone parameter $\kappa \leq 1/4$ and the common designated-rate lower bound $a_* > 0$ first. The slope-specific K_0 is paid solely by the side constant $\mathcal{A}$ and the numerical thresholds in Equation (8.10). After those choices, every designated profile has horizontal decay rate at least $a_*/2$, and its real middle action has the same fixed fractional margin. Its end integrals are bounded by $2/a_*$, while a common choice in $S = K \log |L|$ pays the middle length. All other profiles are bounded by their amplitudes. Thus the integral and tree norm constants are common. On the contour, at most $m \leq C_{a_*}(x+1)$ initial sorted indices need the weak distance bound, and

$$r! \prod_j |s + C_j|^{-1} \leq K^{r+1} (m-1)! (\rho x)^{-m} \leq K_*^{r+1} (K_*/\rho)^{C_{a_*}(x+1)}$$

for $x \geq 1$, with common K_* . Neither $\mathcal{A}$ nor K_0 appears once per vertex. A later budget N affects only its finite cap/low-charge constant and thresholds; increasing side dominance for that request improves the already fixed cone margin.

8.4 Small damping and retirement of its meromorphic coefficients

Here $P = L$, $R = W$, $C = L/W$. The low-charge residue formulas are meromorphic in C . We retain these formulas up to the ratio's fringe, where they can be replaced by Gevrey series in $1/C$.

8.4.1 Retained low-charge residues

For each fixed main charge A , retain the residues with B below a cutoff comparable to $\varepsilon |Z_k|$, with $\varepsilon > 0$ fixed and small. Freeze this cutoff locally in a fixed-factor modulus bin. The multiplicity m_{AB} of an identical pair (A, B) is bounded in terms of A . Expanding the other factors to order $m_{AB} - 1$ yields terms

$$e^{-AL-BW} L^d C^t M_{ABdt} (1/C), \quad (8.17)$$

where $d, t \geq 0$ range over finite sets for fixed A . All factors with equal A and unequal B have a fixed lattice gap in their B difference; their powers of C have been included in C^t . The remaining functions are fixed jets of products with factors

$$(A_j - A + (B_j - B)/C)^{-1}, \quad A_j \neq A. \quad (8.18)$$

For B in the cutoff, every sufficiently large A_j pays a constant multiple of A_j : use $B_j \geq -K_0 A_j$, bounded $|B/C_\bullet|$, and the fact that $1/C_\bullet$ stays away from the negative real direction. The finitely

many remaining A_j have finitely many possible nonzero lattice differences $A_j - A$. Equation (8.18) consequently gives finite-order poles at finitely many positive-multiple lattices in $C_\bullet$, with fixed additional powers of $|C_\bullet|$. For clarity, the fixed residue jets can be estimated directly, without a fixed-radius circle near a real pole. If the unequal- A denominators at the residue are $d_1, \dots, d_s$, then

$$\left. \frac{\partial^q}{\partial t^q} \prod_{\ell=1}^s (t + d_\ell)^{-1} \right|_{t=0} = (-1)^q q! \left(\prod_{\ell=1}^s d_\ell^{-1} \right) \sum_{q_1 + \dots + q_s = q} \prod_{\ell=1}^s d_\ell^{-q_\ell}.$$

Here q is bounded in terms of the fixed main charge A . Only boundedly many d_ℓ can be small, by the sorted-cost estimate. The formula adds at most q inverse-distance powers and a factor bounded by a fixed multiple of $(1+r)^q$. The latter is at most $C_q 2^r$ and so can be absorbed in a single larger geometric base, independent of the charge budget. Thus the coefficient bound is $K^{(A)} K_*^{r+1}/r!$ times the stated finite pole and polynomial losses. Equivalently a Cauchy circle could be taken with radius at most half the smallest $|d_\ell|$; that radius is not fixed near a lattice pole.

8.4.2 Control of the meromorphic coefficients

The following lemma controls these lattice poles throughout the later bands, including the real columns required by the packet definition. It will also apply to the one-sided charts in the fast-damping case.

Lemma 8.6 (Meromorphic losses and their derivatives). *Suppose a fixed outer formula has only finite-order poles on lattices of finitely many positive multiples of $C_\bullet$, with fixed polynomial factors in $C_\bullet$ and retained primary inputs, and its remaining analytic factors are uniformly bounded on the smaller argument boxes. It then has the raw growth and simultaneous safe-column improvements of Definition 3.1, with every prescribed finite number of independent derivatives. This remains true with a scalar detuning β_s , analytic with the same smaller-box margins, satisfying $|\beta_s| \leq K e^{-\omega \Re R}$ before the first of the n, k transitions. At a transition $j \leq k$, unit arguments in $C_\bullet$ may vary on relative radii*

$$\varepsilon \min(1, \Lambda_k / \Lambda_j). \quad (8.19)$$

On a general usable j band replace Λ_j by Λ_{j-1} ; on safe sets a sufficiently small inverse power of $|Z_k|$ also suffices.

Proof. For real $a \neq 0, b$,

$$|a + b/C_\bullet| = |a| \frac{|C_\bullet + b/a|}{|C_\bullet|} \geq |a| \frac{|\Im C_\bullet|}{|C_\bullet|}.$$

The same calculation applies to a fixed multiple of the ratio and to any fixed pole order. Equations (8.7) and the entering angular scale give logarithmic loss $O(f_j(u) + \log(2 + \Lambda_{j-1}))$. The flag estimate $\log(2 + \Lambda_{j-1}) \leq CD_{j-1} + o(\Re Z_j)$ places this within the permitted raw growth. At the far side $|v_{\text{ang}}| \asymp 1/\Lambda_j$, and the same loss is $O(f_j + \log(2 + \Lambda_j))$; there is no exceptional constant there.

For completeness, verify the hypotheses of Lemma 2.9 for the actual pole sweeps. A fixed positive multiple has real value $x = \exp(f_k + \gamma)$ with $\gamma', \gamma'' = O(1)$: the bounded backgrounds have bounded real derivatives and the differentiated strict-small unit corrections tend to zero. Hence

$$\frac{x'}{x} = \Lambda_k + O(1), \quad \frac{x''}{(x')^2} = \frac{1}{x} \left(1 + \frac{\Lambda'_k + \gamma''}{(\Lambda_k + \gamma')^2} \right) = \frac{1 + o(1)}{x}. \quad (8.20)$$

The inverse speed varies by $1 + o(1)$ over a fixed lattice period. Thus a small fixed lattice collar occupies an arbitrarily small proportion of each fixed path interval; incomplete end periods have

vanishing length. For a finite list the sum of the bad proportions is less than one. At the remaining columns real detuning persists at small imaginary parts by Equation (8.8); at larger imaginary parts the imaginary separation suffices.

In the exceptional entering collar put $D = D_{j-1}$ and $F = f_j(u_*)$. If $D > e^F/F^{K+3}$, then $F = o(D)$, while the uncapped term defining D gives $\Lambda_{j-1} \gtrsim e^D$. The window $|\Delta f_{j-1}| \leq C_1 \log(2+D)$ has path width $O(e^{-D} \log(2+D))$. Within it $f_j \leq F + O(\log D) = o(D)$ and the sweeps with $k \geq j$ satisfy $x' = O(Z_j \Lambda_j) = e^{o(D)}$. Their changes are therefore $o(1)$. This verifies the required persistence on the entire exceptional collar, with one choice for a finite list.

Relative variations of the units on Equation (8.19) change $C_\bullet$ by at most a small fixed fraction of its imaginary separation. The actual first- j small arguments are exponentially smaller than these radii. At safe points inverse-power radii preserve real detuning. In the fast-damping construction below the balanced detuning has size $O(e^{-\omega \Re R})$. Before $\min(n, k)$ is passed,

$$\Lambda_{j-1} \leq \Lambda_i = \Lambda_n + \Lambda_k + O(1), \quad |\Im C_\bullet| \gtrsim |C_\bullet| \frac{\Lambda_k}{\Lambda_n + \Lambda_k + O(1)}.$$

The logarithm of the inverse lower bound is $O(\log(2 + \Lambda_n)) = o(\Re R)$. The detuning cannot remove the separation. On safe real columns it is likewise eventually smaller than the chosen lattice gap.

Take frozen holomorphic choices first. The finite triangular formulas permit local independent-variable Cauchy radii $\exp(-C f_j(u))$, multiplied when necessary by the pole-preserving factor just obtained. Fixed differentiation increases only fixed pole powers and these polynomial losses. On both sides of a j transition the loss has logarithm $O(f_j)$, after taking extra angular room; choose deeper undifferentiated accuracy first. The general-band additional loss is precisely the allowed $O(\log(2 + \Lambda_{j-1}))$. Derivatives of the smooth cutoff choices have the same bounds. These arguments apply on the small tubes of controlled high real jets; they do not require a fixed ordinary complex disk around a rotating dependent scale. $\square$

8.4.3 The leading transition and cutoff retirement

We return to the retained low- A residue sum. At the leading fringe, all its pole losses and the fixed powers of L, C have logarithm $o(\Re L)$, by Lemmas 8.2 and 8.6.

For a requested depth N , include all $A \leq 6N$ with these cutoffs. Lemma 8.5 shows that this includes every projected-low pole; the additional included poles have $A + \mu B \geq N'$ and are individually exponentially small at that depth after its fixed losses. There are only boundedly many such low- A indices per simplex. Thus these expressions, summed against the analytic weights, give the leading transition. One may move its finite budget endpoints to avoid charge values.

Choose a positive lattice unit ω for the B values and put

$$D = e^{-\omega W}. \tag{8.21}$$

Since $B \geq -K_0 A$, at fixed A one can first extract a fixed negative power of D . The remaining sum uses nonnegative integer powers of D . The inequality $|B| \leq K_0 M_0$ makes it normally convergent on a fixed small D disk, and even on a somewhat larger disk if the original weights are further reduced. The powers $L^d C^t$ are exact monomial prefactors times units. In particular, once e^{-AL-BW} has been split using the affine primary inputs, it is never recomputed by varying the artificial ratio arguments of a coefficient.

Lemma 8.7 (Small-damping formal replacement). *Before the first of the n, k transitions, changing the moving cutoff in a fixed coefficient changes its actual value, with fixed derivatives, by*

$$O(\exp(-a|Z_k|\Re R)) = O\left(\exp\{-a'e^{\Re f_i} \cos(\Im f_n)\}\right). \tag{8.22}$$

If $n < k$, a fixed Taylor degree in D removes that cutoff at the n transition. At the k fringe the remaining coefficients have Gevrey-1 expansions in $h = 1/C_\bullet$, with optimal error $O(e^{-a|Z_k|})$, uniformly on their smaller analytic argument boxes. If $k \leq n$, the same assertion holds before D is dispersed, after replacing the cutoff sum by its full sum.

Proof. Neighboring cutoffs differ only in positive $B \geq a|Z_k|$. The factors e^{-BW} give Equation (8.22); the finite pole and prefactor losses are absorbed by decreasing a . Equation (8.5) identifies its two costs. If D is dispersed first, any fixed coefficient uses a fixed finite set of B values, so eventually it is independent of the cutoff. Its unequal- A factors retain the pole estimates already proved.

At the k fringe the ratio is transverse to the real axis. For real lattice $a \neq 0, b$, $|a + b/C_\bullet| \gtrsim |a|$. Thus all B can be included, with the same factorial gain; the omitted terms have $M_0 \gtrsim |Z_k|$ and cost $e^{-a|Z_k|}$ in the geometric norm, also on the chosen D disks. For an individual tree, after the explicit powers C^t have been extracted, the functions in Equation (8.18) are analytic for

$$|h| < \frac{\varepsilon_A}{1 + M_0},$$

with bound $K^{(A)} K_*^{r+1}/r!$: the unequal- A lattice gap and $|B_j| + |B| = O(M_0)$ prove this directly. Its h^q coefficient is bounded by the same prefactor times $K_A^q (1 + M_0)^q$. For $0 < \eta < 1$,

$$\sum_{M \geq 0} \eta^M (1 + M)^q \leq K^{q+1} q!;$$

for example compare the sum with the integral of $e^{-ax}(2+x)^q$ and expand the latter polynomial. Summing the tree bounds proves the Gevrey estimate.

Set $Q = |C_\bullet|$ and truncate at $J = \lfloor \varepsilon' Q \rfloor$. For $M_0 \leq \varepsilon'' Q$, the coefficient disk is larger than the used h by a fixed factor; its Taylor remainder is exponentially small in J . For $M_0 > \varepsilon'' Q$, the actual terms still have their uniform imaginary-sector bounds. The truncated terms lose at most

$$(1 + K(1 + M_0)/Q)^J.$$

Choose ε' small: its logarithm is then at most a fixed small multiple of M_0 , and the geometric weight still leaves e^{-aM_0} . Summing this tail proves the optimal remainder. The proof on fixed analytic disks also gives the fixed jets by Cauchy estimates. When D was dispersed first it applies directly to each of its fixed coefficients. $\square$

8.5 Fast damping: one-sided charts and a common normalization

Now $P = W$, $R = L$, and $C = c = W/L \rightarrow \infty$. Write the layer charges as $A = a \cdot p$, $B = b \cdot q$ in a common positive rational lattice $\omega\mathbb{N}$. The layer derivation

$$\Delta = \sum_{\ell} (-a_{\ell} X_{\ell}) \partial_{X_{\ell}} + \sum_{\ell} b_{\ell} Y_{\ell} \partial_{Y_{\ell}}$$

has eigenvalue $B - A$ on $X^p Y^q$; let Π project onto $A = B$. We use different normalizations at the two endpoints. The output graph equation has divisors $c + B - A$, so a bound on A protects its inverse while leaving the right charges unrestricted. At state degree $m \geq 1$, the equation for the logarithm of the input-coordinate derivative has divisors $B - A - mc$, so its inverse is protected by a bound on B . Later coordinate inversions may still have lattice poles, controlled by Lemma 8.6. Their common charge range will identify the two coordinates at the midpoint, where both omitted endpoint layers are small.

Take N a small fixed multiple of $|Z_k|$, locally frozen, so $N \leq \varepsilon|c|$ even on the required variant neighborhoods. The output, input, and joint rings retain respectively $A < N$, $B < N$, and both inequalities. They are quotient rings by monomial ideals. Equip them with absolute coefficient norms on fixed small layer polydisks, and on a state disk strictly inside a larger one. The field norms, including the finite margins needed for its derivatives, are small because $g(0, 0, v) = 0$.

Lemma 8.8 (Compatible layer charts). *In these rings there are an output graph ϕ , an input coordinate Φ , an output coordinate $T = \phi + \sum_{m=1}^M t_m u^m$ for each fixed M , and a balanced polynomial β without constant term. Their constructions are compatible under smaller charge quotients. Keeping a scalar β_s external until the final substitution, their only remaining poles are finite-order poles on finitely many positive-multiple lattices of $c - \beta_s$. In the joint ring, after $\beta_s = \beta$,*

$$\Phi(T(u)) = u \pmod{u^{M+1}}, \quad u' = (-c + \beta)u. \quad (8.23)$$

Proof. In the output ring solve

$$(\Delta + c)\phi = g(X, Y, \phi). \quad (8.24)$$

Here $B - A \geq -N$, and the allowed argument of c gives $|B - A + c| \geq a|c|$; thus the inverse has norm $O(1/|c|)$. The small field norm makes Equation (8.24) a contraction in a fixed small graph ball. Put

$$b = g_v(X, Y, \phi), \quad \beta_o = \Pi b, \quad t_1^0 = \exp(\Delta^{-1}(b - \beta_o)), \quad (8.25)$$

where the inverse is zero on balance. The rational lattice gives a bounded off-balance inverse, so t_1^0 is a bounded unit close to one. Its eventual balanced normalization is specified below.

In the input ring solve

$$(\Delta - cv\partial_v)H_1 = \beta_e - g_v - g\partial_v H_1, \quad \Pi[H_1]_{v^0} = 0. \quad (8.26)$$

Choose β_e to cancel the omitted balanced v^0 projection of the right side. At degree v^m , $m \geq 1$, the inverse is $O((m|c|)^{-1})$, since $B - A \leq N$. In a product $g_p v^p \partial_v (H_l v^l)$ of degree m one has $l = m - p + 1 \leq m + 1$, so the factor l from differentiation divided by m is at most two. At degree zero only $l = 1$ can contribute, and off balance the inverse is bounded by the lattice gap. In the absolute coefficient norm these bounds make the operator on H_1 contractive when the g norm is small. There is no shrinking state radius in successive iterations: the derivative has already been compensated coefficientwise. Thus H_1 is small, $\Psi = e^{H_1}$ is a bounded unit close to one, and β_e is a small balanced polynomial without constant term.

Introduce the external scalar β_s and define

$$\Phi = f + \int_0^v \Psi(X, Y, t) dt, \quad (\Delta + c - \beta_s)f = -g(X, Y, 0)\Psi(X, Y, 0). \quad (8.27)$$

The layer-constant term of f is zero. This final inverse is meromorphic; it is not used in solving Equation (8.26).

In the joint ring, Equation (8.24) and the chain rule in Equation (8.26) give

$$\Delta(H_1(X, Y, \phi)) = \beta_e - b.$$

Balanced projection yields $\beta_e = \Pi b = \beta_o =: \beta$. These are equal as polynomials, since balance in either one-sided ring already lies in the joint ring. Moreover $\Delta \log(\Psi(\phi)t_1^0) = 0$. Lift the balanced reciprocal of $\Psi(\phi)t_1^0$ from the joint ring to the output ring and multiply t_1^0 by it. The resulting bounded unit t_1 satisfies

$$\Psi(\phi)t_1 = 1 \quad (8.28)$$

in the joint ring. Taking this balanced reciprocal and its lift commutes with every smaller charge quotient.

For $m \geq 2$ put $t_m = t_1 s_m$ and solve successively

$$[\Delta - (m-1)(c - \beta_s)]s_m = t_1^{-1}[g(X, Y, T) - g(X, Y, \phi) - b(T - \phi)]_{u^m}. \quad (8.29)$$

The right side uses only lower fiber orders. Keeping β_s scalar makes this one diagonal inversion at each step. For fixed M only finitely many pole factors and pole powers occur; they are at lattice values of the multiples $(m-1)(c - \beta_s)$. The input inverse has the same property. In the joint ring $|B - A| < N$, so all these inverses are uniformly safe for β_s in a fixed scalar disk containing β with a strict margin.

Substitute $\beta_s = \beta$ in the joint ring. To check the input conjugacy explicitly, set

$$F = \Delta\Phi + (-cv + g)\Phi_v - (-c + \beta)\Phi.$$

Equation (8.26) gives $F_v = 0$, and Equation (8.27) gives $F(0) = 0$. Thus $F = 0$. The graph equation, $\Delta t_1 = (b - \beta)t_1$, and Equation (8.29) prove the opposite conjugacy through degree M . If $K(u) = \Phi(T(u))$, these two identities make its constant term annihilated by $\Delta + c - \beta$ and its degree- m coefficient, $m \geq 2$, annihilated by $\Delta - (m-1)(c - \beta)$. Those joint inverses are safe. The constant term is therefore zero, the linear term is one by Equation (8.28), and all degrees two through M vanish. This proves Equation (8.23).

All equations, projections, products, and state substitutions preserve the charge ideals. Uniqueness of the contractions and the diagonal inversions proves compatibility under restriction, first with β_s held scalar and then under its fixed jets or formal substitution. This also proves the asserted uniform joint bounds and the description of the one-sided poles. $\square$

Lemma 8.9 (Fast-damping leading transition). *On the i fringe, for every fixed desired depth in $\Re W$ the passage has the finite expansion*

$$\phi_{\text{out}} + \sum_{m=1}^M e^{-mW} e^{m\beta_s L} t_{m,\text{out}} \Phi_{\text{in}}^m, \quad \beta_s = \beta(X, Y), \quad (8.30)$$

up to that depth, with M sufficiently large. Each fixed- m coefficient is independent of how much larger a comparison order is chosen.

Proof. Use the input coordinate on the first half of the passage, compare the two coordinates in their joint charge range at the midpoint, and then use the output coordinate on the second half. Damping makes the fiber variable small by the midpoint, while the omitted endpoint charges are small on their respective halves.

Along actual profiles, a balanced monomial is constant and has size $O(e^{-\omega \Re L})$. Consequently β_s is constant, $|\beta_s| \leq K e^{-\omega \Re L}$, and $|L\beta_s| \ll 1$. Equation (8.9) gives $|\Im c| \gtrsim |c| \Lambda_k / \Lambda_i$, and this is much larger than $|\beta_s|$ by Lemma 8.6. Put

$$\mathcal{P} = K_0 \left(1 + |\Im(c - \beta_s)|^{-1} \right).$$

At each fixed fiber order the one-sided coefficients and the needed state or layer derivatives have bounds by fixed powers of $\mathcal{P}$ and $1 + |c|$. Here $\log \mathcal{P} = O(\log(2 + \Lambda_n)) = o(\Re L)$.

Split a monotone-action path at its midpoint $L/2$. On the first half all omitted right charges have $B \geq N$ and pay $e^{-N \Re L/2}$. Evaluating the input equations along the actual solution gives

$$\frac{d\Phi}{d\theta} = (-c + \beta_s)\Phi + \epsilon_{\text{in}}, \quad |\epsilon_{\text{in}}| \leq K_M (1 + |c|) \mathcal{P}^{K_M} e^{-N \Re L/2}.$$

To justify this estimate even for the state derivative equation, first use its polynomial balanced β in the quotient; its actual evaluation is exactly β_s . The difference between products taken before or after projection has only omitted right charges. The constant equation uses the scalar directly. Stable variation of constants shows that at the midpoint Φ differs from

$$u(\theta) = e^{(-c+\beta_s)\theta} \Phi_{\text{in}}$$

by the displayed bound times $O(|L|)$. On the second half, $|u| \leq K\mathcal{P}^K e^{-\Re W/2}$.

At the midpoint restriction of both one-sided data to the joint ring changes evaluation by at most $K_M \mathcal{P}^{K_M} e^{-N\Re L/2}$. This holds for projected products and compositions as well. The joint coefficients are analytic on a fixed scalar β_s disk, so substituting the smaller balanced series and then projecting has the same error as projecting before substitution. We can therefore use Equation (8.23) at the midpoint. The unused Taylor orders cost $K_M \mathcal{P}^{K_M} |u|^{M+1}$; an inverse-power $\mathcal{P}$ radius in u keeps T in its state disk. Since $\Phi_v = \Psi$ is uniformly close to one, the actual midpoint state and $T(u)$ differ by the sum of these errors and the preceding path-length loss.

On the second half the approximate solution $T(X, Y, u(\theta))$ satisfies the original equation through degree M , up to omitted left charges, again of size at most $K_M(1+|c|)\mathcal{P}^{K_M} e^{-N\Re L/2}$. Higher degrees cost $K_M \mathcal{P}^{K_M} |u|^{M+1}$. Comparison with the exact stable propagator and the small integral of $|g_v|$ loses only a constant and path-length factors. At this fringe

$$\frac{N\Re L}{\Re W} \asymp \frac{\cos \arg L}{\cos \arg W} \longrightarrow \infty.$$

All polynomial factors have logarithm $o(\Re W)$. Thus the charge errors beat every fixed depth, and choosing the comparison order M beyond twice that depth handles the Taylor error. Evaluating T at the output gives Equation (8.30). The triangular recursion for a fixed t_m does not involve the later comparison orders. $\square$

8.6 Fast-damping dispersion and the full charts

Put $D = e^{-\omega L}$. At the input a right charge contributes $D^{B/\omega}$, and at the output a left charge contributes $D^{A/\omega}$. The balanced construction is pole-free and gives

$$\beta_s = D\tilde{\beta}(D) \tag{8.31}$$

with $\tilde{\beta}$ uniformly bounded and analytic on small D disks. Treat LD as a separate strict-small argument in $e^{m\beta_s L}$; it is a finite sum of monomials times ordinary factors through the primary log formulas, with the same first index n . This prevents the unbounded L from being used as an ordinary parameter of that exponential.

Before $\min(n, k)$ is passed, the actual D is much smaller than the imaginary separation of $C_\bullet$. At an n transition with $n < k$, a disk

$$|D| \leq \varepsilon \min(1, \Lambda_n^{-1}) \tag{8.32}$$

still preserves that separation: by Equation (8.31) its detuning is $O(\Lambda_n^{-1})$, whereas $|C_\bullet| \Lambda_k \rightarrow \infty$. Its Taylor tail of degree J is bounded, up to fixed losses, by

$$\exp(-J\omega\Re L + O(J\log(2 + \Lambda_n))),$$

which supplies arbitrary fixed n depth. The denominator jets after this expansion use $\beta_s = 0$, leaving finite powers of the original lattice poles.

Every fixed D degree uses only bounded output left charges and bounded input right charges. Lemma 8.8 proves that these coefficients are independent of the growing cutoff: restriction first at scalar detuning, balanced normalization on the joint intersection, and finite scalar differentiation all commute with the smaller charge quotients. Before this first dispersal, neighboring cutoffs differ at actual endpoints only by omitted charges or balanced terms of size $e^{-aN\Re L}$. The change in scalar detuning has the same size; finite pole powers and derivatives preserve it after reducing a . This is exactly Equation (8.22), now with $R = L$. In comparing ratio variants keep D independently small on its allowed disk; the already extracted shifts stay fixed.

At the k fringe use the full convergent layer rings, with no charge quotients. On a fixed small ratio-unit disk, transversality gives, for real layer eigenvalues d and integers $l \geq 1$,

$$|d \pm lC_\bullet| \gtrsim |d| + l|C_\bullet|. \quad (8.33)$$

All constructions in Lemma 8.8 now work in these full rings; keep β_s external on a bounded scalar disk and keep D, LD on separate small disks. The same balanced normalization and conjugacy proof applies. If $k \leq n$, the partial charts are their compatible projections. On slightly enlarged amplitude and D boxes their omitted endpoint charges cost $e^{-aN} = e^{-a'|Z_k|}$, also after factoring D from β_s . The bounded analytic substitutions preserve that precision. If $n < k$, the previously obtained fixed D, LD jets agree directly with those of the full charts.

Lemma 8.10 (Gevrey estimates for full charts). *For each fixed output fiber order, the full charts and their bounded analytic endpoint operations have consistent Gevrey-1 formal expansions in $h = 1/C_\bullet$. On the transverse k fringe they agree with sufficiently small proportional optimal truncations to precision $O(e^{-a|Z_k|})$. The assertion is uniform on smaller disks in all the ordinary and small arguments, and holds for their fixed jets.*

Proof. Multiply the graph equation by h ; its linear part is $1 + h\Delta$. At state degree $m \geq 1$, divide the input equation by $-cm$. Its leading inverse is regular and its perturbations are $h\Delta/m$ and the small product operator $h[g\partial_v H_1]_m/m$. At $m = 0$ retain the off-balance Δ^{-1} equation. Only degree one of H_1 can enter that last differentiated product. The f and fixed- s_m equations, after multiplication by h , likewise have regular leading inverses.

Here are uniform coefficient estimates for these formal equations. For a final budget J use absolute formal weights $(\varepsilon/(J+1))^q$ at order h^q , and layer radii whose logarithms decrease linearly with $q/(J+1)$ between two fixed nested disks. Products respect the associated coefficient norm. Applying Δ while advancing the formal degree once costs at most $K(J+1)$ by the one-step radius loss, which the h weight reduces to $K\varepsilon$. Move those terms to the right of the leading inversion. The graph is a contraction for small ε and small g . For H_1 the state derivative is still compensated by $l/m \leq 2$; the $m = 0$ coupling is small at the same fixed state radius and has a bounded off-balance inverse. These estimates give a uniform contraction on the entire finite formal norm, and hence a bound independent of J . The equations determine the same coefficient at order q for every $J \geq q$. Taking $J = q$ gives $K^{q+1}(q+1)^q$, and hence $K_1^{q+1}q!$, for that coefficient.

Balanced projection and the off-balance primitive are bounded in these norms. Analytic operations on units with a fixed nonzero limiting value preserve them on smaller disks; this is also the single-variable case of Lemma 6.4. The balanced normalization and the finite triangular f, t_m operations therefore have the same factorial estimates.

For the actual comparison take $J = \lfloor \varepsilon'|C_\bullet| \rfloor$ with ε' small enough that the formal dummy radius is at least twice the used $|h|$. Formal cancellation and the bounded norm just proved make the residual of the truncated equations $O(e^{-a|C_\bullet|})$ in their scaled norms on smaller layer disks; one extra fixed disk shrink pays a final Δ . Equation (8.33) bounds the actual scaled inverses. The same

small Lipschitz constants compare the actual graph and input fixed point with the approximations, preserving the residual precision. The remaining bounded analytic and triangular operations preserve it in turn. Endpoint substitution, factoring D from β , and the scalar substitution take place on strictly smaller fixed argument boxes and are bounded analytic operations. Cauchy estimates there prove the same assertions for fixed jets. This includes the jets already used when D, LD were dispersed before reaching this fringe. $\square$

8.7 Dispersal and source retirement

The preceding lemmas give the actual leading band and its arbitrary-depth expansion. We now construct the later packet bands. The order of the two unbalanced replacements determines when a coefficient ceases to depend on a discarded free input.

First extract every explicit affine exponential and every monomial prefactor. At each subsequent index Taylor-expand the strict-small arguments whose first index is current, keeping the other arguments as variables and setting the dispersed arguments to zero in coefficient evaluations. Fixed derivatives of already dispersed coefficients obey the same subsequent estimates: their formulas are the corresponding fixed jets, and their Cauchy margins were reserved before that dispersal. In the unbalanced cases, before the ratio's k transition these evaluations use $C_\bullet$ and the radii in Lemma 8.6. At the k fringe fixed ratio disks are available.

In either unbalanced case the moving charge cutoff exists only on bands $i < j \leq \min(n, k)$. If $n < k$, disperse D (and LD when present) first. Every fixed coefficient then has bounded charges and no numerical cutoff. If $k \leq n$, first pass to the full sum or full charts and their formal expansions at k ; disperse D, LD when their first index is reached. If $n = k$, the full/formal replacement is done on the small D, LD disks before the Taylor dispersal at that same transition.

After the formal replacement write, by Equation (8.5),

$$h = C_\bullet^{-1} = \frac{A_R}{A_P} e^{-b_*} e^{-f_k} \frac{1 + \Theta_R^\bullet}{1 + \Theta_P^\bullet}.$$

Absorb its bounded unit into the formal coefficients, preserving their Gevrey bounds. The monomial e^{-f_k} has first index $\sigma(k) > k$. Retain a small proportional optimal polynomial until that index. Its tail from any fixed degree q is bounded by $K_q |e^{-f_k}|^q$, since its terms decrease geometrically up to the chosen cutoff. Thus only fixed sufficiently many degrees remain at the $\sigma(k)$ transition. Beyond it no expression uses the numerical value of that cutoff. These are precisely the finite-argument dispersal operations justified in Proposition 6.6; the estimates here also prove them directly for the additional meromorphic stage.

For clarity, verify every source cost. Before $\min(n, k)$ has passed, neighboring cutoffs have cost

$$U = e^{\Re f_i} \cos(\Im f_n) \asymp |Z_k| \Re Z_n.$$

It is admissible because $i < j$, $\sigma(i) > \min(n, k) \geq j$, and $n \geq j$. Throughout its active band,

$$\frac{U}{\Re Z_j} \gtrsim e^{f_i(u) - f_j(u)} \frac{\cos(\Im f_n)}{\cos(\Im f_j)} \gtrsim e^{f_i(u) - f_j(u)} \longrightarrow \infty. \quad (8.34)$$

This remains true on the thin side. In addition $\Lambda_{j-1} \leq \Lambda_i = \Lambda_n + \Lambda_k + O(1) \leq 2\Lambda_j + O(1)$ there. An exceptional uncapped constant, if present, is consequently only $D_{j-1} = O(\log(2 + \Lambda_j))$ in these active bands. All raw, normalizing, pole, and fixed derivative losses therefore have logarithm $o(U)$. At the owning i fringe $U/\Re Z_i \asymp \cos \arg R / \cos \arg P \rightarrow \infty$, so its error also pays arbitrary leading transition depth.

After full/formal replacement the optimal-polynomial cost is the simple cost $e^{\Re f_k}$; it is used only on bands $k < j \leq \sigma(k)$. The simple source estimate from the flag calculus absorbs all fixed losses there. At the replacement itself $e^{-a|Z_k|}$ pays every fixed k fringe depth, since $\Re Z_k/|Z_k| \rightarrow 0$. Thus neither type of source survives the last index at which its defining logarithm is retained.

Choose frozen orders and cutoffs on fixed-factor bins of $e^{\Re f_k}$ and interpolate smoothly on overlaps. The differences just estimated have the appropriate exponential precision; the partitions and their fixed derivatives have logarithmic losses negligible relative to those costs. These choices use only retained formulas and can be made simultaneously for every finite request. There are no sources in band one. The actual passage through the leading band is genuinely holomorphic by Lemma 8.3.

8.8 Controlled inputs and terminal coefficient germs

Lemma 8.11 (Uniformity for one controlled family). *The mixed constructions and their finite dispersion operations have the controlled-family uniformity of Definition 4.1. For one fixed finite request, their constants and numerical far thresholds depend only on its common analytic boxes and margins, field bounds, finite rate and affine data, and finite quantitative flag and input-jet controls. They do not depend on when an individual input path returns to a smaller neighborhood of its ordinary germ.*

Proof. Fix the finite labels, derivative orders, and Taylor and transition budgets in the request. Choose common compact ordinary, layer, and state boxes with positive distance to the larger analytic domains. The field suprema and their finitely needed Cauchy bounds are then common. The initial amplitude choice gives a fixed contraction margin, say at most $1/2$, in the actual integral equations and in the layer-ring fixed-point equations. For the latter the graph inverse, the coefficientwise $l/m \leq 2$ estimate, and the bounded off-balance inverse give the same margin at every point satisfying the prescribed numerical conditions on $|c|$ and the charge cutoff. These estimates involve values in these boxes, not the subsequent history of the ordinary test in the flag path parameter.

The angular comparisons use the common finite real jets and Taylor remainders only through finite triangular sensitivity bounds. Their thresholds can therefore be chosen as common inequalities in the finitely many rates, gaps, and angular buffers. In the comparable case first fix κ and the side constant as in Equation (8.10); its vanishing remainder has the common modulus provided by the family controls. This gives common normalized action and pole-cone margins. The polygon length constants and stable comparison bounds are then common by Lemma 8.3. A dependence on the fixed limiting slope is permitted here; common amplitude smallness over slopes is the stronger, separately proved assertion above.

For the trees, geometric weights, counting constants, and the sorted-cost constant are determined by the field boxes and finite rate list. For each of the finitely many requested main charges, the finite low-cost separations and derivative orders give one common constant; the tail has the common factorial bound of Lemma 8.5. Its contour remainders require only specified inequalities such as $\Re L > K \log(1/\rho)$ and the finitely many cap-separation thresholds. The small-slope formal replacement uses the same geometric moment estimate at every point. Similarly, the finite formal norms in Lemma 8.10 use two fixed layer radii, a fixed state margin, and a common small formal weight. Taking the minimum of the finitely many permissible optimal proportions makes all factorial and optimal-error constants common. An already fixed convention for a shorter label is preserved by its exponentially small cutoff-comparison estimate; this does not redefine that label on a new test.

On a meromorphic band, the finite lattice lists and maximal pole orders are determined by the labels and their derivative orders. The direct finite jet formula above gives common powers of the explicit pole losses, even when the numerical denominators vary. Lemma 8.6 retains the allowed

D_{j-1} loss on the entire forward domain. Its safe-column proof is also uniform: the common real-jet and flag controls make Equation (8.20) uniform, so a fixed forbidden lattice width and a fixed union bound leave columns in every required fixed-length interval after common numerical thresholds. The exceptional-collar width and sweep-speed estimates use the same constants; no later return to the germ is used to obtain their $o(1)$ variation.

Independent Cauchy radii are a common fixed exponential $\exp(-Cf_j)$ times the explicit pole-preserving factor. Their finite powers have the common logarithmic bounds already proved. Taylor depths can therefore be chosen once for the request, before applying those losses. The moving-cutoff and optimal-order comparisons have the common costs in Equation (8.22) and $e^{\Re f_k}$, respectively. Their absorption uses the common numerical gap and side inequalities in Equation (8.34) and the simple-cost estimate. Fixed-width bins in the retained log modulus have uniformly bounded overlap. The same partitions in those bins have common derivative bounds after the stated triangular losses; hence both the number and precision constants of the source terms are common for this finite request.

Finally, high real-jet agreement preserves these domains, angles, and evaluations of independent derivatives. The differentiated source bounds come from the frozen holomorphic formulas and their proved cutoff overlaps. Holomorphic ordinary inputs contribute no input defect; smooth sheets already constructed by Theorem 4.4 supply their own differentiated defects.

For pullbacks, the finite chain-rule factors have exactly the raw, Cauchy, and controlled-sheet losses just bounded. Their logarithms are $o(U)$ at intrinsic active cutoff sources by the preceding cost comparisons. Inherited sheet sources retain their own permitted costs; the comparisons in Definition 3.1 and Equation (4.7) apply to them as well. At a level- p good join, keep a fixed current loss as $C\Re Z_p$; the remaining later-rate and local Cauchy losses are $o(V_p)$. Lemma 4.7 therefore preserves the source estimates, choosing a transition budget larger than its current losses or a sufficiently vertical collar for an independently established e^{-cV_p} error. The common controls above make these choices uniform for this finite request.

All the preceding estimates hold throughout the part of each full forward domain on which the stipulated common ordinary boxes and numerical inequalities hold. Entering a smaller germ neighborhood may be delayed by a different amount on each path. It affects a later request using that smaller box, not any constant, starting inequality, or source count for this fixed request. This proves the claimed uniformity. $\square$

Proof of Theorem 8.1. We have constructed the actual passage, its transition expansions, and the raw coefficients in every band. It remains to collect their growth and derivative estimates and to verify support and germ invariance.

The growth of a fixed raw label is at most $\exp(C\Re Z_j + CD_{j-1})$: after explicit shift translations, its bounded analytic pieces, factorially summed tree pieces, or finite chart pieces have only the losses of Lemma 8.6. That lemma also supplies the side and safe-column improvements, simultaneously for fixed derivatives and the full exceptional collar. Its local Cauchy estimates applied with deeper initial transition accuracy prove all differentiated transition bounds. Intrinsic antiholomorphic derivatives arise only in the smoothed primitive choices; their differences and all fixed derivatives have exactly the active precision costs. On previously constructed smooth input sheets the inherited input sources are also propagated by the chain rule, as proved in Lemma 8.11. These observations establish the required growth, transition, and source estimates in the independent variables, including controlled input tests.

Finally the support is iterated left-finite. At the first index the main charges form a locally finite positive semigroup or a positive lattice. At a fixed charge there are only finitely many prefactor translations; all further operations take Taylor powers of a finite set of strict-positive monomials,

or the next fixed lattice charges. Fixing each prefix bounds the degrees and translations used at its next coordinate. Gevrey evaluation is used only before its first index is extracted, so it introduces no infinite set below a fixed owning budget. Colliding coefficients are added by these finite recursive rules. At the last band only ordinary analytic operations and normally convergent bounded-argument sums remain.

The terminal recipes are invariant under an allowed saturated refinement of the flag. In the comparable construction they are the residues of the exact convolution formula, with collisions interpreted by the analytic cluster integral Equation (8.16). Splitting a primary affine shift into refined nodes changes only the representation of its exponent, and regrouping a finite fixed-charge extraction leaves these analytic residue identities unchanged. In the small-slope construction each coefficient after cutoff retirement is a coefficient of the same fixed residue product; its formal h coefficients are determined by that product, independently of optimal order. In the fast-damping construction, uniqueness of the graph and input contraction, the stipulated off-balance inverse, the balanced normalization, and the finite triangular fiber recursion determine the coefficient recipes. Compatibility with smaller charge ideals shows that every fixed endpoint jet is the same germ before and after removing its cutoff. The full-ring formal recursions have exactly these jets. Thus all terminal identities are identities of ordinary analytic germs, with the same fixed lateral determinations.

The same argument also covers the small absolute perturbations of old free inputs in Definition 5.6. Keep the selected affine flag formulas, rate lists, limiting slope and residual regime, ordinary field germs, and lateral determinations. Changing the numerical old free inputs within that regime changes the evaluations of the extracted shifts and of the intermediate coefficient formulas, but changes none of the residue identities, charge projections, normalizations, or formal recursions just listed. Their final coefficient formulas contain no numerical free input or cutoff. They consequently give the identical ordinary analytic germs, rather than merely the same values at the limiting ordinary point. A change of modulus bin affects only an intermediate cutoff choice, whose dependence has already disappeared by the proved retirement rules. The estimates apply to the perturbed sequence by the same controlled-test construction, with its preserved regime gaps and smaller neighborhoods as needed.

An inserted unused node contributes an identity transition. Applying the earlier finite ordinary-chart recipes then invokes Lemma 4.13. This proves the stated retesting and refinement invariance without identifying an unspecified flat error with zero, and completes the theorem. $\square$

8.9 Uniformity of the primitive library

Proposition 8.12 (Quantitative primitive input bounds). *For the primitive constructions in Sections 6, 7, and 8, the family requirement in Assumption 4.2 follows from the estimates on common argument boxes proved in those sections. For each fixed finite request their constants and far-regime thresholds depend only on those boxes and margins, the finite field and flag data, and the finite controls in Definition 4.1. The time at which an ordinary test returns to its germ does not enter these constants.*

Proof. We specify the dependence in the estimates used there. This also distinguishes this assertion from merely assuming uniformity in addition to estimates proved one test at a time.

For additive primitives, fix the larger field, profile, and state boxes and their reserved margins. The path integrals in Equations (6.9) and (6.10) are bounded by constants times the amplitude, with constants depending only on the fixed exponents and boxes. The kernel in Equation (6.22) has the same damping bound throughout those boxes. For a fixed coefficient request, its finitely many

exceptional diagonals have common bounds; the infinite opposite tail uses the common estimate

$$\|\mathcal{L}_m^{-1}\Pi_{>N}F\| \leq (C_0/N)\|F\|.$$

The threshold $N > 2C_0\|A_0\|$ is selected once on the larger boxes, as are the summable state and spatial margins. In the formal estimate the radius ε/J pays a derivative of norm CJ ; hence one choice of ε gives the Gevrey constants for that request. The actual residual comparison, its fixed polynomial moments, and the finite charge remainder then use these same constants. The normalized primitive integral in Equation (6.46) is uniformly integrable after choosing its fixed exponent margin. The infinite anchor in that integral is in the independent spatial variable w , while the ordinary arguments are held in their common boxes. It imposes no return-time requirement on an ordinary test $t(s)$. These observations apply to every estimate in Lemmas 6.2, 6.3, and 6.4, and hence to the finite formulas of Propositions 6.5 and 6.6. Lemma 6.7 gives the resulting explicit common jet and angular-error thresholds on the full forward domain.

For the one-rate primitives, the same common-box choice controls the integrated layer norm and the stable variation-of-constants operator. In Lemma 7.3, the nonzero charge gap is fixed and the high-tail inverse again gains C/N . The weighted norm for slow degree has derivative cost CJ multiplied by ε/J . Consequently the pure and tail contraction margins, factorial constants, optimal proportions, and the endpoint inverses in Lemma 7.2 are chosen on these boxes, before any particular ordinary test is supplied. The detailed controlled-input verification is Lemma 7.4.

For mixed primitives, the actual continuation in Lemma 8.3 is controlled by an integrated fast-profile norm and a small Lipschitz norm on the larger state box. The contour bounds of Lemma 8.5 use the fixed finite rate and charge data. Their infinite charge sums have a common geometric majorant η^M with $\eta < 1$, and for a fixed derivative order q ,

$$\sum_{M \geq 0} \eta^M (1+M)^q \leq C^{q+1} q!.$$

For instance, with $a = -\log \eta > 0$, the exponential-series inequality $x^q \leq q! \tau^{-q} e^{\tau x}$, applied with $\tau = a/2$, proves this bound with a constant depending only on η . Thus the fixed moments and Gevrey constants of Lemmas 8.7 and 8.10 are common. The graph and layer contractions in Lemma 8.8 have common diagonal inverse bounds on the stipulated slope ranges. The finite large-rate and detuning inequalities in these arguments are precisely numerical regime thresholds of Definition 4.1; they are independent of the history of the current inputs. Lemma 8.11 records the complete uniform choice through all of the mixed constructions.

The pole estimates also give uniform safe-column existence, rather than requiring one numerical column for all tests. To see this quantitatively, let x_a range over the finite list of positive real lattice sweeps in Lemma 8.6, and let $h_a > 0$ be their lattice spacings. Common first and second real-jet bounds give, beyond common flag thresholds,

$$\frac{x'_a}{x_a} \geq \frac{1}{2} \Lambda_{k(a)}, \quad \left| \frac{x''_a}{(x'_a)^2} \right| \leq \frac{C}{x_a}.$$

On a complete lattice period where $x_a \geq X$, inverse speeds differ by at most $e^{Ch_a/X}$. A forbidden neighborhood of fixed width 2δ therefore occupies at most $(4\delta/h_a)e^{Ch_a/X}$ of that period's traversal time. The two incomplete endpoint periods cost at most $2h_a/\inf x'_a$. First choose δ for the finite list, and then choose common lower thresholds for X and the speeds. The union of the forbidden parts occupies less than a prescribed fixed-length interval, giving a safe column for each test with the same detuning margin. When the exceptional entering collar is needed, its change in each sweep is bounded by $C \exp(-D + o(D)) \log(2 + D)$, with a common modulus in the $o(D)$ from the same

finite jet and flag controls. It is therefore smaller than the chosen detuning margin. This proves the uniform collar assertion while leaving D explicit.

Finally the independent-input Cauchy disks, frozen optimal orders, and modulus-bin cutoffs in all three constructions are local in the current retained tuple. Their radii and loss exponents depend on the common argument margins, finite pole orders, and quantitative flag controls. They are not disks of analytic continuation for the entire test path. Adjacent cutoff formulas have the stated exponential errors on the larger common disks, so their fixed differentiated errors and partition losses have common source constants. A sufficiently high fixed Taylor order preserves these geometric margins uniformly by Lemma 2.7; it does not itself create a source estimate. For source estimates under composition, the inputs are holomorphic or have the already established differentiated defects of an old chart sheet. Every remaining factor in the finite chain rule has one of the listed absorbable losses, uniformly for the controlled family. Lemma 4.7 therefore gives the composed source bounds, including its separate treatment of current good-join transition remainders and independently established join errors.

All these estimates are inequalities at the current arguments and hold at every point of the full forward domain on which the common controls hold. Neither these norms nor the safe-column construction waits for an individual test to reach a smaller ordinary neighborhood. Such a wait is used only for further preliminary requests in Lemma 4.9, whose constants disappear in the final weighted estimate. This proves the proposition. $\square$

9 Finite real subdivision and passage words

We work with all coefficient vectors of polynomial fields $V = (P, Q)$ of degree at most d . The spatial domain is a fixed bounded square. There is no compactness assumption on the coefficient vector. A dilation puts any prescribed finite collection of periodic orbits in this square, without increasing the degree. All constants and all finite lists below are allowed to depend on d and on previously fixed subdivision choices. They are independent of the field and of its periodic orbits.

The finite lists below consist of formulas and labels whose parameter values vary with the field. An orbit selects a word in this alphabet and its endpoint values. The further factorizations used for an absolute-zero test are chosen only after a sequence is fixed and do not enlarge the alphabet.

9.1 Preparation, clocks, and small arguments

Here is the precise external preparation result we use. Coordinates p in this statement include all parameters; x is the last variable.

Theorem 9.1 (Simultaneous globally subanalytic preparation). *Let $X \subset \mathbb{R}^k \times \mathbb{R}$ and $f_1, \dots, f_N : X \rightarrow \mathbb{R}$ be globally subanalytic. There is a finite globally subanalytic cell decomposition of $X \cap \{x \neq 0\}$ such that on each cell the functions have a common center $a(p)$ and representations*

$$f_\nu(p, x) = c_\nu(p) |x - a(p)|^{q_\nu} U_\nu(b_{\nu j}(p) |x - a(p)|^{r_{\nu j}} : 1 \leq j \leq h_\nu).$$

All exponents are rational, all displayed parameter functions are globally subanalytic, and each c_ν is identically zero or nowhere zero. The sign of $x - a(p)$ is constant. Each argument tuple lies in $[-1, 1]^{h_\nu}$, and U_ν is real analytic on a neighborhood of that cube and positive there. Moreover, either $a = 0$ or $|x - a(p)| < \epsilon |x|$ on the cell for some $\epsilon < 1$.

This is the globally subanalytic case of Lion–Rolin preparation: [18, §0.3, Theorem 1 in §0.4, and §1.1(5)]; the stated finite-family formulation is [19, Definition 2.14 and Fact 2.15]. The graph

$x = 0$ is separated before using the theorem. The last center inequality will not be needed. A finite union of argument lists makes the arguments common as well. Exponent-zero arguments are bounded parameter arguments. We freely split signs and zero loci.

We also use finite analytic cell decomposition and uniform finiteness for globally subanalytic families. The analytic refinement can be obtained within the cited preparation theorem: recursively refine the base for all its finitely many coefficients, centers, and cell-boundary functions. On every fixed-sign interval the rational powers and strong units are analytic. Induction on the number of coordinates, putting parameters first, therefore gives compatible analytic cylindrical pieces; see also [18, §1.1(1),(6)]. Uniform bounds for fiber components are supplied by [2, Theorem 3.14], after algebraic compactification for globally subanalytic sets. In particular, a fixed such family of sets of fiber dimension at most one admits a uniform finite decomposition of its fibers into points and injectively parametrized regular analytic arcs. Adding finitely many globally subanalytic functions allows their regularity and signs to be fixed on these pieces. Differentiation on a regular piece, algebraic root selection, finite maxima, and positive rational powers preserve global subanalyticity. These facts concern the preparation data on their regular pieces; they assert no uniform bound on their derivatives.

Every strong unit has uniform real upper and positive lower bounds. Compactness of its argument cube also gives a complex neighborhood on which it is holomorphic and nonvanishing, after decreasing that neighborhood. If a prepared function is bounded, its leading monomial is bounded, since its unit is bounded below. We may therefore include the leading monomial as an additional bounded argument whenever only a bounded analytic representation is needed. Real and imaginary parts of complex algebraic data are prepared separately.

Lemma 9.2 (Compatible change of Euler center). *Suppose that the old clock is $D_0 = (x - a_0(p))\partial_x$. After simultaneous preparation and finite subdivision, all the required data can be represented in an Euler clock*

$$D = (x - a(p))\partial_x = \tau\partial_\tau = hD_0, \quad \tau = |x - a(p)| > 0, \quad 0 < |h| \leq 2.$$

If the initial clock is ∂_x , the same assertion holds without the multiplier bound at this first step. Previously used time data can be included in the preparation.

Proof. For a center a furnished by preparation, keep a on $|x - a| \leq 2|x - a_0|$. The multiplier is $h = (x - a)/(x - a_0)$. Put all zero denominators and centers on the boundary. On the complementary region return to a_0 . With $q = (x - a_0)/(a_0 - a)$ the complementary inequality is $|1 + q| > 2|q|$, which implies

$$-\frac{1}{3} < q < 1, \quad \frac{2}{3} < 1 + q < 2.$$

Consequently, for every rational r occurring in the finite list,

$$|x - a|^r = |a_0 - a|^r(1 + q)^r.$$

The second factor is analytic with a fixed complex neighborhood of the indicated real interval. If $b|x - a|^r$ was bounded, then $b|a_0 - a|^r$ is bounded, with a uniform bound depending only on r . The resulting analytic compositions have compact argument ranges with room. Now $h = 1$. Splitting its sign, when necessary, finishes the proof. $\square$

The next observation supplies the small pure-time profiles used in Section 10. Suppose finitely many bounded arguments $b_j\tau^{r_j}$ occur, with nonzero r_j . Fix a desired smallness $\delta > 0$. On pieces where all are smaller than δ , keep the clock. Otherwise select one with $\delta \leq |b_j\tau^{r_j}| \leq M$. With

$t = |b_j|^{-1/r_j}$, the ratio τ/t belongs to a fixed compact subinterval of $(0, \infty)$. Subdivide that interval into finitely many relative bins. Choose a baseline $\tau_* = ct$ slightly below each bin, so that throughout it

$$\tau = \tau_* + \tilde{\tau}, \quad 0 < \tilde{\tau}/\tau_* < \eta.$$

All old powers become constant multiples of $(1 + \tilde{\tau}/\tau_*)^r$. Previously bounded multipliers $b_j \tau_*^{r_j}$ remain bounded. A small mixed argument $b \tau_*^a u^s$, $s \neq 0$, becomes $b \tau_*^a u^s (1 + \tilde{\tau}/\tau_*)^a$ and remains small after an arbitrarily small fixed adjustment to its tolerance. The new clock multiplier is $\tilde{\tau}/\tau$, and

$$\frac{\tilde{\tau}}{\tau} A \tau^\lambda = A \tau_*^{\lambda-1} \tilde{\tau} (1 + \tilde{\tau}/\tau_*)^{\lambda-1}. \quad (9.1)$$

Thus these *contact bins* have leading time exponent 1. Choose finitely many analytic neighborhoods of the compact tuples of baseline constants first, and then make η small inside their common margins. This ensures that setting the new small arguments to zero stays within the analytic neighborhoods. Narrowing bins changes their number, but introduces no new exponent other than the patterns in Equation (9.1).

9.2 Poles, annuli, and pole-free boxes

Away from $P = 0$ the scalar equation is $\partial_x z = Q(x, z)/P(x, z)$. We need a slightly more general construction, since it will also be used after rational weighting: start with

$$D_0 z = R(p, x, z), \quad (9.2)$$

where R is rational in z of bounded degree and its base coefficients are globally subanalytic and analytic on the selected base pieces. The clock is either the first clock ∂_x or an Euler clock.

Partition by denominator degree, root multiplicities, and coefficient regularity. The complex roots $p_j(p, x)$ can be enumerated on finitely many regular pieces, with multiplicities. Choose a pole whose real part c is horizontally nearest to the real state z , and split by this choice and by the sign of $z - c$. Set $u = \sigma(z - c) > 0$, $\sigma \in \{1, -1\}$. For every denominator root,

$$|z - p_j| \geq |z - \Re p_j| \geq |z - c| = u. \quad (9.3)$$

With no denominator roots use $c = 0$ instead. The transformed equation is $D_0 u = \sigma R(p, x, c + \sigma u) - \sigma D_0 c$. Its numerator remains polynomial in u after using the denominator; the degree is uniformly bounded. In particular the derivative of the center is retained exactly. A numerator that vanishes identically is a constant-state passage. Degree drops and zero coefficients are separated before any ratios are formed. Common numerator and denominator factors cause no problem: retaining a removable pole only adds a boundary.

Write the translated denominator factors as $u - \varpi_j$, where $\varpi_j = \sigma(p_j - c)$, and the numerator as $\sum_{l=0}^N n_l u^l$. The *complete radius list* for the following split is

$$\mathcal{R} = \{\rho_j = |\varpi_j| : \varpi_j \neq 0\} \cup \{|n_l/n_r|^{1/(r-l)} : 0 \leq l < r \leq N, n_l n_r \neq 0\}. \quad (9.4)$$

In particular every positive translated pole modulus is a radius, whether or not any two numerator terms are present. Zero displacements are separated and will contribute exact powers of u . Prepare simultaneously this list, the nonzero coefficients, the real and imaginary parts of the bounded directions ϖ_j/ρ_j , the centers and their needed derivatives, and all earlier time-coordinate data needed to return to physical endpoints. Use Lemma 9.2. For every $\rho \in \mathcal{R}$ the preparation gives a positive pure representative $s_\rho = C_\rho(p) \tau^{\gamma_\rho}$ and, for a single fixed $K \geq 1$ on the finite family,

$$K^{-1} s_\rho \leq \rho \leq K s_\rho. \quad (9.5)$$

There are at most $q + N(N + 1)/2$ entries, where q is the denominator degree. Algebraic root selection, modulus, and positive rational powers make them globally subanalytic on their regular pieces. Repeated or equal radii may be retained. Thus including the pole moduli changes neither the finite-family hypothesis of preparation nor degree control.

Choose a large fixed factor $H > K$. The far domain is the union of the pieces on which, for *every* $\rho \in \mathcal{R}$, either $u > Hs_\rho$ or $u < s_\rho/H$. Fix these alternatives on each piece. Equation (9.5) implies respectively $\rho/u < K/H$ or $u/\rho < K/H$. One numerator term therefore dominates. Indeed the ratio of terms of degrees $l < r$ has absolute value $(u/|n_l/n_r|^{1/(r-l)})^{r-l}$ or its reciprocal; the selected far inequalities make all ratios to the largest term at most K/H once $H > K$. For a nonzero pole displacement the exact extractions are

$$u - \varpi_j = u(1 - \varpi_j/u) \quad \text{or} \quad u - \varpi_j = -\varpi_j(1 - u/\varpi_j),$$

and the chosen correction has modulus less than K/H . The normalized directions have modulus one, so the bounds hold also for nonreal poles and for several poles with equal real part. Poles at displacement zero contribute an exact power of u ; conjugate factors are paired to retain real formulas. Thus these pieces have the exact form

$$Du = A\tau^\lambda u^m F(X), \quad X_j = b_j \tau^{a_j} u^{s_j}, \quad u > 0, \quad (9.6)$$

where $A \neq 0$ is parameter-only, $m \in \mathbb{Z}$, and all other exponents are rational. Constant arguments have been included in a bounded ordinary tuple, suppressed in this notation.

Here and below “exact” refers to equality of the scalar fields on the piece, not to an asymptotic approximation. Prepared units occur in the small ratios just described and are retained in F . After the contact construction and division by the nonzero value at zero profiles, we have $F(0) = 1$ on larger analytic boxes and, for any prescribed fixed $\varepsilon > 0$, can arrange

$$\max_j |X_j| < \varepsilon, \quad |F - 1| + |\partial_{\log u} F| < \varepsilon. \quad (9.7)$$

For the second inequality use $\partial_{\log u} F = \sum s_j X_j F_{X_j}$ and Cauchy bounds on the larger boxes. The leading-term and pole-extraction alternatives form a finite list before H is increased. Their bounded unit arguments have fixed analytic neighborhoods. Hence the required smallness can be chosen from this list before choosing the final far factor. Contact bins contribute bounded baseline constants and fixed power patterns, so their later narrowing does not change that assertion. In particular, one may anticipate any fixed compact power-clock state boxes required in Section 10, whose state factors then cost only fixed constants.

On each remaining piece there is a radius $\rho \in \mathcal{R}$ with $s_\rho/H \leq u \leq Hs_\rho$. Choose one such radius by a finite subdivision, set $s = s_\rho = C\tau^\gamma$, and put $z_1 = u/s$. Thus $H^{-1} \leq z_1 \leq H$, whether the selected radius is a pole modulus or a numerator balance. Equalities in these cuts are boundary pieces. The equation is

$$Dz_1 = s^{-1}Du - \gamma z_1.$$

By Equation (9.3) every pole in the z_1 -plane has distance at least $z_1 \geq 1/H$ from the actual real point. Choose fixed real bins of radius $\delta_1 < 1/(8H)$ and concentric larger complex disks of radius $2\delta_1$. For each bin restrict the base to parameters for which the actual piece meets the bin. The triangle inequality from any such actual point gives pole distance at least $1/H - 3\delta_1 > 1/(2H)$ throughout the larger disk. Thus the lower bound at actual points supplies the required complex disk bound; it is not presumed to hold at arbitrary bin centers. These finitely many projected base sets are globally subanalytic and can be regularly subdivided. On the disks the field is rational with

numerator degree at most $\max(N, q + 1)$ and denominator degree q : the new term $-\gamma z_1$ adds at most z_1 times the denominator. This remains true when several pole radii are comparable, when some other pole radii tend to zero or infinity relative to s , and when several poles have the same nearest real part.

Example 9.3 (A conjugate pole pair on the comparable branch). For $R = 1/(z^2 + \epsilon^2)$, $0 < \epsilon < 1/10$, both poles have real part zero. On $z > 0$ we have $c = 0$, $u = z$, and the radius list is $\{\epsilon, \epsilon\}$, with no numerator balance radius. The point $u = \epsilon$ belongs to the comparable branch $s = \epsilon$. In the original clock,

$$z_1 = u/\epsilon, \quad \partial_x z_1 = \frac{\epsilon^{-3}}{1 + z_1^2}.$$

Near $z_1 = 1$ the poles are $\pm i$, so, for example, the disk $|z_1 - 1| < 1/2$ is uniformly pole-free. On the smaller real interval $|z_1 - 1| < 1/4$, the normalized field is at least $16/41$ and bounded on the larger disk. This is a nonzero box with scale comparable to ϵ^{-3} . Changing to an Euler clock only adds its prepared time factor. A purported far-annulus description at $u = \epsilon$ would instead have $|u/\epsilon| = |\epsilon/u| = 1$; the complete radius list is precisely what sends these points to the box construction.

Lemma 9.4 (Uniform box data). *After the preceding finite subdivision, each nonzero comparable-scale field has, on a fixed larger state disk $\mathcal{D}$ and a smaller real interval $I \Subset \mathcal{D}$, a positive globally subanalytic scale $L(p, x)$ and constants K, m such that*

$$\sup_{z \in \mathcal{D}} |F(p, x, z)| \leq KL(p, x), \quad \sum_{j=0}^m |\partial_z^j F(p, x, z)| \geq L(p, x) \quad (z \in I). \quad (9.8)$$

The normalized rational coefficients belong to bounded sets with uniform analytic room. The constants and degrees are fixed on each of finitely many boxes.

Proof. Normalize the denominator by its value at the disk center. In its factored representation its coefficients and its reciprocal are bounded on a slightly smaller disk, because all poles are uniformly separated. Let L_0 be the maximum modulus of the coefficients of the resulting numerator. Separate $L_0 = 0$; otherwise divide the numerator by L_0 . The normalized numerator and denominator belong to compact coefficient sets, the denominator is uniformly nonzero, and at least one numerator coefficient has modulus one. If the numerator degree is at most m , no member can have its first $m + 1$ jets vanish at a point: multiplication by the nonzero denominator would give a polynomial of degree at most m with a zero of order $m + 1$. Compactness, including $z \in \bar{I}$, gives a positive uniform lower bound on the sum of those jets. Rescaling L_0 by that fixed bound gives Equation (9.8). The supremum estimate follows from the coefficient bounds. All the normalizations use subanalytic operations and nonzero divisors. $\square$

9.3 The finite box induction

The order m in Equation (9.8) is the induction index. During this argument the state disks can be decreased by fixed amounts and then covered by finitely many fixed smaller disks. A “uniform bound” always refers to such specified larger disks.

A box may also carry an absolute bound for its field on its larger state disk. Every nonterminal child below either has lower jet order or carries such a bound; once the bound is present, the next continuing step lowers the order. At each selected index, the possible center branches and their preparations will be fixed before the final lower-jet thresholds.

Choose positive thresholds $\epsilon_0, \dots, \epsilon_m$ with $\sum \epsilon_j < 1$. Their choice is made from higher to lower indices, and lower ones will be made sufficiently small below. Assign a point to the least i with $|\partial_z^i F| \geq \epsilon_i L$; some such i exists by Equation (9.8). Split its sign and the signs of the lower derivative expressions on regular cells. Cauchy estimates on the larger disk bound every fixed further derivative by a constant times L .

If $i = 0$, the field has fixed nonzero sign and a lower bound on the piece. If $i = 1$, the smallness of $|F|/L$ relative to the derivative margin places the point as close as desired to a unique real simple root. For completeness, choose a radius r_0 so that the derivative changes by at most half its lower bound on the corresponding real interval and complex disk. If $|F(z)| < r_0 \epsilon_1 L/4$, the values at $z \pm 2|F(z)|/(\epsilon_1 L)$ bracket zero. Strict monotonicity gives a unique real root c , and $|F_z(c)| \geq \epsilon_1 L/2$. The analytic implicit theorem gives a root chart with a fixed smaller complex neighborhood. This proves both existence and the needed room.

Prepare L and the bounded normalized rational coefficients in these two cases, together with the coordinate data, and perform the final contact refinement. The resulting equation is

$$Dz = B\tau^\lambda f(X, z), \quad X_j = b_j \tau^{r_j}, \quad B > 0, \quad (9.9)$$

where $r_j \neq 0$ are rational, X is as small as required, and f is analytic with uniform bounds on larger boxes. Constant arguments are bounded ordinary parameters. Either f has constant sign with a fixed positive lower bound on the piece, or the state is arbitrarily close to a regular root $z = c(X)$, with a fixed derivative margin. In the latter case the root at $X = 0$ is still regular. To see that these choices do not depend circularly on the final root-tracking tolerance, first prepare the bounded coefficient data on the full pole-free bins. The derivative margin fixes the root neighborhoods there. Only then choose the profile smallness and the order-zero threshold. Finitely many ordinary root charts cover the compact normalized coefficient sets.

Now let $i \geq 2$. Apply the same argument to $\partial_z^{i-1} F$. There is a nearby real root $c(p, x)$ with

$$\partial_z^{i-1} F(p, x, c) = 0, \quad |\partial_z^i F(p, x, c)| \geq \frac{1}{2} \epsilon_i L.$$

Regular root selection and subdivision make c analytic in x and globally subanalytic. Its distance from the actual state can be made arbitrarily small by the lower thresholds. Set $r = z - c$ and write the exact translated field as

$$D_0 r = A_0 + \sum_{l=1}^i A_l r^l + A_i r^{i+1} H(p, x, r), \quad (9.10)$$

$$A_0 = F(p, x, c) - D_0 c, \quad A_l = \frac{\partial_z^l F(p, x, c)}{l!} \quad (l > 0).$$

Here $A_{i-1} = 0$, $A_i \neq 0$, and H has a uniform analytic bound on a fixed disk. It is rational analytic there, with bounded normalized subanalytic coefficient data: subtract the finite Taylor polynomial in the normalized rational representation. For $1 \leq l < i - 1$, $|A_l/A_i|$ can be made arbitrarily small. Indeed the lower derivatives are small at the original point, the center displacement is small, and Taylor's formula with the above Cauchy bounds transfers these estimates to c . No bound has been asserted for $D_0 c$ or for A_0 .

Define

$$S = \max_{0 \leq l < i} |A_l/A_i|^{1/(i-l)}. \quad (9.11)$$

To make the exponent list independent of the final lower-jet thresholds, prepare the full center branches before imposing those thresholds. Fix the current disks, normalized rational coefficient

bounds, and ϵ_i . Choose a compact real interval J containing the actual smaller state interval with room and lying a fixed distance inside the larger disk. The *full candidate-center set* is

$$\mathcal{C}_i = \left\{ (p, x, c) : c \in J, \quad \partial_z^{i-1} F(p, x, c) = 0, \quad |\partial_z^i F(p, x, c)| \geq \frac{1}{2} \epsilon_i L(p, x) \right\}. \quad (9.12)$$

No condition involving a lower-jet threshold is imposed here. Since F is rational of fixed degree, clearing the nonvanishing denominator gives a polynomial equation in c of fixed degree. Its roots in $\mathcal{C}_i$ are simple by the displayed derivative margin. The identically zero polynomial has no roots satisfying that margin. Thus finite subanalytic regular subdivision supplies finitely many candidate root branches. Take the compact ambient normalized coefficient set of Lemma 9.4, including the closure of the attainable coefficient image; its denominator remains uniformly nonzero on the state disk. In its product with J , impose the root equation and the closed derivative margin for the normalized field F/L . This is a compact set, even when the attainable parameter image itself is not closed. The analytic implicit theorem on this compact ambient root set provides the root charts and their uniform neighborhoods before any final lower-jet restriction.

On every full branch form all A_l , including $A_0 = F(c) - D_0 c$, and S from Equation (9.11). Their base functions are globally subanalytic; differentiating the root equation on its regular base gives $c_x = -\partial_x \partial_z^{i-1} F / \partial_z^i F$, so this claim neither assumes nor supplies a bound on c_x . Prepare these full-branch functions, their bounded normalized analytic data and the earlier coordinate data, using Lemma 9.2. Every A_l is multiplied by the same h , so S is unchanged. On each prepared piece with $S > 0$ write $s = C\tau^\gamma \asymp S$ and fix $K_0 \geq 1$ such that $K_0^{-1} S \leq s \leq K_0 S$. The possible exponents γ and comparison constants are now fixed before the lower-jet restrictions.

Fix a small cutoff δ for S . Arrange the positive lower coefficient radii to be less than $\delta/2$. On $S \geq \delta$ the maximum in Equation (9.11) must come from A_0 . On $|r| \leq \eta\delta$, with fixed sufficiently small η , every positive-degree term in Equation (9.10) is a small fraction of $|A_0|$. The remainder has the same property. The actual state lies in a still smaller such disk by the choice of lower thresholds. This is a nonzero box with scale $|hA_0|$.

It remains to consider $S < \delta$. If $S = 0$, discard $r = 0$ as a boundary and use the monomial field of power i . If $S > 0$ and $|r| > Hs$, with H fixed sufficiently large, then

$$|(A_l/A_i)r^{l-i}| \leq (S/|r|)^{i-l} \leq (K_0/H)^{i-l}.$$

The higher remainder is small by the smallness of $|r|$. Thus this is again an annulus, now of state power $m = i \geq 2$, with the prepared unit and contact refinements already described.

For the inner part put $r = sz_2$, $|z_2| \leq H$. On a fixed larger disk we have exactly

$$Dz_2 = L_s \left(z_2^i + \sum_{l < i-1} d_l z_2^l + E(p, x, z_2) \right) - \gamma z_2, \quad (9.13)$$

$$L_s = hA_i s^{i-1}, \quad d_l = (A_l/A_i) s^{l-i}.$$

where $\|E\| \leq Cs$. Its fixed finite jets on smaller disks have the same bound. All d_l are bounded, and at least one is bounded away from zero: choose an index attaining S in Equation (9.11), and use $S/s \in [K_0^{-1}, K_0]$. In particular the polynomial coefficient set in

$$P_0(t) = t^i + \sum_{l < i-1} d_l t^l$$

is compact and excludes t^i .

There are three cases, with fixed thresholds $0 < a < b$.

1. If $|\gamma/L_s| < a$, including $\gamma = 0$, the normalized jet order drops to $i - 1$. To prove it, an i -fold zero of P_0 would give $P_0(t) = (t - t_0)^i$. The missing coefficient of t^{i-1} forces $t_0 = 0$, contrary to its nonzero lower coefficient. Compactness in both coefficients and $|t| \leq H$ therefore gives a positive lower bound for $\sum_{j < i} |P_0^{(j)}(t)|$. Choose a and then δ so small that $-(\gamma/L_s)t$ and E preserve half this lower bound. The supremum bound is uniform as well, with scale $|L_s|$.
2. If $|\gamma/L_s| > b$, divide the field by $|\gamma|$. Its state derivative is $-\text{sgn}(\gamma) + O(1/b)$ on the smaller disk. Increasing b gives a first-jet lower bound and a uniform supremum bound. Its order is therefore at most one.
3. If $a \leq |\gamma/L_s| \leq b$, retain order i . The i th derivative of the normalized polynomial is $i!$ and the remainder is small. Moreover $\gamma \neq 0$ and $|L_s| \leq |\gamma|/a$. Since the possible γ form a finite fixed list, the *unnormalized* field in Equation (9.13) has an absolute uniform bound on its larger disk.

The third case can occur at most once before the order decreases. To verify this, carry that absolute disk bound to the next step. If its selected index is less than i , the order has already decreased. If the selected index is i , its Taylor coefficient A_i at the next center is absolutely bounded by Cauchy's inequality. For each of the finitely many new possible nonzero exponents γ' , choose the new cutoff so small that

$$|h' A_i(s')^{i-1}| < |\gamma'|/b.$$

More explicitly, if the inherited bound is M , the centers have disk margin ρ , the new comparison is $s' \leq K'_0 S'$, and γ'_* is the minimum of the finitely many nonzero $|\gamma'|$, it suffices to impose

$$2M\rho^{-i}(K'_0\delta')^{i-1} < \gamma'_*/b.$$

There is no condition of this kind if the nonzero exponent list is empty. This is possible since $i - 1 > 0$. The inner child is then in the second case. If $\gamma' = 0$, it is in the first case. The large- S' and outer children are already terminal nonzero boxes or annuli. Thus a second retention at order i is impossible.

The choices can now be made in the required order. After the full-branch preparations, choose the outer factor H and then the thresholds a, b from the resulting compact coefficient bounds. Next choose the cutoff δ required for the compact inner disks, small remainders and, when present, the inherited absolute bound. Finally choose the lower jet thresholds small enough to give the center displacement and the large- S nonzero estimate. Any subsequent restriction only restricts these prepared identities and introduces no new exponent. Children of smaller order make their own finite choices.

We have proved the following assertion, with a genuine finite recursion: at a selected index i , every child is terminal, has smaller order, or has order i together with an absolute bound which forces its next nonterminal child to have smaller order. The order starts finite and is a nonnegative integer. Every node has finitely many children by preparation and cell decomposition. Consequently the whole tree is finite. Equivalently, label a box by (m, e) , with $e = 0$ when an inherited absolute bound is available and $e = 1$ otherwise. A retention changes $(m, 1)$ to $(m, 0)$; every other continuing child has smaller order. The nonnegative integer $2m + e$ strictly decreases.

9.4 One weighted redo at a nonzero resonance

Only annuli from the initial pole split can have state power $m = 1$. Annuli created in the box induction have power at least two; contact bins have time exponent one. The remaining special case

for Section 10 is therefore

$$Du = A_0 u F(X), \quad X_j = b_j \tau^{a_j} u^{\beta_j}.$$

All pairs (a_j, β_j) are rational and nonconstant. The nonzero generator resonances are the finite set $\alpha = -a_j/\beta_j \neq 0$, for $\beta_j \neq 0$.

Lemma 9.5 (Termination of the resonant redo). *Around these finitely many resonances, apply the rational weighting $U = u/\tau^\alpha$ and repeat the preceding rational subdivision once. The old resonance widths and old generator smallness can be chosen so that every new annulus with state power one and time exponent zero has a slope in its own small-slope regime. In particular no second nonzero resonance redo is required.*

Proof. In the old Euler clock the exact weighted equation is

$$DU = E(p, x, U) = U(A_0 F - \alpha), \quad E_U = A_0 F - \alpha + A_0 \partial_{\log u} F.$$

On a sufficiently narrow fixed neighborhood of $A_0 = \alpha$ and with Equation (9.7) sufficiently strong, $|E_U| < \varepsilon$, for any predetermined $\varepsilon > 0$. The field E is rational in U , with globally subanalytic base coefficients: rational weighting changes only its base coefficients.

First perform the initial preparations of all these weighted rational fields on their unrestricted domains. They depend on the old exact rational field, its uncontacted clock and state shift, and the finite candidate weights. They do not depend on any width around a resonance. This gives a finite list of new initial uncontacted exponent pairs. For every such list fix a positive small-slope threshold c_* . One possible choice, decreased by a fixed factor to allow units near one, is

$$c_* < \min_{a_j \neq 0} \frac{|a_j|}{8(1 + \max_j |\beta_j|)};$$

when the minimum is empty, take any fixed small positive threshold. Then $|c| < c_*$ gives a uniform nonzero sign margin for all slopes $a_j + \beta_j c$ with $a_j \neq 0$ and excludes all nonzero resonances. Take the minimum of these finitely many thresholds, and choose 4ε below it. Choose the original resonance and generator thresholds to ensure $|E_U| < \varepsilon$.

Consider a new initial unscaled annulus. The state change before extracting its monomial is only $r = \sigma(U - c(p, x))$. For a new Euler clock $D' = hD$, Lemma 9.2 gives $|h| \leq 2$, and

$$D'r = h\sigma(E - Dc), \quad \partial_r(D'r) = hE_U.$$

The unbounded derivative of the moving center cancels from the state derivative. If this annulus has $m' = 1$, $\lambda' = 0$, its field is $A'_0 r F'(X')$. Its own subsequent far and generator cuts arrange

$$|F' + \partial_{\log r} F' - 1| < \frac{1}{2}.$$

It follows that $|A'_0| \leq 4\varepsilon < c_*$. The estimate is needed only at the actual points of the restricted piece; the prepared identities themselves were obtained on the unrestricted family. All other children are boxes, box-induction annuli of power at least two, or contact bins of time exponent one. They cannot initiate a nonzero linear resonance redo. The new analytic smallness choices come after the finite exponent thresholds and preserve their sign margins. This proves the asserted termination without a circular choice of resonance widths. $\square$

9.5 Boundary crossings and uniformly finite words

All cuts constructed so far are globally subanalytic in the physical coordinates and polynomial coefficient parameters. The same is true of the additional real magnitude, state, root, and resonance cuts of Section 10: they use rational powers on sign cells, bounded analytic operations, and fixed comparisons. The one-redo lemma makes their recursion finite. Endpoint choices between alternative passage rules need no planar cut by an unbounded logarithm.

Fix the parameters. By a boundary we mean the actual frontier in the physical plane of the relevant open identity pieces. Equalities which hold on a whole open parameter fiber are not thin boundaries in that fiber. Globally subanalytic cell decomposition, applied with the parameters first, separates these possibilities and ensures that the actual boundary fibers have dimension at most one. Include $P = 0$, all exceptional vertical base locations, and all zero state coordinates excluded during changes of variables. Split the boundary fibers into uniformly finitely many points and simple regular analytic arcs. Further split an arc, with regular oriented tangent T , by the signs of $\det(V, T)$ and by the zeros of V . This is another finite uniform subanalytic subdivision.

The transverse-arc assertion below is the no-contact case of the planar Rolle principle for separating trajectories [17, Lemma 1]. We include its Jordan-curve proof; the tangential-arc assertion follows from uniqueness of the flow.

Lemma 9.6 (Crossings of a boundary arc). *A periodic orbit meets each regular boundary arc on which $\det(V, T)$ has a fixed nonzero sign at most once. A connected nonsingular boundary arc on which this determinant vanishes identically is contained in one trajectory and hence in at most one periodic orbit.*

Proof. A nonconstant periodic orbit of a locally unique planar flow, taken with its least period, is an embedded Jordan curve: a repeated point before that period contradicts uniqueness and minimality. Its orientation is the orientation of V . Traversing a fixed oriented transverse arc, consecutive crossings of a Jordan curve alternate between entry into and exit from its bounded complementary component. Their oriented crossing signs therefore alternate. Here the crossing sign is the fixed sign of $\det(V, T)$ (up to one fixed convention), so two consecutive crossings are impossible. Crossings are isolated by transversality; on any compact subarc they are finite. If there were two crossings on the full arc, a compact subarc containing them would already give the contradiction.

On an identically tangential nonsingular arc, V is a continuous nonzero scalar multiple of its regular tangent. Reparametrizing the arc therefore gives a solution of the field locally. Uniqueness makes the whole connected arc part of a single maximal trajectory. Distinct periodic trajectories cannot share it. $\square$

Arcs on which $V = 0$ cannot meet a periodic orbit. The finite point pieces include isolated tangencies, endpoints and singular points of the boundary. An embedded periodic orbit visits each such point at most once. By Lemma 9.6, the number of periodic orbits containing any nonsingular tangential arc is at most the uniform number of these arcs. Remove those finitely many orbits from consideration. Every remaining orbit has a uniform bound on its number of boundary events, obtained by adding the numbers of point and transverse arc pieces. This argument concerns periodic Jordan curves; no bound on the number of crossings of a nonclosed trajectory is used.

Near each event the field is nonsingular. One of its physical components is nonzero, so a sufficiently short piece of the orbit is a regular graph in that physical clock. Choose these neighborhoods disjoint and their endpoints in the adjacent interior pieces. This produces one short graph connector per event. The complementary closed subarcs are compactly contained in their chosen identity pieces; their positive transformed clock endpoints are finite and strictly positive. The connector

neighborhoods can be as small as needed for the particular orbit. Their number is already uniformly bounded.

A nonconstant periodic orbit cannot avoid every graph-clock boundary: its physical coordinate x attains a maximum and a minimum, so $P = 0$ somewhere on it. If x is constant on the entire orbit, its image lies on a line and cannot be a Jordan curve. This also disposes of fibers with no useful nonvertical clock. An orbit contained in the boundary has a subarc in one of the finitely many regular pieces and is already among the tangential exceptions.

Theorem 9.7 (Finite passage templates). *For each degree bound d , there is a finite collection of scalar identity pieces, coordinate charts and passage rules, and integers N_d, E_d , with the following properties.*

1. *On the square, every polynomial field of degree at most d has the stated finite globally subanalytic subdivision. Its interior scalar fields are constant-state fields, nonzero or simple-root boxes of Equation (9.9), or monomial annuli of Equation (9.6). Their analytic factors have uniform larger neighborhoods and their nonconstant arguments have the prescribed fixed smallness. All changes back to physical coordinates use parameter-only data, positive rational powers, and bounded analytic operations.*
2. *The further real passage refinements in Section 10 are finite; in particular Lemma 9.5 allows at most one weighted redo at a nonzero generator resonance.*
3. *Apart from at most E_d periodic orbits containing tangential boundary arcs, every periodic orbit is a cyclic concatenation of at most N_d interior passages and physical graph connectors. There are only finitely many resulting words, including all choices of chart, sign, orientation, and the finitely many endpoint passage rules.*

The choices and bounds include all coefficient degeneracies.

Proof. The initial rational split is finite, and the box induction has finite depth as proved above. Zero numerators, vanishing coefficients and degree changes were separated before normalization; multiple poles were allowed throughout. Thus the first assertion follows without discarding a special coefficient fiber. The real refinements of the terminal cases use finite sign and magnitude alternatives, and their only recursive operation is the redo covered by Lemma 9.5. The preceding boundary argument gives E_d and bounds the number of connectors and interior links. Each link has a label in a fixed finite alphabet, so words of length at most that bound form a finite set.

Any subdivision into a number of normalized time fractions chosen later for one sequence in the absolute-zero argument is an analytic factorization of a given kernel, not an extra geometric link. It does not enter N_d or the alphabet. Finally, the parameter-only data here are fixed for a field and a word; endpoint-dependent amplitudes and normalizations belong to the endpoint graph variables of Section 11. Thus no choice in this construction depends on an unknown orbit cut when its value is declared parameter-only. $\square$

10 Terminal passages and their independent arguments

The real subdivision supplies identities for scalar fields. We now turn these identities into actual transfer functions to which the packet theorems apply. In particular, a normalization involving a proposed endpoint must remain an identity when that endpoint does not match the computed flow.

Theorem 10.1 (Terminal reduction). *Fix the finite preparation data, exponent lists, and geometric tolerances of Theorem 9.7. After finitely many further real subanalytic subdivisions, every terminal*

passage is represented by one scalar transfer equation and regular, locally unique argument graphs. The transfer is an ordinary analytic transfer, an additive transfer of Theorem 6.1, a one-rate transfer of Theorem 7.1, or a mixed transfer of Theorem 8.1.

More precisely, the following statements hold.

1. These are actual real ODE transfers on open independent-argument domains. The domains have finite lists of strict clock, sign, divisor, analytic-box, amplitude, and state-room margins. A bounded limit at which these margins remain positive is an analytic point of the transfer family. At bounded-clock corners, adding the indicated bounded ratios gives analytic normalized transfers also at zero clock limits, with state room.
2. At fixed physical parameters, on the graph of the argument ties near an actual passage, the equation is the physical mismatch in regular transverse coordinates. This identity holds for independently proposed nearby endpoints, before imposing the transfer equation.
3. For an arbitrary sequence of arguments and any fixed finite collection of independent-argument derivatives, the transfers and those derivatives reduce to the preceding packet primitives and bounded analytic operations. This assertion includes sequences on which any of the imposed strict margins tend to zero; it does not assume the positive-margin condition in the first assertion. When all enlarged normalized arguments stay bounded, the kernel factors have analytic extensions with state room after the finite graph additions, and algebraic divisors can be cleared. If an added argument diverges, its recoverable coordinate is included in the packet test. The reduction may introduce bounded ratios and a fixed finite internal subdivision chosen for that sequence. The old derivatives are obtained by the chain rule for exact transfer identities. The internal state and finite jet graphs are locally unique and preserve isolated solutions. The coefficient determinations are those fixed by the primitive theorems and are preserved under their allowed retests.

The number of geometric passage types and boundary crossings is finite. Internal subdivisions in the last assertion do not enter that number. Geometric smallness is fixed before a limiting slope or any asymptotic or derivative request is selected.

The reductions are organized by the scalar identity from Section 9. Nozero boxes use their integrated displacement bound. Simple-root boxes first linearize the state: the zero time exponent gives the two clocks L, W , while a nonzero time exponent leads to an additive action. For annuli, the pair $(m - 1, \lambda)$ determines the initial power or logarithmic change. The final annular case $(m, \lambda) = (1, 0)$ is split by slope and generator resonance. In every case the ensuing clock limits select the ordinary, one-rate, additive, or mixed transfer.

10.1 Conventions and two elementary analytic facts

All the analytic functions below extend to slightly larger complex boxes than the boxes used for their arguments and trajectories. Ordinary parameters are suppressed from formulas; identities in profile variables hold for all these parameters. At each physical passage the field and preparation parameters are held fixed while time or state is differentiated. Their derivatives as independent arguments of a transfer are retained later. Once an independent-argument kernel has been chosen, its field and all its exogenous arguments are retained when a limiting regime is rewritten. An alternative formula that agrees only after physical ties are imposed is used as a new physical encoding, not as an identity of the old ambient kernel. Old ambient derivatives always use an identity on an open set of the old arguments. Smallness means a sufficiently small fixed geometric constant relative to the

finite exponent lists and these larger analytic boxes. It never means a quantity to be chosen after an expansion order.

Order the positive clock endpoints as $t_- < t_+$ and put $L = \log(t_+/t_-) > 0$. A signed monomial bt^r , $r \neq 0$, is exactly

$$bt^r = \begin{cases} (bt_-^r)e^{-|r|Ls}, & r < 0, \\ (bt_+^r)e^{-|r|L(1-s)}, & r > 0, \end{cases} \quad t = t_-e^{Ls}, \quad 0 \leq s \leq 1. \quad (10.1)$$

The amplitude is taken at the end where its absolute value is largest. Zero amplitudes are allowed and are not subjected to root extraction. Reversing the traversal interchanges the two ends. The orientation used for a transfer need not be positive physical time.

Lemma 10.2 (Rational diagonal primitives). *Let $D_r = \sum_j r_j X_j \partial_{X_j}$, with fixed rational r_j , and let $a(X)$ be analytic near the origin, uniformly in ordinary parameters. For rational λ , write $a = \sum_\nu a_\nu X^\nu$. Then*

$$a = (\lambda + D_r)H + R, \quad H = \sum_{\lambda+r \cdot \nu \neq 0} \frac{a_\nu}{\lambda + r \cdot \nu} X^\nu, \quad R = \sum_{\lambda+r \cdot \nu = 0} a_\nu X^\nu \quad (10.2)$$

is analytic on a smaller common box and $D_r R = -\lambda R$. For $\lambda = 0$, the decomposition is $a = \bar{a} + D_r H$, with $D_r \bar{a} = 0$ and $H(0) = 0$.

Proof. A common denominator for the finitely many rational numbers shows that the nonzero denominators in (10.2) have absolute value at least a fixed positive number. Thus both subseries converge absolutely on every smaller polydisk, with the same property after any fixed number of ordinary or profile derivatives. Termwise differentiation proves the identities. If $X_j = b_j t^{r_j}$, a weight-zero monomial satisfies $X^\nu = b^\nu$; hence $\bar{a}$ is constant for every nearby value of t , at fixed coefficients, not just along a solution of an ODE. $\square$

We will also use this simple real estimate. If a profile X has fixed sign, $|X| \leq \eta$, and its logarithmic slope in an oriented variable t has fixed sign and absolute value at least $a > 0$, then

$$\int |X(t)| dt \leq \eta/a. \quad (10.3)$$

Indeed $|(d/dt)|X| \geq a|X|$, and integration bounds the left side by the total variation of $|X|/a$. The assertion includes the identically zero profile by continuity. For an analytic field vanishing when specified profiles vanish, its norm is bounded by a constant times the sum of their absolute values on smaller boxes. Thus (10.3) controls both its integrated forcing and its integrated state Lipschitz constant. The integral equation, or Gronwall's inequality, then keeps a smaller input disk in the larger state disk. Stable linear terms improve this estimate.

If the subdivision field is identically zero, its transfer is simply $z_{\text{out}} = z_{\text{in}}$. This is an ordinary analytic kernel for every clock length, with its physical endpoint coordinate graphs retained.

10.2 Nozero boxes

The prepared field is

$$Dz = B\tau^\lambda f(X, z), \quad D = \frac{d}{d \log \tau}, \quad X_j = b_j \tau^{r_j}, \quad B > 0, \quad (10.4)$$

where $r_j \neq 0$ are rational, X is tiny, and constant arguments are ordinary parameters. In a nonzero box, $|f| \geq m_f > 0$ with fixed sign. Use small state bins of width h . Monotonicity on a passage contained in one bin gives

$$B \int \tau^\lambda |d \log \tau| \leq h/m_f. \quad (10.5)$$

Choose h sufficiently small against the already fixed analytic bounds.

For $\lambda = 0$, put $b = BL$. The normalized physical equation is $z_s = bf(X, z)$, with b small by (10.5). For a large L , choose the following independent-argument family:

$$z_s = bf(0, z) + Lb'(f(X, z) - f(0, z)), \quad b' = B, \quad b = Lb' \quad \text{on the physical graph.} \quad (10.6)$$

The small base flow $bf(0, z)$ is regular with room, and the perturbation vanishes at $X = 0$ for every independent b, b' . Formula (10.1) therefore gives the regular one-rate model with rate L . On the long-length rule, $b' = b/L$ is bounded and small. If L is bounded, use $z_s = bf(X, z)$ directly; the layer exponentials are bounded analytic functions of (L, s) , including at $L = 0$. This is the physical short rule. At a bounded- L corner of the already chosen independent family (10.6), retain that same displayed field with independent b, b' ; it too is an ordinary normalized analytic field, and hence preserves its original ambient derivatives.

For $\lambda \neq 0$, let τ_e be the end where τ^λ is largest and set $b_0 = B\tau_e^\lambda$. Exactly,

$$B \int \tau^\lambda |d \log \tau| = \frac{b_0}{|\lambda|} (1 - e^{-|\lambda|L}). \quad (10.7)$$

Thus $L \geq 1$ makes b_0 small. The coefficient $B\tau^\lambda$ is itself a nonconstant endpoint layer, so the normalized field has zero base and the form $LW(E, H, z)$, with $W(0, 0, z) = 0$. It is a regular one-rate transfer when L grows. For a bounded short interval use the small bounded prefactor $b_1 = LB\tau_e^\lambda$ and the ordinary equation $z_s = b_1 e^{-|\lambda|Ls_e} f(X, z)$, where $s_e = s$ or $1 - s$. All formulas extend at $L = 0$. The short and long rules may overlap, for example on $1/2 < L < 2$; they are endpoint choices, not cuts defined by an unbounded logarithm in the physical plane.

10.3 Simple boxes

Now $f(X, z)$ in (10.4) has a regular analytic simple root $z = c(X)$; the physical state stays sufficiently close to this root. Write $b(X) = f_z(X, c(X))$, which is nonzero with fixed sign on a root chart. Set

$$\psi(X, z) = (z - c(X)) \exp \left(\int_{c(X)}^z \left\{ \frac{b(X)}{f(X, \zeta)} - \frac{1}{\zeta - c(X)} \right\} d\zeta \right). \quad (10.8)$$

The apparent pole in the integrand is removable. The formula defines an analytic coordinate, $\psi_z(X, c) = 1$, and direct differentiation gives $\psi_z f = b\psi$. With $v = \psi(X, z)$,

$$Dv = B\tau^\lambda b(X)v + g(X, v), \quad g(0, v) = 0. \quad (10.9)$$

Here g is $D_X \psi$ expressed in (X, v) , with $D_X = \sum r_j X_j \partial_{X_j}$; the ordinary parameters are fixed. Its vanishing is an analytic identity on the entire independent profile box. We always traverse the linear term in its damped direction.

10.3.1 Zero time exponent

For $\lambda = 0$, apply Lemma 10.2 to obtain $b = \bar{b} + D_X H$, $D_X \bar{b} = 0$, $H(0) = 0$. The invariant $\bar{b}$ has the sign of $b(0)$. Define

$$\rho = \tau \exp(H/\bar{b}), \quad Y_j = X_j \exp(r_j H/\bar{b}) = b_j \rho^{r_j}. \quad (10.10)$$

The profile map has identity Jacobian at zero, hence an analytic inverse on smaller boxes, and $D \log \rho = b/\bar{b} > 0$. In the damped oriented logarithmic variable $t \in [0, L]$ of ρ , the exact equation is

$$v_t = -cv + \tilde{g}(Y, v), \quad c = B|\bar{b}|, \quad W = cL, \quad \tilde{g}(0, v) = 0. \quad (10.11)$$

The profiles are endpoint layers of fixed positive rational rates. The normalized equation, useful also when either clock vanishes, is $v_s = -Wv + L\tilde{g}(Y, v)$. The complete clock alternatives are

Clock behavior along the test sequence	Exact model used
$L, W \rightarrow \infty$, with $W/L \rightarrow 0$, a positive finite limit, or ∞	Mixed model (I), with the auxiliary ratio prescribed by Theorem 8.1 in the unbalanced cases.
$W \rightarrow \infty$, L bounded, including $L \rightarrow 0$	Stable one-rate model with rate W ; all profiles are slow analytic functions of (L, s) .
$L \rightarrow \infty$, W bounded, including $W \rightarrow 0$	Regular one-rate model with rate L , base $-Wv$.
Both clocks bounded, including zero limits	Ordinary normalized analytic transfer.

For a finite but large limiting bounded clock, subdivide the normalized interval into a fixed number of fractions so its length per fraction is small. This gives bounded base-flow inverses and fixed analytic room. Exact endpoint shifts regenerate the profiles at internal ends; damping and (10.3) keep the real states in the smaller disk. The rigorous status of this internal operation is specified below.

10.3.2 Nonzero time exponent and the additive action

For $\lambda \neq 0$, diagonal inversion gives

$$b = (\lambda + D_X)H + R, \quad D_X R = -\lambda R, \quad H_0 = H(0) = b(0)/\lambda \neq 0, \quad R(0) = 0. \quad (10.12)$$

Solve

$$H = H_0 h^\lambda + R \log h \quad (10.13)$$

for h near 1. The derivative in h at the origin is $\lambda H_0 = b(0) \neq 0$, so this is a bounded analytic operation. Put

$$\begin{aligned} I = B\tau^\lambda R, \quad w = |BH_0|(\tau h)^\lambda, \quad \sigma = \text{sgn}(BH_0), \quad \gamma = \sigma I/\lambda, \\ Y_j = X_j h^{r_j}, \quad \frac{\gamma}{w} = \frac{R}{b(0)h^\lambda}. \end{aligned} \quad (10.14)$$

Here $DI = 0$, Y_j is a constant multiple of $w^{r_j/\lambda}$, and $X \mapsto Y$ is an analytic change tangent to the identity. The exact primitive identity is

$$\begin{aligned} B\tau^\lambda H + I \log \tau &= BH_0(\tau h)^\lambda + I \log(\tau h) \\ &= \sigma(w + \gamma \log w) - \frac{I}{\lambda} \log |BH_0|. \end{aligned} \quad (10.15)$$

The last term is constant in time. Also

$$D \log w = \frac{\lambda b(X)}{b(0)h^\lambda + R}, \quad (10.16)$$

a nonzero analytic factor near λ . Differentiating (10.15) therefore turns (10.9) into

$$v_w = \sigma(1 + \gamma/w)v + w^{-1}\widehat{g}(Y, v), \quad \widehat{g}(0, v) = 0. \quad (10.17)$$

Thus no sign or factor of λ is absorbed in an asymptotic replacement.

On the large-action part, choose q_* sufficiently large once for these finite formulas and use

$$\begin{aligned} q &= \min w \geq q_*, \quad p = \max w, \quad \delta = \gamma/q, \\ A(w) &= w + \gamma \log(w/q), \quad Q = A(p), \quad S = Q - q > 0. \end{aligned} \quad (10.18)$$

The tiny bound on γ/w makes A' positive and bounded away from zero. In particular $p \asymp Q$ and $Q/p = 1 + O(\delta)$. Negative profile powers are anchored at q ; positive powers may be anchored at Q , with their exact factor $(w/Q)^r$. Their amplitudes are still tiny because Q/p is close to 1. Include γ/w as a negative profile. Then (10.17) is exactly additive model (A), in its damped direction.

If $Q/q \rightarrow \infty$, apply Theorem 6.1. For the complementary bounded-ratio case set

$$M = S/q, \quad s = (A - q)/S, \quad y = w/q, \quad y + \delta \log y = 1 + Ms. \quad (10.19)$$

Its implicit derivative is $1 + \delta/y$, a unit on the real interval. The normalized forward equation is exactly

$$v_s = \sigma S v + \frac{M}{y + \delta} \widehat{g}(Y, v). \quad (10.20)$$

Reverse s if required for damping. The profiles are analytic powers of y and $y/(1 + M)$. For bounded M , this is a stable one-rate model if $S \rightarrow \infty$ and an ordinary transfer if S is bounded, including $M = 0$ or $S = 0$ in its normalized extension. A large finite limit of M is treated by the fractions in (10.32) below.

For the remaining bounded range of w , $B\tau^\lambda$ is bounded because h and H_0 have fixed unit bounds. Cut using a tiny positive threshold for the subanalytic quantity $B\tau^\lambda$. Above it, the logarithmic length is uniformly bounded: both endpoint values of $B\tau^\lambda$ lie between two fixed positive constants, and their log ratio equals λ times the clock length. This gives an ordinary transfer. Below it, $B\tau^\lambda$ is an additional tiny nonconstant layer. Both terms of (10.9) then vanish without layers; use a regular one-rate transfer for large log length and the ordinary normalized transfer otherwise. These cuts involve no unbounded logarithm.

10.4 Annuli with at least one nonzero power exponent

The annular identity is

$$Du = A_0 \tau^\lambda u^m F(X), \quad X_j = \chi_j \tau^{a_j} u^{\beta_j}, \quad u > 0, \quad (10.21)$$

where $A_0 \neq 0$ is parameter-only, m is integral, all exponents are rational, and $(a_j, \beta_j) \neq (0, 0)$. The profiles are tiny and $F(0) = 1$, so $F > 0$ and $\partial_{\log u} F$ is small. Pure multipliers in the following changes remain parameter-only until an endpoint normalization is explicitly made.

10.4.1 Two nonzero power exponents

Suppose $m \neq 1$ and $\lambda \neq 0$. Set

$$U = u^{1-m}, \quad T = \tau^\lambda, \quad \frac{dU}{dT} = kF, \quad k = \frac{(1-m)A_0}{\lambda}. \quad (10.22)$$

Split by fixed tiny and huge thresholds of $|k|T/U$. In the tiny range choose U_* at $T_{\max}$. Since F is bounded,

$$|U(T) - U_*| \leq C|k|T_{\max}, \quad |k|T_{\max}/U_* \ll 1.$$

Thus $V = U/U_*$ is close to 1. Its field in $\log T$ is

$$V_{\log T} = (kT/U_*)F.$$

The first factor is a tiny endpoint layer. Every original profile is an end-normalized rational power of T , times an analytic power of V ; a zero time power is a tiny ordinary argument. Use the zero-base regular one-rate model or the bounded-length ordinary model. The defining conjugacy holds for every nearby positive choice of U_* .

In the huge range invert (10.22): $dT/dU = k^{-1}F^{-1}$. The clock U is transverse since $kF \neq 0$, and the new ratio $|k|^{-1}U/T$ is tiny, reducing to the previous case. In the comparable range put $z = U/(|k|T)$. It stays in a fixed compact positive interval, and

$$z_{\log T} = \text{sgn}(k)F - z. \quad (10.23)$$

The profiles become pure time powers times analytic state powers. The state derivative of the right side is $-1 + o(1)$, uniformly on this compact interval after the already permitted generator smallness choice. Choose a fixed small threshold for the right side. Away from zero this is a nonzero box; near zero the nonzero derivative gives a unique simple-root chart with room. Both were treated above.

10.4.2 Exactly one zero power exponent

If $m = 1, \lambda \neq 0$, use

$$v = \log u, \quad w_0 = |A_0/\lambda|\tau^\lambda, \quad \kappa = \text{sgn}(A_0/\lambda).$$

If $m \neq 1, \lambda = 0$, invert the transverse field and use

$$v = \log \tau, \quad w_0 = u^{1-m}/|(1-m)A_0|, \quad \kappa = \text{sgn}((1-m)A_0),$$

replacing F by F^{-1} . In either case the exact equation is

$$v_{w_0} = \kappa F(X), \quad X_i = \tilde{\chi}_i w_0^{p_i} e^{d_i v}, \quad \kappa \in \{-1, 1\}, \quad (10.24)$$

with fixed rational $(p_i, d_i) \neq (0, 0)$. Absolute v is not an argument of a restricted analytic function; the underlying endpoint variable is the positive e^v .

For bounded w_0 , take finitely many small absolute-width bins. Then $v - v_-$ is small, and in $\log w_0$ its derivative is $\kappa w_0 F$. Write each profile using its large time-power end, times $e^{d_i(v-v_-)}$. The amplitudes are tiny, since the state displacement is small and the original endpoint profiles are tiny. Near $w_0 = 0$, the coefficient w_0 is an extra tiny layer; away from zero the logarithmic length is bounded. These give respectively the regular one-rate and ordinary rules, with the bounded-length

overlap. For the ordinary rule keep an explicit integrated-coefficient guard in the independent family. Let $w_e = w_{0,+}$ and $L = \log(w_{0,+}/w_{0,-})$. The exact normalized field is

$$\begin{aligned} z_s &= \kappa L w_e e^{-L(1-s)} F, & z &= v - v_-, \\ \int_0^1 L w_e e^{-L(1-s)} ds &= w_e (1 - e^{-L}) = w_{0,+} - w_{0,-}. \end{aligned} \quad (10.25)$$

Impose $w_e(1 - e^{-L}) < h$, with h chosen so that the forcing and state Lipschitz bounds Ch , even on the closed range, keep solutions strictly inside the larger state disk. This is an analytic inequality in the old independent (L, w_e) and their affine exponential, so it preserves the original kernel and its ambient derivatives. The physical absolute-width bins satisfy this guard. All other profiles retain their independent endpoint amplitudes and the analytic state factor $e^{d_i(v-v_-)}$. Bounded length alone would not supply this estimate. Near zero the tiny coefficient-layer bound already controls the integral by w_e , independently of the logarithmic length.

For large w_0 , separate the pure profiles X^0 , those with $d_i = 0$, and set $F^0 = F(X^0, 0)$. Apply Lemma 10.2 with weight 1:

$$F^0 = (1 + D_p)H + R, \quad D_p R = -R, \quad H(0) = 1, \quad H = h + R \log h, \quad D_p = \sum_{d_i=0} p_i X_i \partial_{X_i}. \quad (10.26)$$

Set $w = w_0 h$, $\gamma = w_0 R$, and $Y_i = X_i^0 h^{p_i}$. The profile change is regular, γ is constant on the time line, and

$$\int F^0 dw_0 = w + \gamma \log w + \text{constant}, \quad \frac{dw}{dw_0} = \frac{F^0}{1 + \gamma/w}. \quad (10.27)$$

Use q, p, δ, A, Q, S from (10.18), choosing the cut high enough that $q \geq q_*$. From the smaller-action end put

$$z = v - v_- - \kappa(A(w) - q), \quad z_w = \kappa \frac{dw_0}{dw} (F - F^0), \quad z(q) = 0. \quad (10.28)$$

For a nonpure nonzero profile, its logarithmic slope in w_0 is

$$\frac{p_i}{w_0} + d_i \kappa F.$$

It has the sign of $d_i \kappa$ and absolute value bounded below by a fixed multiple of $|d_i|$ once the cutoff is large and the generators are small. Equation (10.3) therefore bounds the total residual displacement by a fixed constant times the amplitudes.

For $d_i \kappa < 0$ and $d_i \kappa > 0$, respectively, define

$$\begin{aligned} e_i &= \tilde{\chi}_i q^{p_i} e^{d_i v_-}, & E_i &= e_i (w/q)^{p_i} e^{-|d_i|(A-q)}, \\ D_i &= \tilde{\chi}_i Q^{p_i} e^{d_i v_+ - d_i z_+}, & H_i &= D_i (w/Q)^{p_i} e^{-|d_i|(Q-A)}. \end{aligned} \quad (10.29)$$

In the second line tie the proposed output residual by

$$e^{v_+} = e^{v_-} \exp(\kappa S + z_+). \quad (10.30)$$

Then the original profile is exactly $E_i h(Y)^{-p_i} e^{d_i z}$ or $H_i h(Y)^{-p_i} e^{d_i z}$. In particular the residual field is analytic in pure profiles, the fast profiles, and z , and vanishes when all fast profiles vanish. The rates $|d_i|$ belong to one positive rational lattice. This is additive model (B), with tiny right amplitudes because Q/p and h are bounded units and z_+ is tiny near the physical passage.

For $Q/q \rightarrow \infty$, use its additive packets. For bounded $M = S/q$, the action variable in (10.19) gives instead

$$z_s = S \frac{y}{y + \delta} G(X, E, H, z), \quad E_i = \widehat{e}_i(s) e^{-|d_i| S s}, \quad H_i = \widehat{D}_i(s) e^{-|d_i| S (1-s)}, \quad (10.31)$$

where the hatted amplitudes are the original amplitudes times analytic slow powers of y and $y/(1 + M)$. This is the zero-base regular one-rate model if $S \rightarrow \infty$, or an ordinary transfer if S is bounded. Both bounded-ratio models, (A) and (B), allow $M \rightarrow 0$.

Here are explicit fractions when the finite limit of M is large. Write $A = q + S(j + t)/K$, $0 \leq t \leq 1$, and let y_0 be the value of y at j/K . Put $y = y_0 R$. Subtracting the equation for y_0 from (10.19) gives

$$R + \frac{\delta}{y_0} \log R = 1 + \frac{M}{K y_0} t, \quad y_0 \geq 1. \quad (10.32)$$

Choose one finite K making M/K small on a neighborhood of the chosen finite limit. Then R and every fixed power of R have fixed analytic room near 1. An internal left amplitude is, exactly,

$$e_i^{(j)} = e_i(w_j/q)^{p_i} e^{-|d_i| j S/K},$$

and there is the analogous right formula. Adverse powers are paid by the exponential: the logarithmic derivative in A of the absolute left factor is

$$\frac{p_i}{w + \gamma} - |d_i| \leq \frac{|p_i|}{q_*(1 - |\delta|)} - |d_i| < 0$$

after increasing the fixed q_* . The reversed estimate handles the right factor. Thus internal amplitudes remain tiny, and pure powers are anchored at their local large ends. The total real forcing estimate is unchanged. Every new argument uses bounded regular solves, fixed rational powers, and exponentials of fixed fractions of S .

10.5 The logarithmic annulus and its small-slope boundary

It remains to treat $m = 1, \lambda = 0$. Set

$$\theta = \log \tau, \quad v = \log u, \quad v_\theta = A_0 F(X), \quad X_i = \chi_i e^{a_i \theta + \beta_i v}. \quad (10.33)$$

For huge $|A_0|$, exchange the two transverse logarithmic variables: $d\theta/dv = A_0^{-1} F^{-1}$. This interchanges the rational weights and enters the small-coefficient case below. Hence it suffices to consider bounded $|A_0|$.

10.5.1 Slopes separated from zero and resonance

Exclude a fixed neighborhood of zero and of every nonzero number $\alpha = -a_i/\beta_i$ with $\beta_i \neq 0$. On each resulting coefficient interval, $|a_i + \beta_i A_0|$ has a fixed positive lower bound. Since $F - 1$ is tiny, the physical logarithmic slopes $a_i + \beta_i A_0 F$ have the same signs and fixed lower bounds. It follows that all profiles have small integral in θ . Put $\kappa = \text{sgn } A_0$, $c = |A_0|$, $L = \theta_+ - \theta_- > 0$, $W = cL$, and

$$z(t) = v(\theta_- + t) - v_- - \kappa c t, \quad 0 \leq t \leq L.$$

Its field is $z_t = A_0(F - 1)$, and z stays small. Write $\rho_i = a_i + \beta_i \kappa c$, with its fixed nonzero sign. For $\rho_i < 0$ use a left amplitude; for $\rho_i > 0$ use a right amplitude. The resulting positive rate is $|\rho_i|$,

an affine rational function of the exact occurrence $c = W/L$. The factor in state is $e^{\beta_i z}$, and the endpoint residual tie is

$$e^{v+} = e^{v-} \exp(\kappa W + z_+). \quad (10.34)$$

The same cancellation as in (10.29) gives mixed model (II). Bounded occurrences of A_0 in its field may be independent ordinary parameters, tied separately. If L grows, then W grows and c stays in a compact positive interval; if L is bounded, so is W , and the normalized transfer is ordinary, including zero length. In this independent kernel domain impose the selected bounded slope interval on the exact W/L , with slightly relaxed endpoints still separated from zero and the resonances. These are linear inequalities between W and L , in addition to the fixed positive rate margins. An independent ordinary copy of A_0 does not replace these inequalities.

10.5.2 Straightening the pure-state field

For sufficiently small $|A_0| > 0$, let X^s denote the profiles with $a_i = 0$ and set $F_s = F(X^s, 0)$. The pure-state weights β_i here are nonzero. Apply diagonal inversion to the reciprocal:

$$F_s^{-1} = b + D_\beta H, \quad D_\beta b = 0, \quad b(0) = 1, \quad H(0) = 0, \quad D_\beta = \sum_{a_i=0} \beta_i X_i \partial_{X_i}. \quad (10.35)$$

The following invariance is needed for mismatched endpoint values. If $\beta \cdot \nu = 0$, then for every nearby v , at fixed coefficients,

$$(X^s(v))^\nu = \chi^\nu e^{(\beta \cdot \nu)v} = \chi^\nu. \quad (10.36)$$

Consequently $b(X^s(v))$ is independent of both v and θ on the physical profile surface. One can compute it at either proposed endpoint without imposing a redundant equality between two evaluations. This does not make its derivatives in independent coefficients or profiles zero.

Define

$$\tilde{v} = v + H/b, \quad Y_i = X_i \exp(\beta_i H/b) = \chi_i e^{a_i \theta + \beta_i \tilde{v}}. \quad (10.37)$$

The pure-state part of the profile map has identity Jacobian at zero; invert it analytically first and then invert the remaining profiles by their unit factors. Direct differentiation, using (10.35), gives

$$\tilde{v}_v = 1 + \frac{D_\beta H}{b} = \frac{1}{b F_s}, \quad \frac{d\tilde{v}}{d\theta} = \frac{A_0}{b} \frac{F}{F_s}. \quad (10.38)$$

Thus this is an actual state diffeomorphism. In the positive coordinate $R = e^{\tilde{v}} = u \exp(H/b)$, its derivative is

$$R_u = \frac{\exp(H/b)}{b F_s} > 0.$$

Let $p = A_0/b = \kappa c$, where $c > 0$, and subtract the exact slope:

$$z = \tilde{v} - \tilde{v}_- - \kappa c t, \quad z_t = p \left(\frac{F}{F_s} - 1 \right). \quad (10.39)$$

Under the analytic inverse profile change, $F/F_s - 1$ vanishes identically when all profiles with $a_i \neq 0$ vanish. Hence every nonzero residual Taylor monomial contains such a profile.

Here the small-slope threshold can be fixed from the finite weight list. For example, when there are nonzero a_i , take

$$c_* < \frac{\min_{a_i \neq 0} |a_i|}{4(1 + \max_i |\beta_i|)}.$$

By first bounding b, F, F_s near 1 and choosing $|A_0|$ small enough, both the straight-line rates and the physical logarithmic slopes of these profiles have the sign of a_i and magnitude at least $|a_i|/2$. If no such profile exists, the residual is identically zero and no threshold is needed. Pure-state layers have positive rates $|\beta_i|c$, with their end selected by the sign of $\beta_i\kappa$. The residual has small total integral since every monomial contains a uniformly fast profile; the other profiles stay tiny. More explicitly, on a smaller fixed box the analytic vanishing gives

$$|G| + |G_z| \leq C \sum_{a_i \neq 0} |Y_i|, \quad \int_0^L (|G| + |G_z|) dt \leq \frac{CN\eta}{a_*},$$

where $G = p(F/F_s - 1)$, the N designated fast profiles have amplitudes at most η , and their rates are at least a fixed $a_* > 0$. The bounds are uniform in the remaining tiny profiles and the bounded coefficient p . Thus η can be fixed using only analytic room and the finite fast-rate margin. No integral of an isolated pure-state layer of rate $|\beta_i|c$ is used in this bound. For $L, W = cL \rightarrow \infty$, this is mixed model (II), including $c \rightarrow 0$. A positive but arbitrarily small limiting c requires no new geometric smallness: the designated fast factor has a uniform positive rate, exactly the uniformity clause of Theorem 8.1. The independent domain imposes $0 < W < c_*L$ itself, as well as the bounded range for the independent coefficient p . Thus the exact slope stays in the selected small-slope regime even away from the physical tie $pL = \kappa W$.

For completeness, when $L \rightarrow \infty$ and W is bounded, put $t = Ls$. For $a_i < 0$ the left layer factors exactly as

$$e^{(a_i + \beta_i\kappa c)Ls} = e^{-|a_i|Ls} e^{\beta_i\kappa Ws}, \quad (10.40)$$

whereas for $a_i > 0$ the right layer factors as

$$e^{-(a_i + \beta_i\kappa c)L(1-s)} = e^{-|a_i|L(1-s)} e^{-\beta_i\kappa W(1-s)}. \quad (10.41)$$

These slow factors may grow or decay: their signs are retained. Pure-state layers are $e^{-|\beta_i|Ws}$ or $e^{-|\beta_i|W(1-s)}$, also slow. They are ordinary bounded analytic arguments at a finite W limit. The field $Lp(F/F_s - 1)$ has a uniformly fast layer in each monomial, so it is a zero-base regular one-rate field of rate L , with p an independent bounded coefficient. If necessary, choose fixed fractions making the bounded W -length per fraction small; regenerate amplitudes with the full affine shift $a_iL + \beta_i\kappa W$. This avoids enlarging a tiny fast amplitude by a large finite slow exponential. If L is bounded, W is bounded as well, and the normalized formula is ordinary, including $(L, W) = (0, 0)$. Thus the small-slope alternatives are exhaustive:

Clock behavior	Model
$L, W \rightarrow \infty, 0 \leq \lim W/L \leq c_*$	Mixed (II); at zero slope every residual monomial has a fast factor.
$L \rightarrow \infty$, bounded W , including $W \rightarrow 0$	Regular one-rate, using the signed factorizations above.
Bounded L (hence bounded W), including zero limits	Ordinary normalized transfer.

The residual tie is $R_+ = R_- \exp(\kappa W + z_+)$. It has the same right-amplitude cancellation as (10.34).

10.5.3 Nonzero resonances

For each nonzero rational generator resonance $\alpha = -a_i/\beta_i$, weight the physical state before solving the passage:

$$U = u/\tau^\alpha, \quad DU = U(A_0F - \alpha). \quad (10.42)$$

This is still an exact rational field in U with subanalytic base coefficients. Its state derivative is

$$\partial_U(DU) = A_0 F - \alpha + A_0 \partial_{\log u} F. \quad (10.43)$$

It is as small as prescribed by the resonance width and generator smallness. Apply the initial rational subdivision again. Lemma 9.5 proves that this redo terminates: the finite candidate exponent lists are prepared before choosing those widths; only a new initial unscaled annulus with powers $(m', \lambda') = (1, 0)$ could encounter this issue again, and its leading slope is bounded by a fixed multiple of (10.43). Choose the original widths to place it below the small-slope threshold for every precomputed list. The new formula's own generator cuts then put its units near 1. Contact bins have $\lambda' = 1$, and outer annuli in the box induction have $m' \geq 2$, so neither can restart a nonzero-resonance redo. The huge slope swap already enters the small case and is not used in this redo.

10.6 Endpoint ties, extensions, and differentiated identities

Lemma 10.3 (Identity on the nearby tied endpoint graph). *Every transfer constructed above computes the actual scalar mismatch for all nearby proposed endpoints on its argument graph. The terminal state coordinate has nonzero transverse derivative. Endpoint-dependent normalizations and the pure-state invariant do not destroy this assertion.*

Proof. All time and profile primitives above are identities of analytic functions on the prepared physical piece. Invariants in a time change are constant for every nearby time, by the weight-zero identity in Lemma 10.2. The pure-state invariant is constant for every nearby state by (10.36). The linearizer, power maps, and profile inversions have nonzero Jacobian on their specified domains. Clock swaps use a nonzero scalar velocity.

The only apparent dependence of the computed flow on a proposed final state occurs in right amplitudes. For the additive annulus, the tie (10.30) is equivalent, on its positive real domain, to $z_+ = v_+ - v_- - \kappa S$. If $d_i \kappa = |d_i|$, it gives

$$D_i = \tilde{\chi}_i Q^{p_i} \exp(d_i v_- + |d_i| S). \quad (10.44)$$

This expression is independent of the proposed v_+ . Substitution in (10.29), at every intermediate state z , gives

$$H_i h^{-p_i} e^{d_i z} = \tilde{\chi}_i w_0^{p_i} \exp(d_i [v_- + \kappa(A - q) + z]),$$

exactly the original profile. The left identity is immediate. Using Q , rather than p , in the amplitude is exact because the layer contains $(w/Q)^{p_i}$.

For a mixed right layer with positive $\rho_i = a_i + \beta_i \kappa c$, use

$$D_i = \chi_i \exp(a_i \theta_+ + \beta_i v_+ - \beta_i z_+).$$

The tie $z_+ = v_+ - v_- - \kappa W$ gives

$$D_i = \chi_i \exp(a_i \theta_+ + \beta_i v_- + \beta_i \kappa W).$$

Multiplication by $e^{-\rho_i(L-t)} e^{\beta_i z}$ recovers $\chi_i e^{a_i(\theta_- + t) + \beta_i(v_- + \kappa c t + z)}$. The calculation uses no sign restriction on β_i or κ . For the small-slope case replace v by $\tilde{v}$, which is regular by (10.38); invariance makes its $p = A_0/b$ independent of the proposed other endpoint. Thus the flow side is independent of that proposed endpoint before it is equated with z_+ .

For clarity, an endpoint normalization such as U_* obeys a slightly different but equally exact rule. Let C_a be its regular coordinate map, with a held temporarily independent. The transfer is a conjugacy for every nearby fixed a , so its mismatch is $C_a(U_{\text{flow}}) - C_a(U_+)$. At equality,

$$d\{C_a(U_{\text{flow}}) - C_a(U_+)\} = C'_a(U_+) d(U_{\text{flow}} - U_+).$$

The two a -variations cancel even when a is then chosen from a proposed endpoint. For a clock swap, the nonzero clock velocity likewise determines the hitting-time variation; the remaining state derivative is transverse and nonzero. This proves the assertion. $\square$

Example 10.4 (An exact check of the right-amplitude correction). For $v_w = 1 + \chi e^v$, with no pure profiles and $S = p - q$, the residual equation is $z' = D e^{-(p-w)} e^z$, where $D = \chi e^{v_+ - z_+} = \chi e^{v_- + S}$. Starting with $z(q) = 0$,

$$z(p) = -\log(1 - \chi e^{v_-} (e^S - 1)).$$

Hence the derivative of the mismatch $z(p) - z_+$ in the proposed v_+ is exactly -1 . The bounded correction e^{-z_+} in the right amplitude is necessary for this identity.

Lemma 10.5 (Structural extensions and the physical chain rule). *The terminal kernels can be chosen on independent-argument neighborhoods so that every layer-vanishing identity required by its primitive theorem holds throughout that neighborhood. On their physical graph they give the stated transfer identity. Every fixed finite derivative of the physical identity, or of an old kernel rewritten in a new regime, is a finite expression in derivatives of these kernels, bounded analytic operations, fixed affine exponential shifts, and algebraic factors with nonzero divisors on the domain.*

Proof. Use the displayed structural formulas, not arbitrary extensions of a quantity which vanishes only on the physical graph. For nonzero boxes the choice is (10.6). If its transfer is $K(L, b, b', \eta)$, where η denotes independent amplitudes and other arguments, the physical identity at fixed η is

$$T(L, B, \eta) = K(L, BL, B, \eta), \quad T_B = LK_b + K_{b'}, \quad T_L = K_L + BK_b.$$

When the amplitudes also vary, their chain terms must be added. For simple boxes $g(0, v) = 0$ is an analytic identity in the root-chart parameters. Use damping exactly $-W/L$ in the mixed kernel and tie $W = B|\bar{b}|L$; duplicate bounded coefficients remain independent. The other changes preserve the vanishing identically by analytic substitution and subtraction of their pure-profile restrictions.

In particular, the admissible independent-argument small-slope residual is

$$z_t = p(F/F_s - 1), \tag{10.45}$$

with p an independent bounded parameter. Layer rates use the exact $c = W/L$, and the physical graph includes $p = \kappa W/L$. Its fast-layer vanishing holds for every independent p . The expression $pF/F_s - \kappa W/L$ is a different ambient extension; it has an unwanted constant term away from the graph and is not used. For example, when $F = F_s = 1$, the chosen residual is identically zero, whereas that other expression integrates to $pL - \kappa W$. Physical reconstruction adds κW ; on the tie $W = \kappa pL$, its p -derivative is the correct L . Thus equality on a graph does not assert equality of unrelated ambient partial derivatives.

More generally, suppose a rewriting is the exact identity

$$\Phi(a) = \Psi(a, G(a)), \quad E(a, G(a)) = 0, \quad \det E_y \neq 0$$

on an open neighborhood of the old arguments. Then

$$G_{a_j} = -E_y^{-1} E_{a_j}, \quad \Phi_{a_j} = \Psi_{a_j} + \Psi_y G_{a_j}.$$

Repeated differentiation uses finitely many derivatives and finite powers of $(\det E_y)^{-1}$. These denominators may be cleared while retaining their nonzero domain conditions. For a concrete instance, the bounded-action identity, with all original amplitudes kept independent, is

$$\Phi(Q, q, \delta, \eta) = \Psi(Q - q, (Q - q)/q, \delta, \eta).$$

It implies, holding δ, η fixed,

$$\Phi_Q = \Psi_S + q^{-1}\Psi_M, \quad \Phi_q = -\Psi_S - (Q/q^2)\Psi_M. \quad (10.46)$$

In particular the ratio derivative terms cannot be suppressed. The implicit function in (10.19) has a unit denominator and so obeys the same rule.

Positive rational powers are tied by positive roots of polynomial equations after clearing integer powers. These are regular on the positive domain; signed multipliers are used algebraically. Clock logs are tied by

$$t_- = t_+ e^{-L}, \quad L > 0; \quad q = p e^{-L_w}, \quad Q = p + \delta q L_w. \quad (10.47)$$

Log-state residuals use the positive endpoint ties (10.30) and (10.34), placing the decaying exponential on the appropriate side when necessary. Thus no absolute unbounded state logarithm is an analytic-box argument. Mixed layer exponents are the fixed affine combinations $\alpha L + \beta W$; additive fractions use fixed multiples of $S = Q - q$. Unbounded preparation constants are only algebraic multipliers or graph coordinates. They are not inserted into a bounded analytic argument without normalization.

Internal subdivisions are identities of actual flows on an open old argument neighborhood. The new amplitudes use precisely these powers and affine shifts. Their intermediate states are fixed successively by equations $z_j - \Phi_j(a_j, z_{j-1}) = 0$; the state Jacobian is triangular with diagonal 1. Differentiating the exact composition identity supplies the corresponding finite jet graphs with the same property; see Lemma 11.5. Therefore their addition preserves isolation and gives the claimed derivative formulas. This uses the differentiable packet theorem for each output, not an assertion that its variational equation has the original layer-vanishing property. $\square$

Lemma 10.6 (Boundary corners of the chosen domains). *The independent-argument domains in this section can be chosen so that the following holds for every sequence in any one of them, including a sequence on which imposed strict margins vanish. After passage to a subsequence and the finite graph additions in the displayed regime reductions, either there is a divergent enlarged coordinate to which the packet alternative applies, or the normalized kernel factors extend analytically near all their limiting arguments and retain state room. This applies to the old independent kernel and all its fixed finite derivatives through the exact identities of Lemma 10.5. Auxiliary algebraic graphs and equations obtained by clearing their nonzero divisors are analytic at bounded limits; their regularity is required at the sequence points, not at the limiting boundary point.*

Proof. There are two different kinds of margin. A *working-box margin* restricts data to a smaller part of a region on which the analytic formula and its existence estimates are already valid. A *structural margin* chooses a positive coordinate, a nonzero algebraic divisor, an ordering, or a clock regime. We check their boundaries separately. This is a property of the concrete domains, stronger than completeness for sequences whose margins stay positive.

Analytic boxes, amplitudes, and state room. Choose three nested closed boxes for each bounded argument: an admitted box, a larger working box, and an analytic-extension box, each contained in the interior of the next. The finite preparation and root charts provide the outer boxes before the admitted ones are chosen. Similarly choose an admitted input-state set strictly inside a

working state disk and the working disk strictly inside the field's analytic state disk. Choose the admitted amplitude bound η , coefficient bounds, and any integrated prefactor bounds so that the real existence estimate holds on a slightly larger closed range than the admitted one. These choices are made once. For example a zero-base passage with integrated forcing and Lipschitz bounds $C\eta$ sends $|z_{\text{in}}| \leq r_0$ into

$$|z(t)| \leq (r_0 + C\eta)e^{C\eta} \leq r_1 < r_2,$$

where r_2 is the analytic state radius. Stable damping permits the same bound; the regular base is first removed on its specified larger flow box. Thus a working-box margin tending to zero reaches the edge of the admitted box, which still has uniform analytic room. It does not mean that a trajectory reaches the singular boundary of the field.

Here all needed integrated estimates hold for the independent ambient families. In (10.6), the base has the small coefficient b , while the layer integral of the other term is bounded by $C|b'|\eta$, independently of L . The physical equality $b = Lb'$ is not used for that estimate. In the mixed simple model, $\int |\tilde{g}| dt \leq C\eta$ and the homogeneous term is damped for every $W, L > 0$. The mixed annular fields have a designated fast factor with a fixed rate margin, giving the bound preceding (10.40) for every independent bounded prefactor. The additive bounds integrate nonconstant powers against dw/w in (A), and power-corrected fast layers against dw in (B); they hold uniformly on the closures of the smaller amplitude ranges and for $q \geq q_*$. In the one-rate stable model the fixed lower large-rate threshold may be increased before defining its domain; the bounded slow field is then controlled by damping even at that threshold. The bounded- w_0 ordinary annular kernel has the guard in (10.25); its forcing and Lipschitz integrals are at most Ch on the closed admitted range, even for independent normalized profile arguments. Bounded-clock factors have analytic ODE dependence; if a finite limiting slow interval is too long for one base-flow chart, take one fixed finite subdivision and smaller flow charts. The original trajectory estimate supplies the same state room at its internal states. If it is the damping of an old simple kernel that tends to zero, use its inherited forcing estimate, not a claim about an arbitrary bounded stable remainder. In (10.11), $\int_0^1 |L\tilde{g}| ds \leq C\eta$ independently of $W \geq 0$. In (10.20),

$$\frac{M}{y + \delta} ds = \frac{dy}{y},$$

so the nonconstant-power estimate bounds its forcing and Lipschitz integrals independently of $S \geq 0$, including $S = 0$ by continuity. Thus zero damping is harmless for these particular old kernels; no such assertion is needed for a general stable equation at zero rate. Consequently input, output, amplitude, and artificial state-range inequalities can vanish without loss of these larger existence bounds.

Rate, sign, and unit margins. All nonzero rational one-rate or additive rates are fixed positive numbers. Mixed (I) has fixed positive layer rates and nonnegative damping at its limiting boundary. In the away-from-resonance mixed (II) rule, let $r_* > 0$ be a lower bound for the finitely many actual sign-adjusted rates on the chosen coefficient range after shrinking the generator sizes. Impose in the independent domain, with slack, the inequalities $r_i > r_*/2$. At their boundary the rates are still positive. Here $r_i = r_i(W/L)$ is the actual sign-adjusted rate of the independent kernel, not a rate evaluated at a separately copied ordinary coefficient. Thus no rate tending to zero at a positive limiting slope is admitted, even off the physical coefficient graph. The same choice applies to the designated fast rates in the small-slope rule. Its pure-state rates $|\beta_i|c$ can tend to zero, precisely when $c \rightarrow 0$; the field then still vanishes without the designated fast layers. The both-large case is the zero-slope alternative of the mixed theorem, the bounded W case is (10.40)–(10.41), and

bounded clocks are treated below. Pure slow layers are never used alone to bound the forcing integral.

Likewise fix the domains of prepared units, root linearizers, and profile inversions on the larger working boxes where their required Jacobians and unit divisors are bounded away from zero. This applies to $F, F_s, b, b(0), h$, the denominator in (10.16), and the root-chart transversality constant. Choose $|\delta| \leq \delta_* < 1$ on the working box, so $1 + \delta/y$ and $y + \delta$, $y \geq 1$, are units even on the admitted boundary. A margin choosing the sign of an algebraic prefactor, such as B or p , may instead tend to zero. The kernel formulas are analytic in that prefactor at zero; if a normalization divided by it, that division is an auxiliary graph of the next paragraph, not an uncontrolled analytic argument. Overlapping magnitude rules permit the same conclusion at their finite cutoff boundaries.

Algebraic divisors and positive endpoint variables. A physical positive variable or an algebraic normalization divisor may approach zero. Keep every bounded normalized value as its own graph coordinate; for instance $y = a/d$ uses $dy = a$, with $d \neq 0$ retained as a domain condition. Positive rational powers use polynomial relations with the designated positive branches. These relations are analytic also at zero, although their graph Jacobians may then become singular. At each point of the sequence the original nonzero or positive condition gives the same locally unique graph as before. A fixed finite number of differentiations produces only finite powers of these divisors, which can be cleared without changing the local solution set there. The coefficients of the resulting equations are analytic at a bounded graph limit. If a required normalized value is unbounded, it is instead a new recoverable large coordinate. A vanishing determinant imposed by a later rank restriction behaves in the same way: clear its reciprocal powers in the finite equations and retain its strict domain condition. No uniform inverse bound at the limiting boundary point is asserted or needed for the local graph equivalence at the sequence points.

Zero clocks, ordering boundaries, and ratios. The ordinary normalized formulas explicitly include zero length. The nonzero short field uses its bounded integrated coefficient. The nonzero long field, if it reaches a bounded-clock corner, retains independent b, b' in (10.6). The simple mixed field is $-Wv + L\tilde{g}$, whose layers have exponents affine in L . Mixed (II) has field $Lp\mathcal{G}$ and layers whose exponents are the fixed affine combinations $\alpha L + \beta W$. All are analytic at bounded (L, W) , including a zero value of either clock, without evaluating W/L in a field coefficient. A duplicated bounded coefficient remains independent, with its product tie retained. If a slope interval itself is imposed using a ratio, its inequalities and tie may be cleared by $L > 0$; no ratio is necessary inside these bounded-clock kernels. For example even $L, W \rightarrow 0$ and $W/L \rightarrow \infty$ cause no singularity in the mixed (I) normalized field. A diverging ratio already present as an auxiliary coordinate may still be retained as a large algebraic input, but does not invoke the unbalanced mixed primitive when the two primary clocks are bounded. In particular no auxiliary ratio is introduced merely to express a slope which these bounded-clock formulas have already eliminated. The mixed auxiliary ratio is required only in the specified regimes in which both primary clocks tend to infinity.

For additive kernels the fixed bound $q \geq q_* > 0$ prevents a zero denominator. If Q stays bounded, then so does $M = (Q - q)/q$, and (10.19) is regular, including $Q - q = 0$. For a finite large limit of M , the single fixed K in (10.32) provides analytic fraction charts. If Q is unbounded, the divergent-ratio or bounded-ratio packet alternatives already proved apply. If two positive physical endpoints both approach zero, their ratio is handled by its graph and (10.47): bounded log length has the normalized extension just described, while a divergent log length is an explicit large coordinate. Vanishing physical clock speed similarly affects its algebraic endpoint normalization; the normalized field's unit denominator remains in its larger working box.

This list exhausts the imposed terminal margins. Fractions and other rewritings use an exact identity of the old independent kernel on a neighborhood of each sequence point, with the same exogenous arguments. The fixed finite state and jet graphs therefore preserve isolation there, even if their neighborhoods shrink along the sequence. If all enlarged coordinates are bounded, the preceding checks give analytic kernel factors and analytic divisor-cleared graph equations at their limits. Otherwise the divergent enlarged coordinates are available for the flag and packet alternative. The two conclusions are not conflated with positive-margin completeness. $\square$

Proof of Theorem 10.1. The real cases above exhaust the constant-state pieces, nonzero boxes, simple boxes, and annuli from Theorem 9.7; the nonzero-resonance redo is finite. Every additional real cut used a magnitude of a subanalytic variable, a root or state bin, a coefficient or resonance interval, or a rational weighting. Hence these cuts preserve the finite boundary-word bound. Boundary connectors keep each actual middle passage compactly inside its identity piece. Length alternatives overlap and are selected from endpoint arguments; large-input subsequences and internal fractions are used only at an absolute zero test, not as geometric subdivisions.

Choose all analytic boxes first with larger extension, all finite rate and derivative margins next, and state widths and amplitude ranges finally inside the required margins. The nonzero integrated prefactors are explicitly bounded arguments; the simple transfers use damping and small profile integrals; annular residuals use (10.3). The additive models have the real estimates of Theorem 6.1. These establish real existence and state room on open domains described by finitely many strict inequalities, rather than a condition of unspecified continuation along a mismatched orbit. An actual target satisfies tighter margins. Bounded tuples whose listed margins stay positive remain in the extended analytic boxes and have analytic ODE dependence, which proves the completeness assertion in part (i). For part (iii), no positive lower bound on these margins is assumed: Lemma 10.6 treats all their vanishing limits. Its bounded alternative supplies analytic normalized kernel factors with state room and divisor-cleared graph equations; its unbounded alternative supplies the enlarged coordinates used in the following packet test.

Along an arbitrary test sequence select bounded limits or divergent subsequences for the finitely many clocks and ratios. The tables and the additive ratio alternatives select the asserted primitive. A finite but large bounded clock or ratio needs only one fixed finite subdivision for that subsequence. Its locally unique state and finite jet graphs preserve isolation by Lemma 10.5. The large clocks have their independent affine flag representation; insert the initial log nodes and, for an unbalanced mixed test, the auxiliary ratio required by the corresponding primitive theorem. On that flag the three packet theorems supply all fixed finite independent derivatives; bounded operations and regular pullbacks use Theorem 4.4. No equality between two originally independent clock arguments is imposed inside a packet test unless provided by its stated flag relation.

These are also the fixed determinations needed on a retest. The primary affine clock formulas, lifted logarithms, signs, and analytic profile fields remain the same under sufficiently small free-coordinate variations preserving the regime and ordinary limits. The additive coefficient determinations are their fixed formal recursions, compatible bounded-charge coefficients, and, for retained bounded inner endpoints, the normalized infinite-anchor jets of Theorem 6.1. The one-rate and mixed determinations are the fixed coefficient constructions of their respective theorems. Numerical cutoffs or anchors have been retired by their proved dispersals before they could be discarded from the retained arguments. Subsequent regular graph operations therefore act on the same ordinary coefficient germs by Theorem 4.4; inserting unused nodes uses identity transitions. This verifies the claimed compatibility with primitive retests. Its application to the complete enlarged matching systems is made in Section 11.

Finally, Lemma 10.3 gives the nearby physical mismatch identity, and Lemma 10.5 gives the

derivative assertion with all graph terms retained. The common smallness clauses in the mixed theorem apply because each relevant residual monomial has a designated uniformly fast layer; hence geometric tolerances were fixed independently of a subsequently chosen limiting slope and of all finite asymptotic requests. This proves every assertion. $\square$

11 Projection, matching, and the cycle count

We first prove a finite-dimensional counting principle with its full closure hypothesis. We then construct the analytic systems to which it applies. Throughout this section, a parameter becomes an unknown when an absolute zero test is made. The distinction between these two roles is essential.

Uniform component bounds for relatively compact subanalytic families are classical [6, Corollary 1]; see also Bierstone and Milman [2, Theorem 3.14]. In the Pfaffian setting, Khovanskii similarly reduces component bounds to critical-point equations of proper functions [16, §4, Theorems 3–4]. Here we prove the required implication for open analytic systems with an explicit differential and algebraic closure hypothesis, using proper functions and Sard’s theorem. The passage constructions must verify that hypothesis before the component bound can be applied.

11.1 Open domains and the projection principle

Definition 11.1 (Admissible open data). An open analytic system consists of a specified open set $U \subset \mathbb{R}^a \times \mathbb{R}^n$, real analytic equations $F : U \rightarrow \mathbb{R}^m$, and a finite list of real analytic margins $g_1, \dots, g_s > 0$ specifying its open restrictions. We require the following completeness of the list of margins: if a sequence in U has bounded coordinates and all its margins are bounded below by positive constants, every finite limit of that sequence belongs to U and is a point of analyticity of all the data. Coordinate-chart boundaries are included among these restrictions. A nonzero condition $h \neq 0$ can be written as the margin $h^2 > 0$.

We call this condition *positive-margin completeness*. It concerns bounded limits for which every listed margin stays bounded away from zero; it does not assert analytic extension when a margin tends to zero.

The *finite differential and algebraic closure* of these data allows fixed real constants, additional real coordinates, finite sums and products, finitely many independent-coordinate derivatives, and inverses of functions declared nonzero on the domain. One may clear these denominators while retaining their nonzero restrictions. A system used below retains the original equations, argument graph ties, and domain restrictions, and can append equations and margins from this closure. The operation can be iterated finitely many times. Fixed real values may be assigned to designated parameters at any stage. All such specialized equations and restrictions are retained. Equivalently, a specialization retains those parameters as coordinates and appends the polynomial equations fixing their values.

We say that the data have the *absolute zero property for this closure* if every fixed finite system so obtained has only finitely many isolated real solutions in its specified open domain, when all remaining coordinates are treated as unknowns. This assertion concerns the entire system, including every added coordinate. It is not an assertion merely about isolated points of the original zero set.

For a system with full row rank in x , one can use separate charts for the finitely many nonzero $m \times m$ minors of F_x . Their squares are added to the margin list. Every further determinant restriction in the following proof is treated in the same way.

Theorem 11.2 (Projection count). *Let $F(p, x) = 0$ be admissible open data as in Definition 11.1, with $p \in \mathbb{R}^a$, $x \in \mathbb{R}^n$, and $m \leq n$. Suppose that F_x has row rank m throughout the zero set under*

consideration and that the data have the absolute zero property for their finite differential and algebraic closure. Then there is a finite constant N , depending on this fixed system and its domain, such that every fiber

$$M_p = \{x : (p, x) \in U, F(p, x) = 0\}$$

has at most N connected components. If $m = n$, every fiber has at most N points.

Proof. We first prove the square case by induction on the number of parameters. With the other parameters fixed, allowing one parameter to vary turns its solutions into curves. A fixed generic tilt associates their components with a square critical-point system having one fewer parameter. We then treat positive-dimensional fibers by choosing a tilt for a countable family of witness fibers and applying the square case to their multiplier systems.

We may work on one minor chart and finally sum over the finitely many charts. Include its determinant margin in the list g . On a fiber, and also on a slice in which only some parameters have been fixed, use

$$\rho(p, x) = 1 + |p|^2 + |x|^2 + \sum_{j=1}^s g_j(p, x)^{-2}. \quad (11.1)$$

Its sublevels on the closed equation locus are compact. Indeed the coordinates are bounded there, each g_j is bounded below by a positive constant, and Definition 11.1 puts every limit back in the same domain; continuity then preserves $F = 0$. For any fixed linear function $\ell \cdot y$ of the varying coordinates,

$$|y|^2 + \ell \cdot y \geq \frac{1}{2}|y|^2 - \frac{1}{2}|\ell|^2.$$

Consequently $\rho + \ell \cdot y$ is proper and bounded below as well. Every nonempty connected component of a smooth fiber is closed in that fiber and therefore contains a minimum of this function. The minimum is an interior constrained critical point.

We prove the assertion for square systems first, simultaneously for all systems in the stated closure, by induction on the number a of parameters. If $a = 0$, invertibility of F_x makes every solution isolated, and the absolute zero property is exactly the required finiteness. Suppose $a > 0$ and write $p = (b, t)$ with $b \in \mathbb{R}^{a-1}$ and $t \in \mathbb{R}$. At fixed b , the equation locus is a one-dimensional analytic manifold in $y = (t, x)$, since F_x is invertible. Its tangent vectors satisfy

$$F_t dt + F_x dx = 0, \quad dx = -F_x^{-1} F_t dt.$$

Thus dt never vanishes on a nonzero tangent vector. The function t is injective on every connected component: a connected one-dimensional manifold without boundary is a circle or an interval; a circle would force an extremum of t , and on an interval its nonzero derivative has a constant sign. It follows that a fiber over (b, t) contains at most one point from each of these curve components.

We next choose one tilt, independent of b . Introduce $\lambda \in \mathbb{R}^n$ and $\ell \in \mathbb{R}^{n+1}$, and consider

$$F(b, y) = 0, \quad \nabla_y \rho(b, y) + \ell - F_y(b, y)^T \lambda = 0. \quad (11.2)$$

The universal critical locus is a smooth manifold: first impose $F = 0$, whose derivative in x is surjective, and then solve the second block for ℓ . Its dimension equals that of (b, ℓ) . The kernel of the derivative of projection to (b, ℓ) is exactly the kernel of the square Jacobian of the left sides of (11.2) in (y, λ) . Sard's theorem, applied on countably many charts if necessary, says that the critical values of this projection form a null set [21]. Fubini's theorem therefore gives a fixed ℓ for which, for almost every b , every solution of (11.2) has an invertible Jacobian in (y, λ) .

For clarity, this invertibility is precisely nondegeneracy of a critical point of the restricted tilted function. If $A = F_y$ and $H = D_y^2(\rho + \ell \cdot y - \sum_i \lambda_i F_i)$ is the ambient Lagrangian Hessian, its restriction to $\ker A$ is the constrained Hessian. The relevant bordered matrix is

$$\begin{pmatrix} A & 0 \\ H & -A^\top \end{pmatrix}.$$

Its kernel consists of (v, η) with $Av = 0$ and $Hv = A^\top \eta$. It is trivial exactly when the quadratic form H on $\ker A$ is nondegenerate: testing the second equation against vectors in $\ker A$ proves one direction; the orthogonal complement identity $(\ker A)^\perp = \text{im } A^\top$ proves the other.

Now fix this ℓ . On the open locus where its displayed Jacobian is invertible, (11.2) is a square system with $2n + 1$ unknowns (y, λ) and only $a - 1$ parameters b . It belongs to the closure in Definition 11.1. In fact the derivatives of g_j^{-2} are products of derivatives of g_j and powers of its nonzero inverse; determinant conditions are polynomials in such derivatives. All original equations and restrictions are retained. In subsequent proper functions we include the squares of all new coordinates, including λ , and inverse barriers for all new strict conditions. The induction hypothesis therefore gives a uniform bound N_0 on the regular solutions of this new system for every b .

For almost every b , every curve component has a nondegenerate minimum and hence gives a different solution of the new system. There are at most N_0 components and therefore at most N_0 points in each original fiber over such a b . This bound holds at an exceptional (b_0, t_0) too. Otherwise select $N_0 + 1$ distinct points of that fiber. By the implicit function theorem they extend to disjoint local solution graphs over one common neighborhood of (b_0, t_0) , preserving every strict inequality. Choose b in the full-measure set close to b_0 , with $t = t_0$. All selected points persist, a contradiction. Only finitely many points were selected, so no common neighborhood for an infinite fiber was assumed. This completes the induction for square systems.

Finally allow $m < n$. If no uniform component bound existed, choose a countable collection of parameters p_k with more than k components in M_{p_k} . For each fixed p_k the same universal critical construction, now in (x, λ, ℓ) with $\lambda \in \mathbb{R}^m$ and $\ell \in \mathbb{R}^n$, shows that almost every tilt makes $\rho(p_k, x) + \ell \cdot x$ Morse on the whole fiber. Choose one tilt in the intersection of these countably many full-measure sets. The multiplier equations

$$F(p, x) = 0, \quad \nabla_x \rho(p, x) + \ell - F_x(p, x)^\top \lambda = 0$$

are square in (x, λ) . Restricting to their invertible Jacobian locus gives a system covered by the square assertion just proved. Each component in every witness fiber contributes a regular minimum, contradicting its uniform point bound. The same argument includes $m = 0$, with no multiplier coordinates. This proves the theorem. $\square$

Remark 11.3. The closure hypothesis is hereditary, and is used each time the parameter dimension decreases. For example, the equation $\sin x = 0$ on $0 < x < p$ has no isolated zeros in the full (p, x) space, while its finite fibers have unbounded size. Appending $p = x + 1$ produces infinitely many isolated total zeros. Thus finiteness for the original total zero set alone cannot replace the hypothesis of Theorem 11.2. Conversely, its proof uses only finitely many derivative orders at each of finitely many induction steps; it imposes no estimate uniform over all derivative orders.

11.2 Matching equations and their geometric meaning

Theorem 11.4 (Matching systems). *For each polynomial degree bound d , there are a finite number $T(d)$ and finitely many fixed open analytic systems*

$$F_w(p_w, x_w) = 0, \quad g_w(p_w, x_w) > 0, \quad w \in \mathcal{W}_d,$$

with the following properties.

1. Their domains satisfy Definition 11.1. Every finite differential and algebraic system generated from these data, with their original ties and domain restrictions retained, has the absolute zero property.
2. For each field $V = (P, Q)$ of degree at most d , all but at most $T(d)$ of its periodic orbits contained in the fixed square of Section 9 have a representation in one of these systems. The parameter value $p_w(V)$ is the same for every orbit using the same word and preparation choices; endpoint-dependent quantities are unknowns with graph ties.
3. At a representation of a hyperbolic periodic orbit, F_{w, x_w} has full row rank. In each fixed nonzero-minor chart, different represented hyperbolic orbits lie on different connected components of the fiber at $p_w(V)$.

The minor charts and their additional margins also satisfy the first assertion.

The proof occupies this subsection and the next. We first construct the systems and show why their regular fiber components distinguish hyperbolic cycles. Moving the endpoint cuts along an orbit gives continuous families of representations. After establishing the geometric interpretation, we prove the absolute zero property needed to bound its components.

Finite words and connectors. Theorem 9.7 gives a finite alphabet of prepared passages and a degree-dependent bound on the number of boundary events of a periodic Jordan curve. Orbits containing one of its nonsingular tangential boundary arcs account for at most $T(d)$ exceptions. At every other boundary event insert a small connector, with its two ends in the interiors of the neighboring pieces. There are only finitely many events on the selected orbit, so these connectors can be made disjoint. A periodic orbit cannot remain entirely inside a piece with a nonsingular monotone graph clock. In particular, a nonconstant closed curve has an extremum of its horizontal coordinate. Thus the retained cycles are encoded by closed words of bounded length.

Here is a fixed analytic connector model. Choose one nonzero physical velocity component as graph clock X , let Y be the other coordinate, and order the ends so that $\Delta = X_+ - X_- > 0$. Physical time may run in either direction. Introduce $r > 0$ by $r^2 = \Delta$ and set

$$X = X_- + \Delta s, \quad Y = Y_- + rz, \quad 0 \leq s \leq 1.$$

Writing V_X, V_Y for the two components, the scalar equation is

$$z_s = \frac{\hat{N}(s, z)}{\hat{D}(s, z)}, \quad \hat{D} = \frac{V_X(X, Y)}{V_X(X_-, Y_-)}, \quad \hat{N} = \frac{\Delta}{r} \frac{V_Y(X, Y)}{V_X(X_-, Y_-)}. \quad (11.3)$$

These are polynomials of degree bounded in terms of d in (s, z) . Their coefficients are tied to the original field and the endpoints by polynomial equations, with the divisor $V_X(X_-, Y_-)$ retained as nonzero. As the connector is shortened at a fixed nonsingular event, the coefficient vectors of $(\hat{D}, \hat{N})$ tend to $(1, 0)$. Indeed every nonconstant term of $\hat{D}$ contains a factor Δ or r , and every term of $\hat{N}$ contains the factor $\Delta/r = r$. Also $z_- = 0$ and the true z_+ tends to zero. On one fixed small coefficient box, (11.3) therefore has a transfer analytic on a slightly larger box and with state room. Its scalar matching equation is this transfer from 0 minus $(Y_+ - Y_-)/r$.

A word records the passage cells, real charts, signs, traversal orientations, and endpoint-rule alternatives of Theorem 10.1, together with these connector choices. Each choice is finite and the word length is bounded, so $\mathcal{W}_d$ is finite. Internal subdivisions used only in an absolute zero test are not part of $\mathcal{W}_d$.

Parameters, endpoints, and argument graphs. Use the polynomial coefficients and all parameter-only preparation data for the selected word as external coordinates p_w . Preparation was performed with the field coefficients as parameters, before choosing orbit cuts. Thus, for a fixed field and fixed preparation choices, these values are fixed for every cycle assigned that word. They can vary independently on the extended analytic parameter domain; no global analytic relation between different parameter-only preparation formulas is required there. Any quantity depending on the endpoints, including a normalization by a proposed endpoint or a connector coefficient in (11.3), is instead an unknown determined by its argument graph.

For a word with k links introduce physical endpoints $e_j = (x_j, y_j)$, $j \in \mathbb{Z}/k\mathbb{Z}$, shared by successive links. There is no section equation fixing the tangential position of a cut. The other variables are terminal coordinates, clocks, amplitudes, bounded ordinary arguments, and residuals. We use the following encoding conventions.

1. Earlier time-dependent coordinate formulas needed to recover physical endpoints are included in each clock preparation. Their values are positive rational clock powers, products with parameter-only coefficients, and bounded analytic units at prepared arguments. If $q > 0$ and $\nu = A/B \in \mathbb{Q}$, $B > 0$, the equation $u^B = q^A$, on $u > 0$, defines the selected positive power regularly; negative A is handled by a nonzero denominator. Translations, products, and ratios have their usual regular graph equations. Bounded analytic state changes, simple-root linearizers, diagonal primitives, and their regular inversions use smaller boxes with extensions to larger neighborhoods. Complex conjugate preparation data are paired and expressed through real and imaginary parts, so all equations here are real analytic in real coordinates.
2. A logarithmic clock length is tied by

$$t_- = t_+ e^{-L}, \quad L > 0, \quad 0 < t_- < t_+. \quad (11.4)$$

Its derivative in L has absolute value t_- and is nonzero. For an additive clock we additionally use

$$q = p e^{-L_w}, \quad L_w > 0, \quad Q = p + \delta q L_w, \quad S = Q - q.$$

Products tie scaled actions such as W , slopes, and bounded prefactors. A logarithmic state residual is tied using the positive physical transformed state $R = e^v$:

$$R_+ = R_- \exp(\kappa S + z_+), \quad (11.5)$$

or the corresponding equation with $S = W$ or zero subtracted slope. Place the decaying exponential on the appropriate side when useful. The equation has nonzero derivative in z_+ , and hence is a regular graph. Absolute unbounded state logarithms do not become analytic arguments. Every unbounded exponential in these ties is a fixed affine combination of clock coordinates; a variable scaled action has its own coordinate and product tie.

3. Right amplitudes contain their prescribed bounded residual correction. For example in an additive annulus, if $z_+ = v_+ - v_- - \kappa S$, then

$$D_i = \chi_i Q^{p_i} e^{d_i v_+ - d_i z_+} = \chi_i Q^{p_i} e^{d_i v_- + d_i \kappa S}.$$

Thus the proposed final state cancels from the actual-flow amplitude. The analogous mixed identity is

$$\chi_i e^{a_i \theta_+ + \beta_i v_+ - \beta_i z_+} = \chi_i e^{a_i \theta_+ + \beta_i v_- + \beta_i \kappa W}.$$

These are identities on the tied endpoint graph, including mismatched endpoints. Signed or zero amplitudes are handled by products, without extracting roots from their signs. An

invariant diagonal expression is evaluated at one end only. Its weight-zero identity makes it constant under the relevant nearby endpoint variation at fixed field parameters. No redundant second equality is added for that identity.

4. Each link has exactly one scalar transfer equation in addition to its argument ties. A mixed kernel uses independent normalized clock coordinates (L, W) , with exponents $\alpha L + \beta W$ for fixed α, β . Its other bounded slope or coefficient copies are separate ordinary arguments with their own ties. The additive kernels have independent allowed (Q, q, δ) and layer data. Duplicate arguments may be given duplicate coordinates, with regular equality ties. All kernels are the actual real analytic ODE transfers of the preceding sections. The residual extensions with independent arguments are those of Lemma 10.5, preserving the required layer-vanishing identities.

Each auxiliary coordinate has one regular determination in this construction. Ordered in the construction order, the graph Jacobian is block triangular with invertible diagonal. In particular its solutions are locally unique graphs over physical endpoints and fixed parameters.

Figure 4 separates the independent kernel, the tied endpoint graph, and its matching locus. Only the last step imposes the scalar transfer equation.

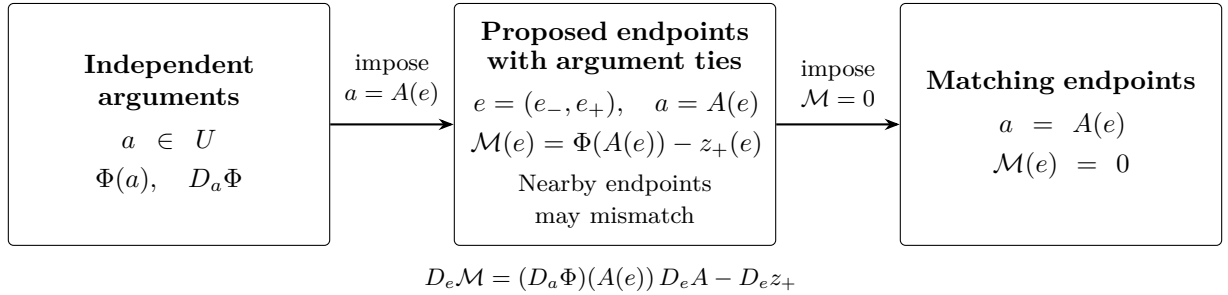


Figure 4: One link near an actual passage, with physical parameters fixed. The arrows impose successive equations. Regular argument ties express all independent kernel arguments, including coefficient copies, in terms of nearby proposed endpoints before the scalar transfer equation is imposed. Here $z_+(e)$ is the prescribed output coordinate in the chosen traversal. On this local tied graph, $\mathcal{M}$ is the physical mismatch in regular transverse coordinates, including at mismatched pairs. Ambient partial derivatives are taken in a ; endpoint derivatives use the displayed chain rule. This is a local statement, not a claim that every point of the extended matching domain is a physical passage.

The specified open domains. We list all domain restrictions as strict margins: positive clock and root conditions; ordering; all nonzero divisors; coordinate-box and smallness restrictions for bounded analytic arguments; coefficient signs and ranges; and the state and rate margins ensuring that the real transfer exists in its larger state disk. The finite analytic charts are chosen with a larger extension about their closed smaller argument boxes. The target smallness thresholds from the geometric reduction are chosen strictly inside these boxes.

For additive transfers this includes $Q > q > q_*$ for the fixed large threshold q_* , a sufficiently small $|\delta|$, and small layer amplitudes and state inputs. The action is increasing. The real estimates of Theorem 6.1 give state room: model (A) has damping and a small integral of the nonconstant-power forcing; in model (B) the large q_* makes the adverse fixed power factors harmless in the integrals of the fast layers. For mixed transfers, fixed rate-sign margins, small amplitudes, and smaller state

boxes give the corresponding room from Theorem 8.1; stable orientation is used in the damped model. The regular and bounded-action alternatives use the integrated bounds of Theorems 7.1 and 10.1. For example their small integrated coefficients are explicitly tied arguments of the form BL , $B\tau_e^\lambda$ on a long-length alternative, or $LB\tau_e^\lambda$ on a short-length alternative. Alternative length ranges overlap about their cutoffs. These finite conditions are coordinate, polynomial/product, or bounded analytic inequalities, with the same affine clock exponentials already allowed in the argument graphs.

Thus a bounded sequence whose listed margins stay bounded away from zero has a limit in the same boxes, sign and ordering domains, and real existence regime. Every formula is analytic at that limit and the limit is admissible. No additional condition that a long mismatched arc remain in a physical subdivision cell is imposed. The analytic systems are the normalized kernels on the stated domains; their identification with physical motion is needed locally at the target representations. This distinction supplies exactly the domain completeness required in Definition 11.1. It also remains valid after adding a determinant margin: a bounded sequence with that margin bounded below cannot leave its minor chart.

Local physical validity and rank. At a target representation and its fixed actual parameter value, Lemma 10.3 identifies each tied transfer equation with a true scalar-flow endpoint mismatch, up to a regular state coordinate and a nonzero factor. The connector has the same property by (11.3). The subarcs are interior and their clocks have nonzero velocity, so this identity holds for all nearby proposed endpoints, not only for matching pairs.

One can see directly why a normalization chosen from the proposed endpoint does not spoil the differential. For a held normalization a , let $H(e_-, e_+, a)$ be the transformed mismatch. In local coordinates consisting of a physical scalar mismatch Δ_0 and the remaining variables, regularity gives $H = U(e_-, e_+, a)\Delta_0$ with $U \neq 0$. This follows, for example, by integrating $\partial_{\Delta_0} H$ from 0 to Δ_0 . After substituting $a = a(e_-, e_+)$, its differential on $\Delta_0 = 0$ is still $U d\Delta_0$. If the terminal clock is also transformed, its nonzero clock velocity determines the hitting-time variation; the regular coordinate pair leaves a nonzero transverse mismatch. The exact residual-amplitude identities displayed above and the weight-zero invariance of the diagonal terms ensure the premise for all the terminal transformations used here.

Choose local flow coordinates (s_j, r_j) at the k endpoints, with s_j tangential and r_j transverse. Up to nonzero row factors, the link differentials at a matching cycle are

$$dr_{j+1} - h_j dr_j, \quad j \in \mathbb{Z}/k\mathbb{Z}, \quad (11.6)$$

where $h_j \neq 0$ is the derivative of the corresponding section holonomy. Reversing the orientation used to write one link merely changes this equivalent equation by a nonzero factor. The kernel of the cyclic transverse matrix satisfies $dr_1 = (\prod_j h_j) dr_1$. For a hyperbolic orbit its return multiplier is $\prod_j h_j \neq 1$, so that kernel is zero and the k link rows have rank k . The tangential variables remain free, as intended. Adding the auxiliary graph equations adds an invertible block in their own variables. Equivalently, eliminating that block by row operations leaves exactly (11.6). Hence the complete equation Jacobian has full row rank, and at least one of its finitely many maximal minors is nonzero.

For a word with k links and n_{aux} auxiliary unknowns, the fiber has $2k + n_{\text{aux}}$ unknowns and $k + n_{\text{aux}}$ equations. Near a hyperbolic target its dimension is therefore k , accounted for by the tangential positions of the cuts. Hyperbolicity gives full row rank of this matching system. The following argument shows that a connected component cannot contain targets from two different cycles.

Injection into analytic components. Fix a word, its actual parameter tuple, and one nonzero-minor chart. Let M be a connected component of its analytic fiber and let $E : M \rightarrow \mathbb{R}^2$ map a representation to its first physical endpoint. Suppose M contains a target on a hyperbolic cycle C . In a neighborhood of that representation, exact local matching and uniqueness of the nearby fixed point of the return map imply $E \in C$. The free tangential cuts only move points along C .

We claim that $E(M) \subset C$. The periodic trajectory C is a compact embedded nonsingular real analytic curve: it is analytic by analytic ODE dependence, regular because V has no zero on a nonconstant periodic trajectory, and embedded by uniqueness and use of a minimal period. Put $A = \text{int}_M E^{-1}(C)$. This is nonempty and open. If $q \in \overline{A}$, closedness of C gives $E(q) \in C$. Near $E(q)$ a regular analytic defining function h cuts out exactly C . Choose a connected analytic neighborhood B of q with $E(B)$ in that neighborhood. The analytic function $h \circ E$ vanishes on the nonempty open set $A \cap B$, so the analytic identity theorem makes it zero on B . Thus $q \in A$; A is closed, and connectedness gives $A = M$.

Different periodic trajectories are disjoint by uniqueness of the ODE. They therefore cannot have target representations on the same component M . This argument permits remote points of M that are not physical passage representations. It uses physical validity on the initial neighborhood and analytic continuation on the entire component, which is precisely why no additional global physical-cell restriction was imposed.

For each word we have now constructed a fixed open analytic system. Its domains are positive-margin complete, every retained cycle has a representation, and different hyperbolic cycles occupy different components on each maximal-minor chart. To apply the projection theorem, it remains to prove absolute isolated-zero finiteness for every finite system generated from these data.

11.3 Absolute zeros of the matching systems

Fix the matching systems just constructed, including their independent kernel arguments, graph ties, and strict domain restrictions. A system in the counting closure may also contain derivatives, inverse margins, and multiplier variables. When its arguments approach a boundary, we will rewrite its kernels by exact identities and retain every added graph equation. The next two lemmas state what these rewritings preserve; the recovery lemma records the consequence used when a free flag input is varied in the final contradiction.

Lemma 11.5 (Graph and finite-jet lifting). *Suppose an analytic transfer $\Phi(u)$, on an open set of its original independent arguments u , has an exact analytic decomposition into finitely many transfers, with auxiliary arguments and intermediate states determined by regular, locally unique graph equations. A finite system involving Φ and finitely many of its independent derivatives can be replaced, locally at any admissible tuple, by a system involving those transfers and finitely many of their derivatives. The replacement, with the graph equations retained, is locally analytically isomorphic to the original solution set. In particular it preserves isolated solutions. This conclusion also holds along a sequence with shrinking graph neighborhoods, provided one fixed finite decomposition is used along the sequence.*

Proof. If $G(u, v) = 0$ has G_v invertible, its selected solution is a locally unique analytic graph $v = v(u)$. Projection of that graph to u is an analytic isomorphism. Substituting $v(u)$ in any other equations proves the first assertion about their solution sets. Denominator clearing preserves this assertion on the retained nonzero domain.

For the derivative assertion, differentiate the identity expressing Φ in terms of the other transfers *after* their arguments have been put on their graphs. Implicit derivatives of $v(u)$ are obtained successively from G_v^{-1} and finite derivatives of G . The finite chain rule expresses each requested

derivative of Φ using only finitely many such quantities. They may either be substituted, or be given their own graph coordinates. In the latter description, the equations for derivatives of order j are linear with coefficient G_v in the order- j unknowns and have previously determined lower-order terms. Their complete finite-jet Jacobian is block triangular with invertible diagonal.

In particular, for a subdivision into K consecutive subflows, write

$$z_j = \Phi_j(a_j(u), z_{j-1}), \quad 1 \leq j < K, \quad \Phi(u) = \Phi_K(a_K(u), z_{K-1}).$$

The intermediate-state equations have a triangular Jacobian with unit diagonal. If jets are adjoined, each derivative of z_j is determined from the earlier z_i and their jets; these graph equations again have unit diagonal. Thus the augmented system has precisely the original germ of solutions and no additional local degree of freedom. This is a pointwise assertion at each member of a sequence; no uniform radius of the isomorphism is needed to preserve isolation. $\square$

Two consequences will be used below. First, partial derivatives of two independently enlarged kernels cannot be identified just because their values agree on a graph. Only the differentiated composition identity in Lemma 11.5 is used. Second, a finite K chosen after taking a subsequence is legitimate in an absolute zero contradiction: it produces one fixed finite system for that sequence. It does not assert a degree-uniform value of K and does not add links to the geometric word count.

Lemma 11.6 (Cleared finite differential closure). *Consider one fixed finite collection of primitive functions, their argument graphs, and a finite differential and algebraic construction from them. Assume that, after one fixed exact graph enlargement, the primitive functions and defining graph equations extend analytically near a finite limiting tuple. The selected graph Jacobians and every declared divisor need be nonzero only at the points of the sequence, not at its limit. Then the constructed equations can be replaced by analytic equations near that limit, retaining all graph equations and all original open restrictions, with exactly the same solution germ at each selected point. Additional multiplier coordinates, polynomial equations, and parameter copies or specializations preserve this assertion.*

Proof. First differentiate every exact rewriting as an identity on its old open argument domain. For a graph $G(u, v(u)) = 0$ this gives

$$v_{u_i} = -G_v^{-1} G_{u_i}.$$

Cramer's rule and induction on derivative order express every requested implicit derivative as a quotient of polynomials in finite jets of G , with denominator a power of $\det G_v$. The same argument applies successively to a finite stack of graphs and to the finite composition identities in Lemma 11.5.

Every further algebraic or differential operation preserves a rational expression in a finite collection of analytic jets. For example,

$$\partial_i(A/B) = \frac{B\partial_i A - A\partial_i B}{B^2}, \quad (A/B)^{-1} = B/A$$

on the retained nonzero domains. An inverse of a new expression therefore adds its nonzero numerator to the possible denominator factors; a determinant is a polynomial in its entries. Thus each final equation is $A/B = 0$, where A, B are analytic near the limiting tuple and B is nonzero at every selected point. Replacing it by $A = 0$ leaves its solution germ unchanged there. This replacement does not require a bound on B^{-1} , nor does it append its value as a new coordinate. In particular a margin or a graph determinant tending to zero causes no difficulty at this step. Finite jet coordinates, if retained instead of substituted, have the differentiated analytic graph equations of Lemma 11.5; their Jacobians need be invertible only at the selected points.

New multipliers are coordinate factors in these expressions. A parameter copy has a polynomial equation $p' - p = 0$, and a fixed specialization has an equation $p - p_0 = 0$. Unrestricted added coordinates likewise enter the closure through polynomial and rational operations. These operations introduce no new primitive function or evaluation outside its original argument domain. Their finite derivatives are of the same form, proving the last assertion. $\square$

Lemma 11.7 (Recovery after graph enlargement). *Let u_i be a sequence of isolated solutions of a fixed system, where u denotes its full tuple, including parameters and multipliers. Suppose an exact finite enlargement appends v_i by equations $G(u, v) = 0$ whose selected graph Jacobian $G_v(u_i, v_i)$ is invertible at every point. Retain all these equations. An old flag input recoverable from (u, v) cannot vary along the local lifted solution set while u stays fixed, provided the variation is taken within the selected graph neighborhood. These neighborhoods may shrink arbitrarily along the sequence.*

Proof. At each i the implicit function theorem supplies an open neighborhood N_i in which $v = g_i(u)$ is the unique selected solution of the complete graph equations. Let B_i be any proposed isolation neighborhood of u_i . Intersect $N_i \cap \pi^{-1}(B_i)$, where $\pi(u, v) = u$, with the local real flag chart and all strict domain neighborhoods at that point. This is an open neighborhood of the lifted point. By continuity of the chart map, all sufficiently small variations of a free chart input remain in it. The size is chosen separately for every i ; it can also be made small enough to preserve the prescribed comparisons and ordinary limits for the retest in Theorem 5.7.

Write $S_* = R(u, v)$ for the retained old free input, recovered by the locally invertible affine and lifted logarithmic flag changes. For any such varied solution, $u' = u_i$ would imply $v' = g_i(u_i) = v_i$ and hence $S'_* = S_*$, a contradiction if this input was changed. Thus a nonzero sufficiently small permitted variation gives a different original tuple inside B_i . Vanishing graph determinants at the limiting tuple affect only the sizes of the individual neighborhoods, not this argument. $\square$

Completion of the proof of Theorem 11.4. Fix an arbitrary finite system in the closure required by Theorem 11.2, including its equations and every original graph and strict domain restriction. Suppose it has a sequence of distinct isolated total solutions. Multipliers and coordinates formerly called parameters are now ordinary unknowns. Clear all declared nonzero denominators, retaining their restrictions. Only finitely many derivatives of finitely many transfer kernels, argument graphs, and margins occur, together with algebraic factors in the coordinates. This assertion is hereditary under the projection construction: gradients of inverse margins, products with new multiplier coordinates, and determinants of their derivatives have exactly this form by Lemma 11.6. Parameter copies and specializations add polynomial equalities. Thus an unbounded new coordinate is included in the large-coordinate analysis below, while a bounded new coordinate is just an additional ordinary argument of these polynomial expressions. No new transfer family is introduced.

At every transfer choose a subsequence regime of Theorem 10.1; there are finitely many transfers in this fixed system. Append any bounded ratios or auxiliary coordinates required by that regime by their unique regular graph ties. For example, the bounded-ratio additive corner is normalized by

$$S = Q - q, \quad M = S/q, \quad y + \delta \log y = 1 + Ms. \quad (11.7)$$

If M tends to a finite value, a fixed sufficiently large finite subdivision into K fractions gives regular analytic normalized pieces. The value of K is chosen once for this limiting corner. Its internal states and amplitudes are regular graphs. Internal amplitudes are products of bounded positive ratios and exponentials of fixed affine clock fractions; for instance a left additive layer at fraction j/K has amplitude

$$e_j = e(w_j/q)^\beta e^{-ajS/K}.$$

The real layer estimates keep these amplitudes in the permitted boxes. The mixed unbalanced auxiliary ratio $C_{\text{aux}} = P/R$ is also appended where required by the mixed packet construction; here P, R denote that construction's positive scales, not the components of the polynomial vector field.

All derivatives of the original kernel are obtained by differentiating the exact identities on its original open argument domain. For example if the decomposition in (11.7) reads $\Phi(Q, q, \delta, a) = \Psi(S, M, \delta, a)$, then

$$\Phi_Q = \Psi_S + q^{-1}\Psi_M, \quad \Phi_q = -\Psi_S - (Q/q^2)\Psi_M.$$

Higher derivatives use finitely many rational factors and kernel derivatives, with the same nonzero denominators. The other bounded clock and subdivision identities are handled by the same chain rule, as proved in Lemma 10.5. Lemma 11.5 therefore lifts the isolated solutions, including any equations containing derivatives, to isolated solutions of one fixed finite enlarged system. Using a different K at every term of the sequence would not give this conclusion; the fixed finite choice just described does.

Bounded enlarged tuples. Suppose first that every coordinate of the enlarged tuple remains bounded, and pass to a finite limit. Some strict margins may vanish. Positive-margin domain completeness, used for the proper barrier, does not prove analytic extension at such a limit. Lemma 10.6 instead gives the required extension of each *old independent-argument* normalized kernel: closed smaller argument and amplitude boxes lie within larger analytic boxes, and the integrated-field bounds leave a positive state margin on their closures. The bounds use independent coefficient copies, before any physical slope or normalization ties are imposed. At a bounded clock corner these normalized fields therefore define analytic transfers near the limiting data, including when a clock is zero.

For example, the normalized mixed equations have the forms

$$v_s = -Wv + Lg(X, Y, v), \quad z_s = LG(X, Y, z),$$

with layer exponents $\alpha L + \beta W$ and other coefficient copies retained independently. They contain no singular W/L in these positions and are analytic also at $(L, W) = (0, 0)$. A slope needed elsewhere is its own coordinate, tied by $Lc - W = 0$. If it stays bounded, this polynomial graph is analytic at the corner. If a necessary new ratio diverges, the enlarged tuple has a large coordinate and belongs to the large-coordinate case below; boundedness of the *old* tuple alone is not the criterion for the present case.

The remaining possible degenerations of the endpoint graphs are equally explicit. A ratio has an equation $bv - a = 0$. A positive rational root has an equation $u^B - q^A = 0$, made polynomial by clearing negative powers when necessary. Clock and residual ties have the forms

$$t_- - t_+ e^{-L} = 0, \quad R_+ - R_- e^{\kappa S + z_+} = 0.$$

They extend analytically when all their coordinates remain finite, even if their derivatives in L or z_+ tend to zero with the positive endpoints. If $t_+/t_- \rightarrow \infty$, then $L \rightarrow \infty$, which is a large-coordinate case. Connector coefficients have polynomial graph equations after multiplication by $V_X(X_-, Y_-)$ and r , and $r^2 = \Delta$ is polynomial. Their bounded normalized coefficient arguments remain in the larger analytic connector box when either divisor tends to zero. Regular endpoint inversions may likewise retain their analytic defining equations. Their graph Jacobians need only be nonzero at the selected points. Finite implicit derivatives then have exactly the denominator description of Lemma 11.6.

Consequently all primitive transfers and defining graph equations extend analytically at the finite limiting tuple of the enlarged system. That lemma clears the finite derivative and multiplier

equations without asserting invertibility of a limiting graph Jacobian. Only equations are cleared; no inequality is replaced by an unverified numerator sign. The old strict inequalities and all later determinant restrictions remain as open restrictions at each selected point. Their limiting values may vanish, and their differentiated forms need not themselves extend across that boundary. This is sufficient here, in contrast to the positive-margin completeness used in the projection theorem. Local finiteness of components of a real analytic set excludes an accumulating sequence of its isolated zeros [2, Corollary 2.7]. Each point of the original strict domain is interior to that domain, so it cannot become an isolated zero merely by intersecting a nonisolated germ with those open restrictions. This excludes the sequence of isolated solutions when its enlarged tuple has a finite limit.

Divergent enlarged tuples. Suppose instead that at least one enlarged coordinate diverges. Split all coordinates of the enlarged sequence into bounded convergent coordinates and signed large coordinates. Apply the affine flag construction of Section 2 to the latter, keeping bounded remainders as ordinary coordinates. Convert the initial basis scales, including scales used in polynomial factors, to dependent logarithmic nodes, and insert the further logarithms required by the primitive constructions. Saturate the resulting finite flag. These real coordinate changes are locally invertible. The kernel rates are affine in the primary clock coordinates and bounded remainders. Polynomial powers of the initial scales are now shift monomials times bounded analytic factors. Under an inserted logarithm, old independent derivatives become finite sums of packet independent derivatives multiplied by constants, signs, and reciprocal powers of the logged dependent scale. These are precisely operations of Theorem 4.4. If only a multiplier or an algebraic auxiliary coordinate diverges while a kernel's clocks remain bounded, that kernel keeps its ordinary analytic normalized formula. It is a packet constant in the extra free inputs before pullback; the new polynomial coordinate factors are treated by the preceding logarithmic conversion. In particular a large auxiliary ratio at bounded primary clocks does not invoke an unbalanced mixed theorem that requires both clocks large.

Theorem 10.1, with the differentiated packet theorems, supplies the packet data for this finite derivative request on independent flag inputs. Graph ties remain equations; they are not assumed to hold when taking independent tests of a primitive kernel. In particular a bounded coefficient copy and a clock ratio remain separate arguments until their graph identity is used by the chain rule. The layer-vanishing residual extensions of Lemma 10.5 are essential here. For each nonzero equation, iterated left-finiteness permits a leading shift normalization making both sign trees nonnegative in the series sense. A formally zero equation is handled by separation. All normalizing monomials are nonzero on the real domain and use the current flag backgrounds. The finitely many real normalized equations can then be combined by a sum of squares, exactly as in Theorem 5.7; the resulting positive operations are covered by Theorem 4.4.

Retesting and return to the original solutions. We verify the retestability of this assembled library in the precise sense of Definition 5.6. For the primitive transfers this is the coefficient-germ assertion of Theorems 6.1, 7.1, and 8.1, carried through the exact terminal identities by Theorem 10.1. The affine and logarithmic flag identities, fixed analytic fields, sector signs, and ordinary germs are unchanged by sufficiently small absolute perturbations preserving the selected comparisons. Terminal coefficients are the fixed formal recursions and, where an infinite anchor is used, its fixed normalized jets. Cutoff and finite-anchor choices have already dispersed before these terminal leaves are taken. Subsequent algebra and finite derivatives preserve those same defining germ identities. The implicit inversion of Theorem 4.4 is used only when the limiting exponent-zero Jacobian is invertible. A root, logarithmic, or ratio graph that is regular at every selected point but degenerates

at the limit is instead retained as an equation in its own coordinates; it is not inverted by the packet implicit theorem. Its finitely many old-coordinate derivatives are expressed by Cramer's rule and their nonzero denominators are cleared, leaving finite algebraic expressions in independent jets of the defining equations. At large scales these equations use the already admitted affine shifts and bounded analytic fields. The ordinary elimination theorem then treats them together with all the other equations, including any multiple limiting zero. Thus only actual unit-Jacobian ordinary chart operations invoke the implicit part of Theorem 4.4; no packet inverse at a degenerate limiting graph is assumed. Further flag refinements leave the old free inputs free. At least one such input is recoverable from the tuple before the flag changes whenever a large scale remains. The sequence-wise graph enlargement retained all its defining equations, so Lemma 11.7 makes it recoverable for the original isolation test as well. More explicitly, at each anchor choose the free variation inside that anchor's graph-uniqueness neighborhood, the real flag chart, and the pullback of the proposed original isolation neighborhood. Such variations may be chosen arbitrarily small, with no common radius required along the sequence. All the hypotheses of Theorem 5.7 therefore hold. It excludes the isolated sequence in that case; the small free variations used there can be made inside every strict open condition at the selected point.

The two alternatives prove the absolute zero property for the arbitrary finite system under consideration. All additional determinant margins and multiplier systems use the same finite operations, so the conclusion holds for them as well.

Together with the construction and geometric arguments above, this proves all three assertions of Theorem 11.4, including the assertion about maximal-minor charts. $\square$

11.4 Rotation and the bound in the whole plane

Lemma 11.8 (Hyperbolic realization of a finite cycle count). *For any finite collection of limit cycles of a planar polynomial field $V = (P, Q)$, a sufficiently small perturbation $V_\mu = V + \mu(-Q, P)$, of one common nonzero sign, has at least as many distinct hyperbolic periodic orbits in the union of disjoint neighborhoods of that collection. The degree bound is preserved.*

Proof. Let $z(t)$ parametrize one selected cycle with period T , and start the perturbed solution at the same section point. Set $u(t) = \partial_\mu z_\mu(t)|_{\mu=0}$ and $J(a, b) = (-b, a)$. Then

$$u' = DV(z(t))u + JV(z(t)), \quad u(0) = 0.$$

For $w(t) = \det(V(z(t)), u(t))$, the identity $\det(Aa, b) + \det(a, Ab) = \text{tr}(A) \det(a, b)$ gives

$$w' = (\text{div } V)(z(t))w + |V(z(t))|^2, \quad w(T) = \int_0^T e^{\int_s^T (\text{div } V)(z(\tau)) d\tau} |V(z(s))|^2 ds > 0. \quad (11.8)$$

The return-time correction adds a multiple of V to the terminal variation and does not change this determinant. Thus the local analytic Poincaré displacement $D(r, \mu)$ has $D_\mu(0, 0) \neq 0$ in a transverse section coordinate r centered at the cycle.

The analytic implicit function theorem writes its zero set as $\mu = \psi(r)$ near $(0, 0)$, with $\psi(0) = 0$. Isolation of the cycle makes $D(r, 0)$ a nonzero analytic germ, so

$$\psi(r) = ar^m + O(r^{m+1}), \quad a \neq 0, \quad m \geq 1.$$

For odd m , both sufficiently small signs of μ give one nearby nonzero root; for even m , the sign of a gives two. All these roots are simple because $\psi'(r) = mar^{m-1} + O(r^m) \neq 0$ for sufficiently small $r \neq 0$. Differentiating $D(r, \psi(r)) = 0$ then gives $D_r = -D_\mu \psi' \neq 0$ at these roots. A planar return

map has positive derivative (as follows from its scalar variational equation), so a simple fixed point has multiplier different from 1 and is hyperbolic.

Choose disjoint tubular neighborhoods of the finitely many original cycles and one common sufficiently small magnitude of μ . If O cycles have odd order and E_+ and E_- have even order favoring the two signs, the resulting numbers are at least $O + 2E_+$ and $O + 2E_-$. Their maximum is at least $O + E_+ + E_-$, the original selection size. The count is taken over the union: an even-order cycle favoring the other sign need not persist for the chosen rotation. The new cycles are distinct because their neighborhoods are disjoint (and the two roots in one sufficiently small return section give different periodic orbits). Finally V_μ is obtained by constant linear combinations of P, Q , and still has degree at most d . $\square$

Proof of Theorem 1.1. Fix d . For every word w of Theorem 11.4 and every one of its maximal-minor charts I , apply Theorem 11.2. Let $N_{w,I} < \infty$ be the resulting component bound. It is independent of the numerical value of every polynomial coefficient and preparation parameter. The component injection gives, in the fixed square, the uniform bound

$$B_*(d) = T(d) + \sum_{w \in \mathcal{W}_d} \sum_I N_{w,I} \quad (11.9)$$

for hyperbolic periodic orbits. An orbit represented in more than one chart is assigned to any one of them; overcounting in this sum is harmless. Finiteness of the sums is geometric template finiteness, and does not involve the internal K used in a sequence-dependent absolute zero test.

Now select any finite collection of limit cycles of any field of degree at most d in the whole plane. A spatial affine scaling $z = a + Ru$, $R > 0$, puts all of them strictly inside the fixed square. The transformed field is $\dot{u} = R^{-1}V(a + Ru)$, again polynomial of degree at most d ; isolation and geometric distinctness are preserved. Apply Lemma 11.8 there. The perturbation can be chosen small enough that its retained hyperbolic cycles remain in the square. Their number is at least the size of the selected collection and is at most $B_*(d)$ by (11.9). Hence every finite collection has size at most $B_*(d)$. If the original field had more limit cycles, one could select $B_*(d) + 1$ of them, a contradiction. Taking any integer upper bound for $B_*(d)$ defines the required finite $B(d)$, without an assumption of prior individual finiteness. $\square$

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